SM-A546B
Service Schematic

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A54 5G EUR_REV04C

2022.12

Sheet01 :	GSM / WCDMA / LTE / NR(N41) RF
Sheet02 :	TRANSCEIVER, SUB ANT
Sheet03 :	CONNECTIVITY(BT/WIFI/FM), NFC
Sheet04 :	AP(Quartz-INTERFACE,GPIO)
Sheet05 :	AP(POWER), MEM(uMCP)
Sheet06 :	PMIC/SUB PMIC, IF PMIC, DC IC
Sheet07 :	AUDIO AMP, SENSOR, SIM, SD
Sheet08 :	CAMERA
Sheet09 :	DISPLAY PMIC, C2C

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:
TITLE: A54
Size:
Changed by:
Date Changed:
Time Changed:
REV:
Drawing Number:
Page:

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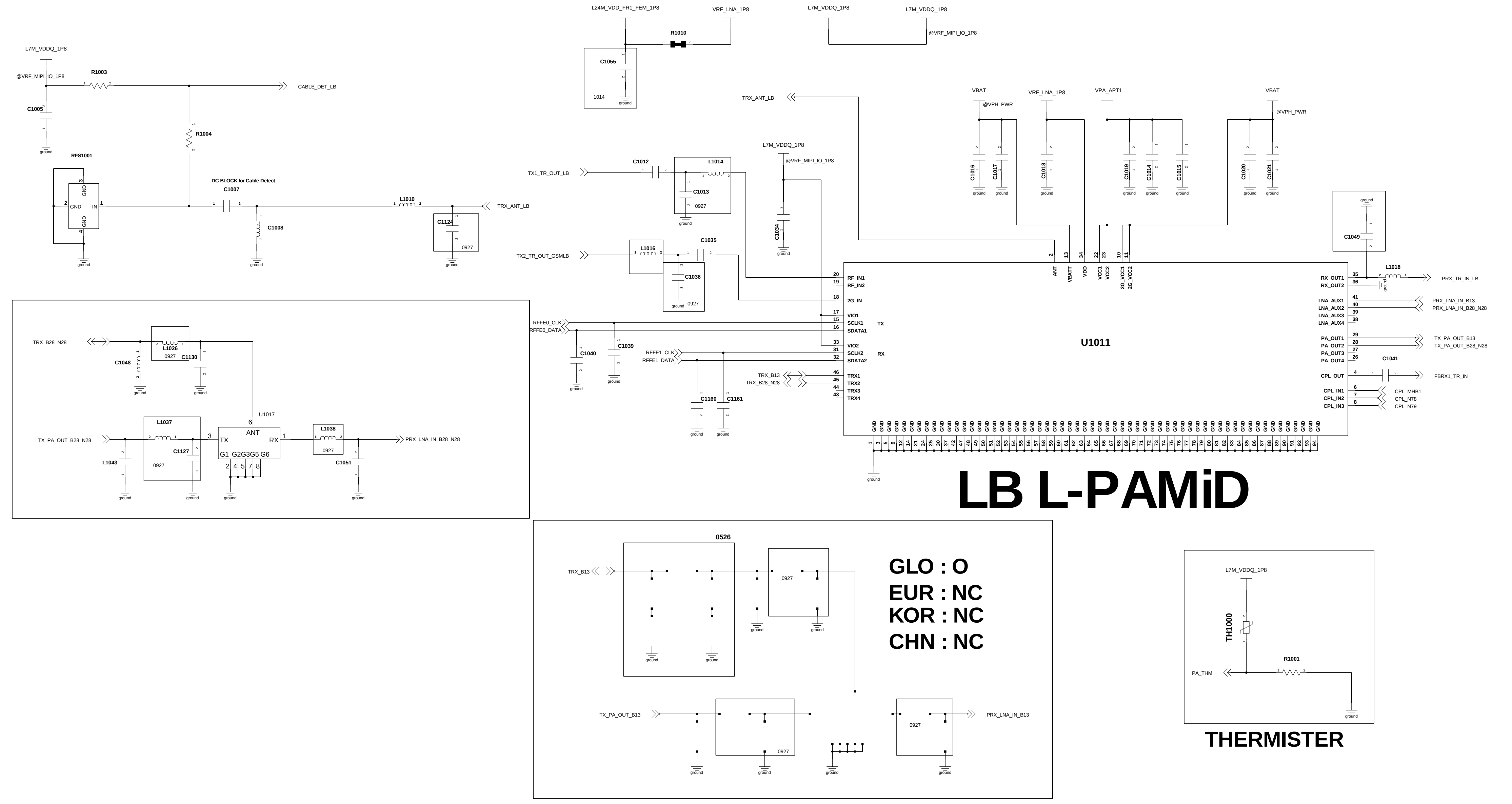
RF PART 1

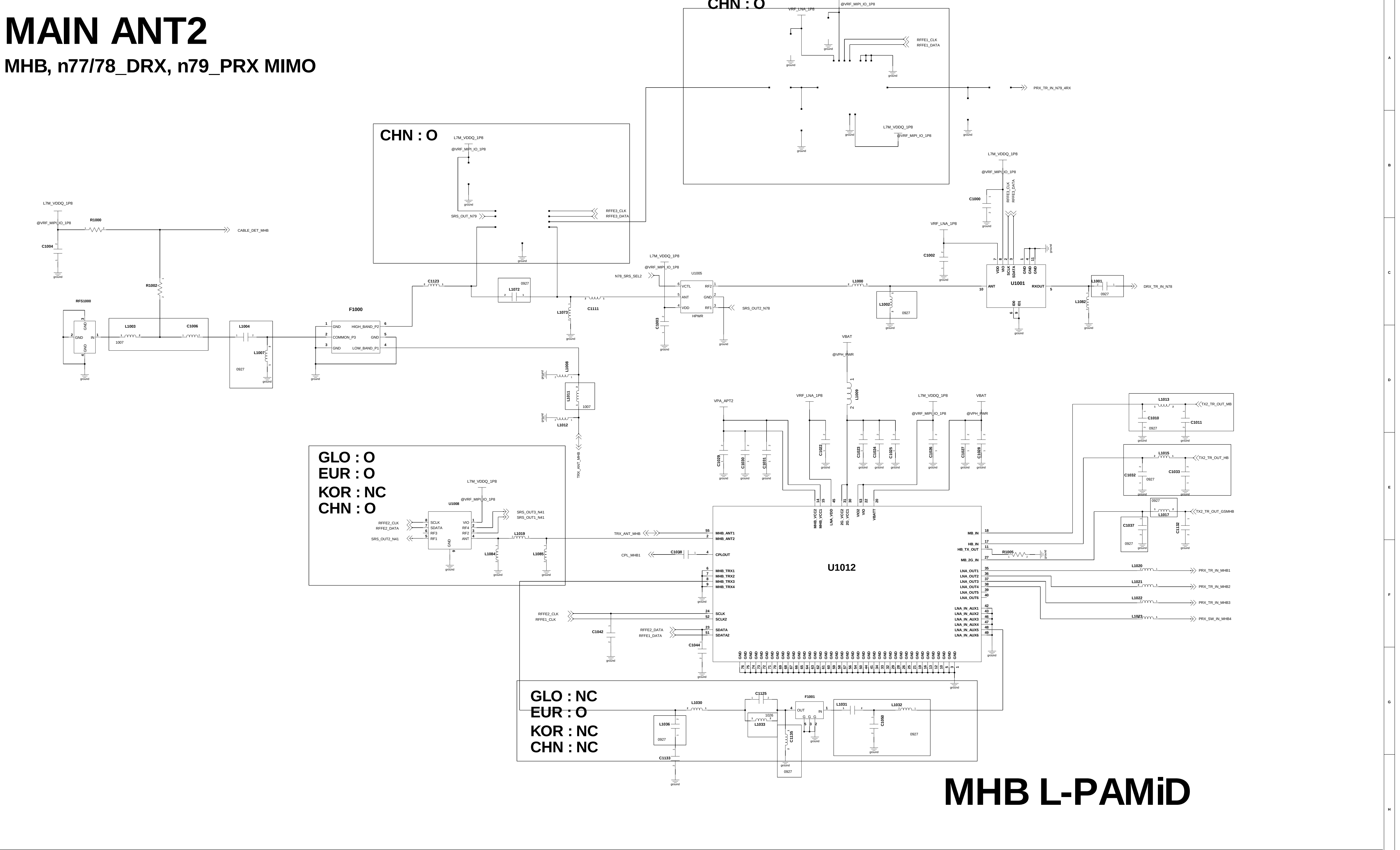
MAIN ANT1

LB


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1003
CABLE_DET_LB
C1005
R1004
RFS1001
DC BLOCK for Cable Detect
C1007
C1008
L1010
C1124
TRX_ANT_LB
L24M_VDD_FR1_FEM_1P8
VRF_LNA_1P8
R1010
C1055
1014
TX1_TR_OUT_LB
C1012
L1014
C1013
TX2_TR_OUT_GSMLB
L1016
C1035
C1036
C1034
RFFE0_CLK
RFFE0_DATA
C1040
C1039
RFFE1_CLK
RFFE1_DATA
TRX_B13
TRX_B28_N28
C1160
C1161
VBAT
@VPH_PWR
VPA_APT1
C1016
C1017
C1018
C1019
C1014
C1015
C1020
C1021
C1049
U1011
RF_IN1
RF_IN2
2G_IN
VIO1
SCLK1
SDATA1
TX
VIO2
SCLK2
SDATA2
RX
TRX1
TRX2
TRX3
TRX4
ANT
VBATT
VDD
VCC1
VCC2
2G_VCC1
2G_VCC2
RX_OUT1
RX_OUT2
L1018
PRX_TR_IN_LB
LNA_AUX1
LNA_AUX2
LNA_AUX3
LNA_AUX4
PRX_LNA_IN_B13
PRX_LNA_IN_B28_N28
PA_OUT1
PA_OUT2
PA_OUT3
PA_OUT4
TX_PA_OUT_B13
TX_PA_OUT_B28_N28
CPL_OUT
C1041
FBRX1_TR_IN
CPL_IN1
CPL_IN2
CPL_IN3
CPL_MHB1
CPL_N78
CPL_N79
GND
LB L-PAMiD
TRX_B28_N28
L1026
C1130
C1048
U1017
TX_PA_OUT_B28_N28
L1037
C1127
L1043
TX
ANT
RX
G1 G2G3G5 G6
L1038
C1051
PRX_LNA_IN_B28_N28
0526
TRX_B13
TX_PA_OUT_B13
PRX_LNA_IN_B13
GLO : O
EUR : NC
KOR : NC
CHN : NC
L7M_VDDQ_1P8
TH1000
R1001
PA_THM
THERMISTER
0927
ground

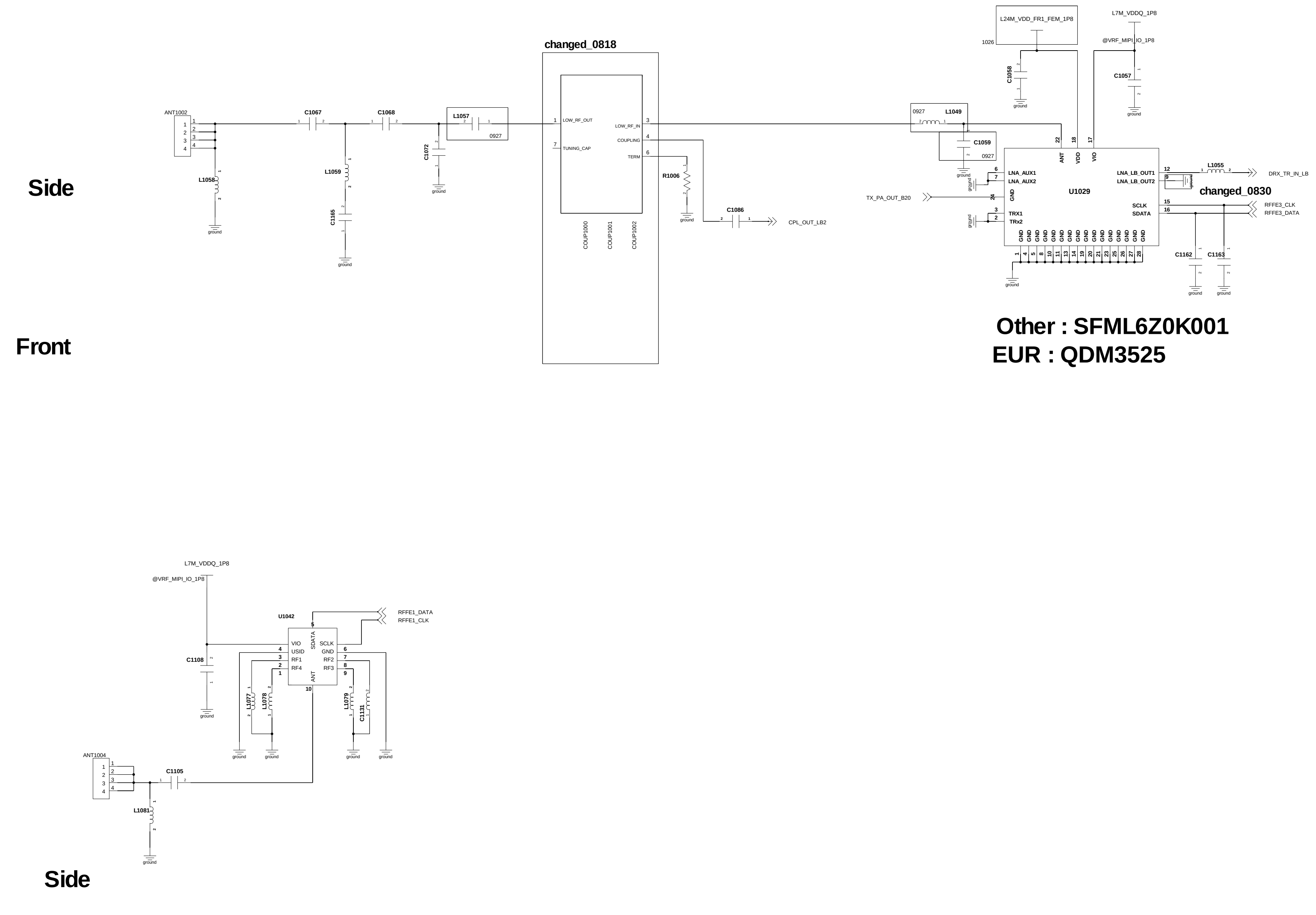
MAIN ANT2

MHB, n77/78_DRX, n79_PRX MIMO


CHN : O
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
RFFE1_CLK
RFFE1_DATA
PRX_TR_IN_N79_4RX
CHN : O
SRS_OUT_N79
RFFE3_CLK
RFFE3_DATA
R1000
CABLE_DET_MHB
C1004
R1002
RFS1000
L1003
C1006
1007
L1004
L1007
F1000
GND
HIGH_BAND_P2
COMMON_P3
LOW_BAND_P1
C1123
L1072
L1073
C1111
U1005
N78_SRS_SEL2
VCTL
ANT
VDD
RF2
RF1
HPWR
SRS_OUT2_N78
C1003
L1000
L1002
C1000
C1002
RFFE3_CLK
RFFE3_DATA
U1001
RXOUT
ID0
ID1
L1001
L1082
DRX_TR_IN_N78
L1008
L1011
L1012
TRX_ANT_MHB
GLO : O
EUR : O
KOR : NC
CHN : O
U1008
SCLK
SDATA
RF3
RF4
VIO
RFFE2_CLK
RFFE2_DATA
SRS_OUT2_N41
SRS_OUT3_N41
SRS_OUT1_N41
L1019
L1084
L1085
VBAT
@VPH_PWR
L1009
VPA_APT2
C1029
C1030
C1031
C1022
C1023
C1024
C1025
C1026
C1027
C1028
U1012
MHB_ANT1
MHB_ANT2
CPLOUT
C1038
CPL_MHB1
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
SCLK2
SDATA2
C1042
C1044
MHB_VCC2
MHB_VCC1
LNA_VDD
2G_VCC2
2G_VCC1
VIO2
VBATT
MB_IN
HB_IN
HB_TX_OUT
R1005
MB_2G_IN
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
L1020
L1021
L1022
L1023
PRX_TR_IN_MHB1
PRX_TR_IN_MHB2
PRX_TR_IN_MHB3
PRX_SW_IN_MHB4
L1013
C1010
C1011
TX2_TR_OUT_MB
L1015
C1032
C1033
TX2_TR_OUT_HB
L1017
C1037
C1132
TX2_TR_OUT_GSMHB
GLO : NC
EUR : O
KOR : NC
CHN : NC
L1030
L1036
C1133
C1125
1026
L1033
C1135
F1001
OUT
IN
L1031
L1032
C1050
MHB L-PAMiD
0927
ground

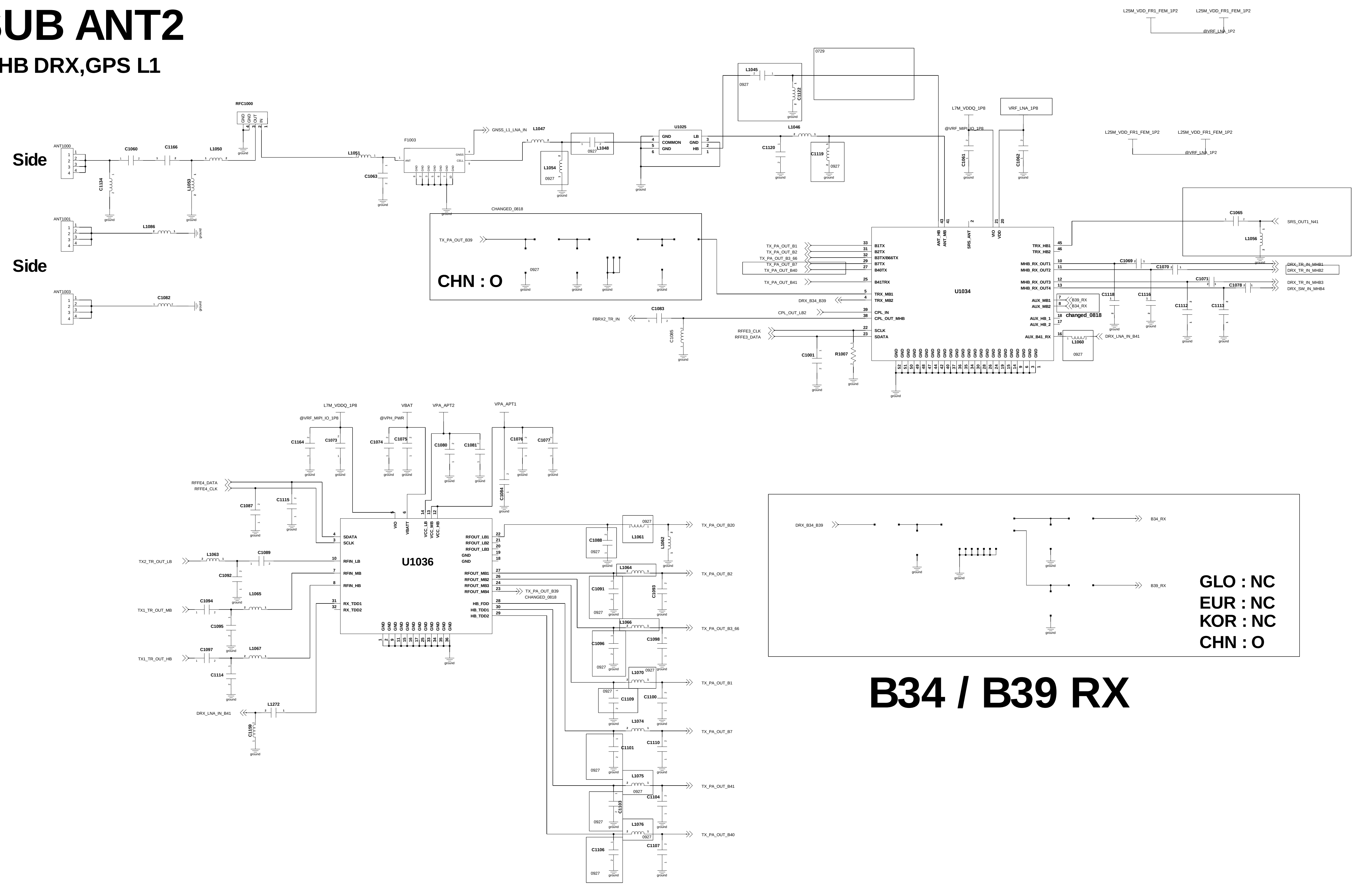
SUB ANT1

LB DRX


changed_0818
ANT1002
C1067
C1068
L1057
C1072
L1058
L1059
C1165
LOW_RF_OUT
TUNING_CAP
LOW_RF_IN
COUPLING
TERM
COUP1000
COUP1001
COUP1002
R1006
C1086
CPL_OUT_LB2
L24M_VDD_FR1_FEM_1P8
1026
C1058
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1057
L1049
C1059
U1029
ANT
VDD
VIO
LNA_AUX1
LNA_AUX2
TX_PA_OUT_B20
TRX1
TRx2
LNA_LB_OUT1
LNA_LB_OUT2
L1055
DRX_TR_IN_LB
changed_0830
SCLK
SDATA
RFFE3_CLK
RFFE3_DATA
C1162
C1163
Other : SFML6Z0K001
EUR : QDM3525
Side
Front
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
U1042
VIO
USID
RF1
RF4
SDATA
SCLK
GND
RF2
RF3
RFFE1_DATA
RFFE1_CLK
C1108
L1077
L1078
L1079
C1131
ANT1004
C1105
L1081
Side
0927
ground

SUB ANT2

MHB DRX,GPS L1


L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
RFC1000
GND
OUT
IN
Side
ANT1000
C1060
C1166
L1050
C1134
L1053
ANT1001
L1086
Side
ANT1003
C1082
L1051
C1063
F1003
GNSS
CELL
GNSS_L1_LNA_IN
L1047
L1048
L1054
U1025
COMMON
LB
HB
0729
L1045
C1122
L1046
C1120
C1119
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
C1061
C1062
CHANGED_0818
TX_PA_OUT_B39
CHN : O
FBRX2_TR_IN
C1083
C1085
U1034
TX_PA_OUT_B1
TX_PA_OUT_B2
TX_PA_OUT_B3_66
TX_PA_OUT_B7
TX_PA_OUT_B40
TX_PA_OUT_B41
B1TX
B2TX
B3TX/B66TX
B7TX
B40TX
B41TRX
TRX_MB1
TRX_MB2
DRX_B34_B39
CPL_OUT_LB2
CPL_IN
CPL_OUT_MHB
RFFE3_CLK
RFFE3_DATA
SCLK
SDATA
C1001
R1007
ANT_HB
ANT_MB
SRS_ANT
VIO
VDD
TRX_HB1
TRX_HB2
MHB_RX_OUT1
MHB_RX_OUT2
MHB_RX_OUT3
MHB_RX_OUT4
AUX_MB1
AUX_MB2
AUX_HB_1
AUX_HB_2
AUX_B41_RX
B39_RX
B34_RX
changed_0818
L1060
DRX_LNA_IN_B41
C1065
L1056
SRS_OUT1_N41
C1069
C1070
C1071
C1078
DRX_TR_IN_MHB1
DRX_TR_IN_MHB2
DRX_TR_IN_MHB3
DRX_SW_IN_MHB4
C1118
C1116
C1112
C1113
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VBAT
@VPH_PWR
VPA_APT2
VPA_APT1
C1164
C1073
C1074
C1075
C1080
C1081
C1076
C1077
C1084
RFFE4_DATA
RFFE4_CLK
C1087
C1115
U1036
SDATA
SCLK
RFIN_LB
RFIN_MB
RFIN_HB
RX_TDD1
RX_TDD2
VBATT
VCC_LB
VCC_MB
VCC_HB
RFOUT_LB1
RFOUT_LB2
RFOUT_LB3
RFOUT_MB1
RFOUT_MB2
RFOUT_MB3
RFOUT_MB4
HB_FDD
HB_TDD1
HB_TDD2
TX2_TR_OUT_LB
L1063
C1089
C1092
TX1_TR_OUT_MB
C1094
C1095
L1065
TX1_TR_OUT_HB
C1097
C1114
L1067
DRX_LNA_IN_B41
L1272
C1159
C1088
L1061
L1062
TX_PA_OUT_B20
L1064
C1091
C1093
TX_PA_OUT_B2
L1066
C1096
C1098
TX_PA_OUT_B3_66
L1070
C1109
C1100
TX_PA_OUT_B1
L1074
C1101
C1110
TX_PA_OUT_B7
L1075
C1103
C1104
TX_PA_OUT_B41
L1076
C1106
C1107
TX_PA_OUT_B40
DRX_B34_B39
B34_RX
B39_RX
GLO : NC
EUR : NC
KOR : NC
CHN : O
B34 / B39 RX
0927
ground

Engineer:	A54	
Drawn by:		
R&D CHK:	TITLE: RF PAM/FEM	Size:
DOC CTRL CHK:		
MFG ENGR CHK:		

Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:

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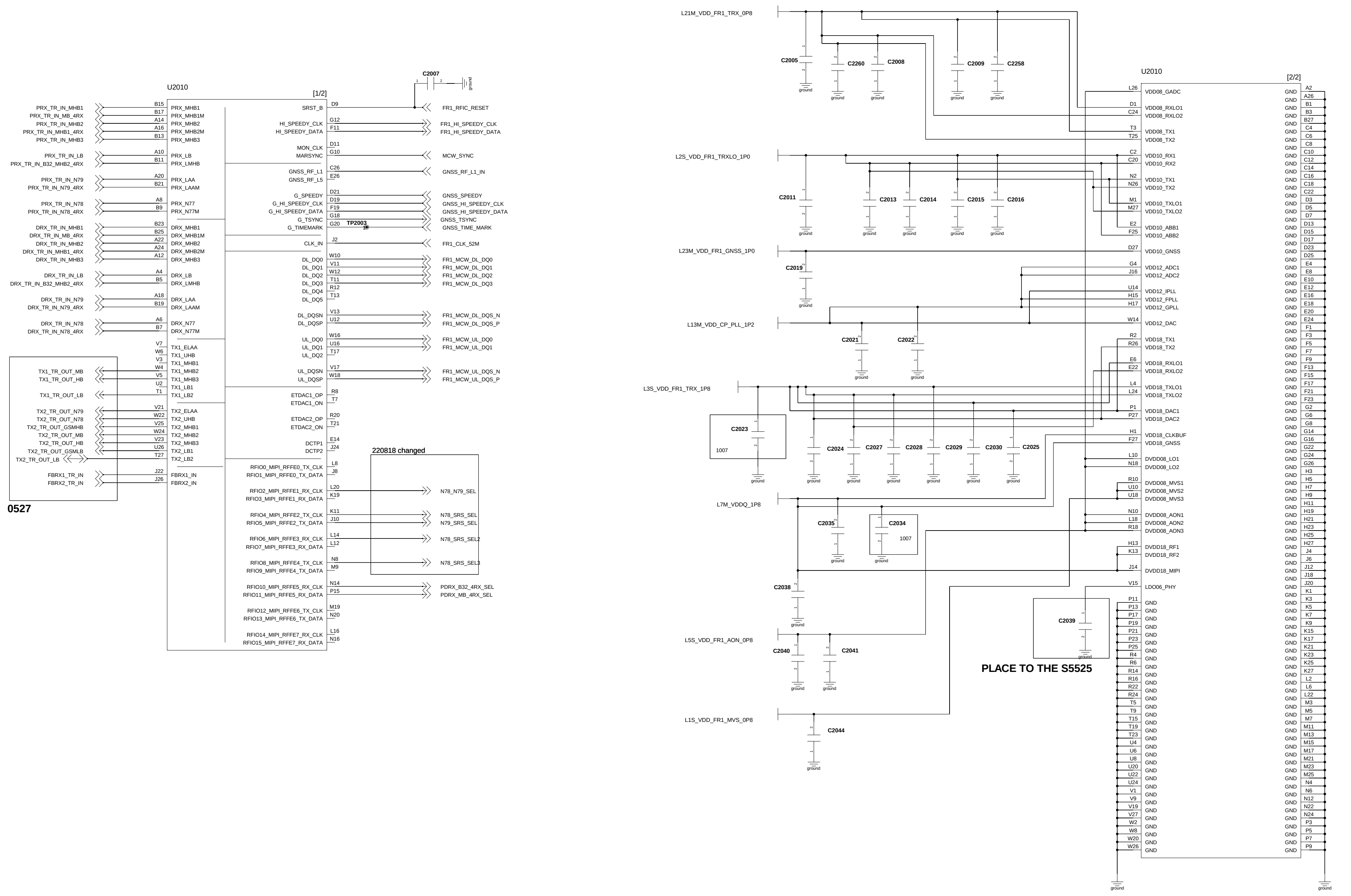
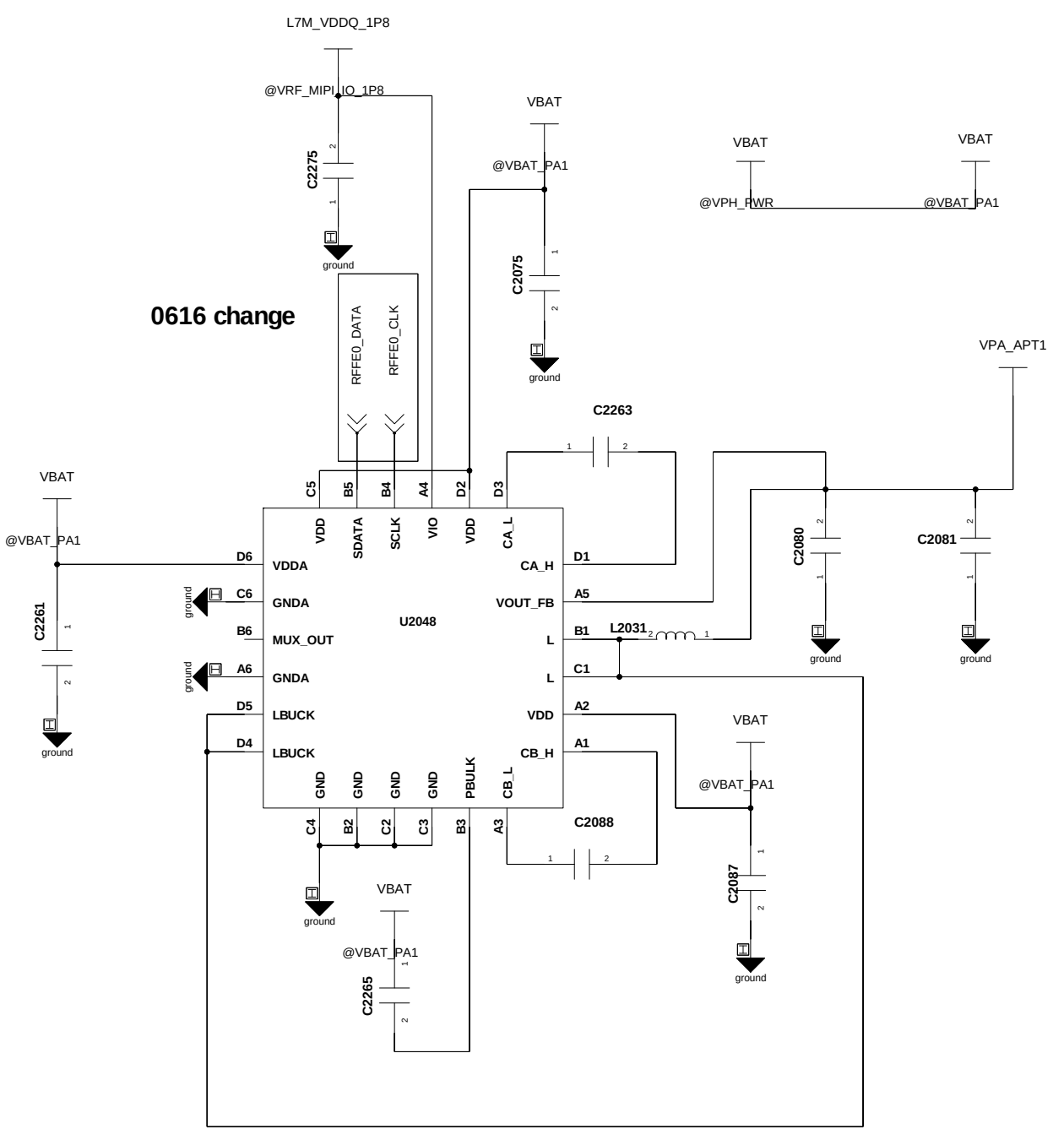
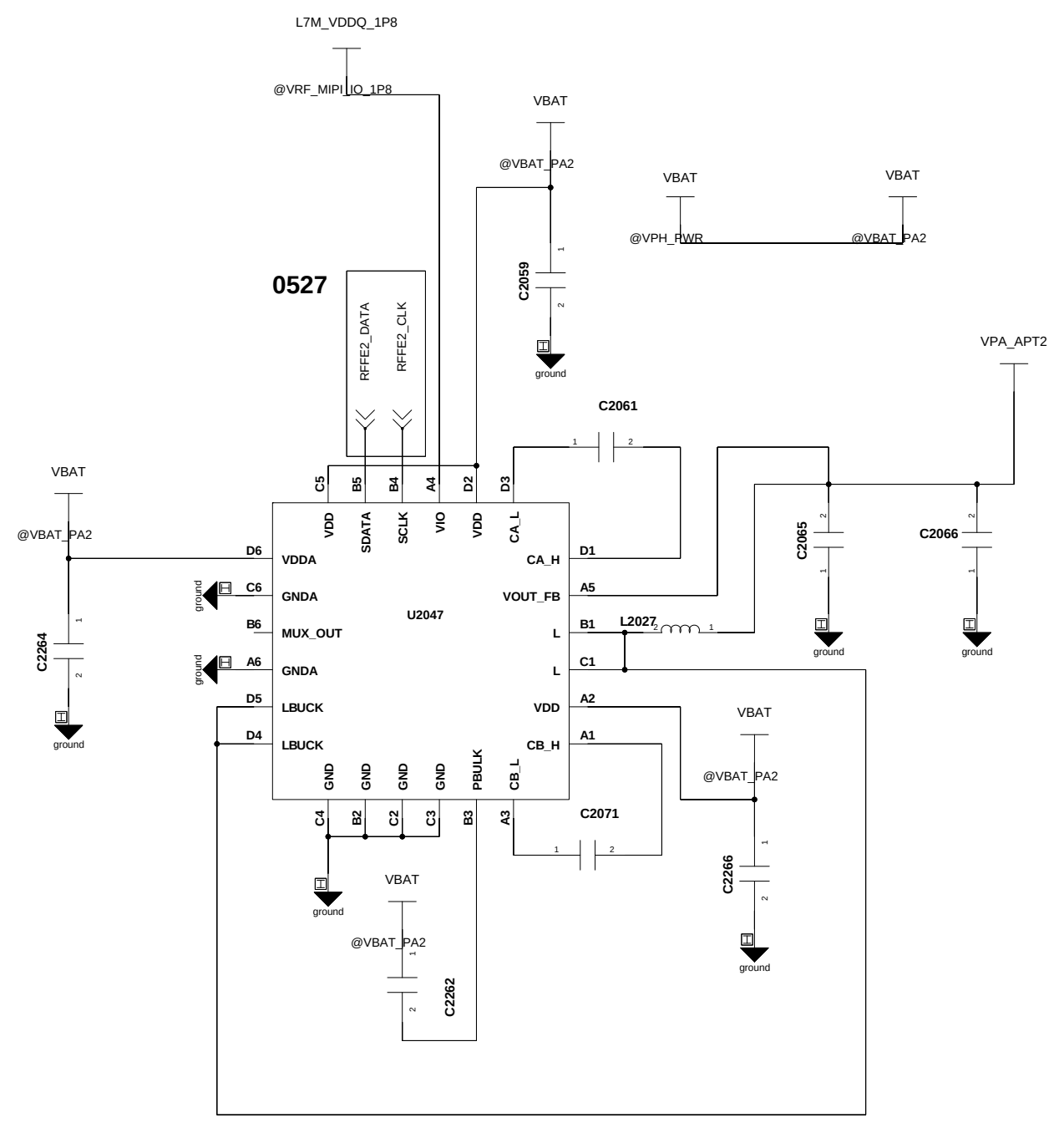
TRANSCEIVER (S5525)


U2010
[1/2]
PRX_TR_IN_MHB1
PRX_TR_IN_MB_4RX
PRX_TR_IN_MHB2
PRX_TR_IN_MHB1_4RX
PRX_TR_IN_MHB3
PRX_TR_IN_LB
PRX_TR_IN_B32_MHB2_4RX
PRX_TR_IN_N79
PRX_TR_IN_N79_4RX
PRX_TR_IN_N78
PRX_TR_IN_N78_4RX
DRX_TR_IN_MHB1
DRX_TR_IN_MB_4RX
DRX_TR_IN_MHB2
DRX_TR_IN_MHB1_4RX
DRX_TR_IN_MHB3
DRX_TR_IN_LB
DRX_TR_IN_B32_MHB2_4RX
DRX_TR_IN_N79
DRX_TR_IN_N79_4RX
DRX_TR_IN_N78
DRX_TR_IN_N78_4RX
TX1_TR_OUT_MB
TX1_TR_OUT_HB
TX1_TR_OUT_LB
TX2_TR_OUT_N79
TX2_TR_OUT_N78
TX2_TR_OUT_GSMHB
TX2_TR_OUT_MB
TX2_TR_OUT_HB
TX2_TR_OUT_GSMLB
TX2_TR_OUT_LB
FBRX1_TR_IN
FBRX2_TR_IN
0527
SRST_B
HI_SPEEDY_CLK
HI_SPEEDY_DATA
MON_CLK
MARSYNC
GNSS_RF_L1
GNSS_RF_L5
G_SPEEDY
G_HI_SPEEDY_CLK
G_HI_SPEEDY_DATA
G_TSYNC
G_TIMEMARK
TP2003
CLK_IN
DL_DQ0
DL_DQ1
DL_DQ2
DL_DQ3
DL_DQ4
DL_DQ5
DL_DQSN
DL_DQSP
UL_DQ0
UL_DQ1
UL_DQ2
UL_DQSN
UL_DQSP
ETDAC1_OP
ETDAC1_ON
ETDAC2_OP
ETDAC2_ON
DCTP1
DCTP2
RFIO0_MIPI_RFFE0_TX_CLK
RFIO1_MIPI_RFFE0_TX_DATA
RFIO2_MIPI_RFFE1_RX_CLK
RFIO3_MIPI_RFFE1_RX_DATA
RFIO4_MIPI_RFFE2_TX_CLK
RFIO5_MIPI_RFFE2_TX_DATA
RFIO6_MIPI_RFFE3_RX_CLK
RFIO7_MIPI_RFFE3_RX_DATA
RFIO8_MIPI_RFFE4_TX_CLK
RFIO9_MIPI_RFFE4_TX_DATA
RFIO10_MIPI_RFFE5_RX_CLK
RFIO11_MIPI_RFFE5_RX_DATA
RFIO12_MIPI_RFFE6_TX_CLK
RFIO13_MIPI_RFFE6_TX_DATA
RFIO14_MIPI_RFFE7_RX_CLK
RFIO15_MIPI_RFFE7_RX_DATA
C2007
FR1_RFIC_RESET
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
MCW_SYNC
GNSS_RF_L1_IN
GNSS_SPEEDY
GNSS_HI_SPEEDY_CLK
GNSS_HI_SPEEDY_DATA
GNSS_TSYNC
GNSS_TIME_MARK
FR1_CLK_52M
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
220818 changed
N78_N79_SEL
N78_SRS_SEL
N79_SRS_SEL
N78_SRS_SEL2
N78_SRS_SEL3
PDRX_B32_4RX_SEL
PDRX_MB_4RX_SEL
L21M_VDD_FR1_TRX_0P8
C2005
C2260
C2008
C2009
C2258
L2S_VDD_FR1_TRXLO_1P0
C2011
C2013
C2014
C2015
C2016
L23M_VDD_FR1_GNSS_1P0
C2019
L13M_VDD_CP_PLL_1P2
C2021
C2022
L3S_VDD_FR1_TRX_1P8
C2023
1007
C2024
C2027
C2028
C2029
C2030
C2025
L7M_VDDQ_1P8
C2035
C2034
C2038
L5S_VDD_FR1_AON_0P8
C2040
C2041
L1S_VDD_FR1_MVS_0P8
C2044
C2039
PLACE TO THE S5525
U2010
[2/2]
VDD08_GADC
VDD08_RXLO1
VDD08_RXLO2
VDD08_TX1
VDD08_TX2
VDD10_RX1
VDD10_RX2
VDD10_TX1
VDD10_TX2
VDD10_TXLO1
VDD10_TXLO2
VDD10_ABB1
VDD10_ABB2
VDD10_GNSS
VDD12_ADC1
VDD12_ADC2
VDD12_IPLL
VDD12_FPLL
VDD12_GPLL
VDD12_DAC
VDD18_TX1
VDD18_TX2
VDD18_RXLO1
VDD18_RXLO2
VDD18_TXLO1
VDD18_TXLO2
VDD18_DAC1
VDD18_DAC2
VDD18_CLKBUF
VDD18_GNSS
DVDD08_LO1
DVDD08_LO2
DVDD08_MVS1
DVDD08_MVS2
DVDD08_MVS3
DVDD08_AON1
DVDD08_AON2
DVDD08_AON3
DVDD18_RF1
DVDD18_RF2
DVDD18_MIPI
LDO06_PHY
GND
ground

APT1


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2275
0616 change
RFFE0_DATA
RFFE0_CLK
VBAT
@VBAT_PA1
C2075
@VPH_PWR
VPA_APT1
C2263
C2080
C2081
C2261
U2048
VDDA
GNDA
MUX_OUT
LBUCK
VDD
SDATA
SCLK
VIO
CA_L
CA_H
VOUT_FB
L
CB_L
CB_H
PBULK
L2031
C2088
C2087
C2265
ground

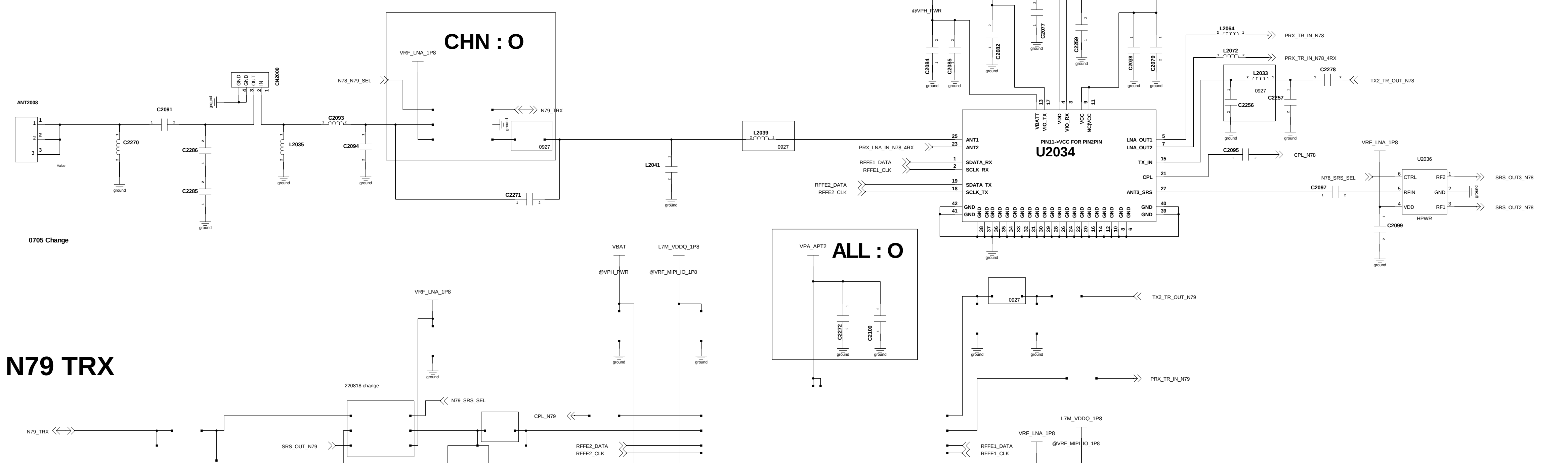
APT2


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
0527
RFFE2_DATA
RFFE2_CLK
VBAT
@VBAT_PA2
C2059
@VPH_PWR
VPA_APT2
C2061
C2065
C2066
C2264
U2047
VDDA
GNDA
MUX_OUT
LBUCK
CA_H
VOUT_FB
L
VDD
CB_H
PBULK
L2027
C2071
C2266
C2262
ground

SUB ANT3

N78 TRX

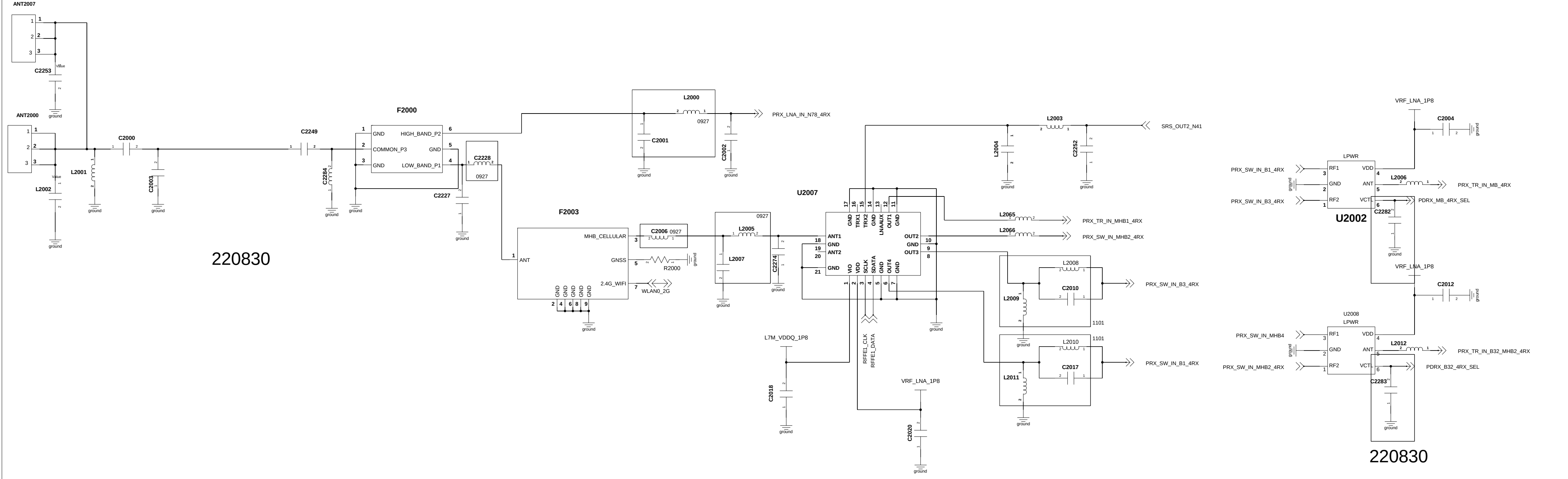
N79 TRX


CHN : O
ANT2008
C2091
C2270
C2286
C2285
CN2000
C2093
L2035
C2094
N78_N79_SEL
VRF_LNA_1P8
N79_TRX
0927
C2271
L2041
L2039
PRX_LNA_IN_N78_4RX
RFFE1_DATA
RFFE1_CLK
RFFE2_DATA
RFFE2_CLK
U2034
PIN11->VCC FOR PIN2PIN
ANT1
ANT2
SDATA_RX
SCLK_RX
SDATA_TX
SCLK_TX
VBATT
VIO_TX
VDD
VIO_RX
VCC
NC/VCC
LNA_OUT1
LNA_OUT2
TX_IN
CPL
ANT3_SRS
GND
VBAT
@VPH_PWR
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
VPA_APT2
C2084
C2085
C2082
C2077
C2259
C2078
C2079
L2064
PRX_TR_IN_N78
L2072
PRX_TR_IN_N78_4RX
L2033
C2256
C2257
C2278
TX2_TR_OUT_N78
C2095
CPL_N78
C2097
N78_SRS_SEL
U2036
CTRL
RFIN
RF2
RF1
HPWR
C2099
SRS_OUT3_N78
SRS_OUT2_N78
0705 Change
ALL : O
C2272
C2100
TX2_TR_OUT_N79
PRX_TR_IN_N79
220818 change
N79_SRS_SEL
CPL_N79
SRS_OUT_N79
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

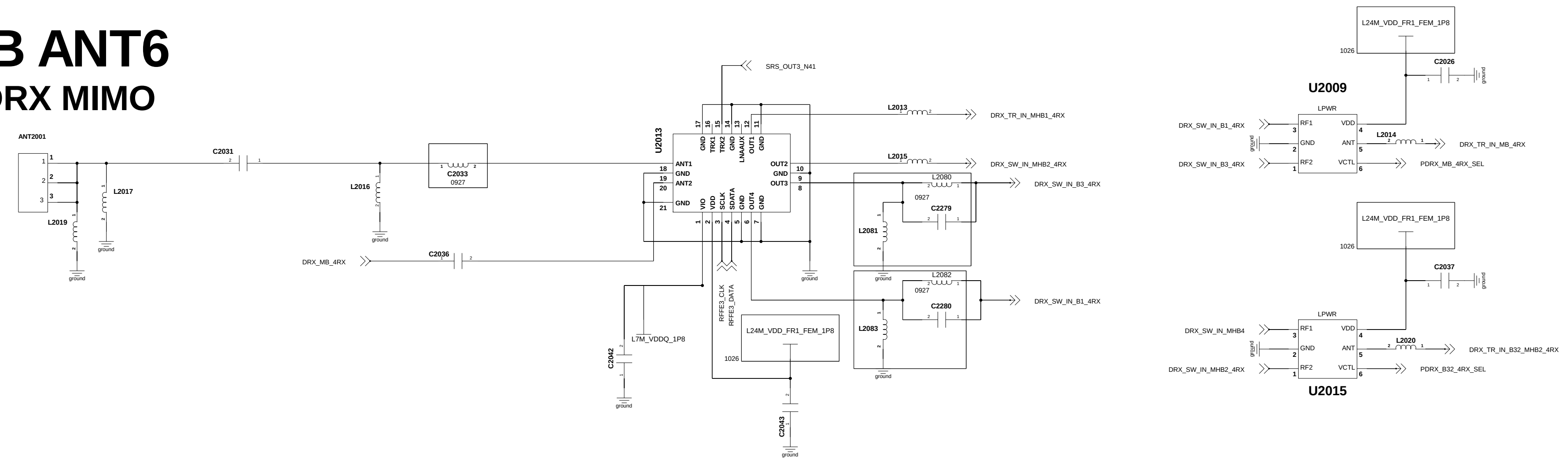
SUB ANT5

MHB PRX MIMO, n77/78 PRX MIMO, WIFI 2.4G


ANT2007
C2253
ANT2000
L2001
L2002
C2000
C2003
C2249
C2284
F2000
GND
COMMON_P3
HIGH_BAND_P2
LOW_BAND_P1
C2228
0927
C2227
L2000
C2001
C2002
PRX_LNA_IN_N78_4RX
220830
F2003
ANT
MHB_CELLULAR
GNSS
2.4G_WIFI
C2006
R2000
WLAN0_2G
L2005
L2007
C2274
U2007
ANT1
ANT2
TRX1
TRX2
LNAAUX
OUT1
OUT2
OUT3
OUT4
VIO
VDD
SCLK
SDATA
RFFE1_CLK
RFFE1_DATA
L7M_VDDQ_1P8
C2018
VRF_LNA_1P8
C2020
L2003
SRS_OUT2_N41
L2004
C2252
L2065
PRX_TR_IN_MHB1_4RX
L2066
PRX_SW_IN_MHB2_4RX
L2008
C2010
L2009
PRX_SW_IN_B3_4RX
1101
L2010
C2017
L2011
PRX_SW_IN_B1_4RX
VRF_LNA_1P8
C2004
LPWR
PRX_SW_IN_B1_4RX
PRX_SW_IN_B3_4RX
RF1
RF2
VDD
ANT
VCTL
L2006
PRX_TR_IN_MB_4RX
PDRX_MB_4RX_SEL
C2282
U2002
C2012
U2008
PRX_SW_IN_MHB4
PRX_SW_IN_MHB2_4RX
L2012
PRX_TR_IN_B32_MHB2_4RX
PDRX_B32_4RX_SEL
C2283
220830
ground

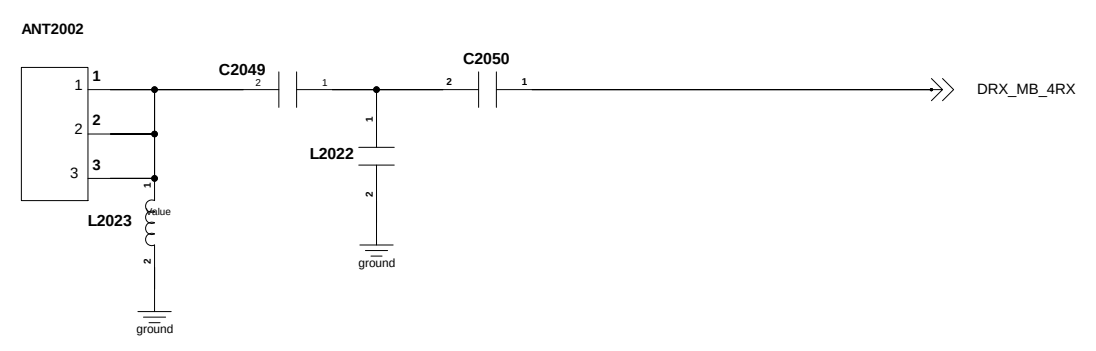
SUB ANT6

HB DRX MIMO


ANT2001
L2017
L2019
C2031
L2016
C2033
0927
DRX_MB_4RX
C2036
SRS_OUT3_N41
U2013
L2013
DRX_TR_IN_MHB1_4RX
L2015
DRX_SW_IN_MHB2_4RX
L2080
C2279
L2081
DRX_SW_IN_B3_4RX
L2082
C2280
L2083
DRX_SW_IN_B1_4RX
RFFE3_CLK
RFFE3_DATA
L7M_VDDQ_1P8
C2042
L24M_VDD_FR1_FEM_1P8
1026
C2043
U2009
LPWR
C2026
DRX_SW_IN_B1_4RX
DRX_SW_IN_B3_4RX
L2014
DRX_TR_IN_MB_4RX
PDRX_MB_4RX_SEL
C2037
U2015
DRX_SW_IN_MHB4
DRX_SW_IN_MHB2_4RX
L2020
DRX_TR_IN_B32_MHB2_4RX
PDRX_B32_4RX_SEL
ground

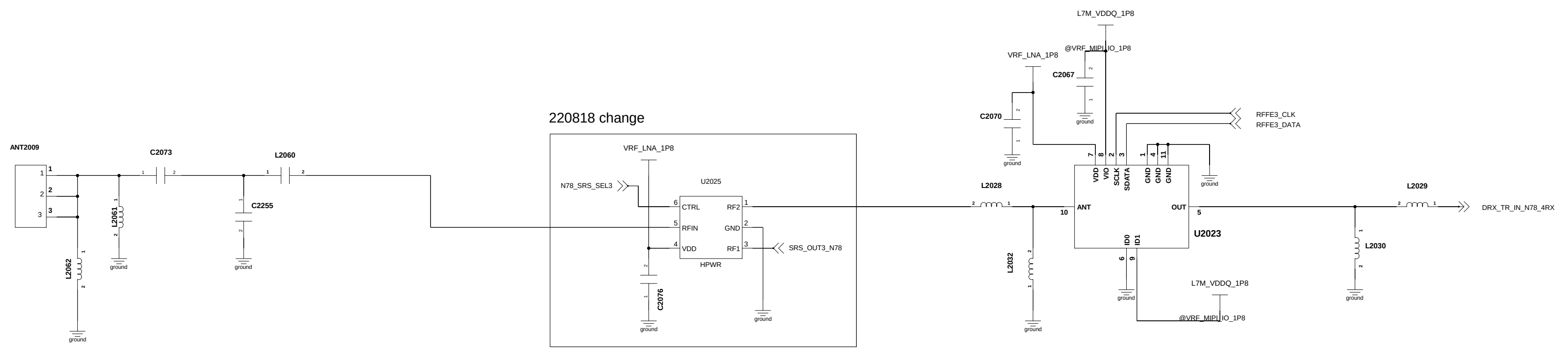
SUB ANT7

MB DRX MIMO


ANT2002
C2049
C2050
L2022
L2023
DRX_MB_4RX
ground

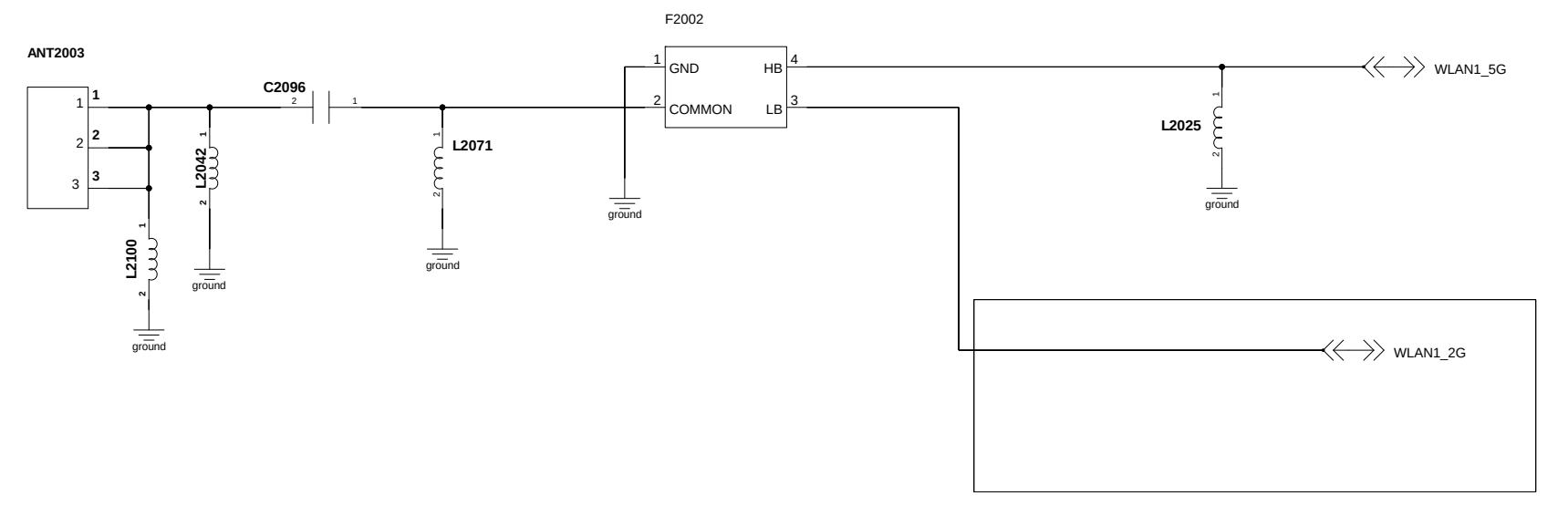
SUB ANT8

N78 DRX MIMO


ANT2009
C2073
L2060
L2061
L2062
C2255
220818 change
VRF_LNA_1P8
U2025
N78_SRS_SEL3
CTRL
RFIN
VDD
RF2
GND
RF1
HPWR
SRS_OUT3_N78
C2076
L2028
L2032
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2067
C2070
RFFE3_CLK
RFFE3_DATA
U2023
ANT
OUT
ID0
ID1
L2029
L2030
DRX_TR_IN_N78_4RX
ground

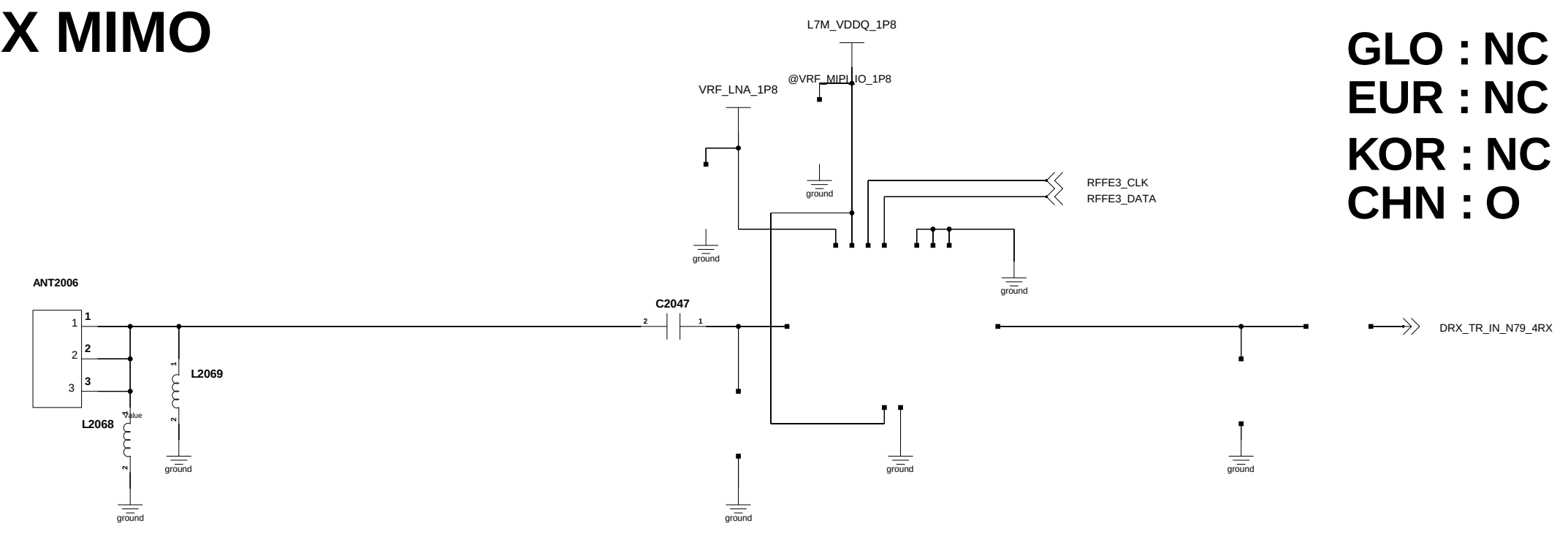
SUB ANT9

WIFI_2.4G+WIFI_5G


ANT2003
C2096
L2042
L2100
L2071
F2002
GND
COMMON
HB
LB
WLAN1_5G
L2025
WLAN1_2G
ground

SUB ANT11

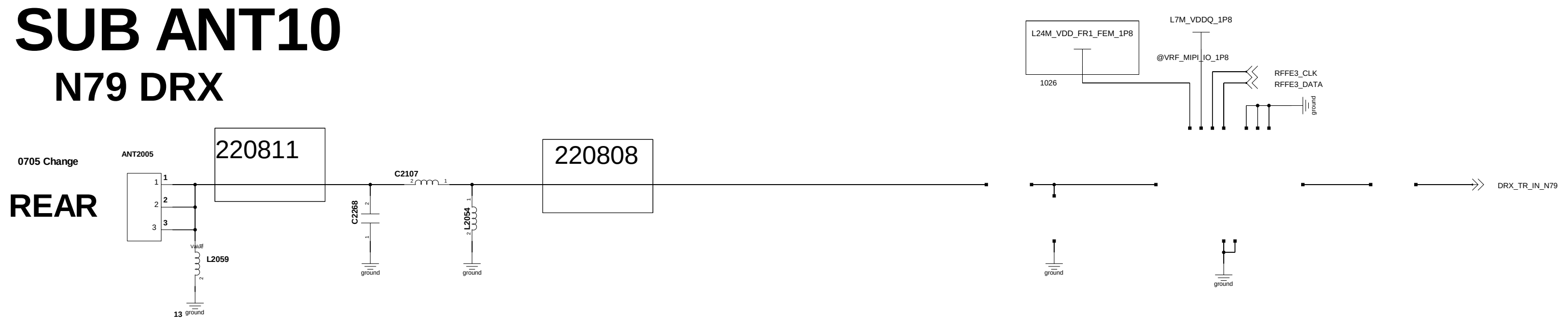
n79 DRX MIMO


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
RFFE3_CLK
RFFE3_DATA
ANT2006
L2069
L2068
C2047
DRX_TR_IN_N79_4RX
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

SUB ANT10

N79 DRX


0705 Change
REAR
ANT2005
220811
L2059
C2268
C2107
L2054
220808
L24M_VDD_FR1_FEM_1P8
1026
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
RFFE3_CLK
RFFE3_DATA
DRX_TR_IN_N79
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: RF TRCV/PAM/APT
DOC CTRL CHK:	
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number:

Changed by: Date Changed: Time Changed: Size:

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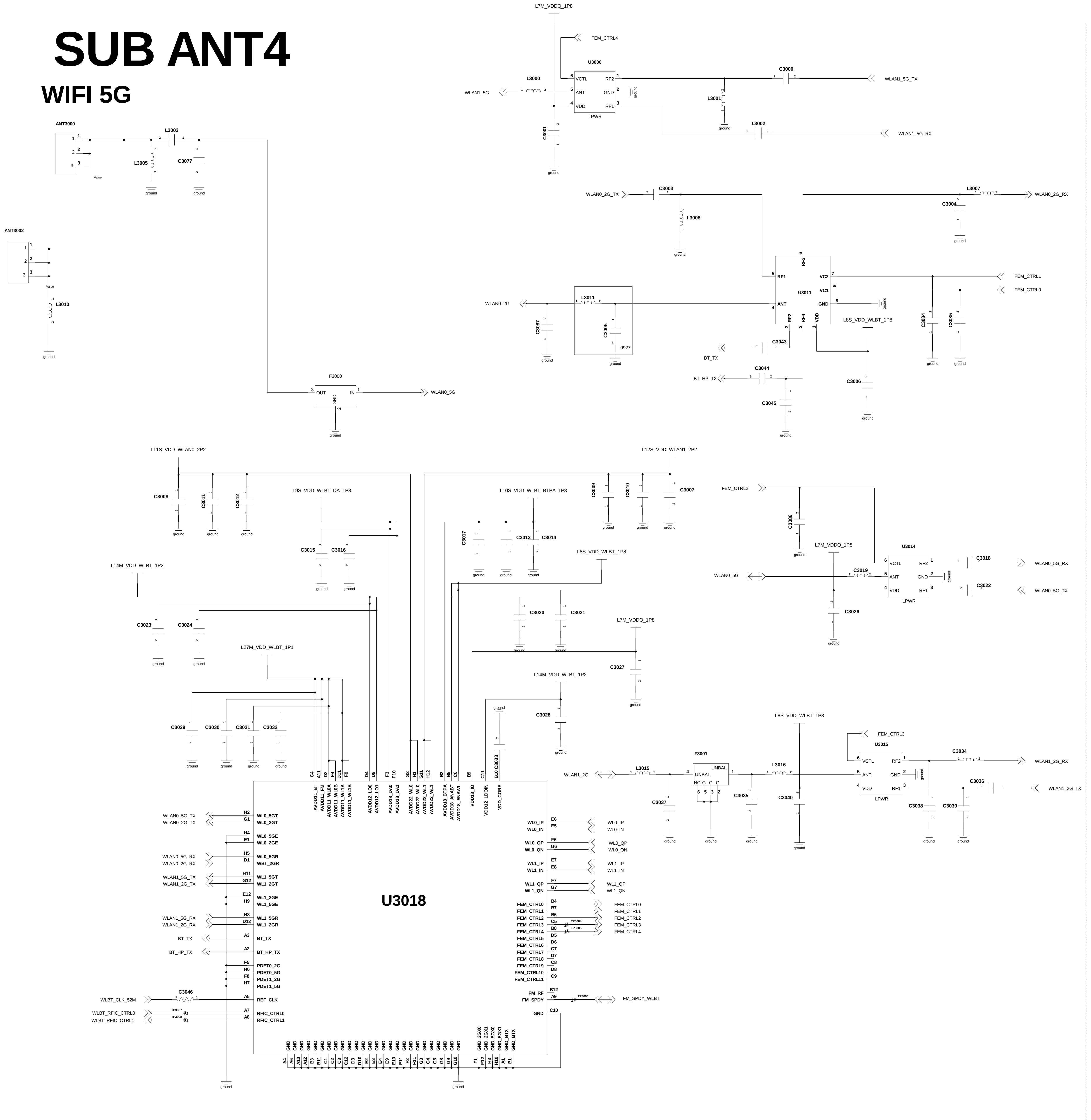
WIFI/BT/FM

WIFI 2.4G - sub5/sub9
WIFI 5G - sub4/sub9

SUB ANT4

WIFI 5G

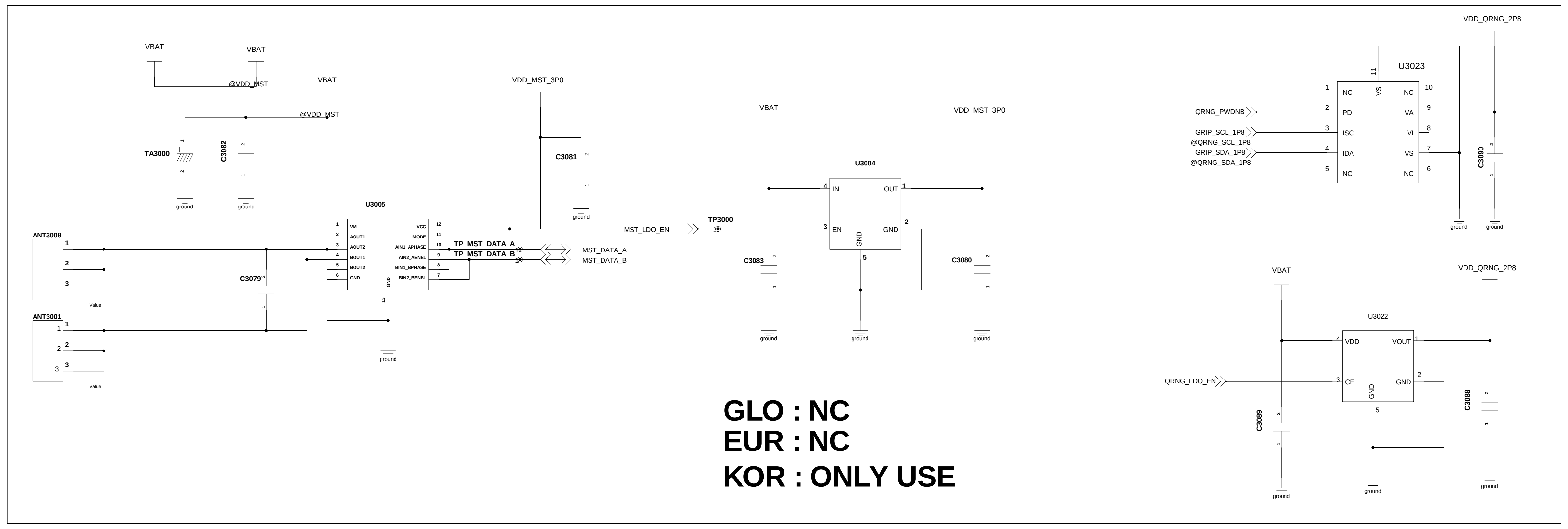

ANT3000
ANT3002
L3003
L3005
C3077
L3010
Value
ground
F3000
OUT
IN
GND
WLAN0_5G
L7M_VDDQ_1P8
FEM_CTRL4
U3000
VCTL
ANT
VDD
RF2
GND
RF1
LPWR
L3000
WLAN1_5G
C3001
L3001
C3000
WLAN1_5G_TX
L3002
WLAN1_5G_RX
WLAN0_2G_TX
C3003
L3008
L3007
WLAN0_2G_RX
C3004
U3011
RF1
RF2
RF3
RF4
VC2
VC1
VDD
FEM_CTRL1
FEM_CTRL0
WLAN0_2G
L3011
C3087
C3005
0927
BT_TX
BT_HP_TX
C3043
C3044
C3045
L8S_VDD_WLBT_1P8
C3006
C3084
C3085
L11S_VDD_WLAN0_2P2
C3008
C3011
C3012
L9S_VDD_WLBT_DA_1P8
C3015
C3016
L10S_VDD_WLBT_BTPA_1P8
C3017
C3013
C3014
L12S_VDD_WLAN1_2P2
C3009
C3010
C3007
L8S_VDD_WLBT_1P8
C3020
C3021
L7M_VDDQ_1P8
C3027
L14M_VDD_WLBT_1P2
C3028
C3033
C3023
C3024
L27M_VDD_WLBT_1P1
C3029
C3030
C3031
C3032
FEM_CTRL2
C3086
L7M_VDDQ_1P8
U3014
C3019
C3026
C3018
C3022
WLAN0_5G
WLAN0_5G_RX
WLAN0_5G_TX
FEM_CTRL3
U3015
F3001
UNBAL
NC G G G
L3015
L3016
C3035
C3037
C3040
C3034
C3036
C3038
C3039
WLAN1_2G
WLAN1_2G_RX
WLAN1_2G_TX
U3018
AVDD11_BT
AVDD11_FM
AVDD11_WL0A
AVDD11_WL0B
AVDD11_WL1A
AVDD11_WL1B
AVDD12_LO0
AVDD12_LO1
AVDD18_DA0
AVDD18_DA1
AVDD22_WL0
AVDD22_WL1
AVDD18_BTPA
AVDD18_ANABT
AVDD18_ANAWL
VDD18_IO
VDD12_LDOIN
VDD_CORE
WLAN0_5G_TX
WLAN0_2G_TX
WL0_5GT
WL0_2GT
WL0_5GE
WL0_2GE
WLAN0_5G_RX
WLAN0_2G_RX
WL0_5GR
WBT_2GR
WLAN1_5G_TX
WLAN1_2G_TX
WL1_5GT
WL1_2GT
WL1_2GE
WL1_5GE
WLAN1_5G_RX
WLAN1_2G_RX
WL1_5GR
WL1_2GR
BT_TX
BT_HP_TX
PDET0_2G
PDET0_5G
PDET1_2G
PDET1_5G
WLBT_CLK_52M
C3046
REF_CLK
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
TP3007
TP3008
RFIC_CTRL0
RFIC_CTRL1
WL0_IP
WL0_IN
WL0_QP
WL0_QN
WL1_IP
WL1_IN
WL1_QP
WL1_QN
FEM_CTRL0
FEM_CTRL1
FEM_CTRL2
FEM_CTRL3
FEM_CTRL4
FEM_CTRL5
FEM_CTRL6
FEM_CTRL7
FEM_CTRL8
FEM_CTRL9
FEM_CTRL10
FEM_CTRL11
TP3004
TP3005
FM_RF
FM_SPDY
TP3006
FM_SPDY_WLBT
GND
GND_2GX0
GND_2GX1
GND_5GX0
GND_5GX1
GND_BTX
GND_BTX

MST

KOR : ONLY USE

QRNG

KOR : ONLY USE


VBAT
@VDD_MST
TA3000
C3082
VDD_MST_3P0
C3081
U3005
VM
AOUT1
AOUT2
BOUT1
BOUT2
GND
VCC
MODE
AIN1_APHASE
AIN2_AENBL
BIN1_BPHASE
BIN2_BENBL
TP_MST_DATA_A
TP_MST_DATA_B
MST_DATA_A
MST_DATA_B
ANT3008
ANT3001
C3079
Value
ground
MST_LDO_EN
TP3000
U3004
IN
OUT
EN
GND
C3083
C3080
U3023
NC
PD
ISC
IDA
VS
VA
VI
QRNG_PWDNB
GRIP_SCL_1P8
@QRNG_SCL_1P8
GRIP_SDA_1P8
@QRNG_SDA_1P8
VDD_QRNG_2P8
C3090
U3022
VDD
CE
VOUT
QRNG_LDO_EN
C3089
C3088

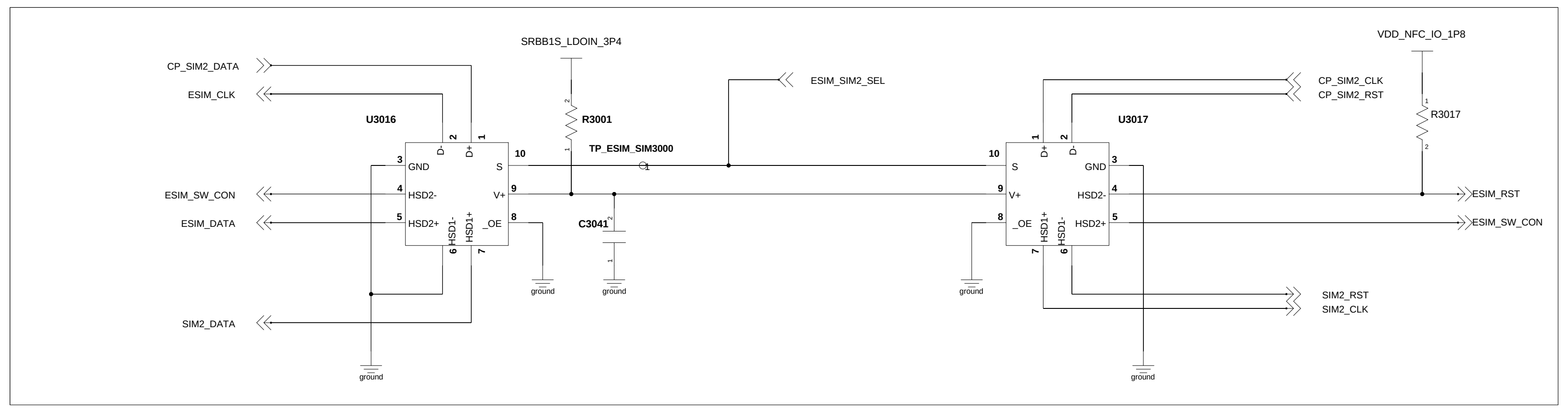
GLO : NC
EUR : NC
KOR : ONLY USE

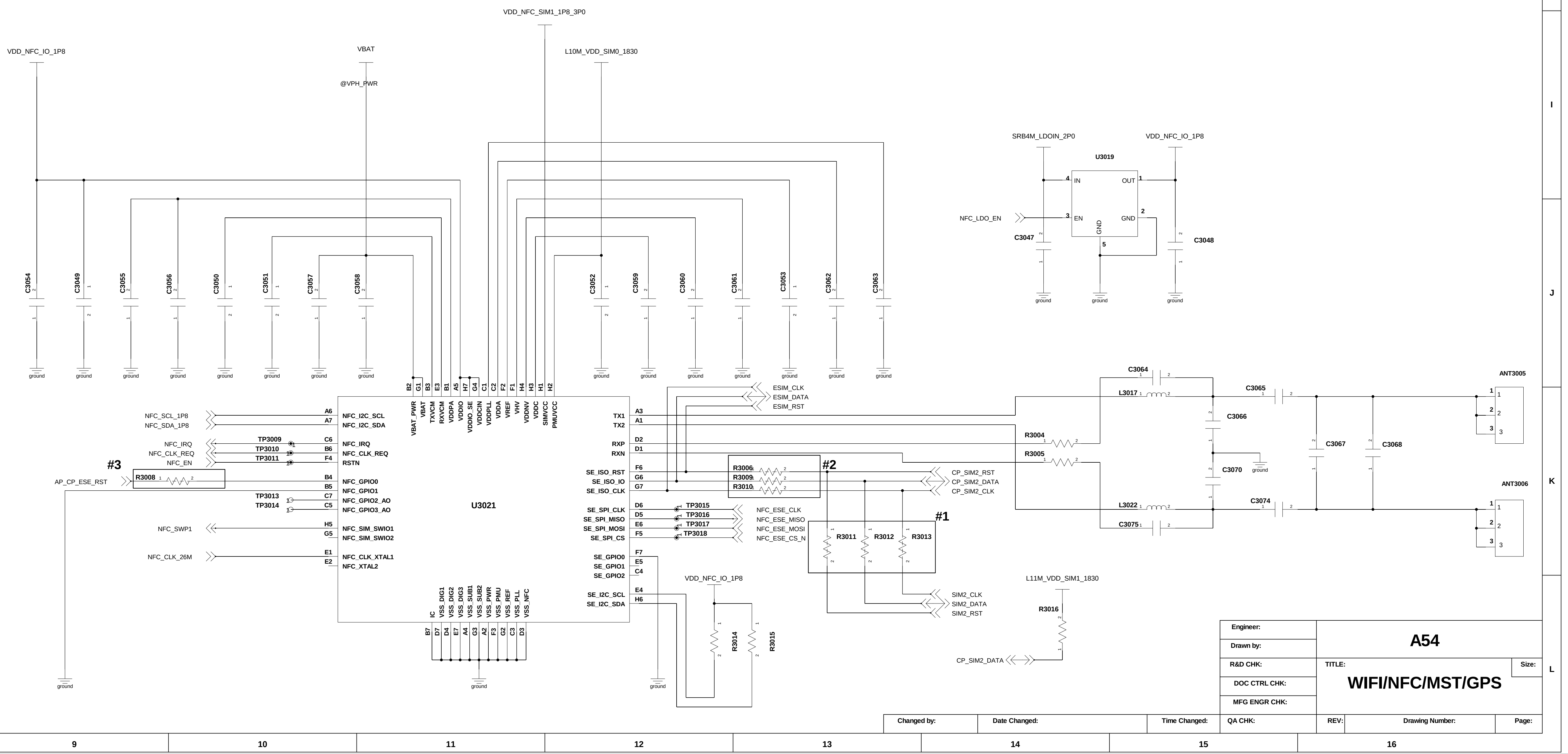
GND CCP

ANT3003
ANT3007
Value
ground

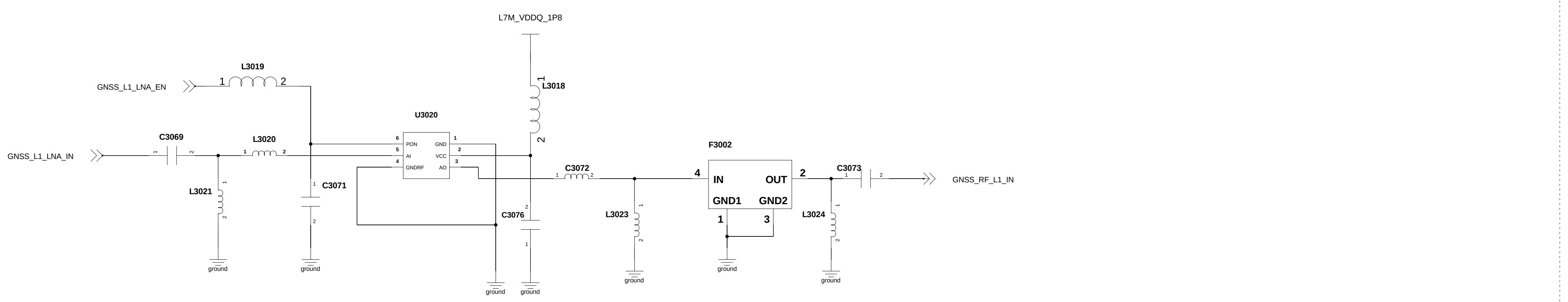
NFC

		NFC IC	#1	#2	#3		Switch IC	ESIM_SIM2_SEL
EUR	2-Nano + eSIM	SN220UV2	OPEN	OPEN	SHORT		SMD	HSD1(H)_SIM2 HSD2(L)_eSIM
KOR/NA	Nano + eSIM	SN220UV2	OPEN	SHORT	SHORT		NC	
JPN(Felica)	Nano + eSIM	SN220PV2	OPEN	SHORT	SHORT		NC	
Other	eSIM X	PN560	SHORT	OPEN	OPEN		NC	
CHN	eSIM X(eSE O)	SN220T	SHORT	OPEN	OPEN		NC	


CP_SIM2_DATA
ESIM_CLK
U3016
ESIM_SW_CON
ESIM_DATA
SIM2_DATA
GND
HSD2-
HSD2+
HSD1-
HSD1+
_OE
S
V+
D-
D+
SRBB1S_LDOIN_3P4
R3001
TP_ESIM_SIM3000
C3041
ESIM_SIM2_SEL
U3017
CP_SIM2_CLK
CP_SIM2_RST
VDD_NFC_IO_1P8
R3017
ESIM_RST
ESIM_SW_CON
SIM2_RST
SIM2_CLK
ground


VDD_NFC_IO_1P8
VBAT
@VPH_PWR
VDD_NFC_SIM1_1P8_3P0
L10M_VDD_SIM0_1830
C3054
C3049
C3055
C3056
C3050
C3051
C3057
C3058
C3052
C3059
C3060
C3061
C3053
C3062
C3063
U3021
VBAT_PWR
VBAT
TXVCM
RXVCM
VDDPA
VDDIO
VDDIO_SE
VDDCIN
VDDPLL
VDDA
VREF
VHV
VDDNV
VDDC
SIMVCC
PMUVCC
NFC_SCL_1P8
NFC_SDA_1P8
NFC_I2C_SCL
NFC_I2C_SDA
NFC_IRQ
NFC_CLK_REQ
NFC_EN
TP3009
TP3010
TP3011
RSTN
#3
R3008
AP_CP_ESE_RST
NFC_GPIO0
NFC_GPIO1
NFC_GPIO2_AO
NFC_GPIO3_AO
TP3013
TP3014
NFC_SWP1
NFC_SIM_SWIO1
NFC_SIM_SWIO2
NFC_CLK_26M
NFC_CLK_XTAL1
NFC_XTAL2
IC
VSS_DIG1
VSS_DIG2
VSS_DIG3
VSS_SUB1
VSS_SUB2
VSS_PWR
VSS_PMU
VSS_REF
VSS_PLL
VSS_NFC
TX1
TX2
RXP
RXN
SE_ISO_RST
SE_ISO_IO
SE_ISO_CLK
SE_SPI_CLK
SE_SPI_MISO
SE_SPI_MOSI
SE_SPI_CS
SE_GPIO0
SE_GPIO1
SE_GPIO2
SE_I2C_SCL
SE_I2C_SDA
ESIM_CLK
ESIM_DATA
ESIM_RST
R3006
R3009
R3010
#2
CP_SIM2_RST
CP_SIM2_DATA
CP_SIM2_CLK
TP3015
TP3016
TP3017
TP3018
NFC_ESE_CLK
NFC_ESE_MISO
NFC_ESE_MOSI
NFC_ESE_CS_N
#1
R3011
R3012
R3013
SIM2_CLK
SIM2_DATA
SIM2_RST
VDD_NFC_IO_1P8
R3014
R3015
L11M_VDD_SIM1_1830
R3016
CP_SIM2_DATA
R3004
R3005
C3064
L3017
C3065
C3066
C3070
C3067
C3068
L3022
C3074
C3075
ANT3005
ANT3006
SRB4M_LDOIN_2P0
VDD_NFC_IO_1P8
U3019
IN
OUT
EN
GND
NFC_LDO_EN
C3047
C3048
ground

GPS L1 / GNSS/BEIDOU


L7M_VDDQ_1P8
L3019
GNSS_L1_LNA_EN
L3018
U3020
PON
AI
GNDRF
GND
VCC
AO
C3069
GNSS_L1_LNA_IN
L3020
L3021
C3071
C3076
C3072
L3023
F3002
IN
OUT
GND1
GND2
C3073
L3024
GNSS_RF_L1_IN
ground

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: WIFI/NFC/MST/GPS Size:
DOC CTRL CHK:	
MFG ENGR CHK:	

Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:	Page:

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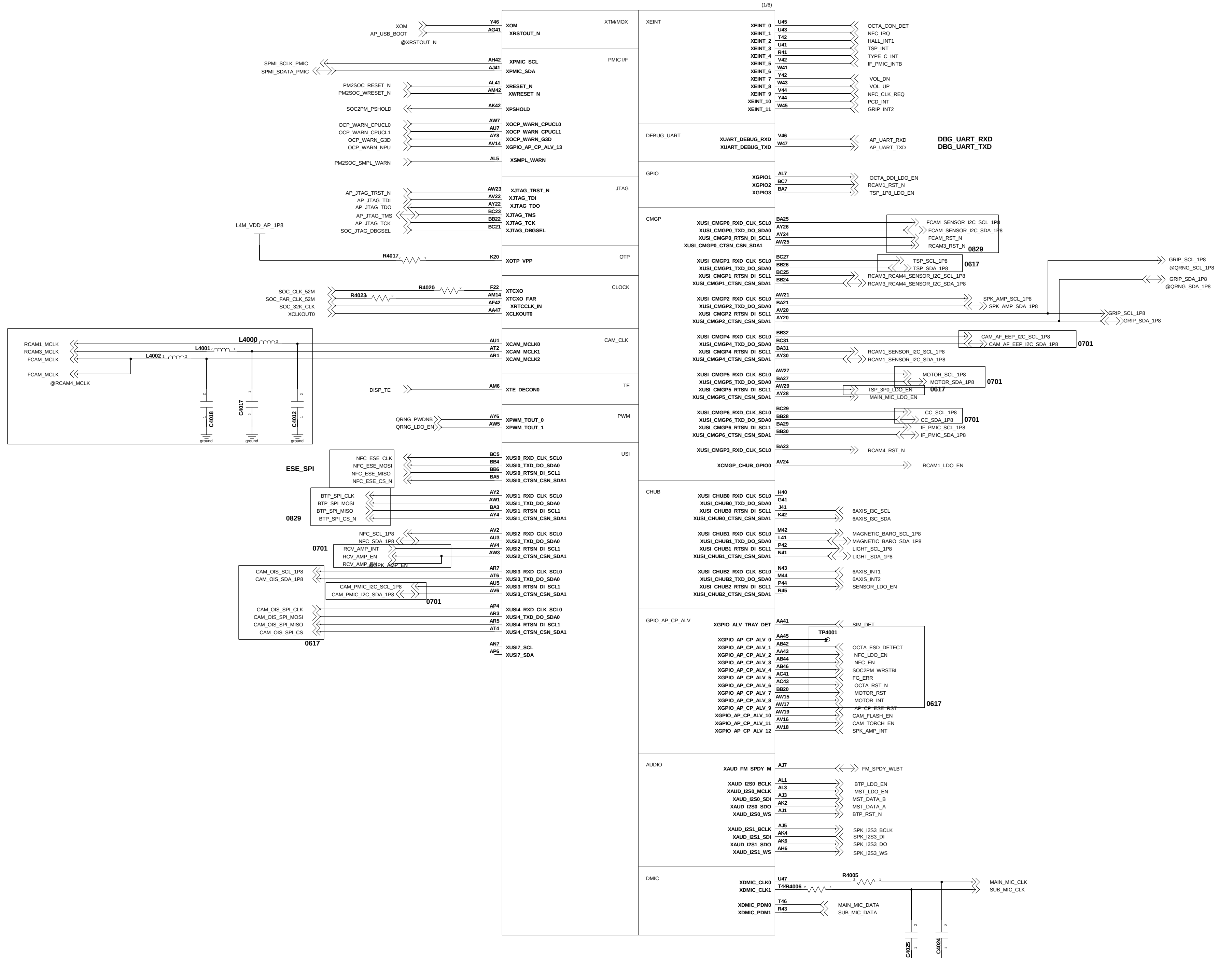
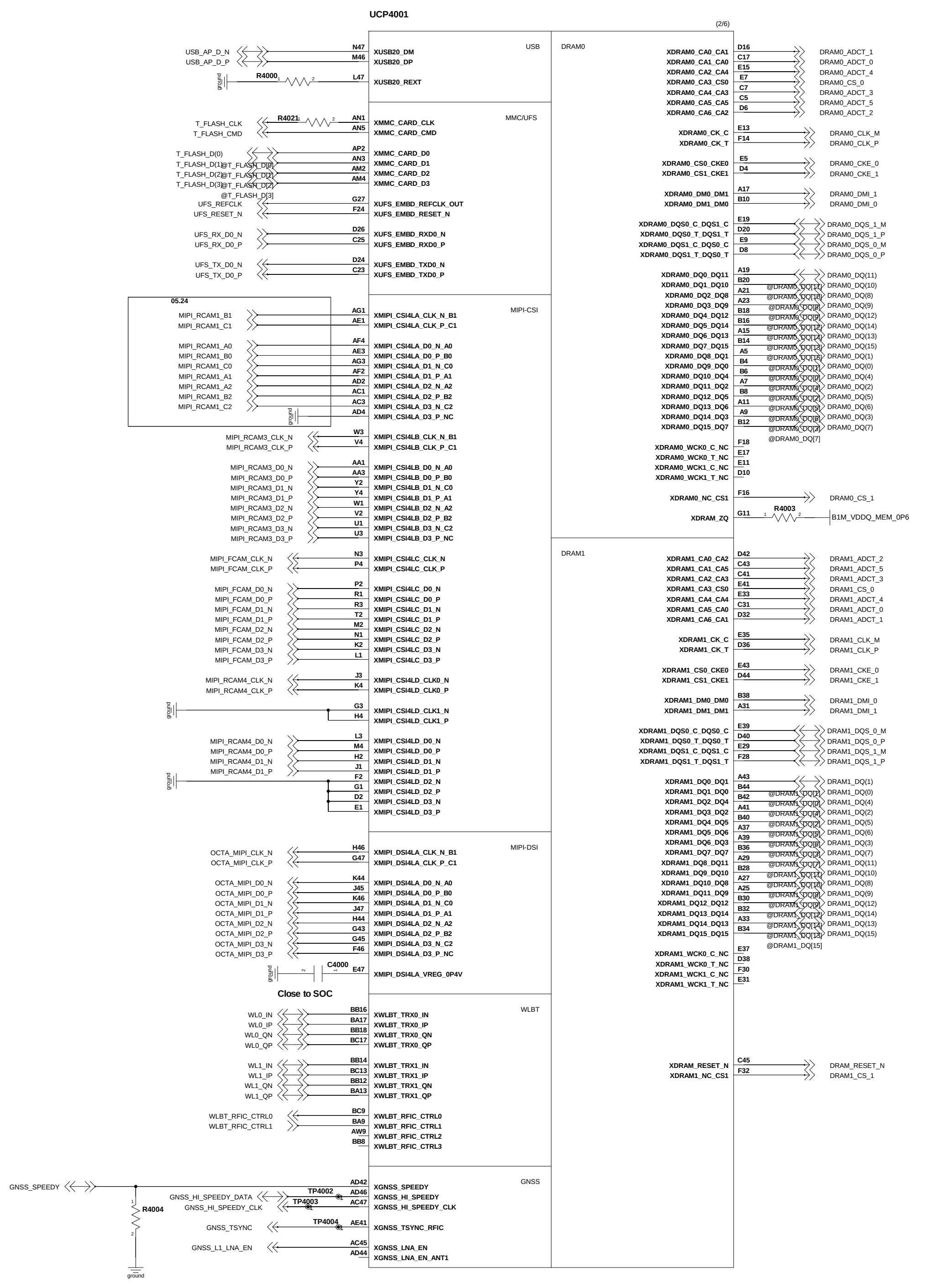
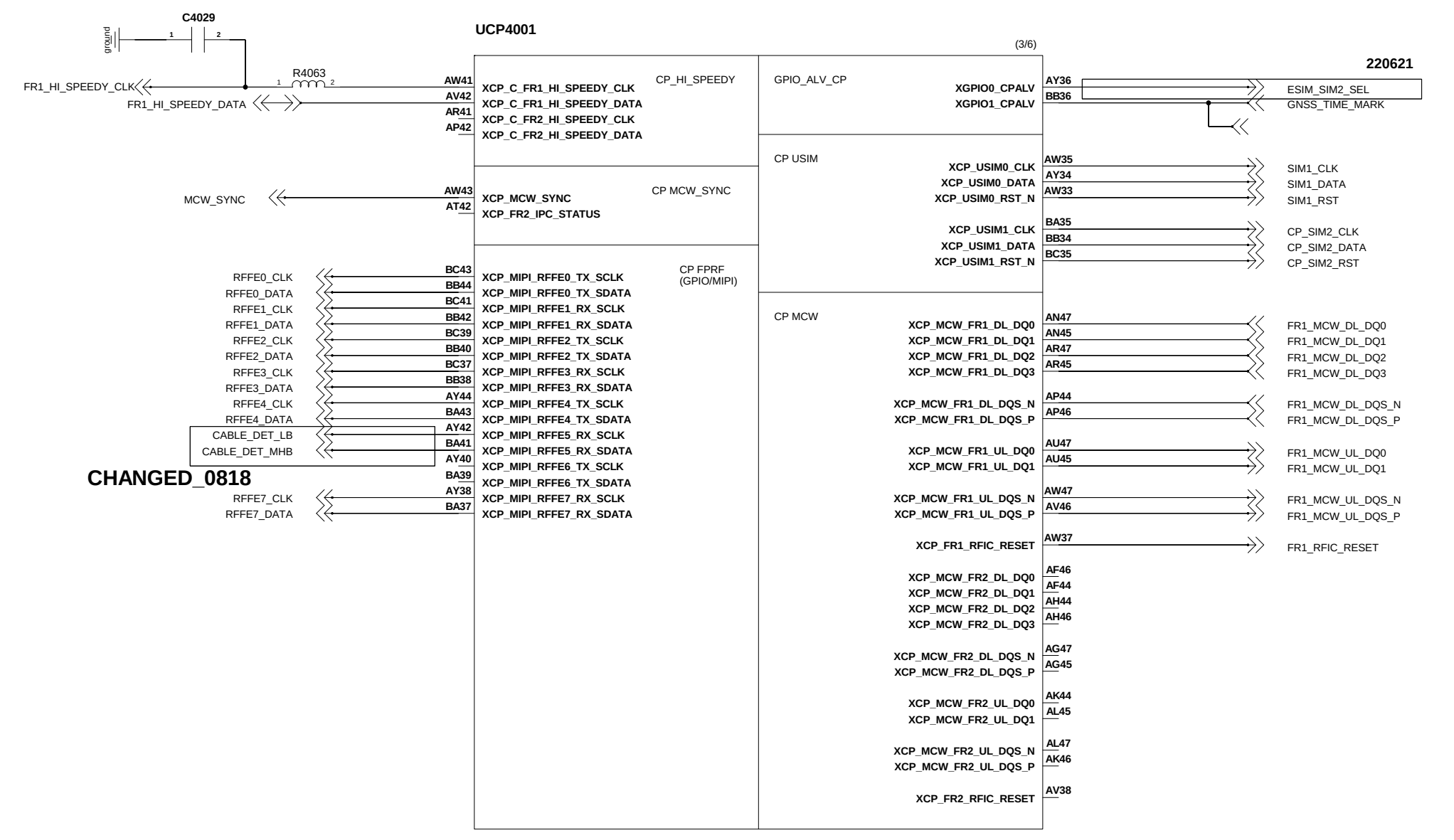
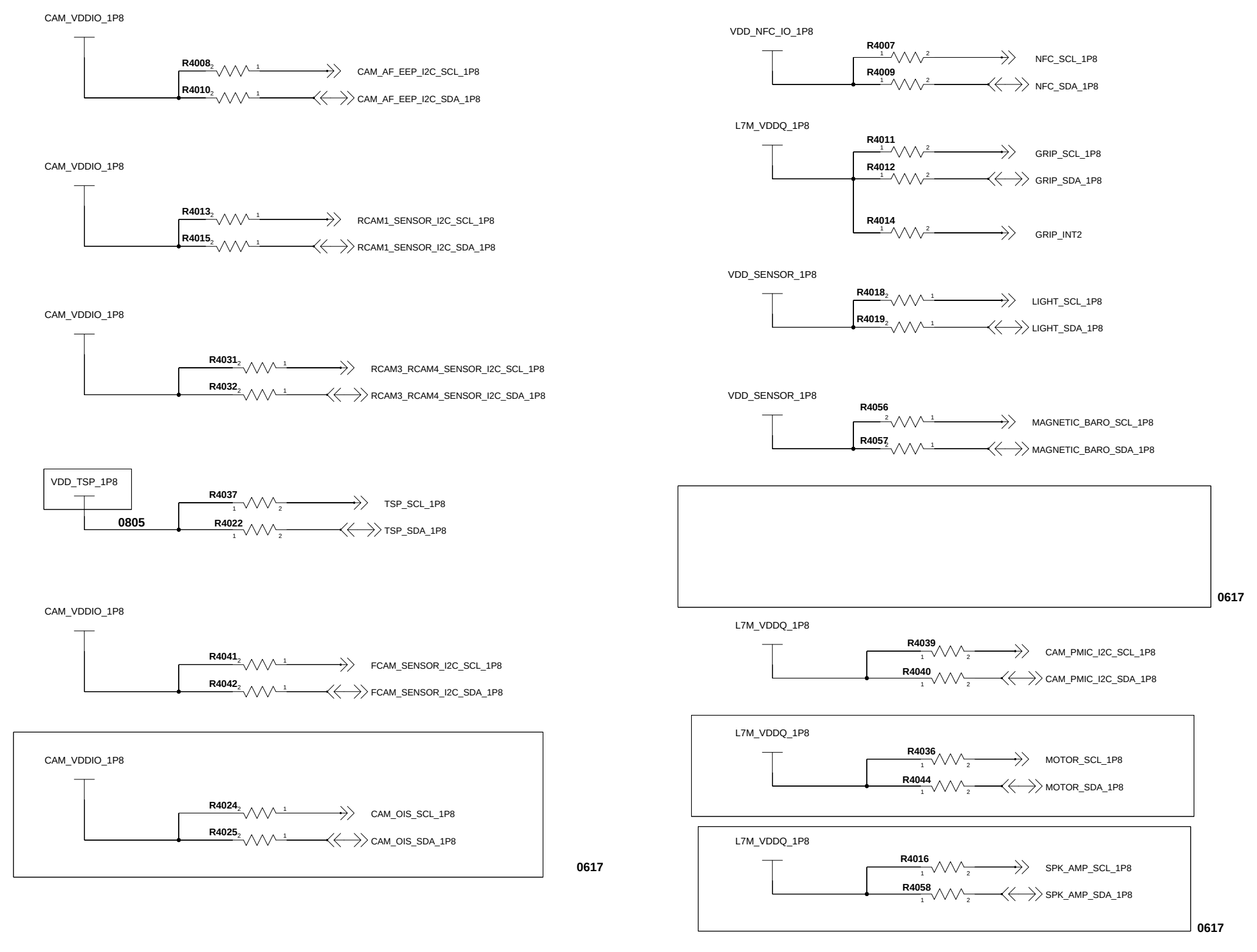
Quartz

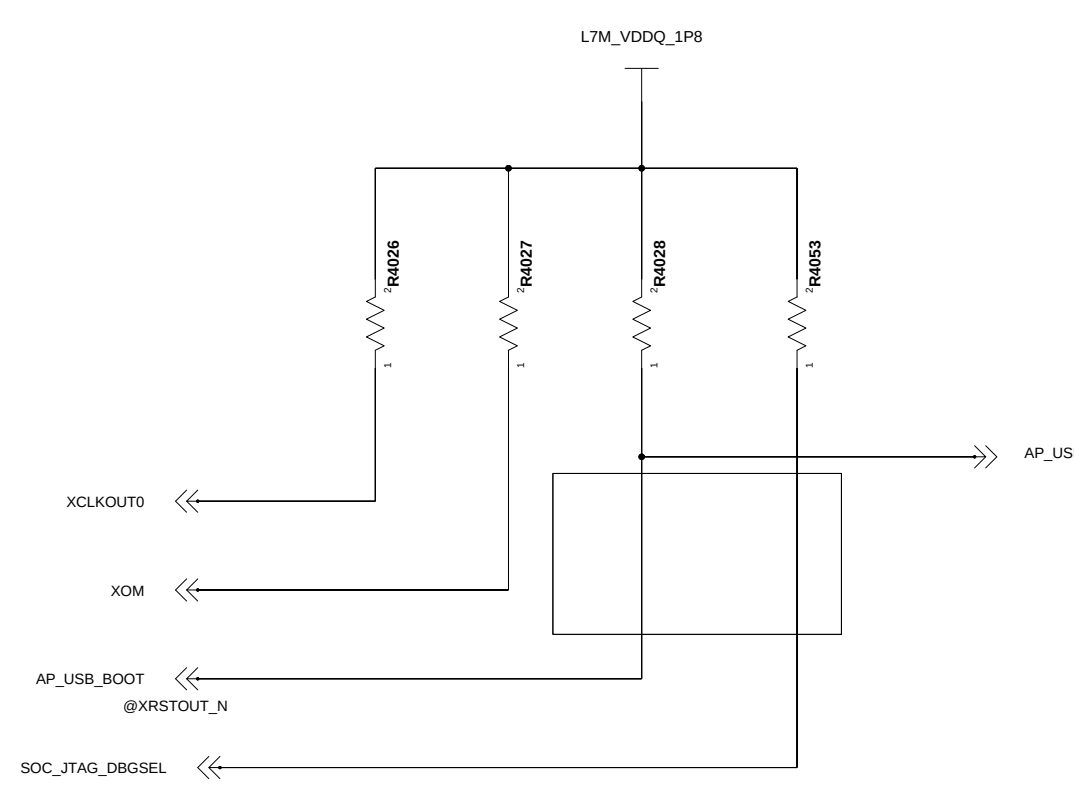

UCP4001
(1/6)
XTM/MOX
XOM
XRSTOUT_N
AP_USB_BOOT
@XRSTOUT_N
PMIC I/F
XPMIC_SCL
XPMIC_SDA
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
PM2SOC_RESET_N
PM2SOC_WRESET_N
SOC2PM_PSHOLD
OCP_WARN_CPUCL0
OCP_WARN_CPUCL1
OCP_WARN_G3D
OCP_WARN_NPU
PM2SOC_SMPL_WARN
JTAG
AP_JTAG_TRST_N
AP_JTAG_TDI
AP_JTAG_TDO
AP_JTAG_TMS
AP_JTAG_TCK
SOC_JTAG_DBGSEL
L4M_VDD_AP_1P8
R4017
OTP
XOTP_VPP
CLOCK
R4020
R4023
SOC_CLK_52M
SOC_FAR_CLK_52M
SOC_32K_CLK
XCLKOUT0
CAM_CLK
L4000
L4001
L4002
RCAM1_MCLK
RCAM3_MCLK
FCAM_MCLK
@RCAM4_MCLK
C4018
C4017
C4012
ground
TE
DISP_TE
PWM
QRNG_PWDNB
QRNG_LDO_EN
USI
ESE_SPI
NFC_ESE_CLK
NFC_ESE_MOSI
NFC_ESE_MISO
NFC_ESE_CS_N
0829
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
BTP_SPI_CS_N
0701
NFC_SCL_1P8
NFC_SDA_1P8
RCV_AMP_INT
RCV_AMP_EN
SPK_AMP_EN
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
CAM_OIS_SPI_CLK
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
0617
XEINT
OCTA_CON_DET
NFC_IRQ
HALL_INT1
TSP_INT
TYPE_C_INT
IF_PMIC_INTB
VOL_DN
VOL_UP
NFC_CLK_REQ
PCD_INT
GRIP_INT2
DEBUG_UART
AP_UART_RXD
AP_UART_TXD
DBG_UART_RXD
DBG_UART_TXD
GPIO
OCTA_DDI_LDO_EN
RCAM1_RST_N
TSP_1P8_LDO_EN
CMGP
FCAM_SENSOR_I2C_SCL_1P8
FCAM_SENSOR_I2C_SDA_1P8
FCAM_RST_N
RCAM3_RST_N
TSP_SCL_1P8
TSP_SDA_1P8
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
GRIP_SCL_1P8
GRIP_SDA_1P8
@QRNG_SCL_1P8
@QRNG_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
RCAM1_SENSOR_I2C_SCL_1P8
RCAM1_SENSOR_I2C_SDA_1P8
MOTOR_SCL_1P8
MOTOR_SDA_1P8
TSP_3P0_LDO_EN
MAIN_MIC_LDO_EN
CC_SCL_1P8
CC_SDA_1P8
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
RCAM4_RST_N
RCAM1_LDO_EN
CHUB
6AXIS_I3C_SCL
6AXIS_I3C_SDA
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
LIGHT_SCL_1P8
LIGHT_SDA_1P8
6AXIS_INT1
6AXIS_INT2
SENSOR_LDO_EN
GPIO_AP_CP_ALV
TP4001
SIM_DET
OCTA_ESD_DETECT
NFC_LDO_EN
NFC_EN
SOC2PM_WRSTBI
FG_ERR
OCTA_RST_N
MOTOR_RST
MOTOR_INT
AP_CP_ESE_RST
CAM_FLASH_EN
CAM_TORCH_EN
SPK_AMP_INT
AUDIO
FM_SPDY_WLBT
BTP_LDO_EN
MST_LDO_EN
MST_DATA_B
MST_DATA_A
BTP_RST_N
SPK_I2S3_BCLK
SPK_I2S3_DI
SPK_I2S3_DO
SPK_I2S3_WS
DMIC
R4005
R4006
MAIN_MIC_CLK
SUB_MIC_CLK
MAIN_MIC_DATA
SUB_MIC_DATA
C4025
C4024


UCP4001
(2/6)
USB
USB_AP_D_N
USB_AP_D_P
R4000
XUSB20_DM
XUSB20_DP
XUSB20_REXT
MMC/UFS
R4021
T_FLASH_CLK
T_FLASH_CMD
T_FLASH_D(0)
T_FLASH_D(1)
T_FLASH_D(2)
T_FLASH_D(3)
UFS_REFCLK
UFS_RESET_N
UFS_RX_D0_N
UFS_RX_D0_P
UFS_TX_D0_N
UFS_TX_D0_P
MIPI-CSI
05.24
MIPI_RCAM1_B1
MIPI_RCAM1_C1
MIPI_RCAM3_CLK_N
MIPI_RCAM3_CLK_P
MIPI_FCAM_CLK_N
MIPI_FCAM_CLK_P
MIPI_RCAM4_CLK_N
MIPI_RCAM4_CLK_P
MIPI-DSI
OCTA_MIPI_CLK_N
OCTA_MIPI_CLK_P
C4000
Close to SOC
WLBT
WL0_IN
WL0_IP
WL0_QN
WL0_QP
WL1_IN
WL1_IP
WL1_QN
WL1_QP
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
GNSS
GNSS_SPEEDY
GNSS_HI_SPEEDY_DATA
GNSS_HI_SPEEDY_CLK
GNSS_TSYNC
GNSS_L1_LNA_EN
TP4002
TP4003
TP4004
R4004
DRAM0
DRAM1
R4003
B1M_VDDQ_MEM_0P6
DRAM_RESET_N
DRAM1_CS_1


UCP4001
(3/6)
C4029
R4063
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
CP_HI_SPEEDY
MCW_SYNC
CP MCW_SYNC
CP FPRF (GPIO/MIPI)
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
RFFE4_CLK
RFFE4_DATA
CABLE_DET_LB
CABLE_DET_MHB
CHANGED_0818
RFFE7_CLK
RFFE7_DATA
GPIO_ALV_CP
220621
ESIM_SIM2_SEL
GNSS_TIME_MARK
CP USIM
SIM1_CLK
SIM1_DATA
SIM1_RST
CP_SIM2_CLK
CP_SIM2_DATA
CP_SIM2_RST
CP MCW
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
FR1_RFIC_RESET

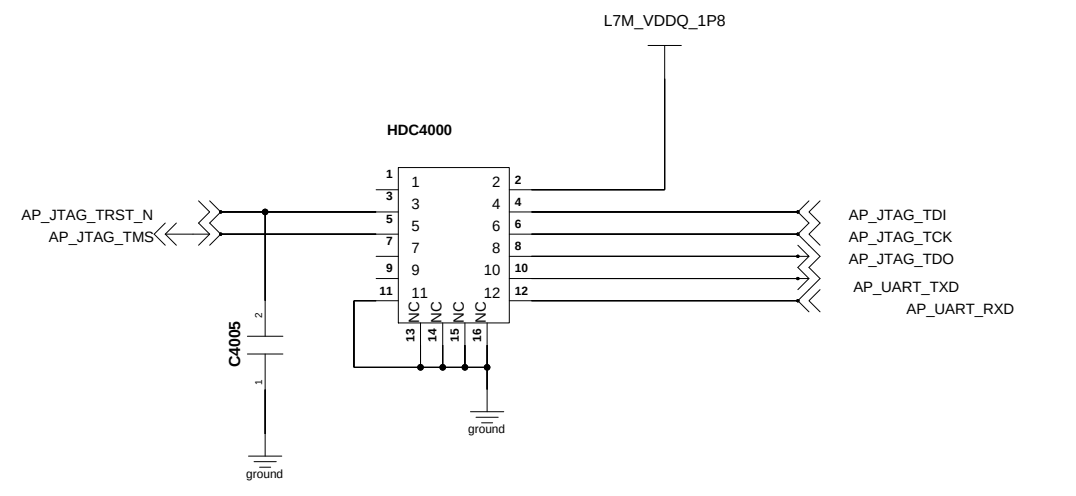
I2C PU RESISTOR


CAM_VDDIO_1P8
R4008
R4010
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
R4013
R4015
RCAM1_SENSOR_I2C_SCL_1P8
RCAM1_SENSOR_I2C_SDA_1P8
R4031
R4032
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
VDD_TSP_1P8
0805
R4037
R4022
TSP_SCL_1P8
TSP_SDA_1P8
R4041
R4042
FCAM_SENSOR_I2C_SCL_1P8
FCAM_SENSOR_I2C_SDA_1P8
R4024
R4025
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
0617
VDD_NFC_IO_1P8
R4007
R4009
NFC_SCL_1P8
NFC_SDA_1P8
L7M_VDDQ_1P8
R4011
R4012
R4014
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
VDD_SENSOR_1P8
R4018
R4019
LIGHT_SCL_1P8
LIGHT_SDA_1P8
R4056
R4057
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
R4039
R4040
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
R4036
R4044
MOTOR_SCL_1P8
MOTOR_SDA_1P8
R4016
R4058
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8


L7M_VDDQ_1P8
R4026
R4027
R4028
R4053
AP_USB_BOOT
XCLKOUT0
XOM
@XRSTOUT_N
SOC_JTAG_DBGSEL

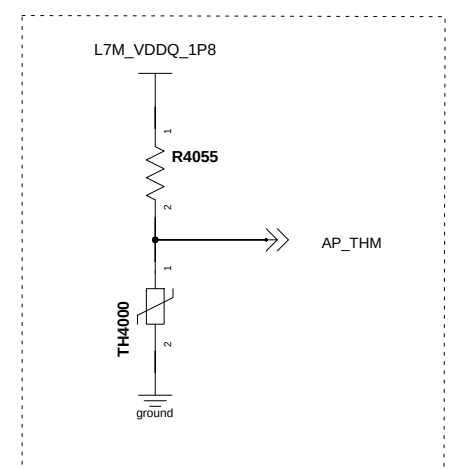
PIN			Function	
XCLKOUT0	XRSTOUT_N	XOM	1st Boot	2nd Boot
0	0	0	UFS	USB
0	1	X	USB	----
1	0	0	SD	USB

BOOT CONFIGURATIONS

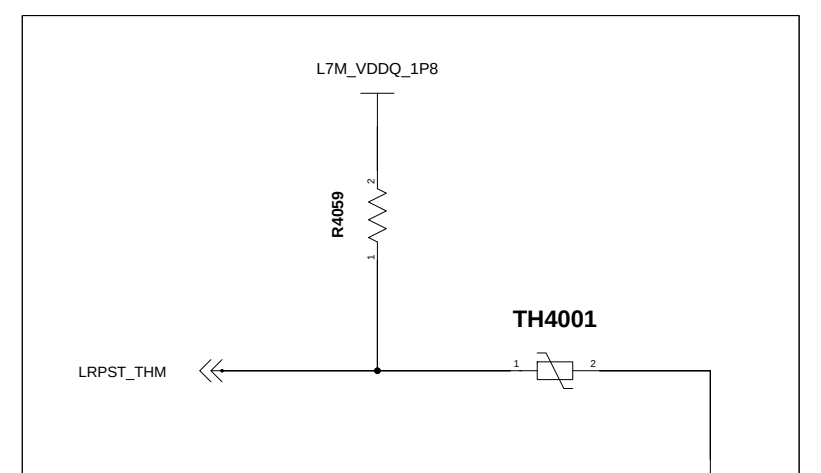

L7M_VDDQ_1P8
HDC4000
AP_JTAG_TRST_N
AP_JTAG_TMS
AP_JTAG_TDI
AP_JTAG_TCK
AP_JTAG_TDO
AP_UART_TXD
AP_UART_RXD
C4005
NC
ground

AP JTAG

SOC THERMISTOR

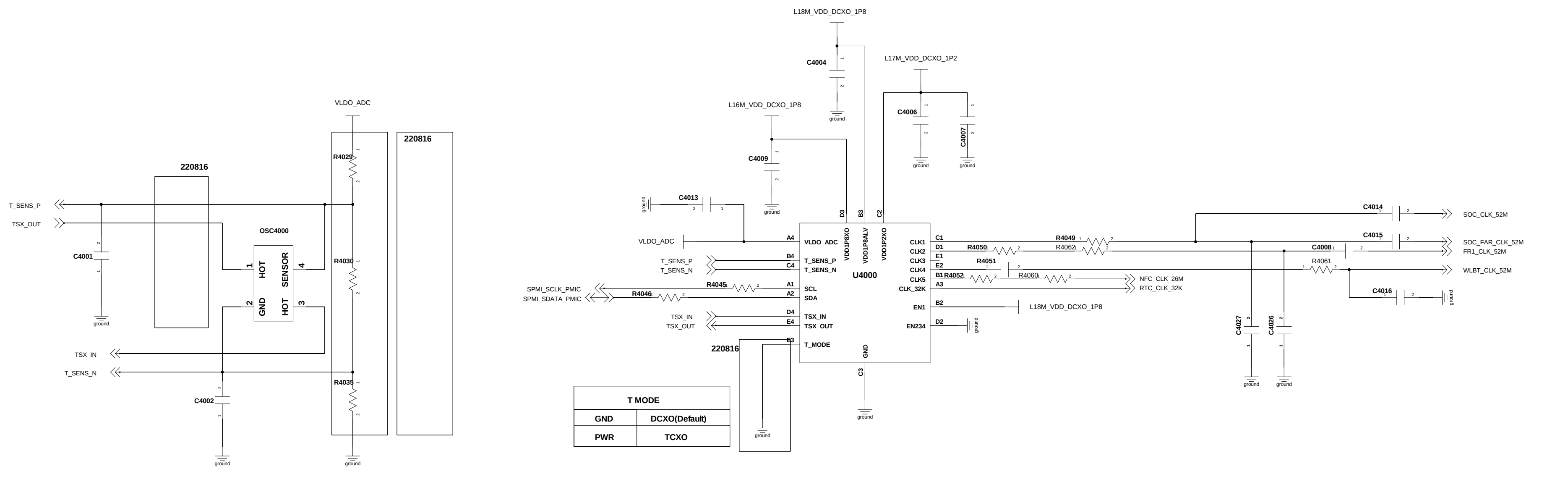

L7M_VDDQ_1P8
R4055
AP_THM
TH4000
ground

AP_THM


L7M_VDDQ_1P8
R4059
TH4001
LRPST_THM
ground

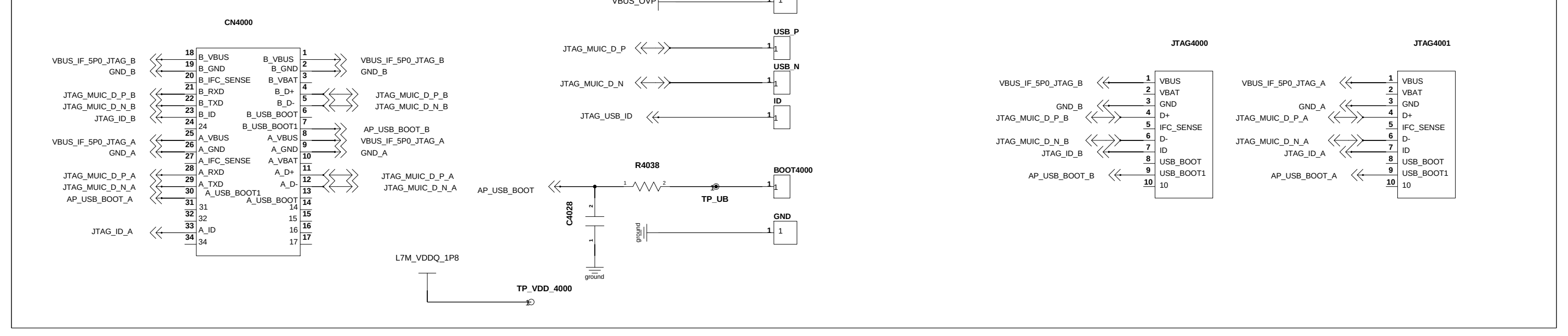
LRPST_THM

SYSTEM CLOCK


VLDO_ADC
220816
T_SENS_P
TSX_OUT
C4001
OSC4000
HOT
SENSOR
GND
HOT
TSX_IN
T_SENS_N
C4002
R4029
R4030
R4035
L18M_VDD_DCXO_1P8
L17M_VDD_DCXO_1P2
L16M_VDD_DCXO_1P8
C4004
C4006
C4007
C4009
C4013
U4000
VLDO_ADC
T_SENS_P
T_SENS_N
SCL
SDA
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
R4045
R4046
TSX_IN
TSX_OUT
T_MODE
VDD1P8XO
VDD1P8ALV
VDD1P2XO
CLK1
CLK2
CLK3
CLK4
CLK5
CLK_32K
EN1
EN234
GND
R4049
R4050
R4051
R4052
R4060
R4061
R4062
C4014
C4015
C4008
C4016
C4027
C4026
SOC_CLK_52M
SOC_FAR_CLK_52M
FR1_CLK_52M
WLBT_CLK_52M
NFC_CLK_26M
RTC_CLK_32K
L18M_VDD_DCXO_1P8
ground

T MODE	
GND	DCXO(Default)
PWR	TCXO

PCB JTAG


CN4000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
AP_USB_BOOT_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
AP_USB_BOOT_A
JTAG_ID_A
VBUS_OVP
@VBUS_OVP_JTAG
VBUS
USB_P
USB_N
ID
JTAG_MUIC_D_P
JTAG_MUIC_D_N
JTAG_USB_ID
R4038
AP_USB_BOOT
C4028
TP_UB
BOOT4000
GND
L7M_VDDQ_1P8
TP_VDD_4000
JTAG4000
JTAG4001
VBAT
IFC_SENSE
USB_BOOT
USB_BOOT1

Engineer:	A54		
Drawn by:			
R&D CHK:	TITLE:		Size:
DOC CTRL CHK:			
MFG ENGR CHK:			
QA CHK:	REV:	Drawing Number:	Page:

Changed by: | Date Changed: | Time Changed:

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UCP4001
(4/6)
(5/6)
(6/6)
UME5000
B1M_VDDQ_MEM_0P6
@RSP1M
@RSN1M
C5007
C5000
C5125
C5126
220608
VDDQ06_DRAM0
VDDQ06_DRAM1
B2_3M_VDD_CAM_1P3
@RSP2_3M
@RSN2_3M
C5132
C5016
C5012
C5001
VDD_CAM
B4_5_6M_VDD_CPUCL1_BIG_1P3
@RSP4_5_6M
@RSN4_5_6M
C5229
C5228
C5138
C5139
C5140
C5219
C5220
C5227
C5221
C5222
C5223
C5224
C5225
C5226
VDD_CPUCL1_BIG
B7M_VDD_CPUCL0_LIT_1P3
@RSP7M
@RSN7M
C5043
C5035
VDD_CPUCL0_LIT
B8M_VDD2_MEM_1P1
C5213
C5053
VDDQ11_DRAM0_CKE
VDDQ11_DRAM1_CKE
L1M_VDD_ALIVE_0P75
C5215
C5216
VDD_ALIVE
VDD_ALIVE_JSC
VDD_CHUBVTS
AVDD075_USB20
VDD075_DLL_GNSS
VDD_GNSS_ALV
VDD_ALV_CP
C5145
C5144
C5143
C5142
L2M_VDD_AP_1833
VDDQ1833_MMC_CARD
C5146
ground
AVDD085_UFS_EMBD
AVDD085_MIPI_DSI
AVDD085_MIPI_CSI
VDD_ICPU_CSILINK
C5006
C5147
C5148
C5149
C5150
L3M_VDD_AP_0P85
AVDD18_UFS_EMBD
AVDD18_MIPI_DSI
AVDD18_MIPI_CSI
AVDD18_OTP
AVDD18_TS
C5151
C5152
C5153
C5154
L4M_VDD_AP_1P8
AVDD12_PLL_AUD
AVDD12_PLL_CPU
AVDD12_PLL_SHARED
AVDD12_PLL_MIF
AVDD12_MIPI_DSI
VDDQ12_UFS_EMBD
C5155
C5156
C5157
L5M_VDD_AP_PLL_1P2
AVDD33_USB20
C5004
L6M_VDD_USB20_3P0
AVDD18_TCXO_FAR
AVDD18_TCXO
AVDD18_USB20
VDDQ18_GPIO_ALV_CP
VDDQ18_MCW_BIG
VDDQ18_MCW_CP
VDDQ1218_GNSS
VDDQ18_WLBT
VDDQ18_GNSS
VDDQ18_GPIO_B
VDDQ18_GPIO_T
VDDQ18_CMGP
C5158
C5159
C5160
C5161
L7M_VDDQ_1P8
VDD_MCW_CP
C5046
C5162
L8M_VDD_MCW_0P75
VDD06_MCW_FR12_CP
C5164
C5165
L9M_VDD_MCW_0P6
VDDQ1830_USIM0_CP
C5163
L10M_VDD_SIM0_1830
VDDQ1830_USIM1_CP
C5166
L11M_VDD_SIM1_1830
VDD_GNSS
C5167
C5168
L12M_VDD_GNSS_0P75
AVDD12_CDPLL_CP
AVDD12_IMPLL_CP
C5169
C5170
L13M_VDD_CP_PLL_1P2
AVDD12_WLBT
C5171
L14M_VDD_WLBT_1P2
VDD_WLBT_M
C5172
L20M_VDD_WLBT_M_0P75
VDD_SRAM_PPC_LIT
C5173
C5174
L34M_VDD_SRAM_PPC_LIT_0P75
VDD_SRAM_PPC_BIG
C5175
C5176
C5177
L35M_VDD_SRAM_PPC_BIG_0P75
VSS
B8M_VDD2_MEM_1P1
@RSP8M
@RSN8M
C5207
C5047
C5050
C5051
C5206
C5208
220801
L30M_VDD_UFS_1P8
C5060
C5061
C5062
L7M_VDDQ_1P8
C5063
C5064
C5065
C5066
C5078
C5212
C5211
B1M_VDDQ_MEM_0P6
C5069
C5070
C5071
C5072
C5073
C5074
C5075
L29M_VDD_UFS_2P95
C5076
C5077
R5000
R5001
VCC
VCCQ
VCCQ2
VDD1
VDD2
VDDI
VDDIQ
VDDIQ2
VDDQ
ZQ0_A
ZQ1_A
CA0_A
CA1_A
CA2_A
CA3_A
CA4_A
CA5_A
DRAM0_ADCT_0
DRAM0_ADCT_1
DRAM0_ADCT_2
DRAM0_ADCT_3
DRAM0_ADCT_4
DRAM0_ADCT_5
CK_C_A
CK_T_A
DRAM0_CLK_M
DRAM0_CLK_P
CKE0_A
CKE1_A
DRAM0_CKE_0
DRAM0_CKE_1
CS0_A
CS1_A
DRAM0_CS_0
DRAM0_CS_1
DMI0_A
DMI1_A
DRAM0_DMI_0
DRAM0_DMI_1
DQ0_A … DQ15_A
DRAM0_DQ(0) … DRAM0_DQ(15)
DQS0_C_A
DQS0_T_A
DQS1_C_A
DQS1_T_A
DRAM0_DQS_0_M
DRAM0_DQS_0_P
DRAM0_DQS_1_M
DRAM0_DQS_1_P
ODT_A
DIN0_C
DIN0_T
DIN1_C
DIN1_T
UFS_TX_D0_N
UFS_TX_D0_P
TP5003
TP5004
DOUT0_C
DOUT0_T
DOUT1_C
DOUT1_T
UFS_RX_D0_N
UFS_RX_D0_P
TP5005
TP5006
REF_CLK
UFS_REFCLK
TP5007
RFU
NC
VSS_G
CA0_B
CA1_B
CA2_B
CA3_B
CA4_B
CA5_B
DRAM1_ADCT_0
DRAM1_ADCT_1
DRAM1_ADCT_2
DRAM1_ADCT_3
DRAM1_ADCT_4
DRAM1_ADCT_5
CK_C_B
CK_T_B
DRAM1_CLK_M
DRAM1_CLK_P
CKE0_B
CKE1_B
DRAM1_CKE_0
DRAM1_CKE_1
CS0_B
CS1_B
DRAM1_CS_0
DRAM1_CS_1
DMI0_B
DMI1_B
DRAM1_DMI_0
DRAM1_DMI_1
DQ0_B … DQ15_B
DRAM1_DQ(0) … DRAM1_DQ(15)
DQS0_C_B
DQS0_T_B
DQS1_C_B
DQS1_T_B
DRAM1_DQS_0_M
DRAM1_DQS_0_P
DRAM1_DQS_1_M
DRAM1_DQS_1_P
ODT_B
RESET_N
RST_N
TP5001
TP5002
DRAM_RESET_N
UFS_RESET_N
DNU
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
TP5008
TP5009
B1S_VDD_MIF_0P75
@RSP1S
@RSN1S
C5209
C5130
C5068
VDD_MIF
B2S_VDD_WLBT_0P75
@RSP2S
@RSN2S
C5214
C5217
VDD_WLBT
B3S_VDD_NPU_0P75
@RSP3S
@RSN3S
C5087
C5210
VDD_NPU
B4_5S_VDD_G3D_0P75
@RSP4_5S
@RSN4_5S
C5178
C5088
C5089
VDD_G3D
B6S_VDD_CPU_CP_0P75
@RSP6S
@RSN6S
C5184
C5180
VDD_CPU_CP
VDD_CP
B7_8S_VDD_CP_0P75
@RSP7_8S
@RSN7_8S
C5185
C5186
C5191
C5190
VDD_INT
B9S_VDD_INT_0P75
@RSP9S
@RSN9S
C5192
C5193
C5198
C5200
VDD_SRAM
B10S_VDD_SRAM_0P75
@RSP10S
@RSN10S
C5201
C5202
C5203
AVDD18_WLBT
C5204
L8S_VDD_WLBT_1P8
VDD_SRAM_PPC_NPU
C5205
L15S_VDD_SRAM_PPC_NPU_0P75

6GB + 128GB uMCP : 1108-000714
8GB + 128GB uMCP : 1108-000721
8GB + 256GB uMCP : 1108-000722

Engineer:		A54			
Drawn by:					
R&D CHK:		TITLE:		Size:	
DOC CTRL CHK:		AP/CP/uMCP			
MFG ENGR CHK:					
Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:	Page:

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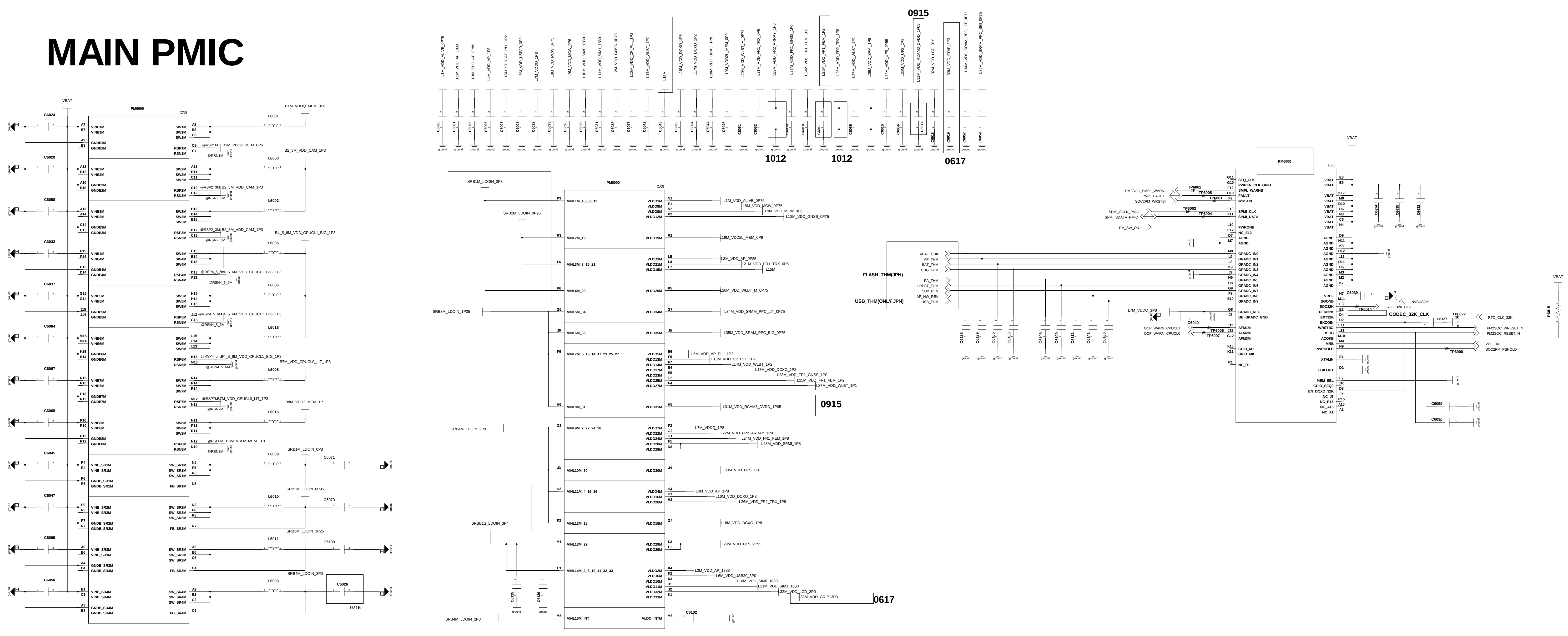
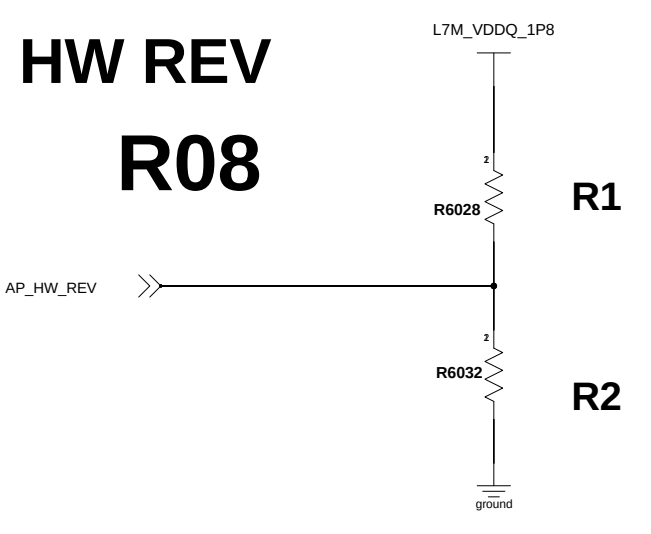
MAIN PMIC


PM6000
(2/3)
(1/3)
(3/3)
VBAT
L6001
L6000
L6002
L6005
L6006
L6018
L6008
L6015
L6009
L6010
L6011
L6003
B1M_VDDQ_MEM_0P6
B2_3M_VDD_CAM_1P3
B4_5_6M_VDD_CPUCL1_BIG_1P3
B7M_VDD_CPUCL0_LIT_1P3
B8M_VDD2_MEM_1P1
SRB1M_LDOIN_0P8
SRB2M_LDOIN_0P95
SRB3M_LDOIN_1P25
SRB4M_LDOIN_2P0
SRBB1S_LDOIN_3P4
C6026
0715
1012
1012
0915
0617
0915
0617
FLASH_THM(JPN)
USB_THM(ONLY JPN)
CODEC_32K_CLK

REV	R1(Kohm)	R2(Kohm)	AP	
0	NC	0	R00	Bringup00
1	430	30	R01	Bringup01
2	390	62	R02	Real REV00
3	360	91	R03	Real REV01
4	330	120	R04	Real REV02
5	300	150	R05	Real REV03
6	270	180	R06	Real REV04
7	270	240	R07	Real REV04B
8	240	270	R08	Real REV04C
9	180	270		
10	150	300		

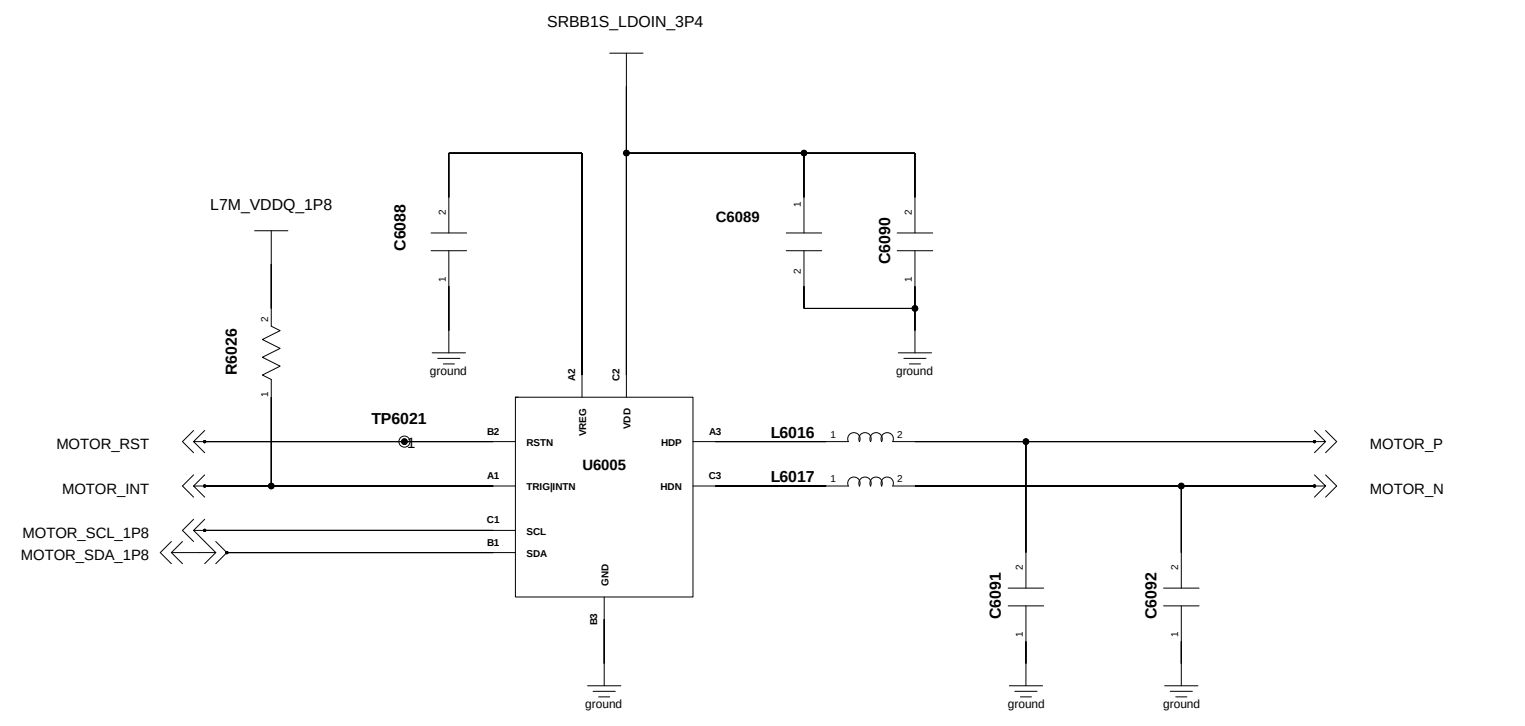
AP HW REV

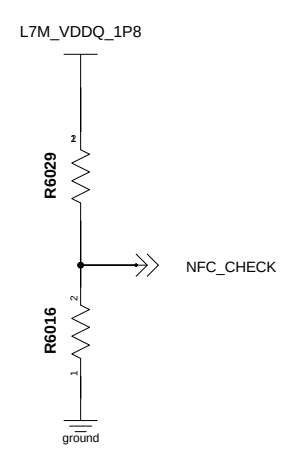
R08


L7M_VDDQ_1P8
R6028
R1
AP_HW_REV
R6032
R2
ground

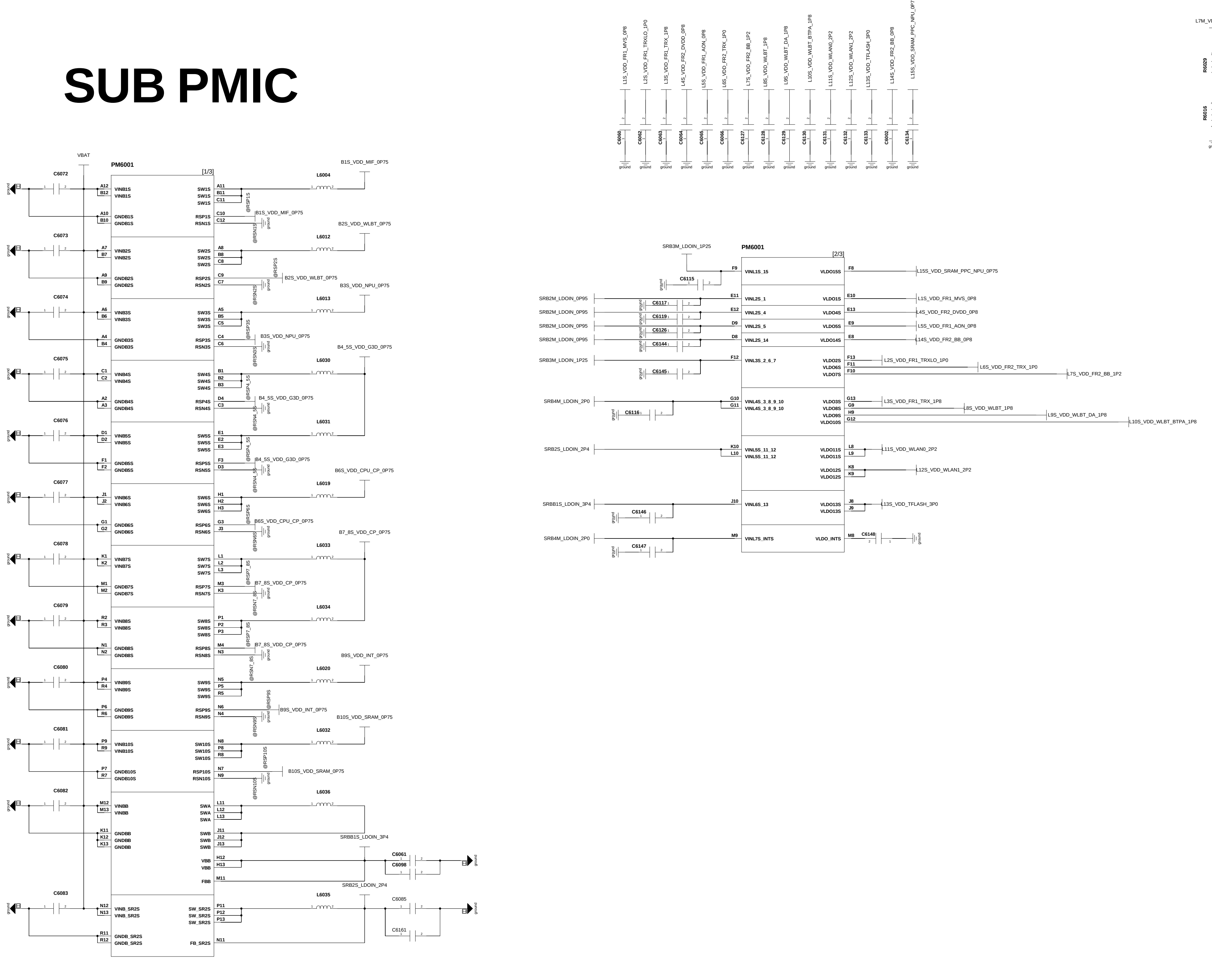
220620

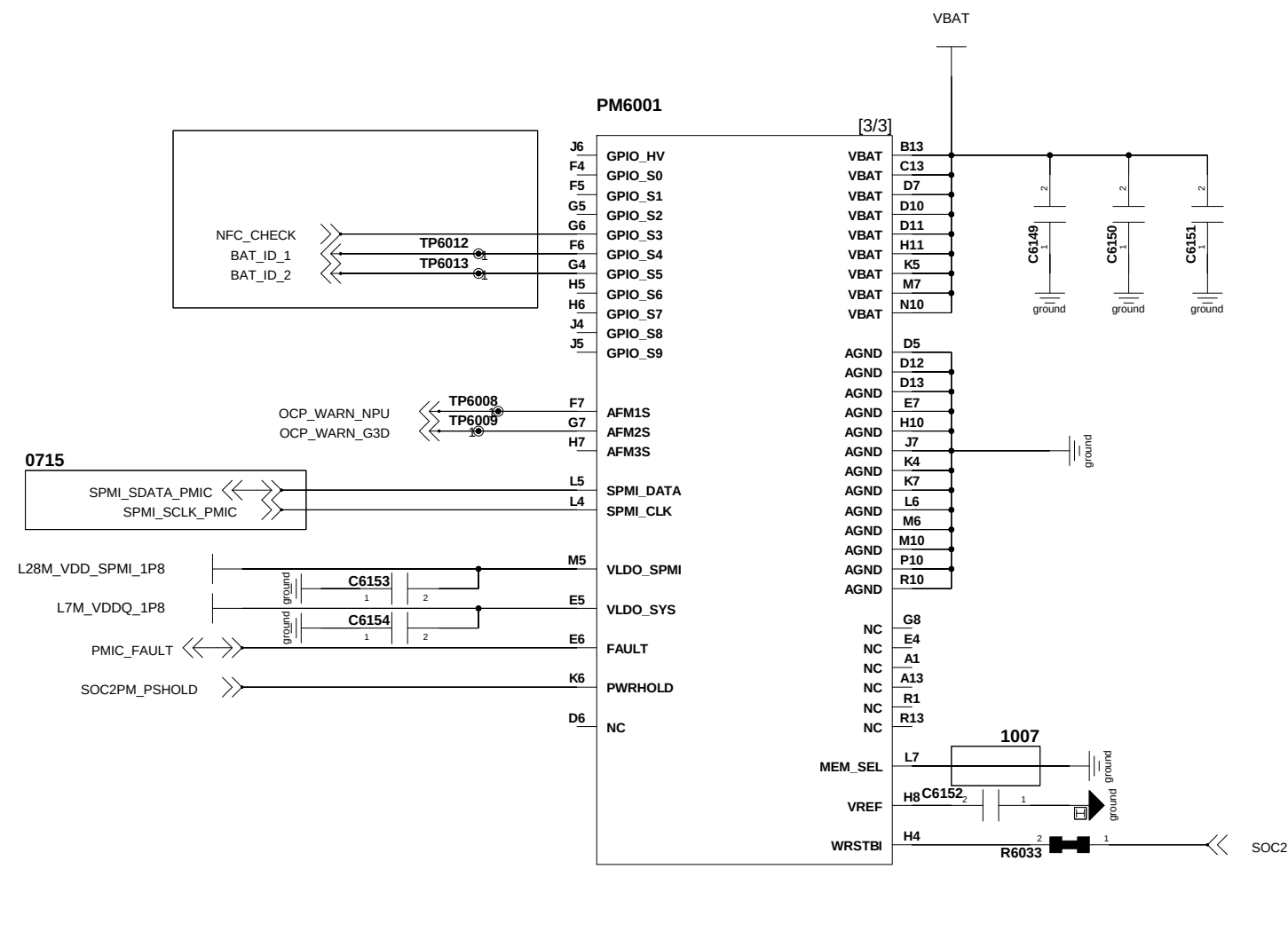
MOT DRV


SRBB1S_LDOIN_3P4
L7M_VDDQ_1P8
R6026
C6088
C6089
C6090
TP6021
U6005
L6016
L6017
MOTOR_RST
MOTOR_INT
MOTOR_SCL_1P8
MOTOR_SDA_1P8
MOTOR_P
MOTOR_N
C6091
C6092

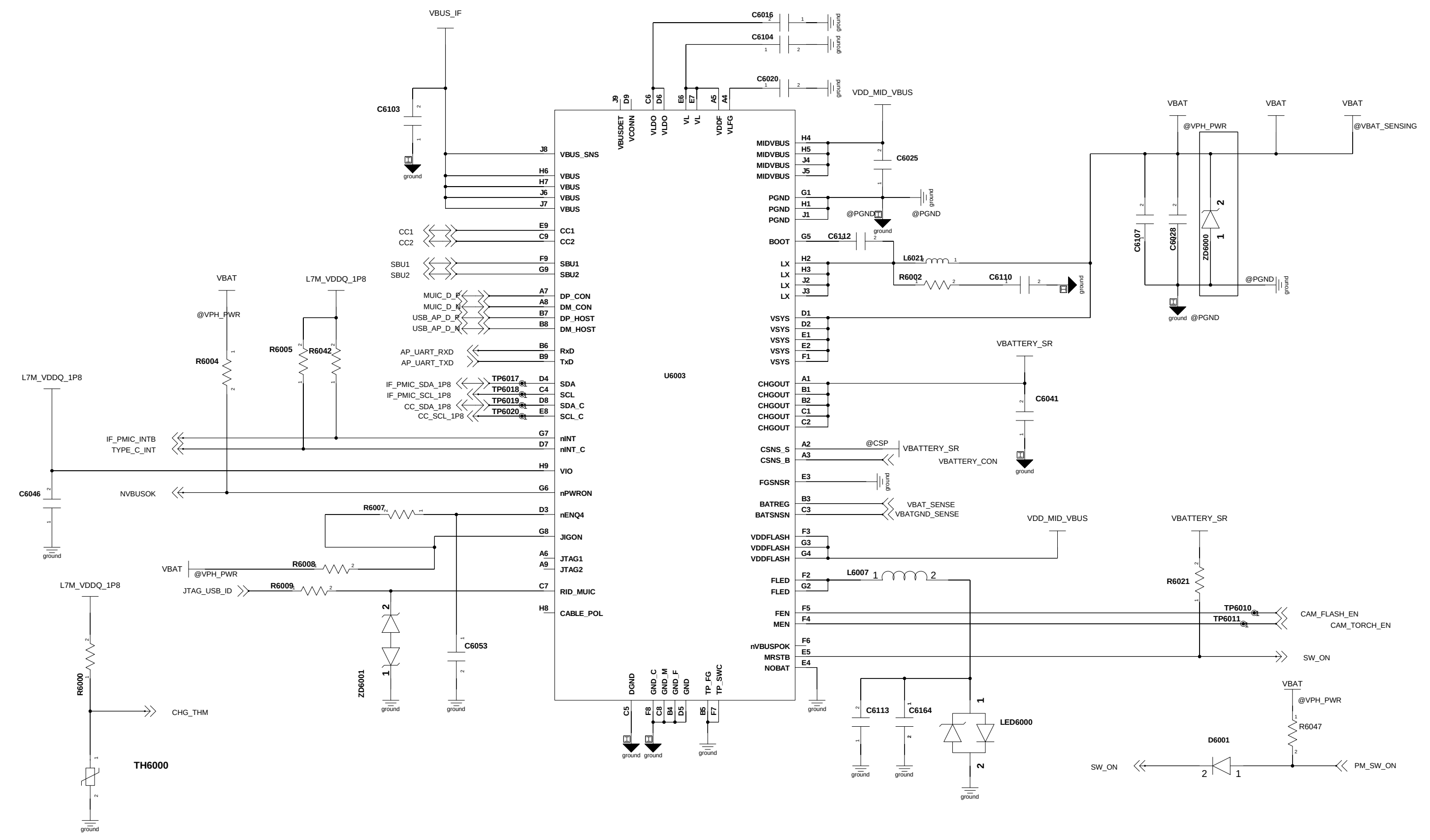

L7M_VDDQ_1P8
R6029
NFC_CHECK
R6016
ground

SUB PMIC


PM6001
[1/3]
[2/3]
VBAT
L6004
L6012
L6013
L6030
L6031
L6019
L6033
L6034
L6020
L6032
L6036
L6035
B1S_VDD_MIF_0P75
B2S_VDD_WLBT_0P75
B3S_VDD_NPU_0P75
B4_5S_VDD_G3D_0P75
B6S_VDD_CPU_CP_0P75
B7_8S_VDD_CP_0P75
B9S_VDD_INT_0P75
B10S_VDD_SRAM_0P75
SRBB1S_LDOIN_3P4
SRB2S_LDOIN_2P4

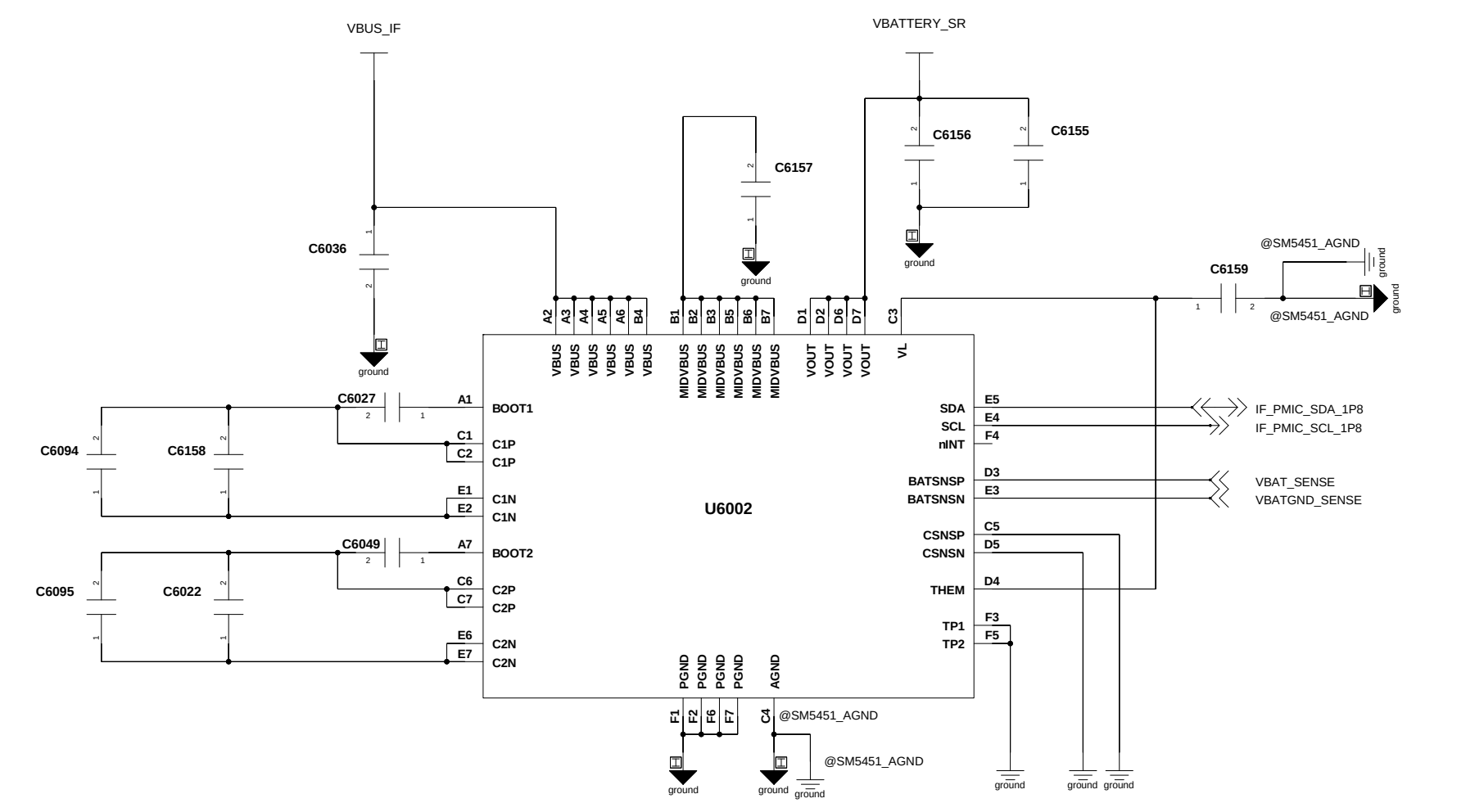

PM6001
[3/3]
VBAT
NFC_CHECK
BAT_ID_1
BAT_ID_2
OCP_WARN_NPU
OCP_WARN_G3D
0715
SPMI_SDATA_PMIC
SPMI_SCLK_PMIC
L28M_VDD_SPMI_1P8
L7M_VDDQ_1P8
PMIC_FAULT
SOC2PM_PSHOLD
1007
R6033
SOC2PM_WRSTBI

IF PMIC(SM5714)

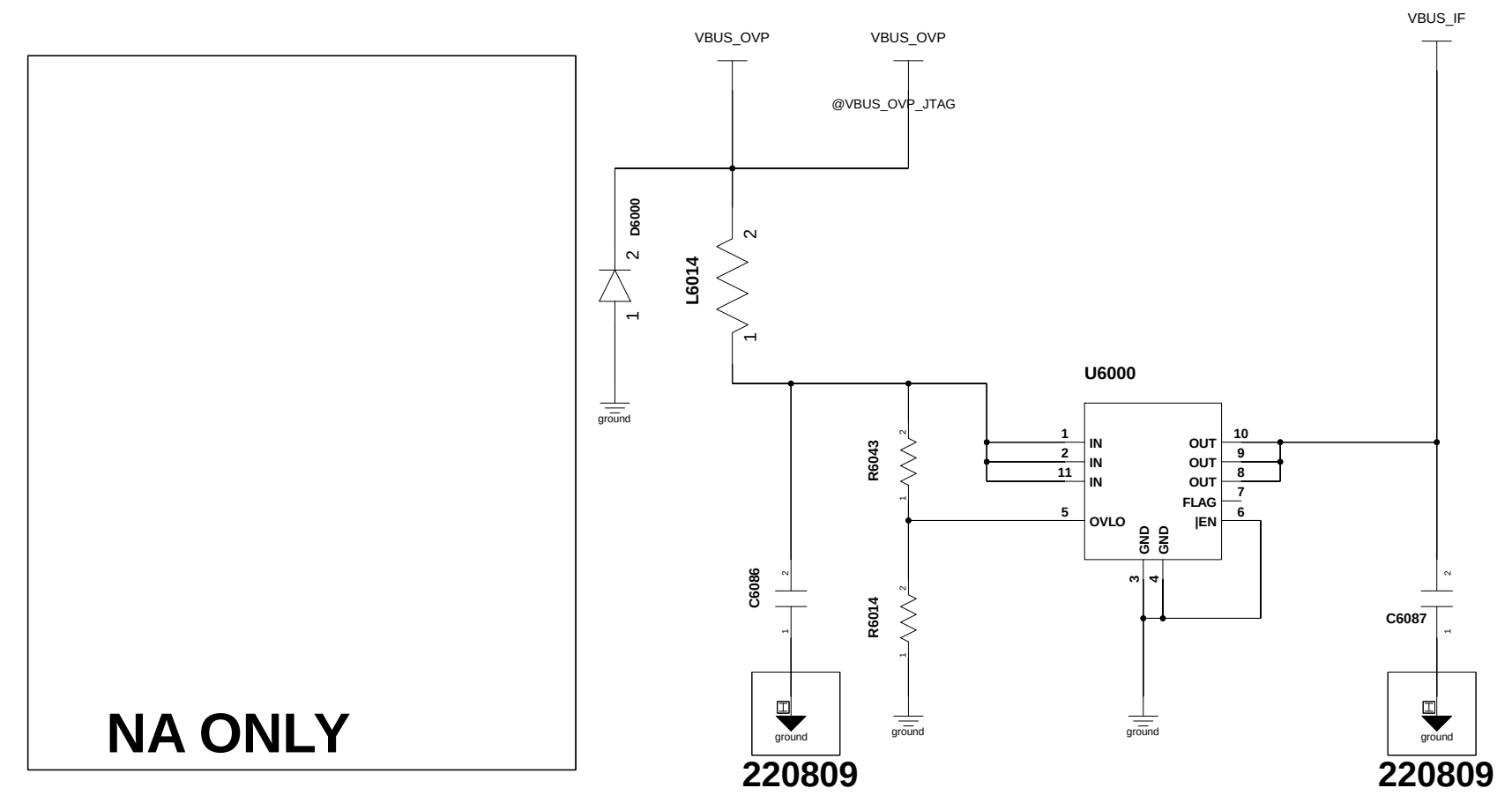

VBUS_IF
U6003
VDD_MID_VBUS
VBAT
@VPH_PWR
@VBAT_SENSING
ZD6000
L6021
R6002
C6110
VBATTERY_SR
CAM_FLASH_EN
CAM_TORCH_EN
SW_ON
PM_SW_ON
D6001
R6047
LED6000
L6007
TH6000
CHG_THM
ZD6001
IF_PMIC_INTB
TYPE_C_INT
NVBUSOK
JTAG_USB_ID

DIRECT CHARGER

(SM5451)

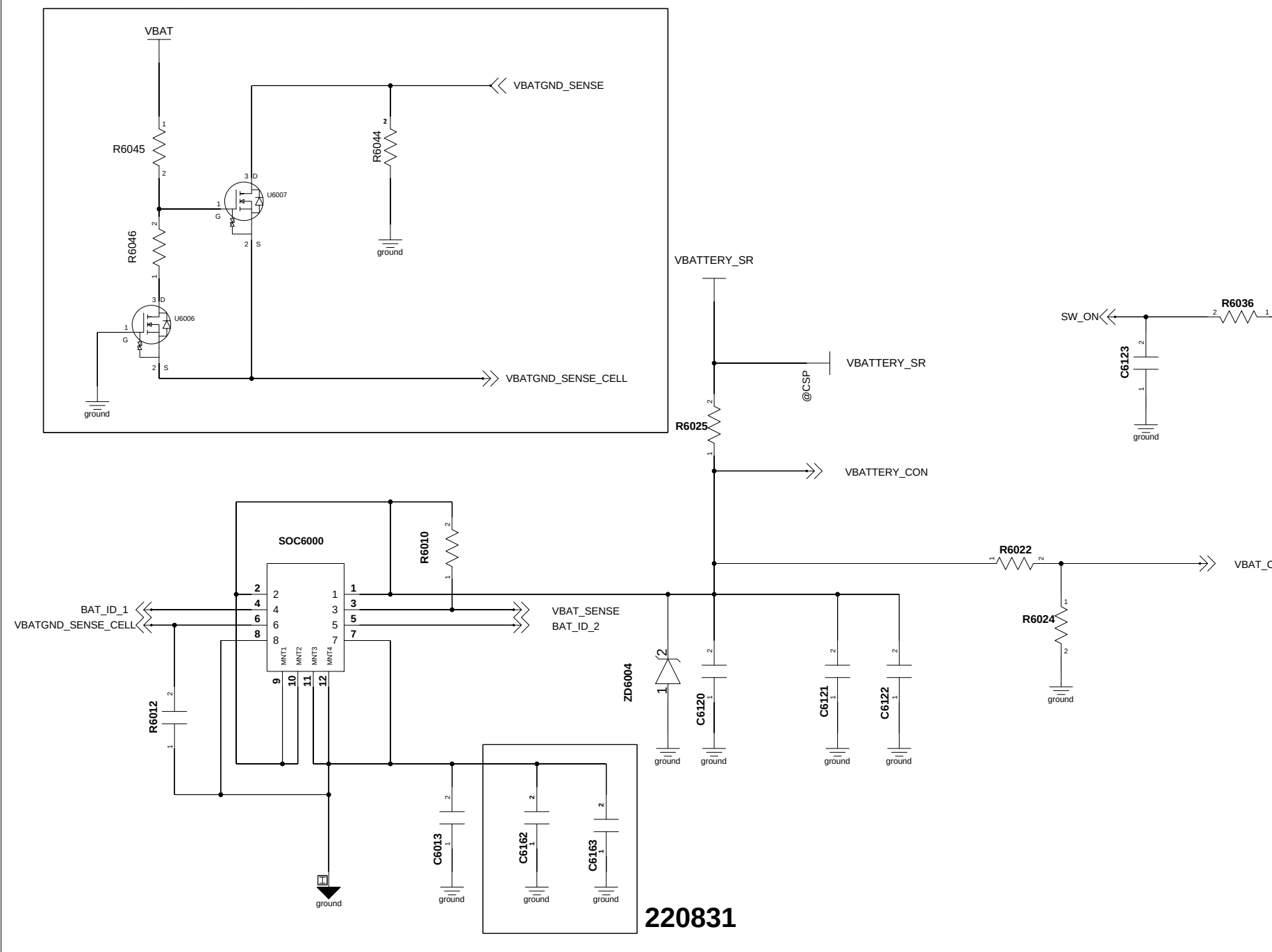

VBUS_IF
VBATTERY_SR
U6002
@SM5451_AGND
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
VBAT_SENSE
VBATGND_SENSE

OVP

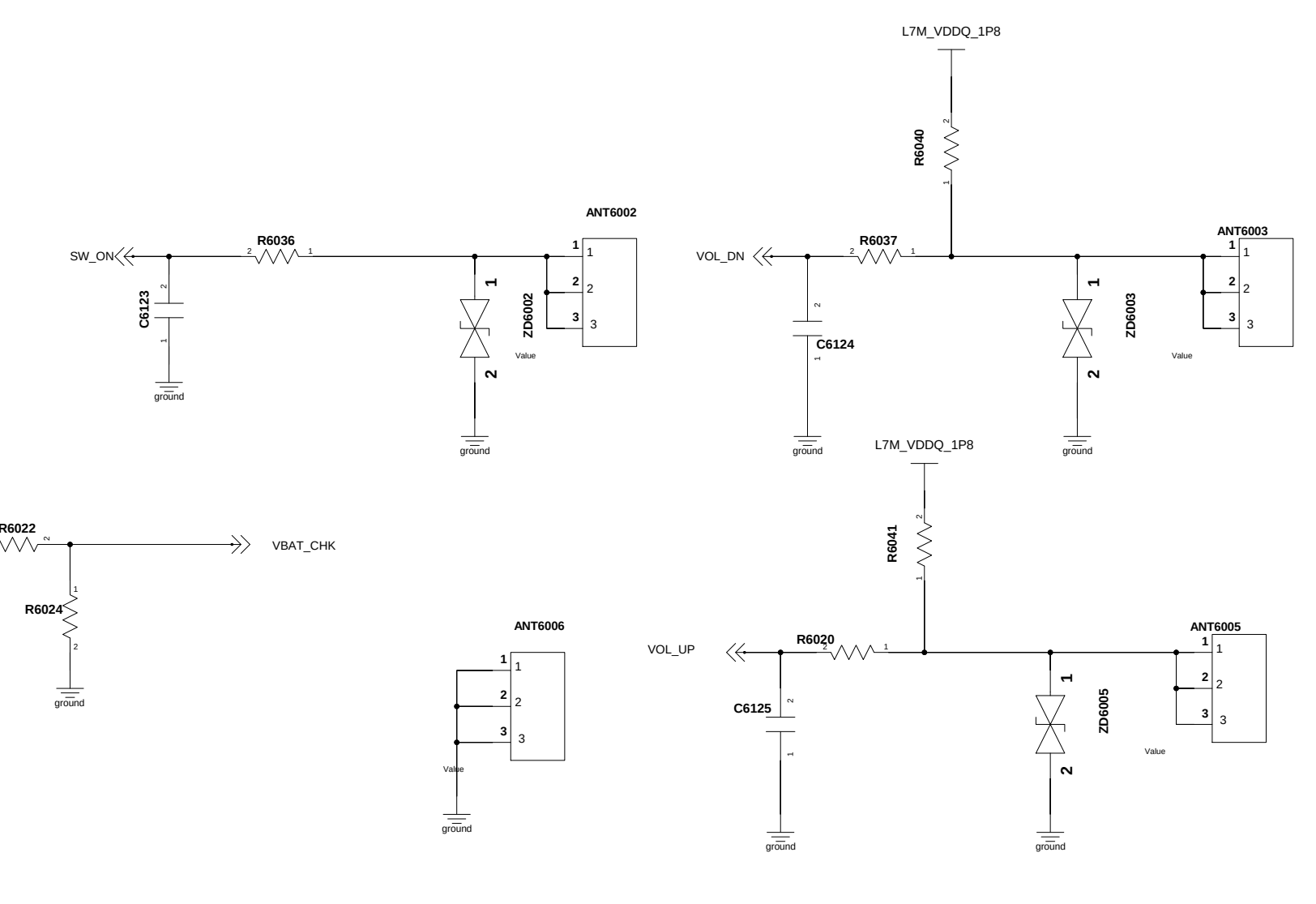

VBUS_OVP
VBUS_OVP
@VBUS_OVP_JTAG
VBUS_IF
D6000
L6014
U6000
R6043
R6014
C6086
C6087
NA ONLY
220809
220809

Cell leakage Sensing 1006

BATTERY CON


VBAT
R6045
R6046
U6007
U6006
R6044
VBATGND_SENSE
VBATGND_SENSE_CELL
VBATTERY_SR
@CSP
R6025
VBATTERY_CON
R6022
VBAT_CHK
R6024
SOC6000
BAT_ID_1
VBATGND_SENSE_CELL
VBAT_SENSE
BAT_ID_2
R6010
R6012
ZD6004
C6120
C6121
C6122
C6013
C6162
C6163
220831

POWER & VOL KEY


SW_ON
R6036
ANT6002
ZD6002
C6123
L7M_VDDQ_1P8
R6040
VOL_DN
R6037
ANT6003
ZD6003
C6124
R6041
VOL_UP
R6020
ANT6005
ZD6005
C6125
ANT6006

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: POWER/MOT/KEY
DOC CTRL CHK:	Size:
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number:

Changed by: Date Changed: Time Changed:

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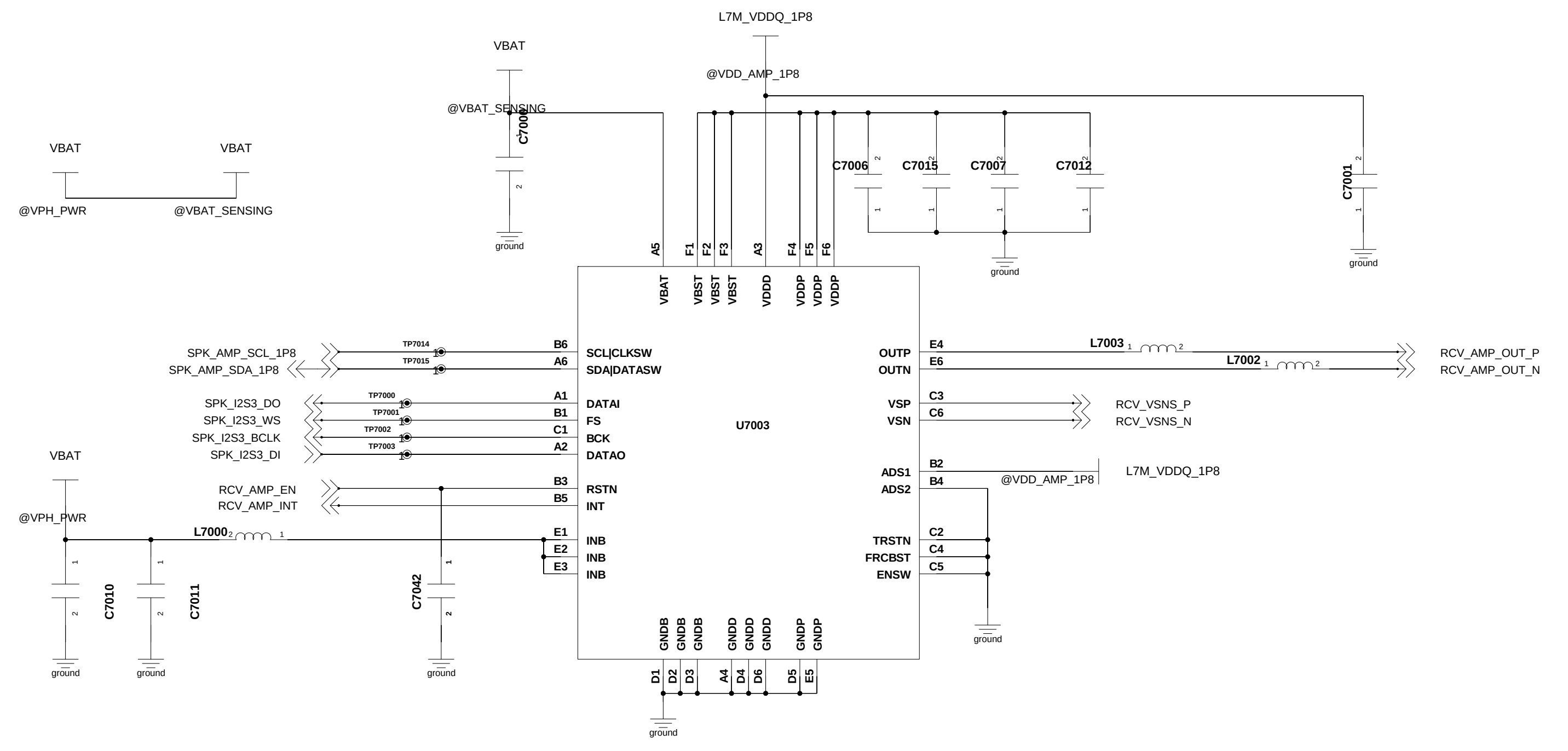
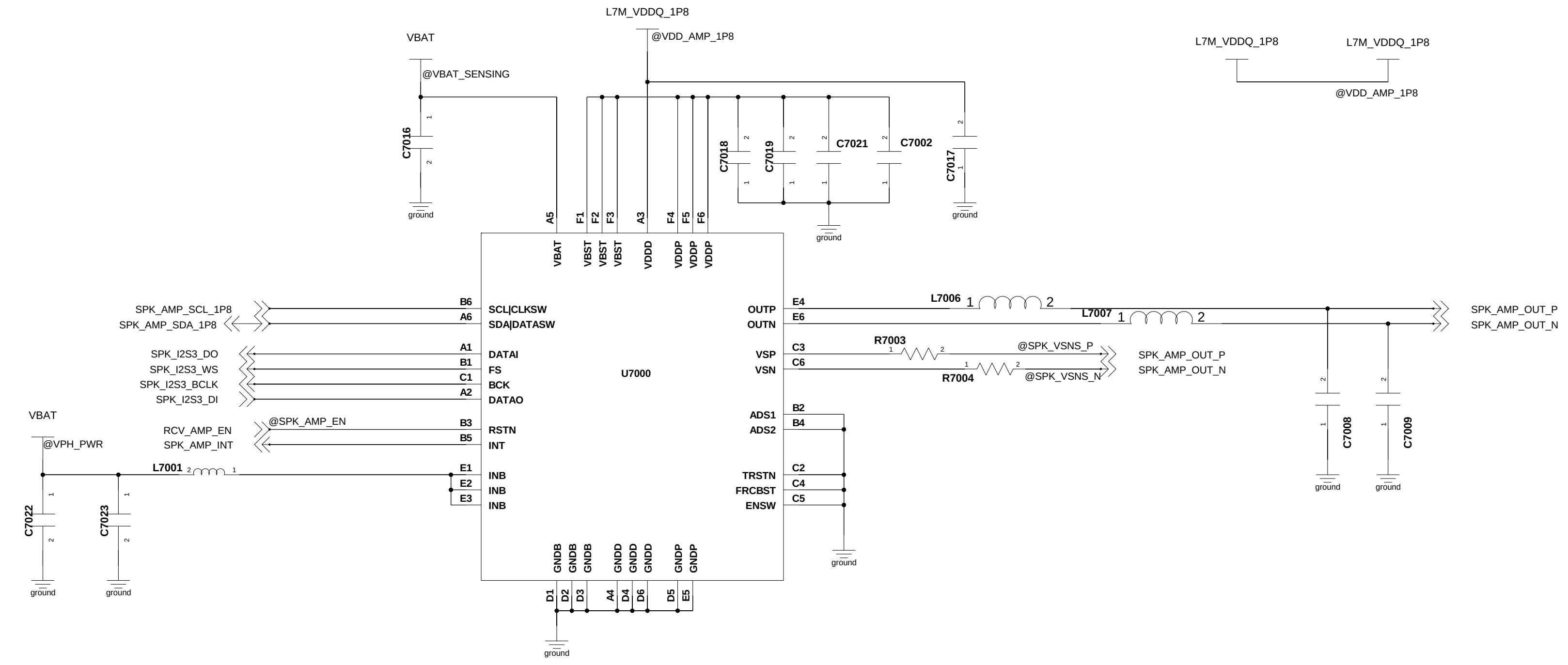
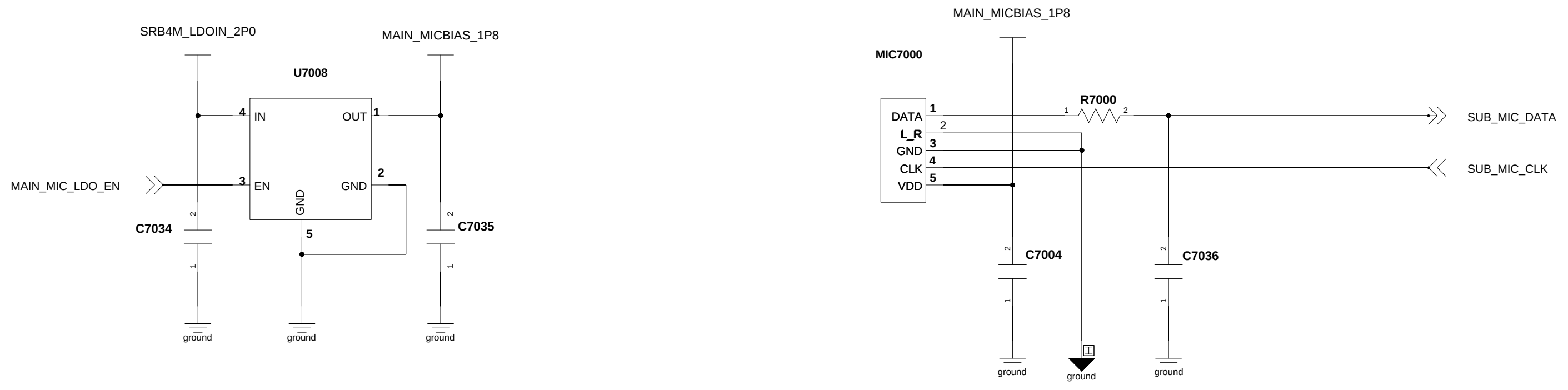
AUDIO & SENSOR PART


L7M_VDDQ_1P8
@VDD_AMP_1P8
VBAT
@VBAT_SENSING
C7000
VBAT
VBAT
@VPH_PWR
@VBAT_SENSING
C7006
C7015
C7007
C7012
C7001
U7003
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
TP7014
TP7015
TP7000
TP7001
TP7002
TP7003
RCV_AMP_EN
RCV_AMP_INT
VBAT
@VPH_PWR
L7000
C7010
C7011
C7042
SCL|CLKSW
SDA|DATASW
DATAI
FS
BCK
DATAO
RSTN
INT
INB
OUTP
OUTN
VSP
VSN
ADS1
ADS2
TRSTN
FRCBST
ENSW
L7003
L7002
RCV_AMP_OUT_P
RCV_AMP_OUT_N
RCV_VSNS_P
RCV_VSNS_N
@VDD_AMP_1P8
L7M_VDDQ_1P8
ground

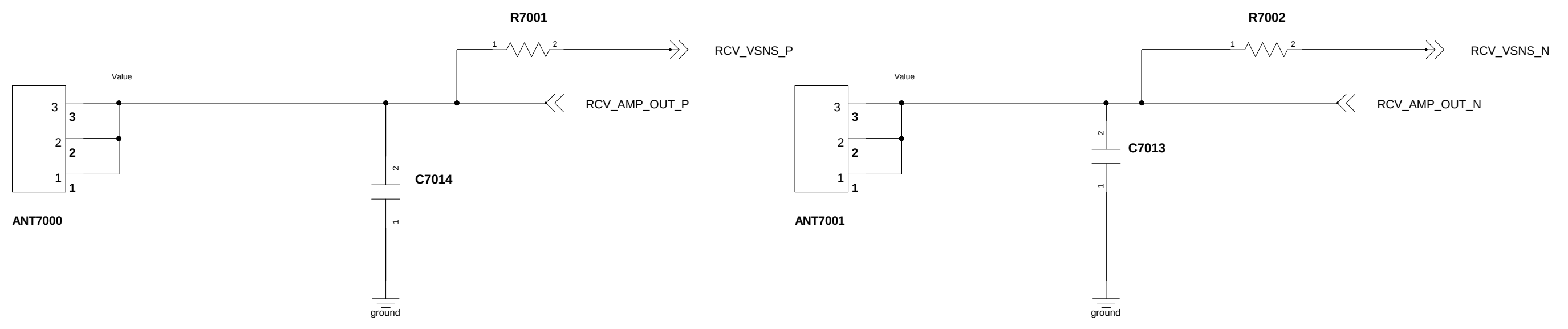
RCV AMP


L7M_VDDQ_1P8
@VDD_AMP_1P8
VBAT
@VBAT_SENSING
L7M_VDDQ_1P8
L7M_VDDQ_1P8
@VDD_AMP_1P8
C7016
C7018
C7019
C7021
C7002
C7017
U7000
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
@SPK_AMP_EN
RCV_AMP_EN
SPK_AMP_INT
VBAT
@VPH_PWR
L7001
C7022
C7023
L7006
L7007
R7003
R7004
@SPK_VSNS_P
@SPK_VSNS_N
SPK_AMP_OUT_P
SPK_AMP_OUT_N
C7008
C7009
ground

SPK AMP

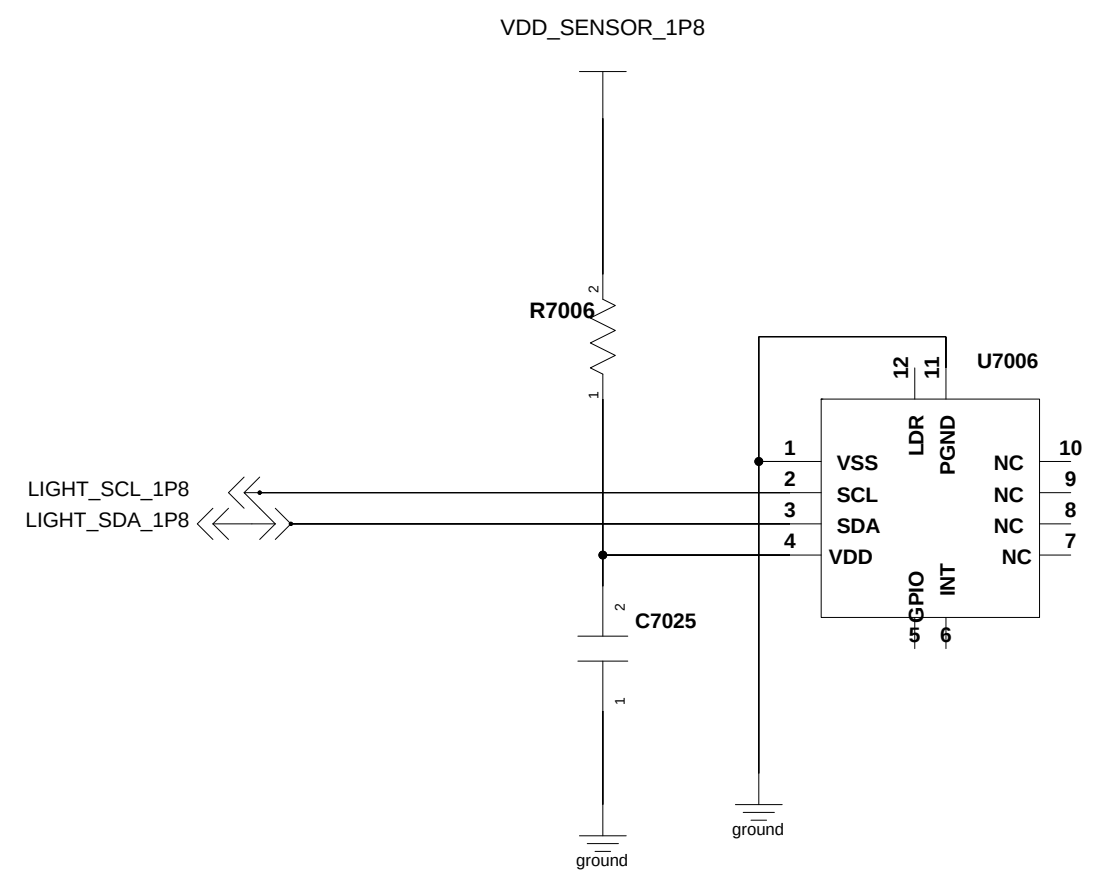

SRB4M_LDOIN_2P0
MAIN_MICBIAS_1P8
U7008
IN
OUT
EN
GND
MAIN_MIC_LDO_EN
C7034
C7035
MAIN_MICBIAS_1P8
MIC7000
DATA
L_R
GND
CLK
VDD
R7000
SUB_MIC_DATA
SUB_MIC_CLK
C7004
C7036
ground

2nd MIC


R7001
RCV_VSNS_P
RCV_AMP_OUT_P
ANT7000
C7014
R7002
RCV_VSNS_N
RCV_AMP_OUT_N
ANT7001
C7013
Value
ground

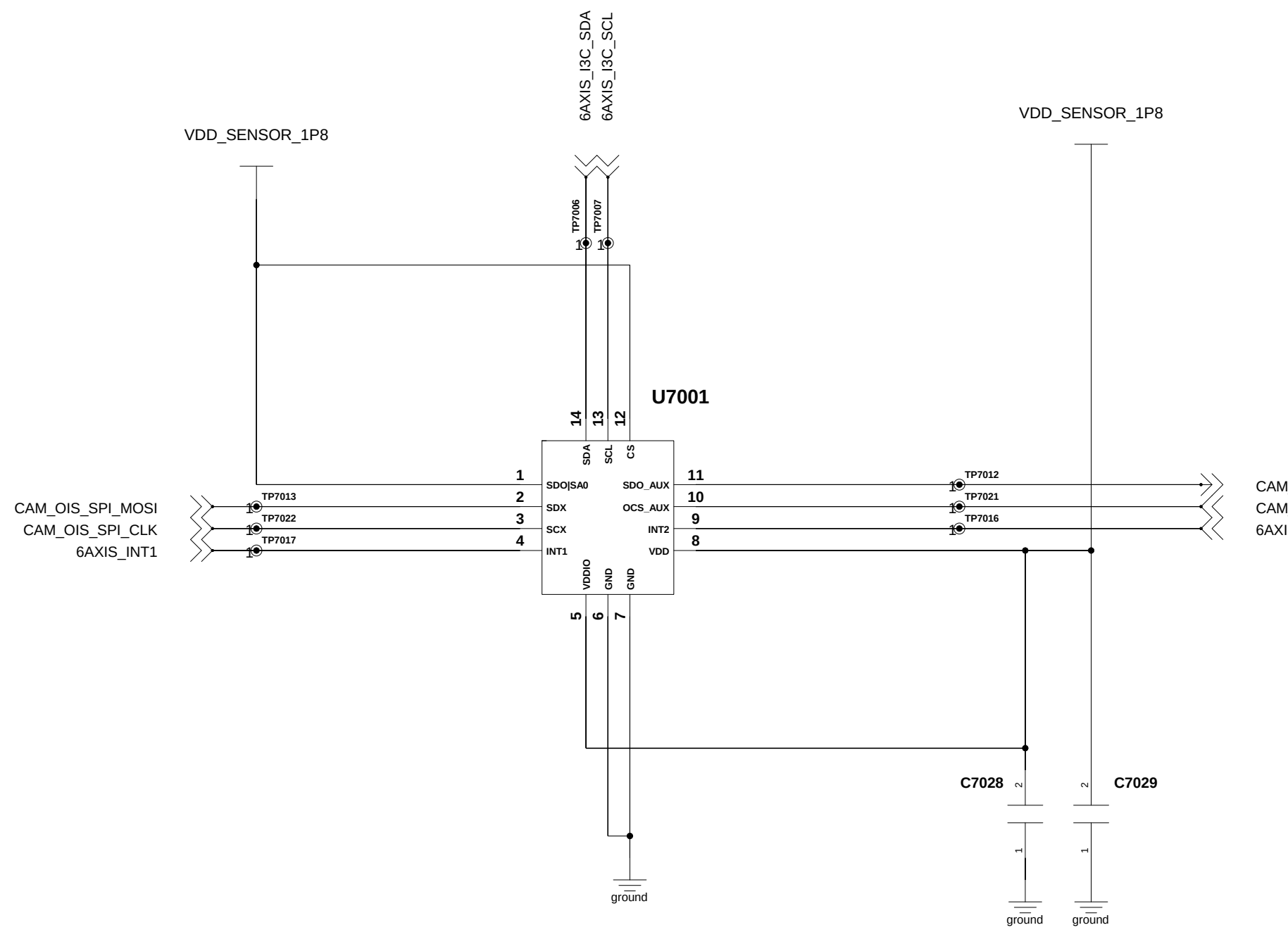
RCV CONTACT

SENSOR

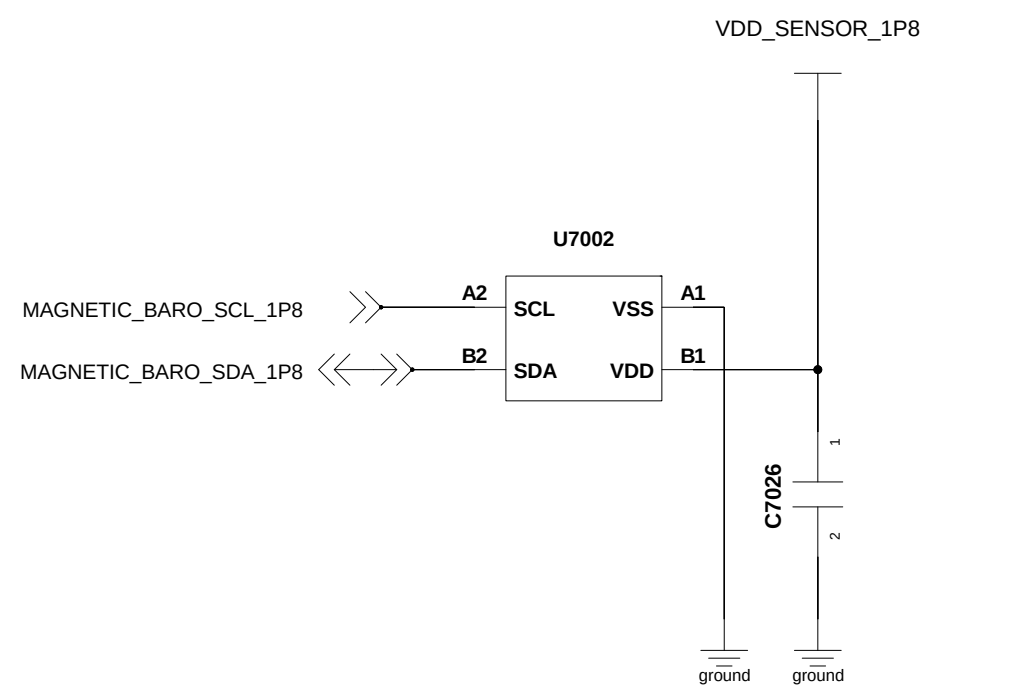

VDD_SENSOR_1P8
R7006
U7006
LIGHT_SCL_1P8
LIGHT_SDA_1P8
VSS
SCL
SDA
VDD
LDR
PGND
NC
GPIO
INT
C7025
ground

AMBIENT SENSOR

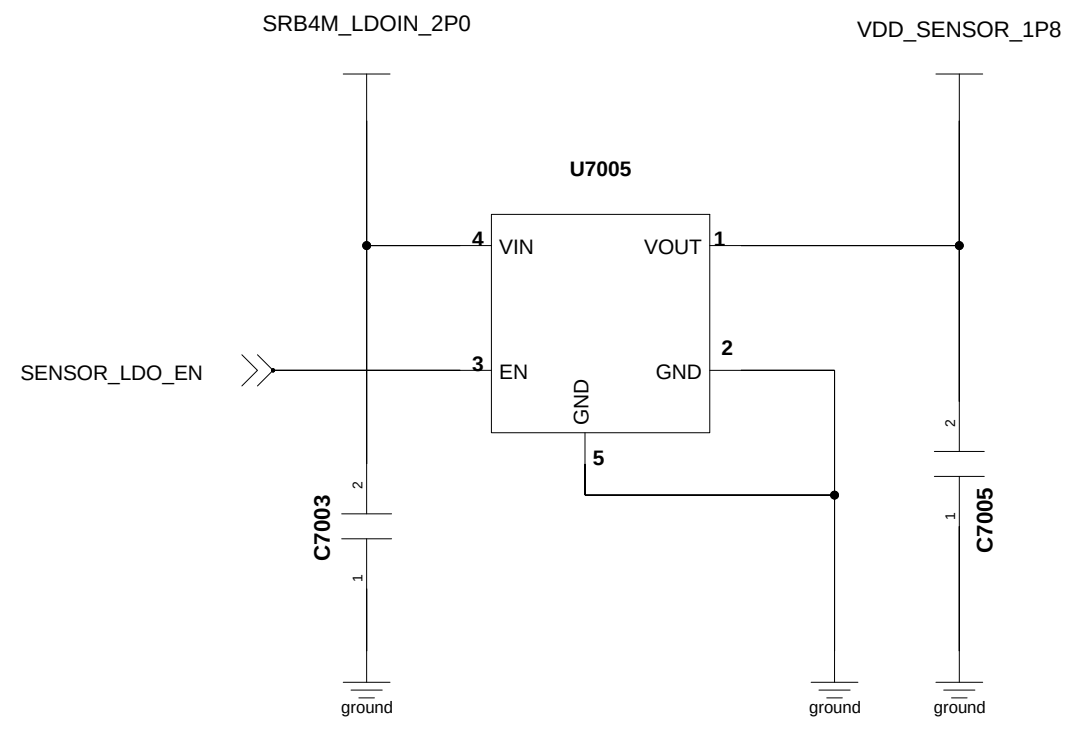
HALL IC


6AXIS_I3C_SDA
6AXIS_I3C_SCL
VDD_SENSOR_1P8
VDD_SENSOR_1P8
TP7006
TP7007
U7001
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_CLK
6AXIS_INT1
TP7013
TP7022
TP7017
SDO|SA0
SDX
SCX
INT1
SDO_AUX
OCS_AUX
INT2
VDD
VDDIO
GND
TP7012
TP7021
TP7016
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
6AXIS_INT2
C7028
C7029
ground

ACC&GYRO SENSOR

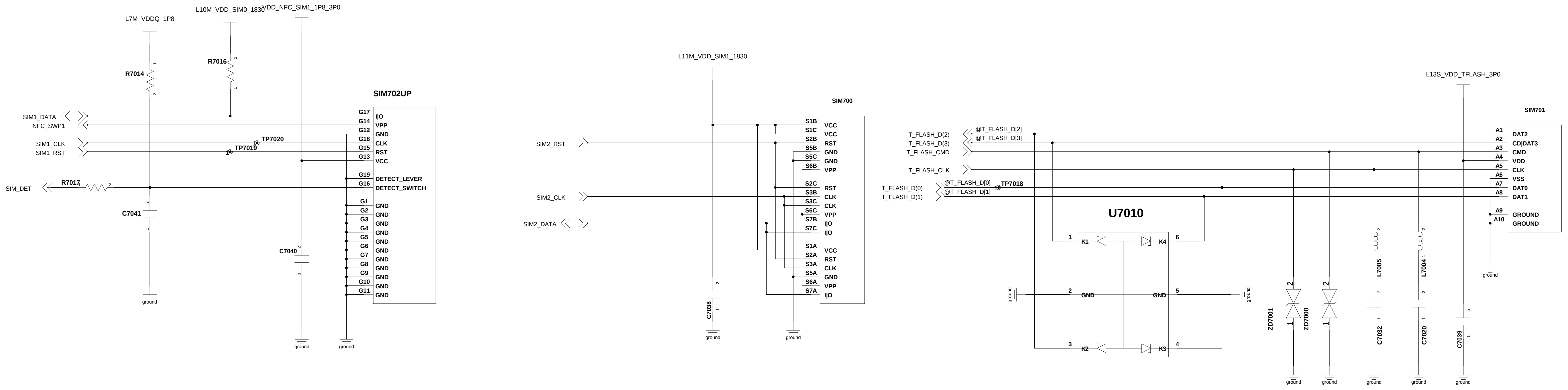

VDD_SENSOR_1P8
U7002
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
SCL
SDA
VSS
VDD
C7026
ground

MAGNETIC SENSOR


SRB4M_LDOIN_2P0
VDD_SENSOR_1P8
U7005
VIN
VOUT
EN
GND
SENSOR_LDO_EN
C7003
C7005
ground

NA ONLY

DUAL SIM / T-FLASH (3in2)


L7M_VDDQ_1P8
L10M_VDD_SIM0_1830
VDD_NFC_SIM1_1P8_3P0
R7014
R7016
SIM702UP
SIM1_DATA
NFC_SWP1
SIM1_CLK
SIM1_RST
TP7020
TP7019
SIM_DET
R7017
C7041
C7040
I|O
VPP
GND
CLK
RST
VCC
DETECT_LEVER
DETECT_SWITCH
L11M_VDD_SIM1_1830
SIM700
SIM2_RST
SIM2_CLK
SIM2_DATA
C7038
T_FLASH_D(2)
T_FLASH_D(3)
T_FLASH_CMD
T_FLASH_CLK
T_FLASH_D(0)
T_FLASH_D(1)
@T_FLASH_D[2]
@T_FLASH_D[3]
@T_FLASH_D[0]
@T_FLASH_D[1]
TP7018
U7010
K1
K2
K3
K4
GND
ZD7001
ZD7000
L7005
L7004
C7032
C7020
C7039
L13S_VDD_TFLASH_3P0
SIM701
DAT2
CD|DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
GROUND
ground

Stacked Dual SIM SOCKET (Reverse)

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	
MFG ENGR CHK:	

Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:	Page:

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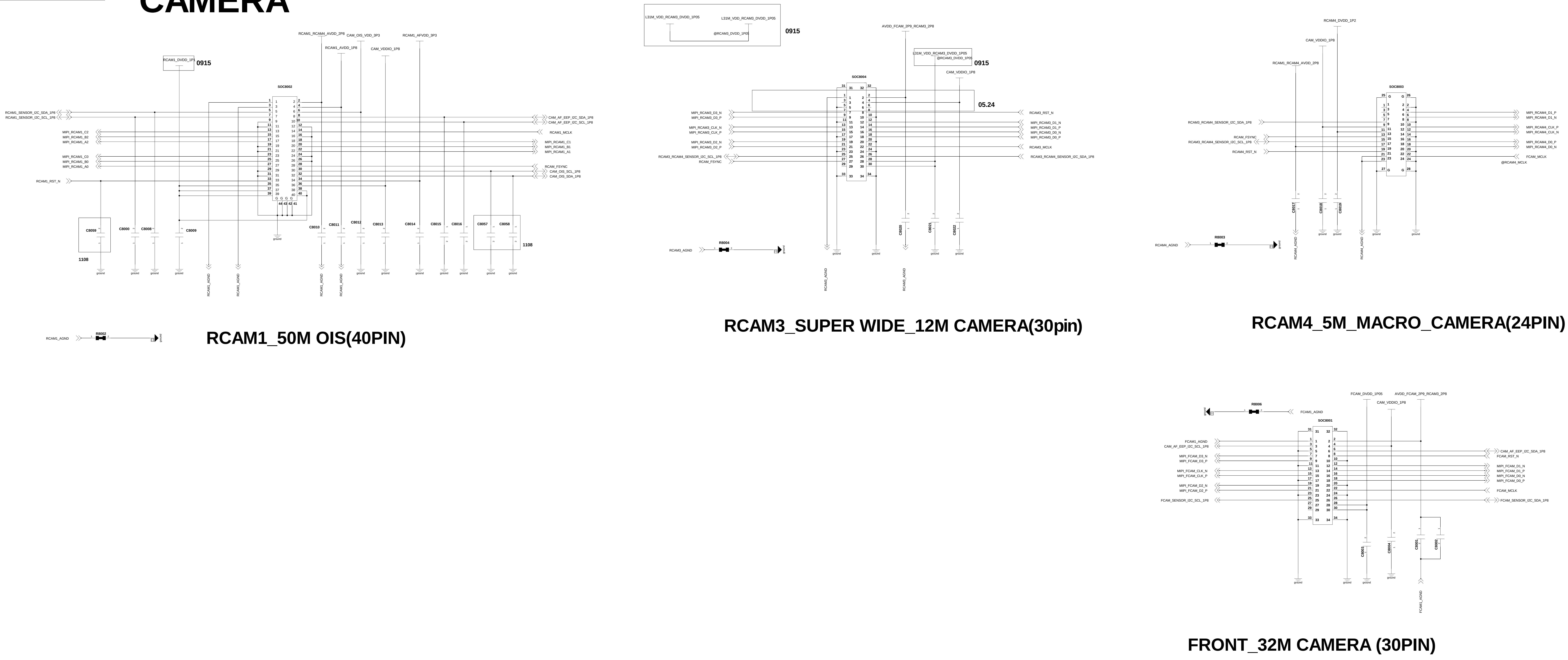
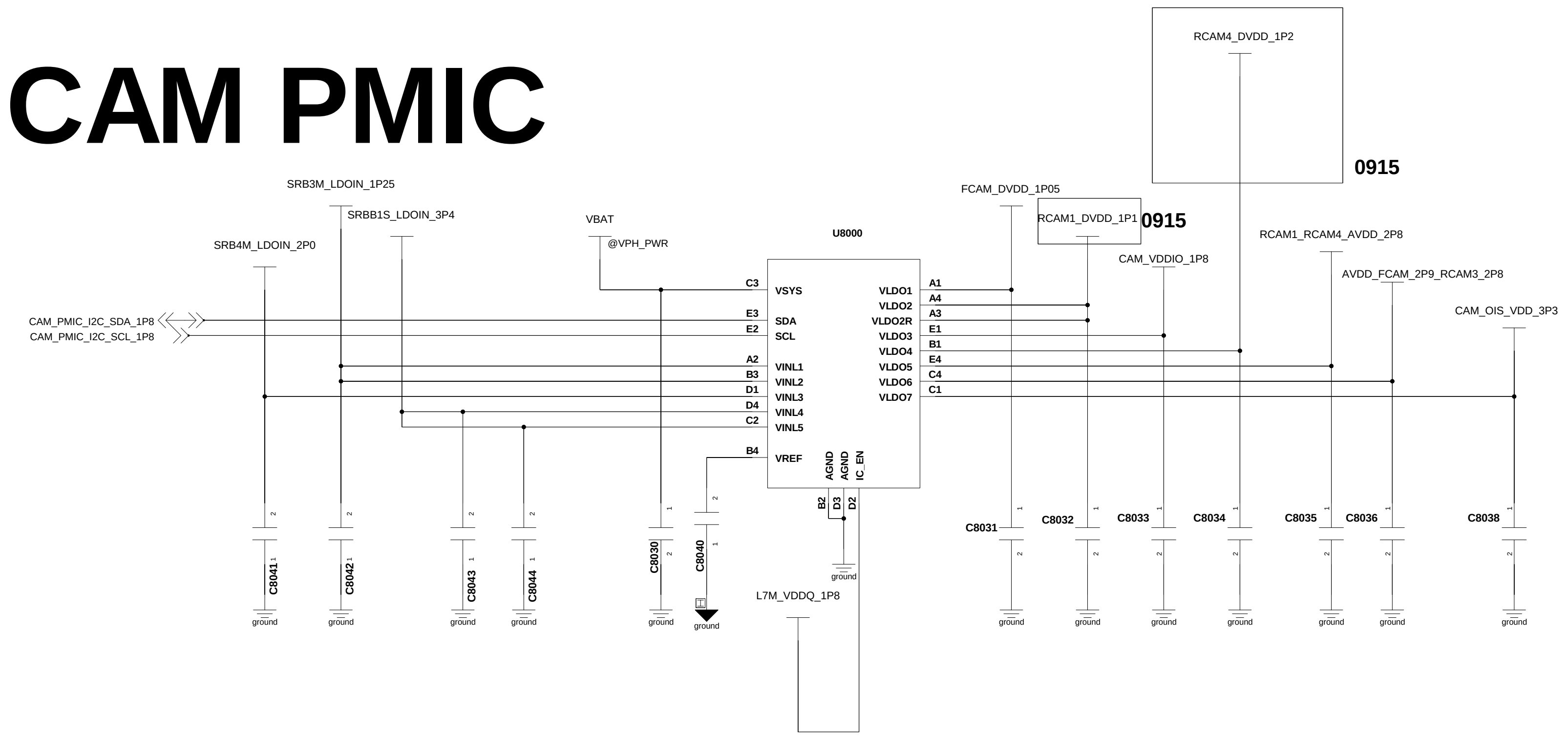
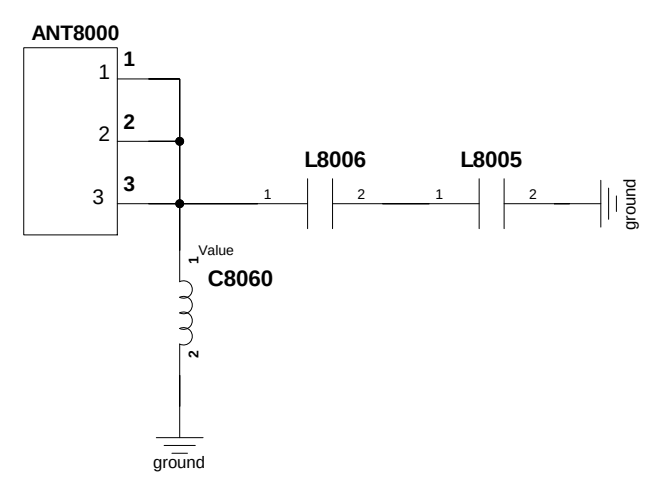
CAMERA


SOC8002
RCAM1_DVDD_1P1
0915
RCAM1_RCAM4_AVDD_2P8
CAM_OIS_VDD_3P3
RCAM1_AFVDD_3P3
RCAM1_AVDD_1P8
CAM_VDDIO_1P8
RCAM1_SENSOR_I2C_SDA_1P8
RCAM1_SENSOR_I2C_SCL_1P8
MIPI_RCAM1_C2
MIPI_RCAM1_B2
MIPI_RCAM1_A2
MIPI_RCAM1_C0
MIPI_RCAM1_B0
MIPI_RCAM1_A0
RCAM1_RST_N
CAM_AF_EEP_I2C_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
RCAM1_MCLK
MIPI_RCAM1_C1
MIPI_RCAM1_B1
MIPI_RCAM1_A1
RCAM_FSYNC
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
C8059
1108
C8000
C8008
C8009
C8010
C8011
C8012
C8013
C8014
C8015
C8016
C8057
C8058
1108
RCAM1_AGND
ground
R8002
RCAM1_AGND
RCAM1_50M OIS(40PIN)
L31M_VDD_RCAM3_DVDD_1P05
L31M_VDD_RCAM3_DVDD_1P05
@RCAM3_DVDD_1P05
0915
AVDD_FCAM_2P9_RCAM3_2P8
L31M_VDD_RCAM3_DVDD_1P05
@RCAM3_DVDD_1P05
0915
CAM_VDDIO_1P8
SOC8004
05.24
MIPI_RCAM3_D3_N
MIPI_RCAM3_D3_P
MIPI_RCAM3_CLK_N
MIPI_RCAM3_CLK_P
MIPI_RCAM3_D2_N
MIPI_RCAM3_D2_P
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM_FSYNC
RCAM3_RST_N
MIPI_RCAM3_D1_N
MIPI_RCAM3_D1_P
MIPI_RCAM3_D0_N
MIPI_RCAM3_D0_P
RCAM3_MCLK
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
C8020
C8021
C8022
R8004
RCAM3_AGND
RCAM3_AGND
RCAM3_AGND
RCAM3_SUPER WIDE_12M CAMERA(30pin)
RCAM4_DVDD_1P2
CAM_VDDIO_1P8
RCAM1_RCAM4_AVDD_2P8
SOC8003
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
RCAM_FSYNC
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM4_RST_N
MIPI_RCAM4_D1_P
MIPI_RCAM4_D1_N
MIPI_RCAM4_CLK_P
MIPI_RCAM4_CLK_N
MIPI_RCAM4_D0_P
MIPI_RCAM4_D0_N
FCAM_MCLK
@RCAM4_MCLK
C8017
C8018
C8019
R8003
RCAM4_AGND
RCAM4_AGND
RCAM4_AGND
RCAM4_5M_MACRO_CAMERA(24PIN)
FCAM_DVDD_1P05
AVDD_FCAM_2P9_RCAM3_2P8
CAM_VDDIO_1P8
R8006
FCAM1_AGND
SOC8001
FCAM1_AGND
CAM_AF_EEP_I2C_SCL_1P8
MIPI_FCAM_D3_N
MIPI_FCAM_D3_P
MIPI_FCAM_CLK_N
MIPI_FCAM_CLK_P
MIPI_FCAM_D2_N
MIPI_FCAM_D2_P
FCAM_SENSOR_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
FCAM_RST_N
MIPI_FCAM_D1_N
MIPI_FCAM_D1_P
MIPI_FCAM_D0_N
MIPI_FCAM_D0_P
FCAM_MCLK
FCAM_SENSOR_I2C_SDA_1P8
C8003
C8004
C8001
C8002
FCAM1_AGND
FRONT_32M CAMERA (30PIN)

CAM PMIC


RCAM4_DVDD_1P2
0915
SRB3M_LDOIN_1P25
SRBB1S_LDOIN_3P4
SRB4M_LDOIN_2P0
VBAT
@VPH_PWR
U8000
FCAM_DVDD_1P05
RCAM1_DVDD_1P1
0915
CAM_VDDIO_1P8
RCAM1_RCAM4_AVDD_2P8
AVDD_FCAM_2P9_RCAM3_2P8
CAM_OIS_VDD_3P3
CAM_PMIC_I2C_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
C3 VSYS
E3 SDA
E2 SCL
A2 VINL1
B3 VINL2
D1 VINL3
D4 VINL4
C2 VINL5
B4 VREF
AGND
AGND
IC_EN
B2
D3
D2
VLDO1 A1
VLDO2 A4
VLDO2R A3
VLDO3 E1
VLDO4 B1
VLDO5 E4
VLDO6 C4
VLDO7 C1
C8041
C8042
C8043
C8044
C8030
C8040
L7M_VDDQ_1P8
C8031
C8032
C8033
C8034
C8035
C8036
C8038
ground

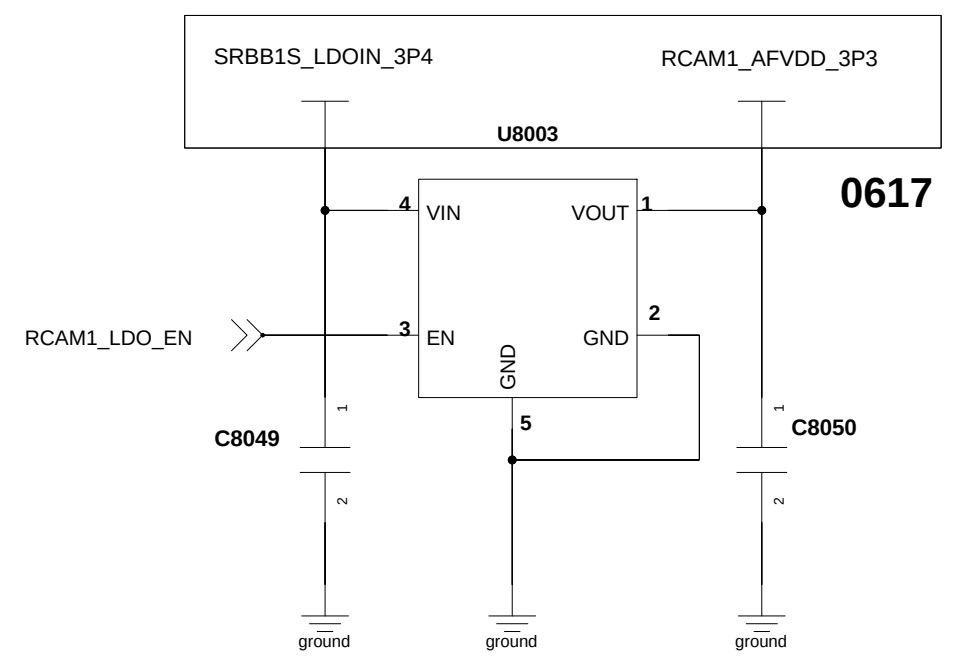
DECO CONTACT


ANT8000
L8006
L8005
Value
C8060
ground

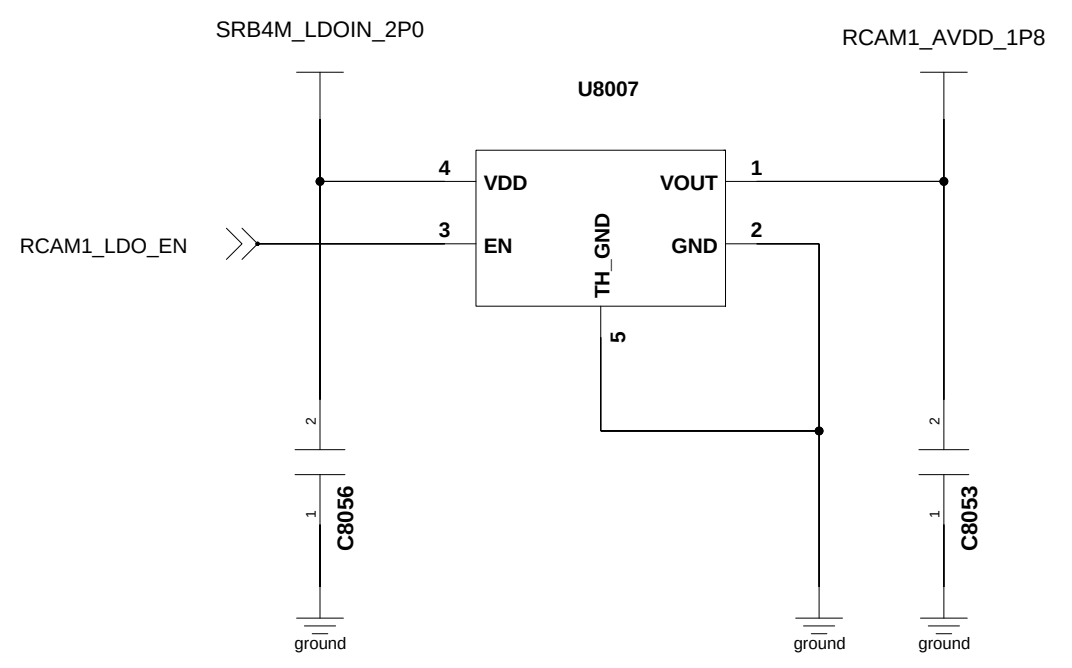
220915

RCAM4_AVDD

RCAM3_AVDD


SRBB1S_LDOIN_3P4
RCAM1_AFVDD_3P3
U8003
0617
VIN
VOUT
RCAM1_LDO_EN
EN
GND
GND
C8049
C8050
ground

RCAM1_AF


SRB4M_LDOIN_2P0
U8007
RCAM1_AVDD_1P8
VDD
VOUT
RCAM1_LDO_EN
EN
TH_GND
GND
C8056
C8053
ground

RCAM1_AVDD

CAM MCU_DVDD

0617

CAM MCU

RCAM4_DVDD

220915

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:

A54

TITLE:
CAMERA

Size:

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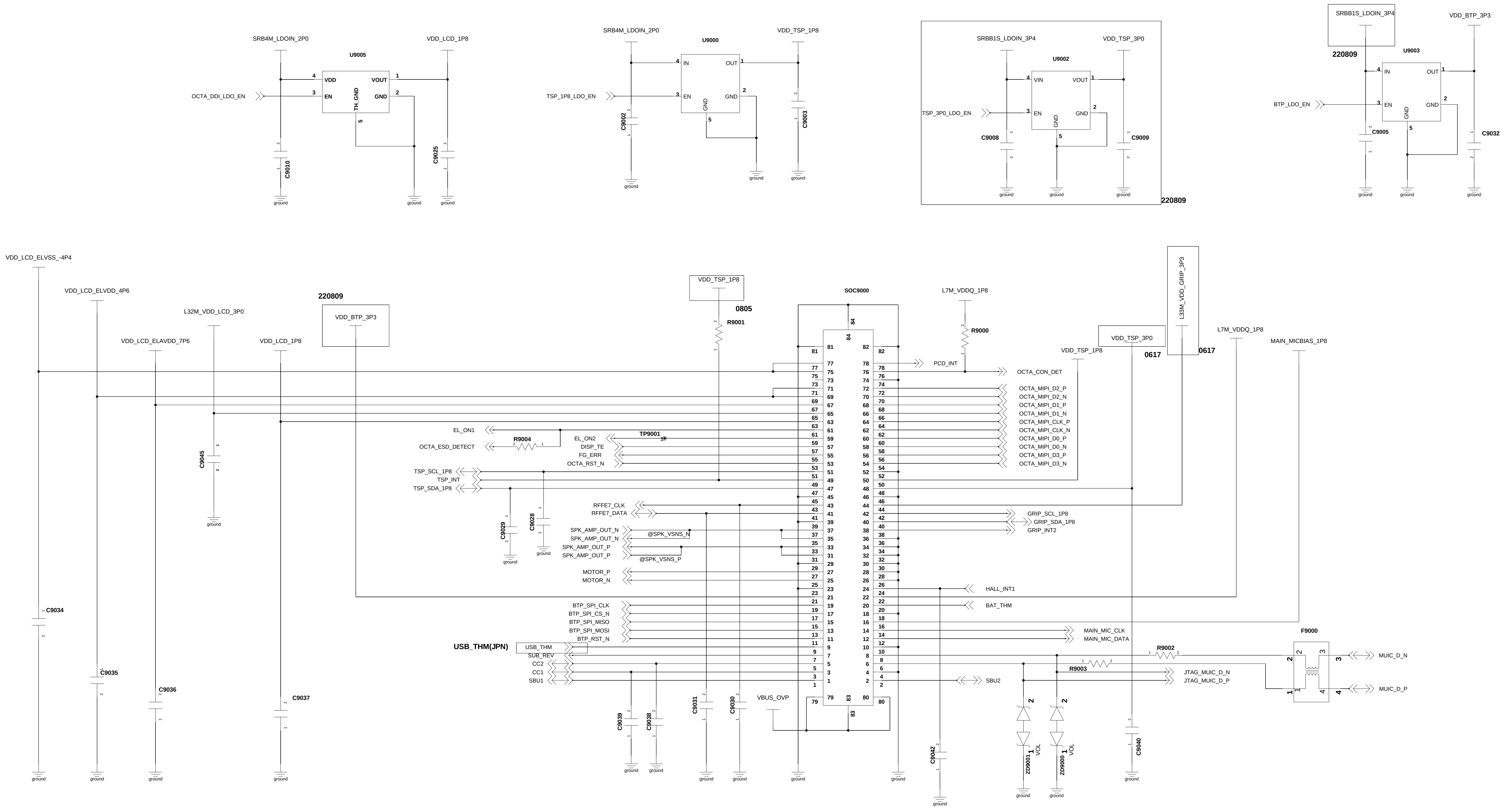
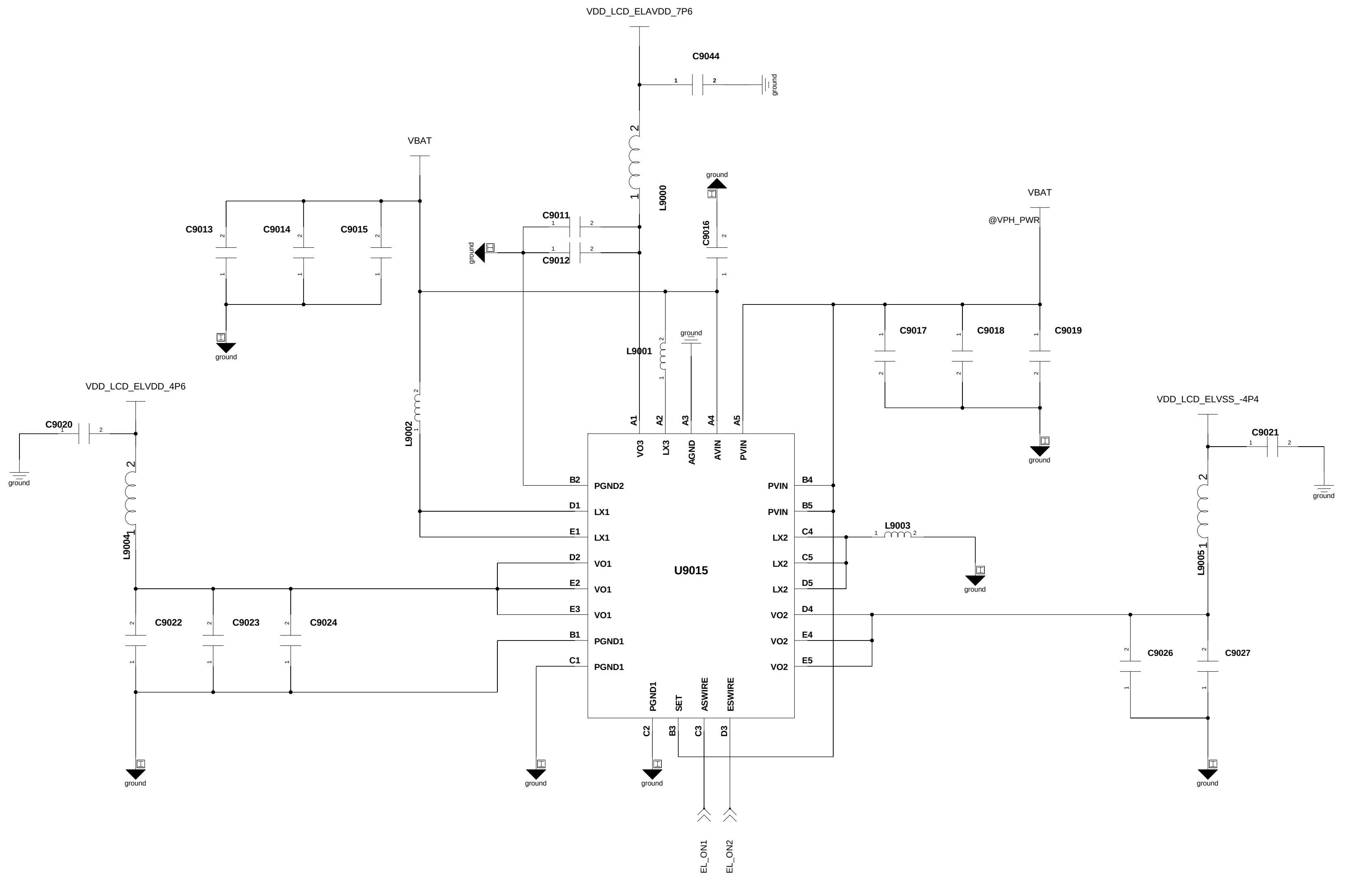
C2C (Conn to Conn FPCB)


SRB4M_LDOIN_2P0
VDD_LCD_1P8
U9005
VDD
VOUT
EN
GND
TH_GND
OCTA_DDI_LDO_EN
C9010
C9025
U9000
IN
OUT
GND
TSP_1P8_LDO_EN
C9002
VDD_TSP_1P8
C9003
SRBB1S_LDOIN_3P4
VDD_TSP_3P0
U9002
VIN
TSP_3P0_LDO_EN
C9008
C9009
220809
SRBB1S_LDOIN_3P4
220809
U9003
VDD_BTP_3P3
BTP_LDO_EN
C9005
C9032
ground
VDD_LCD_ELVSS_-4P4
VDD_LCD_ELVDD_4P6
L32M_VDD_LCD_3P0
VDD_LCD_ELAVDD_7P6
VDD_LCD_1P8
220809
VDD_BTP_3P3
VDD_TSP_1P8
0805
R9001
SOC9000
L7M_VDDQ_1P8
R9000
L33M_VDD_GRIP_3P3
VDD_TSP_3P0
0617
0617
VDD_TSP_1P8
L7M_VDDQ_1P8
MAIN_MICBIAS_1P8
C9045
EL_ON1
R9004
OCTA_ESD_DETECT
EL_ON2
TP9001
DISP_TE
FG_ERR
OCTA_RST_N
TSP_SCL_1P8
TSP_INT
TSP_SDA_1P8
C9029
C9028
RFFE7_CLK
RFFE7_DATA
SPK_AMP_OUT_N
SPK_AMP_OUT_N
SPK_AMP_OUT_P
SPK_AMP_OUT_P
@SPK_VSNS_N
@SPK_VSNS_P
MOTOR_P
MOTOR_N
BTP_SPI_CLK
BTP_SPI_CS_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_RST_N
USB_THM(JPN)
USB_THM
SUB_REV
CC2
CC1
SBU1
C9034
C9035
C9036
C9037
C9039
C9038
C9031
C9030
VBUS_OVP
PCD_INT
OCTA_CON_DET
OCTA_MIPI_D2_P
OCTA_MIPI_D2_N
OCTA_MIPI_D1_P
OCTA_MIPI_D1_N
OCTA_MIPI_CLK_P
OCTA_MIPI_CLK_N
OCTA_MIPI_D0_P
OCTA_MIPI_D0_N
OCTA_MIPI_D3_P
OCTA_MIPI_D3_N
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
HALL_INT1
BAT_THM
MAIN_MIC_CLK
MAIN_MIC_DATA
R9002
R9003
JTAG_MUIC_D_N
JTAG_MUIC_D_P
SBU2
F9000
MUIC_D_N
MUIC_D_P
C9042
ZD9001
ZD9000
VOL
C9040
ground

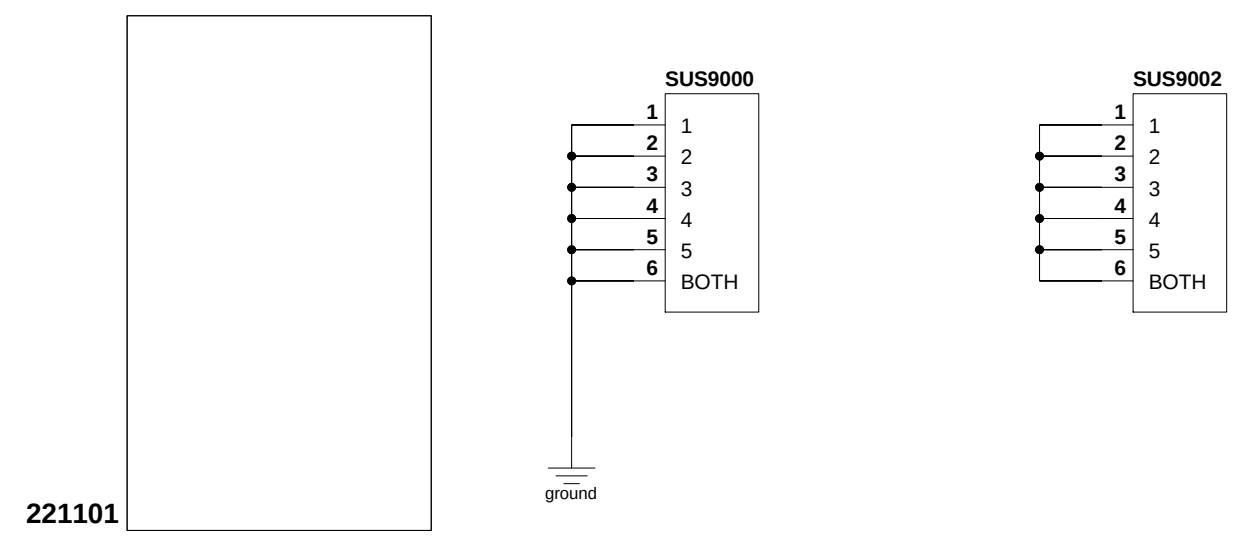
DISPLAY


VDD_LCD_ELAVDD_7P6
C9044
VBAT
L9000
C9013
C9014
C9015
C9011
C9012
C9016
VBAT
@VPH_PWR
L9001
C9017
C9018
C9019
L9002
VDD_LCD_ELVDD_4P6
C9020
VDD_LCD_ELVSS_-4P4
C9021
U9015
VO3
LX3
AGND
AVIN
PVIN
PGND2
LX1
LX1
VO1
VO1
VO1
PGND1
PGND1
PVIN
PVIN
LX2
LX2
LX2
VO2
VO2
VO2
PGND1
SET
ASWIRE
ESWIRE
L9003
L9004
L9005
C9022
C9023
C9024
C9026
C9027
EL_ON1
EL_ON2
ground

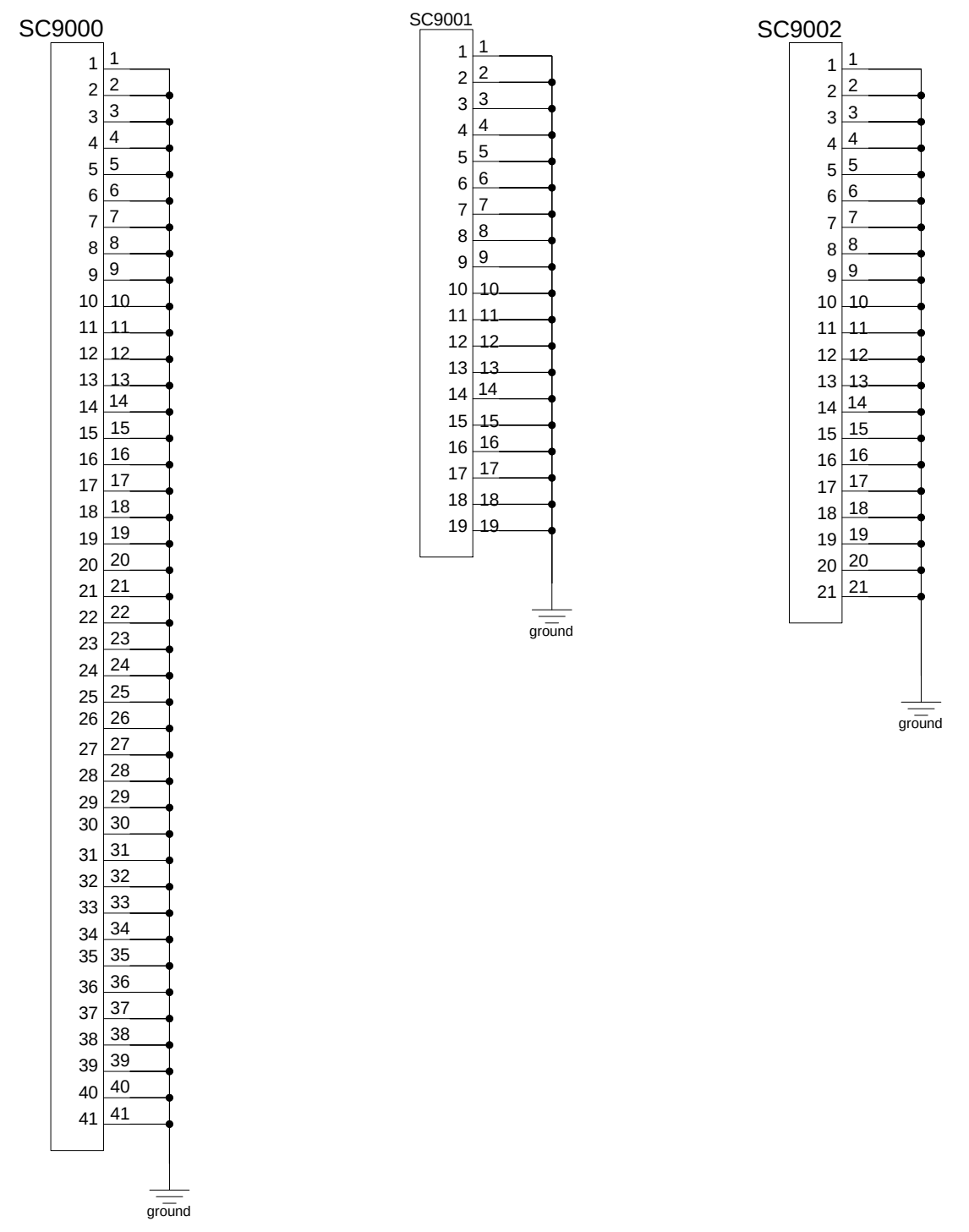
DISPLAY PMIC

2D Barcode

CN9000
CN9001
1


221101
SUS9000
SUS9002
BOTH
ground

SHIELD CAN


SC9000
SC9001
SC9002
ground

REGION		MEMORY		
R9013	B : EUR	R9019	6/128	
R9014	0 : HK CHN	R9015	8/128	
R9018	S : KOR SKT	R9017	8/256	
R9021	E : SEA, MEA, SWA E : AFR, LA, CIS			

mmW_ANT_CONN (Right)

NA ONLY

ES2.0 : OPEN
ES3.0 : 0 ohm

5G IFIC

NA ONLY

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

Changed by: Date Changed: Time Changed:

SM-A546E
Service Schematic

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A54 5G EUR_REV04C

2022.12

Sheet01 : GSM / WCDMA / LTE / NR(N41) RF
Sheet02 : TRANSCEIVER, SUB ANT
Sheet03 : CONNECTIVITY(BT/WIFI/FM), NFC
Sheet04 : AP(Quartz-INTERFACE,GPIO)
Sheet05 : AP(POWER), MEM(uMCP)
Sheet06 : PMIC/SUB PMIC, IF PMIC, DC IC
Sheet07 : AUDIO AMP, SENSOR, SIM, SD
Sheet08 : CAMERA
Sheet09 : DISPLAY PMIC, C2C

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:
TITLE: A54
Size:
Changed by:
Date Changed:
Time Changed:
REV:
Drawing Number:
Page:

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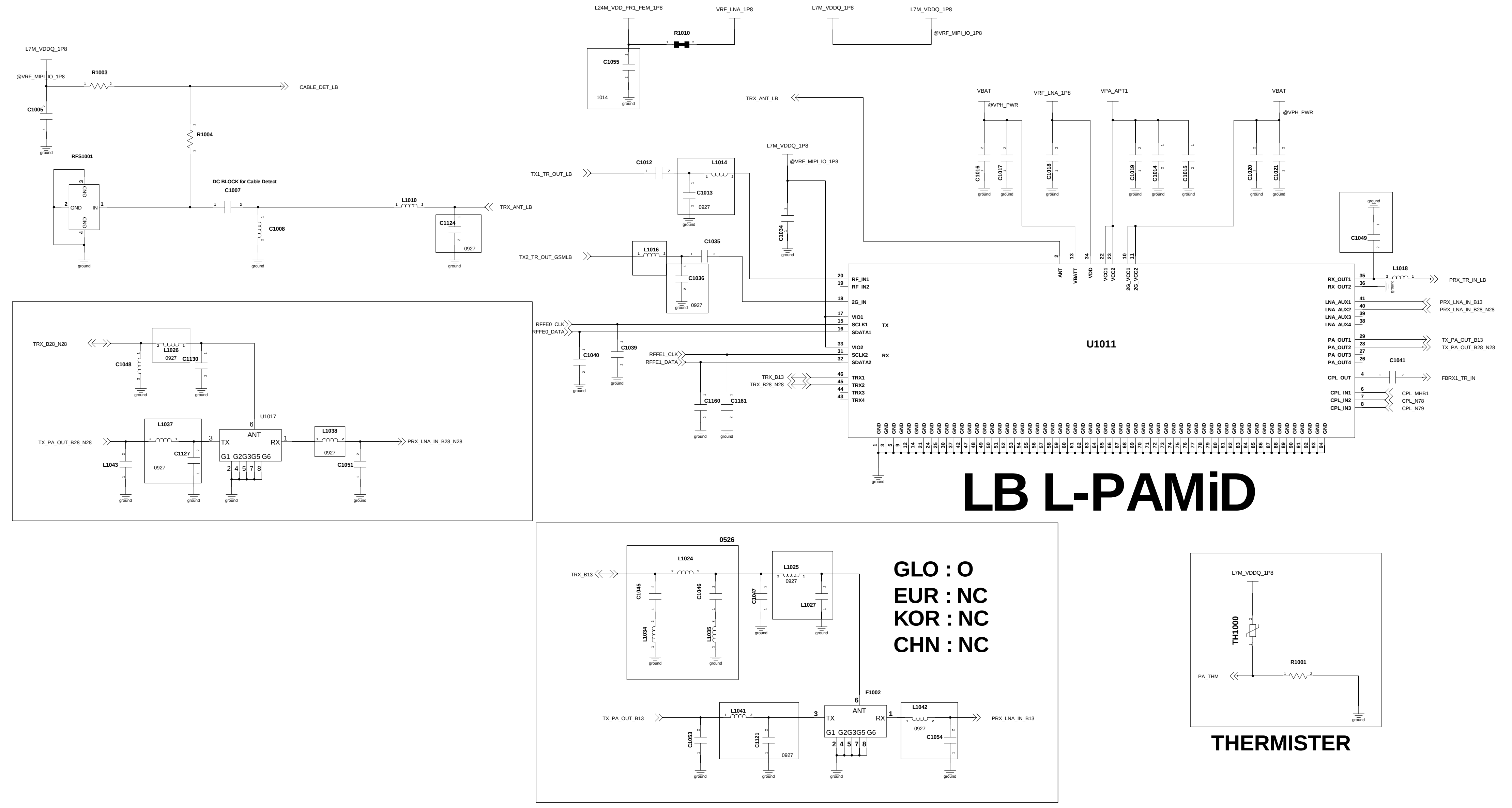
RF PART 1

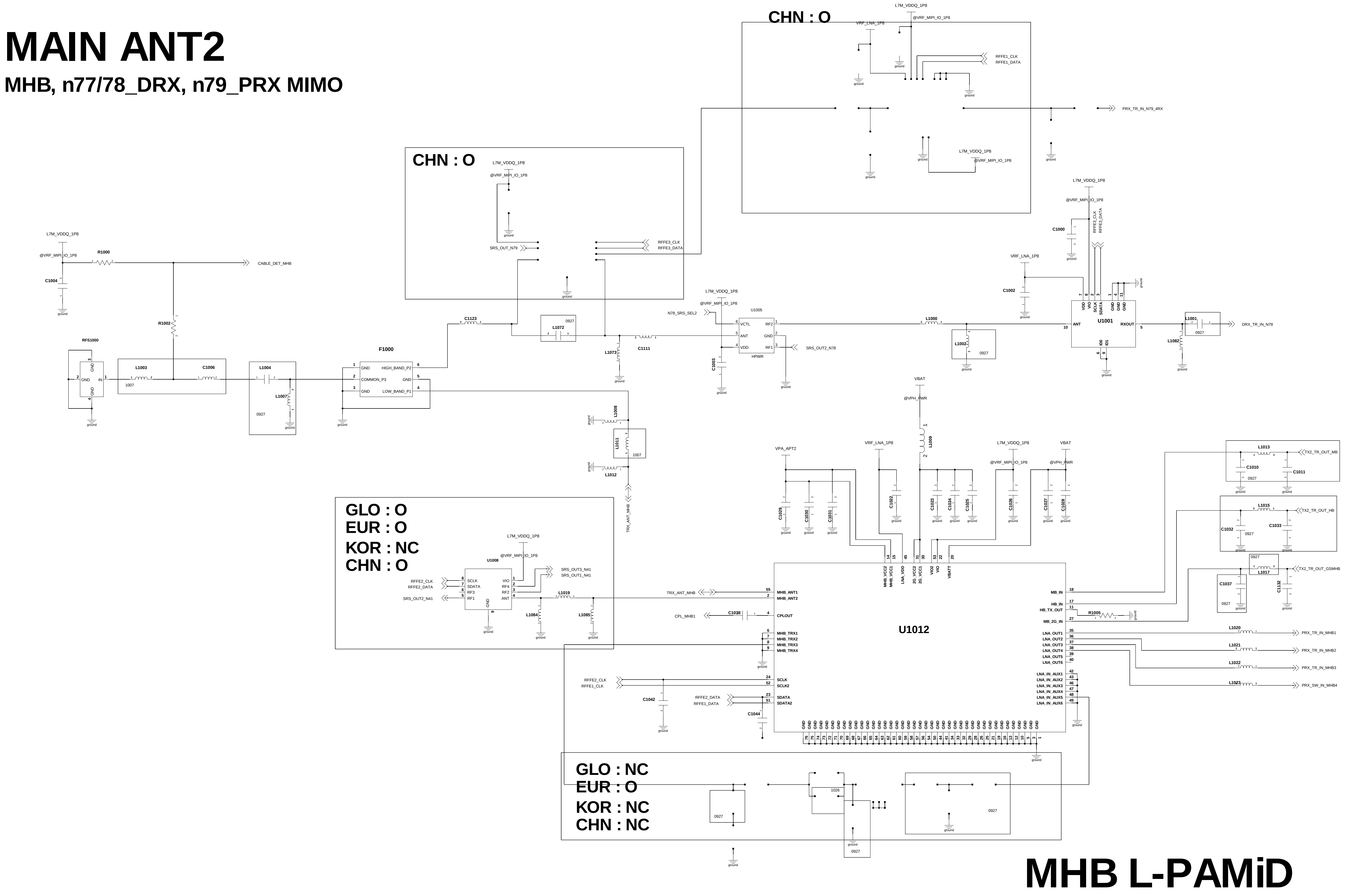
MAIN ANT1

LB


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1003
CABLE_DET_LB
C1005
R1004
RFS1001
DC BLOCK for Cable Detect
C1007
C1008
L1010
TRX_ANT_LB
C1124
0927
L24M_VDD_FR1_FEM_1P8
VRF_LNA_1P8
R1010
C1055
1014
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
TRX_ANT_LB
TX1_TR_OUT_LB
C1012
L1014
C1013
TX2_TR_OUT_GSMLB
L1016
C1035
C1036
C1034
RFFE0_CLK
RFFE0_DATA
C1040
C1039
RFFE1_CLK
RFFE1_DATA
C1160
C1161
TRX_B13
TRX_B28_N28
VBAT
@VPH_PWR
VPA_APT1
C1016
C1017
C1018
C1019
C1014
C1015
C1020
C1021
C1049
RF_IN1
RF_IN2
2G_IN
VIO1
SCLK1
SDATA1
TX
VIO2
SCLK2
SDATA2
RX
TRX1
TRX2
TRX3
TRX4
ANT
VBATT
VDD
VCC1
VCC2
2G_VCC1
2G_VCC2
U1011
RX_OUT1
RX_OUT2
L1018
PRX_TR_IN_LB
LNA_AUX1
LNA_AUX2
LNA_AUX3
LNA_AUX4
PRX_LNA_IN_B13
PRX_LNA_IN_B28_N28
PA_OUT1
PA_OUT2
PA_OUT3
PA_OUT4
TX_PA_OUT_B13
TX_PA_OUT_B28_N28
CPL_OUT
C1041
FBRX1_TR_IN
CPL_IN1
CPL_IN2
CPL_IN3
CPL_MHB1
CPL_N78
CPL_N79
LB L-PAMiD
TRX_B28_N28
L1026
C1130
C1048
U1017
TX_PA_OUT_B28_N28
L1037
C1127
L1043
TX
ANT
RX
G1 G2G3G5 G6
L1038
C1051
PRX_LNA_IN_B28_N28
0526
TRX_B13
L1024
L1025
C1045
C1046
C1047
L1027
L1034
L1035
GLO : O
EUR : NC
KOR : NC
CHN : NC
F1002
TX_PA_OUT_B13
L1041
C1053
C1121
L1042
C1054
PRX_LNA_IN_B13
L7M_VDDQ_1P8
TH1000
PA_THM
R1001
THERMISTER

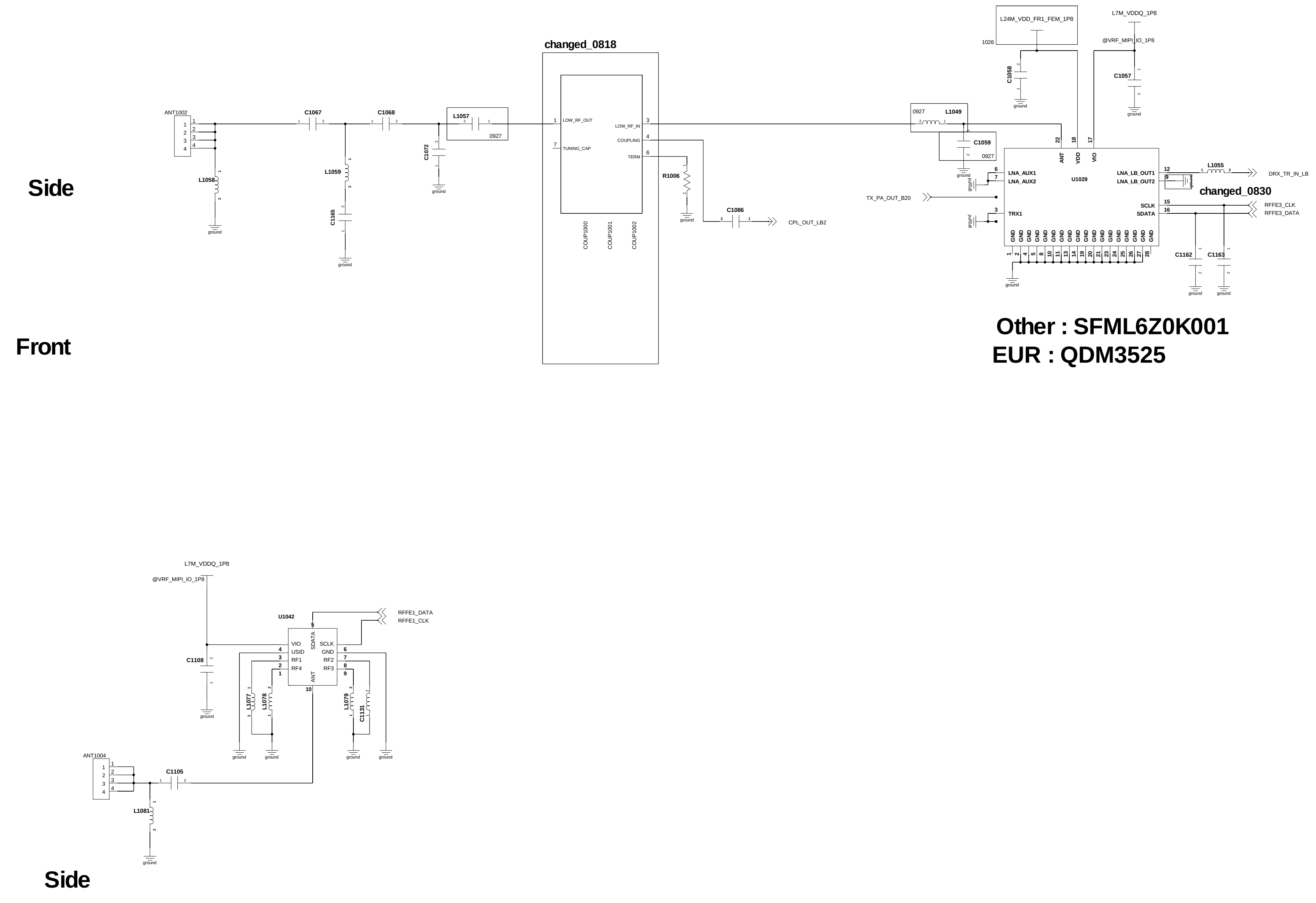
MAIN ANT2

MHB, n77/78_DRX, n79_PRX MIMO


CHN : O
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
RFFE1_CLK
RFFE1_DATA
PRX_TR_IN_N79_4RX
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
CHN : O
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
SRS_OUT_N79
RFFE3_CLK
RFFE3_DATA
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1000
CABLE_DET_MHB
C1004
R1002
RFS1000
L1003
C1006
1007
L1004
L1007
0927
F1000
GND
HIGH_BAND_P2
COMMON_P3
GND
LOW_BAND_P1
C1123
L1072
0927
L1073
C1111
N78_SRS_SEL2
U1005
VCTL
ANT
VDD
RF2
GND
RF1
HPWR
C1003
SRS_OUT2_N78
L1000
L1002
0927
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
RFFE3_CLK
RFFE3_DATA
C1000
VRF_LNA_1P8
C1002
U1001
ANT
RXOUT
VDD
VIO
SCLK
SDATA
GND
ID0
ID1
L1001
0927
DRX_TR_IN_N78
L1082
L1008
L1011
1007
L1012
TRX_ANT_MHB
GLO : O
EUR : O
KOR : NC
CHN : O
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
U1008
RFFE2_CLK
RFFE2_DATA
SRS_OUT2_N41
SCLK
SDATA
RF3
RF1
VIO
RF4
RF2
ANT
GND
SRS_OUT3_N41
SRS_OUT1_N41
L1019
L1084
L1085
VBAT
@VPH_PWR
L1009
VPA_APT2
VRF_LNA_1P8
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VBAT
@VPH_PWR
C1029
C1030
C1031
C1022
C1023
C1024
C1025
C1026
C1027
C1028
MHB_VCC2
MHB_VCC1
LNA_VDD
2G_VCC2
2G_VCC1
VIO2
VIO
VBATT
U1012
TRX_ANT_MHB
MHB_ANT1
MHB_ANT2
CPL_MHB1
C1038
CPLOUT
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
RFFE2_CLK
RFFE1_CLK
SCLK
SCLK2
C1042
RFFE2_DATA
RFFE1_DATA
SDATA
SDATA2
C1044
MB_IN
HB_IN
HB_TX_OUT
MB_2G_IN
R1005
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
L1020
L1021
L1022
L1023
PRX_TR_IN_MHB1
PRX_TR_IN_MHB2
PRX_TR_IN_MHB3
PRX_SW_IN_MHB4
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
L1013
C1010
C1011
TX2_TR_OUT_MB
L1015
C1032
C1033
TX2_TR_OUT_HB
L1017
C1037
C1132
TX2_TR_OUT_GSMHB
GLO : NC
EUR : O
KOR : NC
CHN : NC
1026
0927
MHB L-PAMiD

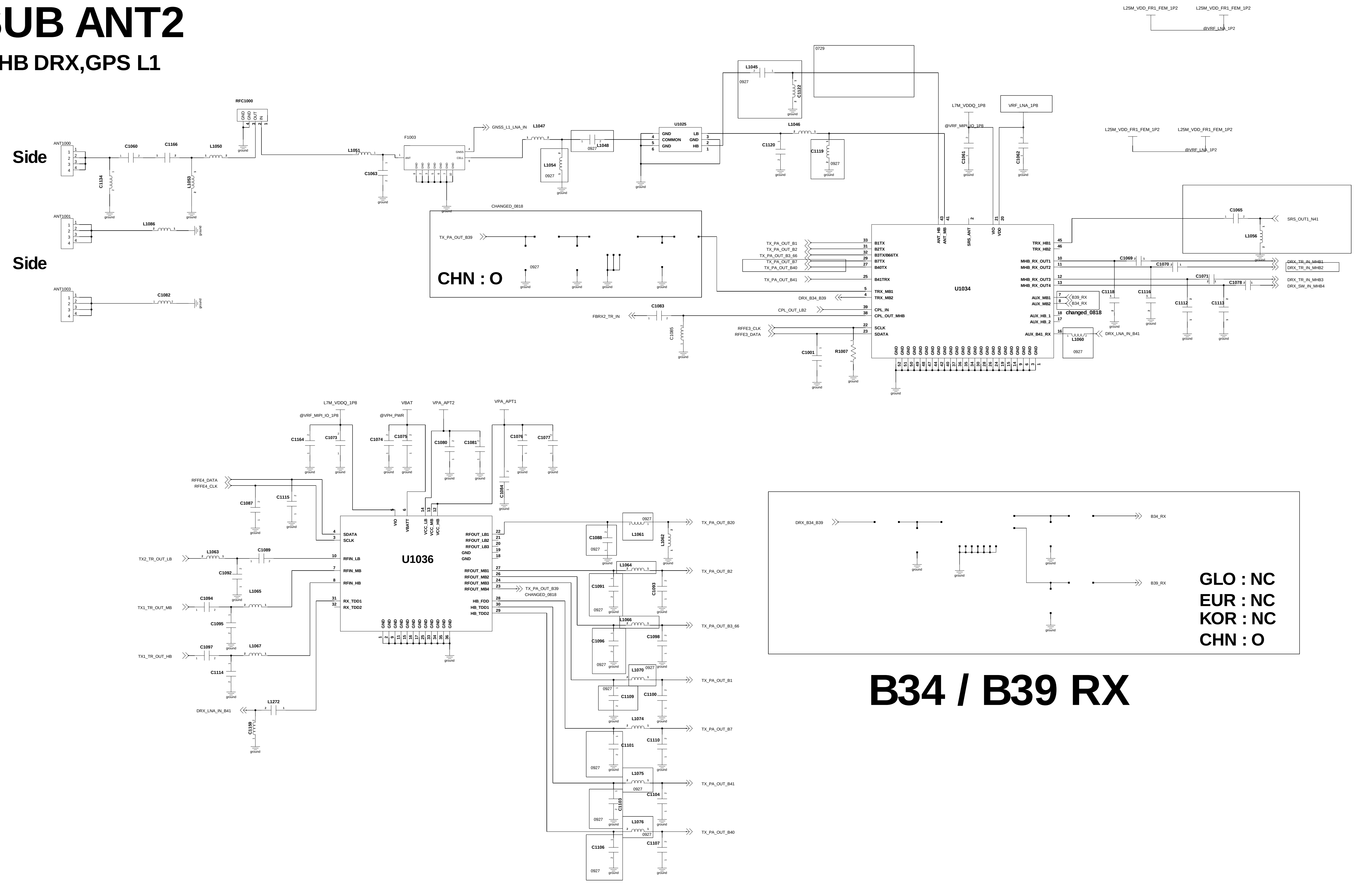
SUB ANT1

LB DRX


changed_0818
ANT1002
C1067
C1068
L1057
0927
L1058
L1059
C1165
C1072
LOW_RF_OUT
TUNING_CAP
LOW_RF_IN
COUPLING
TERM
COUP1000
COUP1001
COUP1002
R1006
C1086
CPL_OUT_LB2
L24M_VDD_FR1_FEM_1P8
1026
C1058
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1057
L1049
C1059
0927
ANT
VDD
VIO
LNA_AUX1
LNA_AUX2
U1029
TX_PA_OUT_B20
TRX1
LNA_LB_OUT1
LNA_LB_OUT2
L1055
DRX_TR_IN_LB
changed_0830
SCLK
SDATA
RFFE3_CLK
RFFE3_DATA
C1162
C1163
Side
Front
Other : SFML6Z0K001
EUR : QDM3525
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
U1042
RFFE1_DATA
RFFE1_CLK
VIO
USID
RF1
RF4
SDATA
SCLK
GND
RF2
RF3
ANT
C1108
L1077
L1078
L1079
C1131
ANT1004
C1105
L1081
Side

SUB ANT2

MHB DRX,GPS L1


L25M_VDD_FR1_FEM_1P2
L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
Side
ANT1000
C1060
C1166
L1050
C1134
L1053
RFC1000
GND
OUT
IN
L1051
C1063
F1003
ANT
GNSS
CELL
GNSS_L1_LNA_IN
L1047
L1048
0927
L1054
0927
U1025
GND
COMMON
LB
HB
L1045
C1122
0729
L1046
C1120
C1119
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
C1061
C1062
L25M_VDD_FR1_FEM_1P2
L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
ANT1001
L1086
Side
ANT1003
C1082
CHANGED_0818
TX_PA_OUT_B39
CHN : O
0927
FBRX2_TR_IN
C1083
C1085
TX_PA_OUT_B1
TX_PA_OUT_B2
TX_PA_OUT_B3_66
TX_PA_OUT_B7
TX_PA_OUT_B40
TX_PA_OUT_B41
B1TX
B2TX
B3TX/B66TX
B7TX
B40TX
B41TRX
TRX_MB1
TRX_MB2
DRX_B34_B39
CPL_OUT_LB2
CPL_IN
CPL_OUT_MHB
RFFE3_CLK
RFFE3_DATA
SCLK
SDATA
C1001
R1007
ANT_HB
ANT_MB
SRS_ANT
VIO
VDD
U1034
TRX_HB1
TRX_HB2
MHB_RX_OUT1
MHB_RX_OUT2
MHB_RX_OUT3
MHB_RX_OUT4
AUX_MB1
AUX_MB2
AUX_HB_1
AUX_HB_2
AUX_B41_RX
B39_RX
B34_RX
changed_0818
L1060
0927
DRX_LNA_IN_B41
C1065
SRS_OUT1_N41
L1056
C1069
C1070
C1071
C1078
DRX_TR_IN_MHB1
DRX_TR_IN_MHB2
DRX_TR_IN_MHB3
DRX_SW_IN_MHB4
C1118
C1116
C1112
C1113
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VBAT
@VPH_PWR
VPA_APT2
VPA_APT1
C1164
C1073
C1074
C1075
C1080
C1081
C1076
C1077
C1084
RFFE4_DATA
RFFE4_CLK
C1087
C1115
SDATA
SCLK
VIO
VBATT
VCC_LB
VCC_MB
VCC_HB
U1036
TX2_TR_OUT_LB
L1063
C1089
C1092
RFIN_LB
RFIN_MB
RFIN_HB
TX1_TR_OUT_MB
C1094
C1095
L1065
TX1_TR_OUT_HB
C1097
C1114
L1067
RX_TDD1
RX_TDD2
DRX_LNA_IN_B41
L1272
C1159
RFOUT_LB1
RFOUT_LB2
RFOUT_LB3
RFOUT_MB1
RFOUT_MB2
RFOUT_MB3
RFOUT_MB4
HB_FDD
HB_TDD1
HB_TDD2
TX_PA_OUT_B39
CHANGED_0818
C1088
L1061
L1062
TX_PA_OUT_B20
L1064
C1091
C1093
TX_PA_OUT_B2
L1066
C1096
C1098
TX_PA_OUT_B3_66
L1070
C1109
C1100
TX_PA_OUT_B1
L1074
C1101
C1110
TX_PA_OUT_B7
L1075
C1103
C1104
TX_PA_OUT_B41
L1076
C1106
C1107
TX_PA_OUT_B40
0927
DRX_B34_B39
B34_RX
B39_RX
GLO : NC
EUR : NC
KOR : NC
CHN : O
B34 / B39 RX

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: RF PAM/FEM Size:
DOC CTRL CHK:	
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number:

Changed by: Date Changed: Time Changed:

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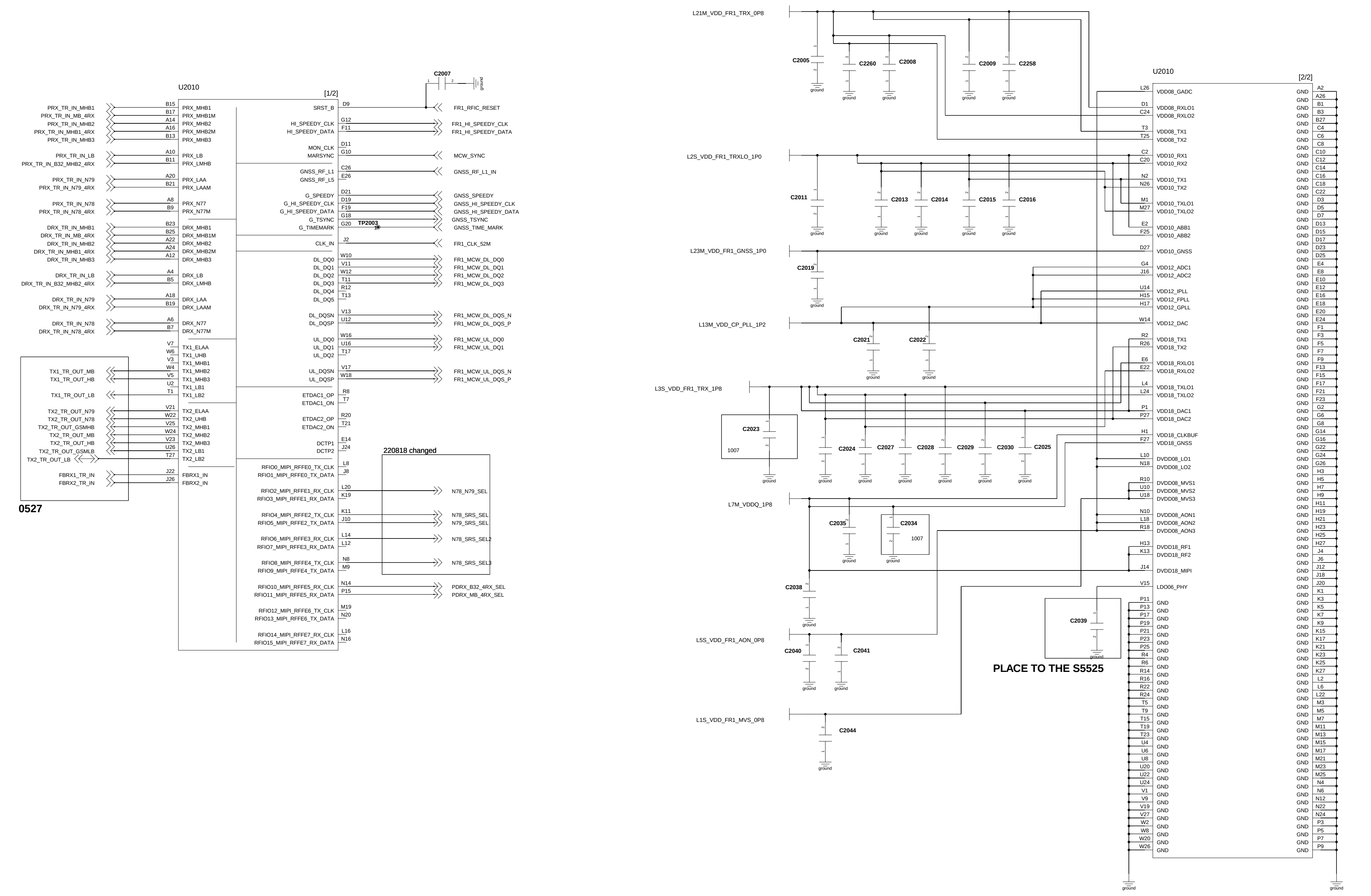
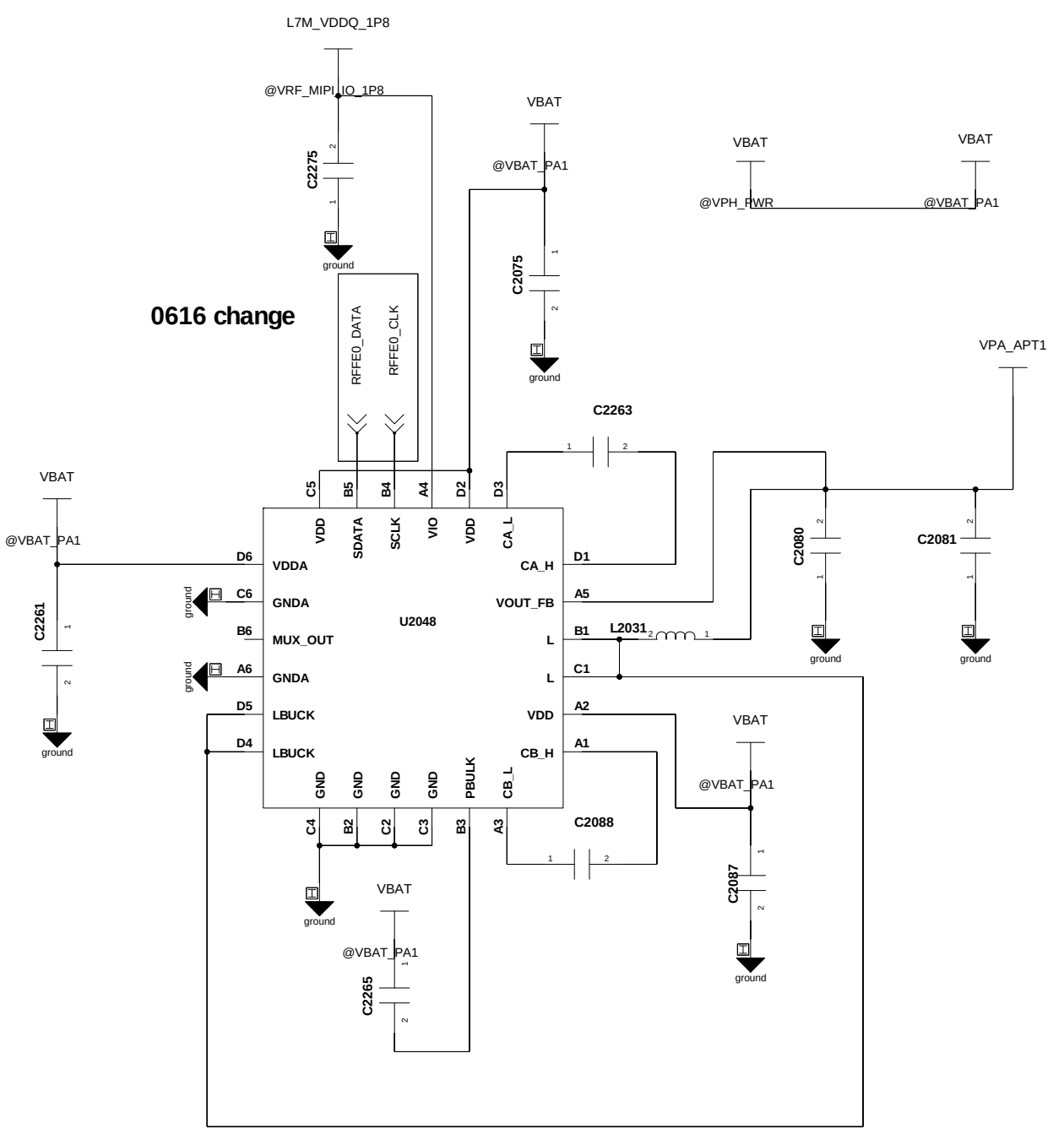
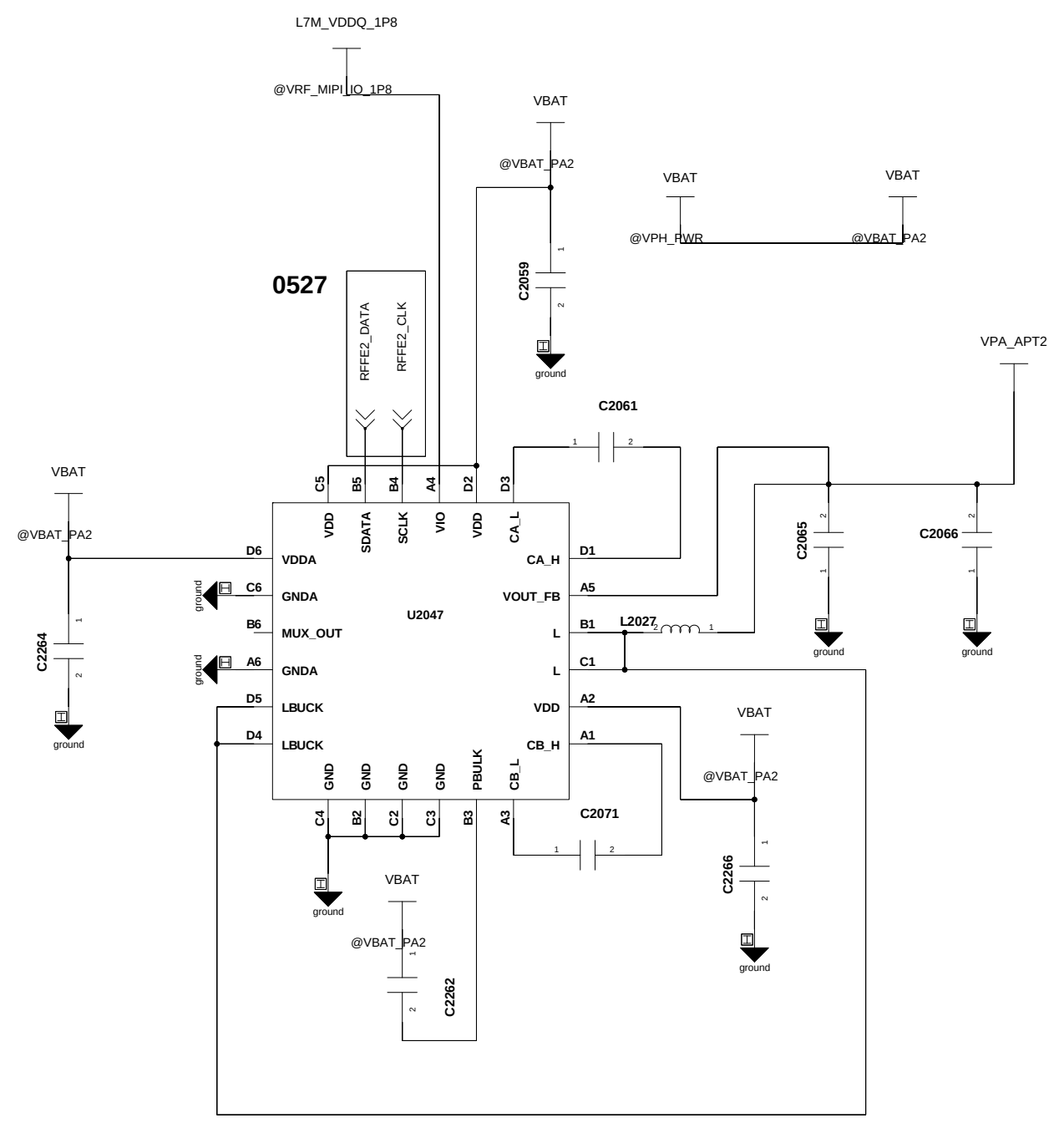
TRANSCEIVER (S5525)


U2010
[1/2]
PRX_TR_IN_MHB1
PRX_TR_IN_MB_4RX
PRX_TR_IN_MHB2
PRX_TR_IN_MHB1_4RX
PRX_TR_IN_MHB3
PRX_TR_IN_LB
PRX_TR_IN_B32_MHB2_4RX
PRX_TR_IN_N79
PRX_TR_IN_N79_4RX
PRX_TR_IN_N78
PRX_TR_IN_N78_4RX
DRX_TR_IN_MHB1
DRX_TR_IN_MB_4RX
DRX_TR_IN_MHB2
DRX_TR_IN_MHB1_4RX
DRX_TR_IN_MHB3
DRX_TR_IN_LB
DRX_TR_IN_B32_MHB2_4RX
DRX_TR_IN_N79
DRX_TR_IN_N79_4RX
DRX_TR_IN_N78
DRX_TR_IN_N78_4RX
TX1_TR_OUT_MB
TX1_TR_OUT_HB
TX1_TR_OUT_LB
TX2_TR_OUT_N79
TX2_TR_OUT_N78
TX2_TR_OUT_GSMHB
TX2_TR_OUT_MB
TX2_TR_OUT_HB
TX2_TR_OUT_GSMLB
TX2_TR_OUT_LB
FBRX1_TR_IN
FBRX2_TR_IN
0527
C2007
FR1_RFIC_RESET
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
MCW_SYNC
GNSS_RF_L1_IN
GNSS_SPEEDY
GNSS_HI_SPEEDY_CLK
GNSS_HI_SPEEDY_DATA
GNSS_TSYNC
GNSS_TIME_MARK
TP2003
FR1_CLK_52M
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
220818 changed
N78_N79_SEL
N78_SRS_SEL
N79_SRS_SEL
N78_SRS_SEL2
N78_SRS_SEL3
PDRX_B32_4RX_SEL
PDRX_MB_4RX_SEL
L21M_VDD_FR1_TRX_0P8
L2S_VDD_FR1_TRXLO_1P0
L23M_VDD_FR1_GNSS_1P0
L13M_VDD_CP_PLL_1P2
L3S_VDD_FR1_TRX_1P8
L7M_VDDQ_1P8
L5S_VDD_FR1_AON_0P8
L1S_VDD_FR1_MVS_0P8
C2005
C2260
C2008
C2009
C2258
C2011
C2013
C2014
C2015
C2016
C2019
C2021
C2022
C2023
1007
C2024
C2027
C2028
C2029
C2030
C2025
C2035
C2034
C2038
C2040
C2041
C2044
C2039
PLACE TO THE S5525
U2010
[2/2]
ground

APT1


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2275
0616 change
RFFE0_DATA
RFFE0_CLK
VBAT
@VBAT_PA1
C2075
@VPH_PWR
VPA_APT1
C2263
C2080
C2081
C2261
U2048
L2031
C2088
C2087
C2265
ground

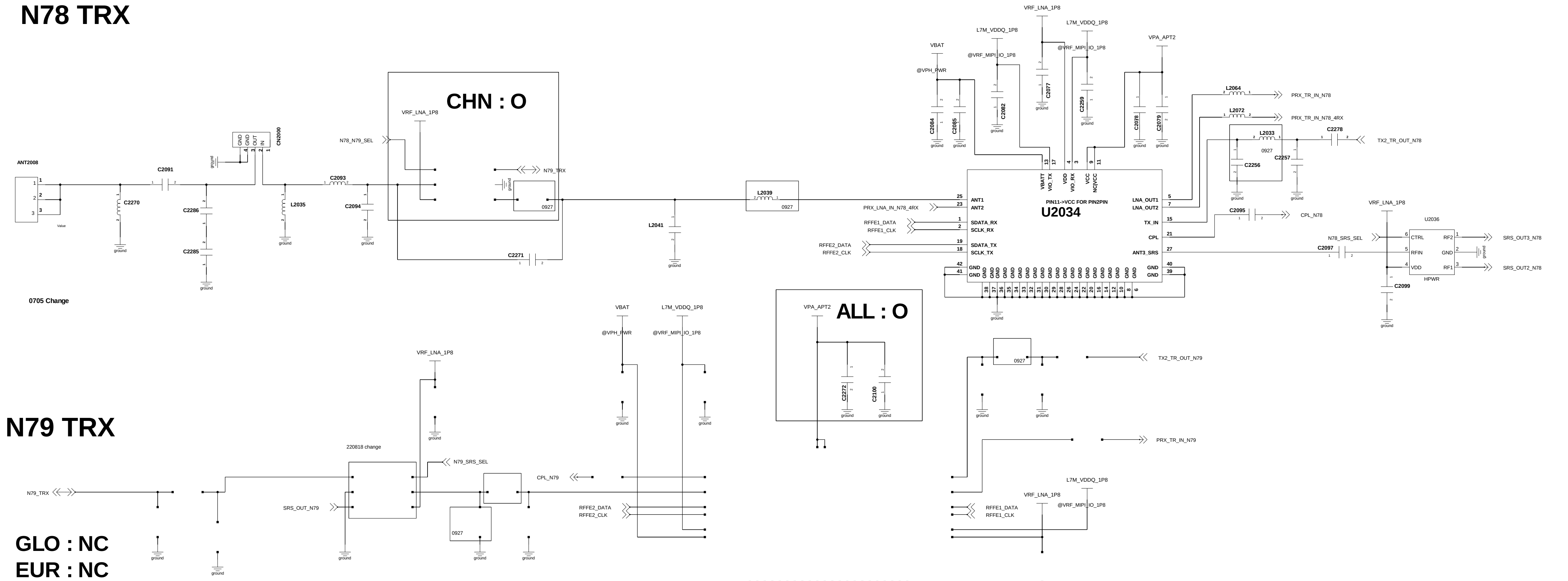
APT2


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
0527
RFFE2_DATA
RFFE2_CLK
VBAT
@VBAT_PA2
C2059
@VPH_PWR
VPA_APT2
C2061
C2065
C2066
C2264
U2047
L2027
C2071
C2266
C2262
ground

SUB ANT3

N78 TRX

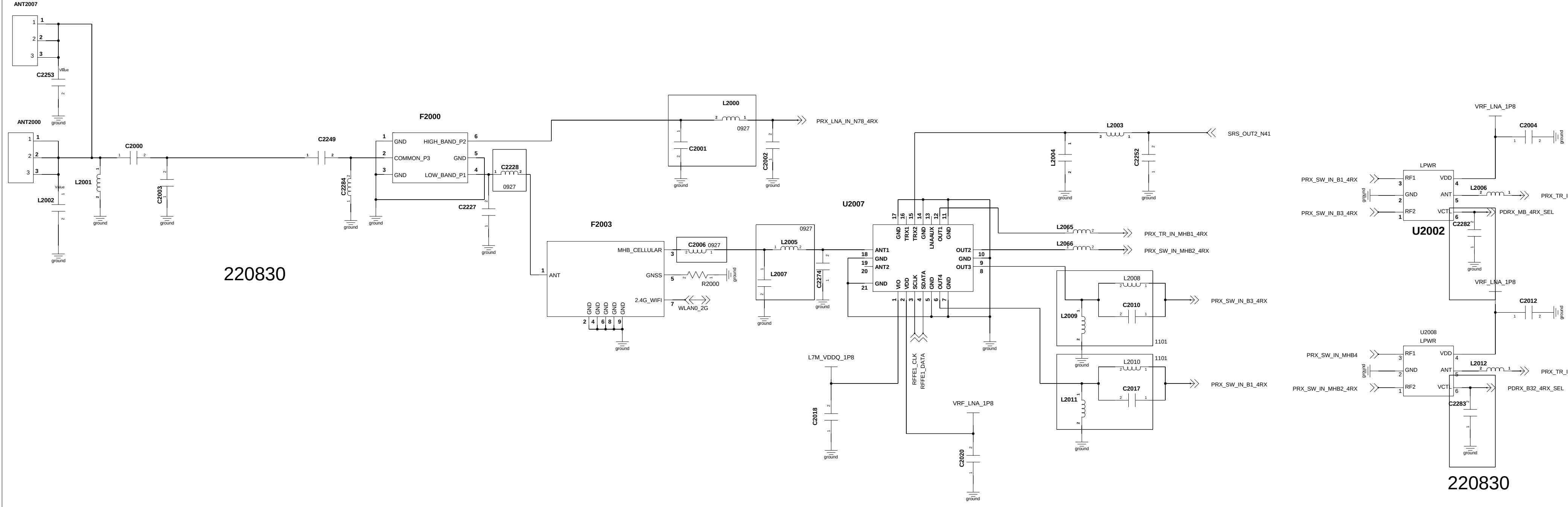
N79 TRX


ANT2008
C2270
C2091
C2286
C2285
CN2000
L2035
C2093
C2094
CHN : O
VRF_LNA_1P8
N78_N79_SEL
N79_TRX
0927
C2271
L2041
L2039
PRX_LNA_IN_N78_4RX
RFFE1_DATA
RFFE1_CLK
RFFE2_DATA
RFFE2_CLK
U2034
PIN11->VCC FOR PIN2PIN
C2084
C2085
C2082
C2077
C2259
C2078
C2079
VPA_APT2
L2064
PRX_TR_IN_N78
L2072
PRX_TR_IN_N78_4RX
L2033
C2278
TX2_TR_OUT_N78
C2256
C2257
C2095
CPL_N78
C2097
N78_SRS_SEL
U2036
SRS_OUT3_N78
SRS_OUT2_N78
HPWR
C2099
0705 Change
ALL : O
C2272
C2100
TX2_TR_OUT_N79
PRX_TR_IN_N79
220818 change
N79_SRS_SEL
SRS_OUT_N79
CPL_N79
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

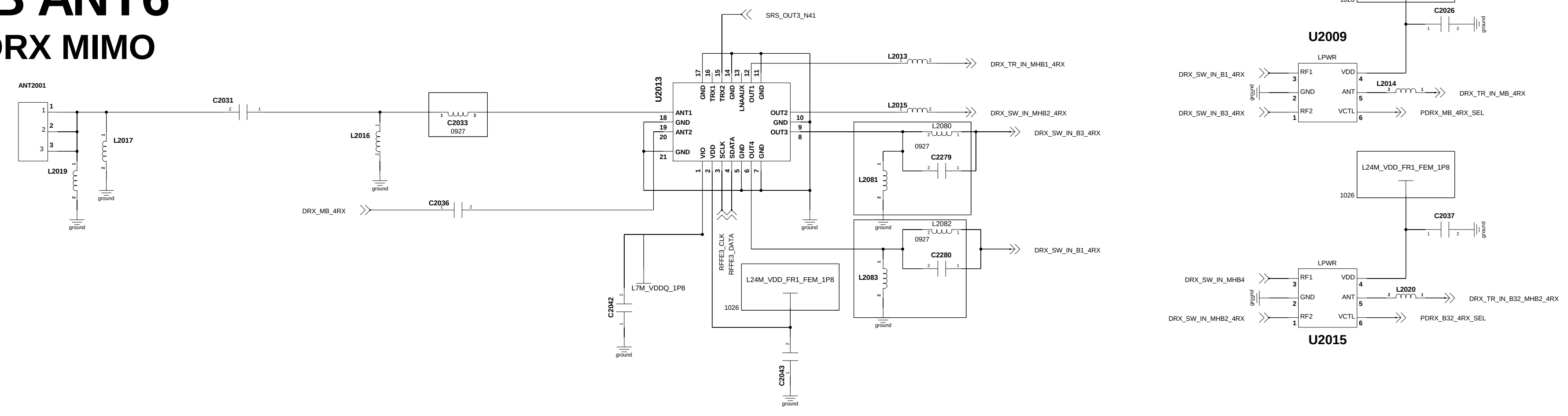
SUB ANT5

MHB PRX MIMO, n77/78 PRX MIMO, WIFI 2.4G


ANT2007
C2253
ANT2000
L2001
L2002
C2000
C2003
C2249
C2284
F2000
HIGH_BAND_P2
COMMON_P3
LOW_BAND_P1
C2228
0927
C2227
L2000
C2001
C2002
PRX_LNA_IN_N78_4RX
220830
F2003
MHB_CELLULAR
GNSS
2.4G_WIFI
C2006
R2000
WLAN0_2G
L2005
L2007
C2274
U2007
L7M_VDDQ_1P8
C2018
RFFE1_CLK
RFFE1_DATA
VRF_LNA_1P8
C2020
L2003
SRS_OUT2_N41
L2004
C2252
L2065
PRX_TR_IN_MHB1_4RX
L2066
PRX_SW_IN_MHB2_4RX
L2008
C2010
L2009
PRX_SW_IN_B3_4RX
1101
L2010
C2017
L2011
PRX_SW_IN_B1_4RX
VRF_LNA_1P8
C2004
LPWR
PRX_SW_IN_B1_4RX
PRX_SW_IN_B3_4RX
L2006
PRX_TR_IN_MB_4RX
PDRX_MB_4RX_SEL
C2282
U2002
C2012
U2008
PRX_SW_IN_MHB4
PRX_SW_IN_MHB2_4RX
L2012
PRX_TR_IN_B32_MHB2_4RX
PDRX_B32_4RX_SEL
C2283
220830
ground

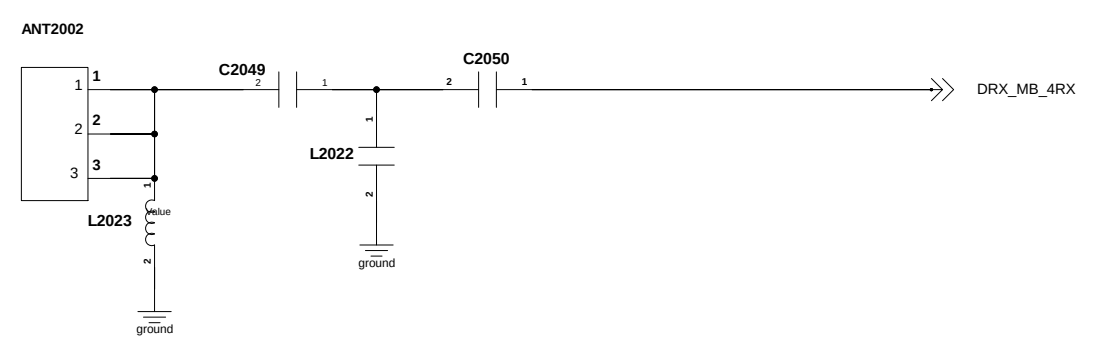
SUB ANT6

HB DRX MIMO


ANT2001
L2017
L2019
C2031
L2016
C2033
0927
DRX_MB_4RX
C2036
SRS_OUT3_N41
U2013
L2013
DRX_TR_IN_MHB1_4RX
L2015
DRX_SW_IN_MHB2_4RX
L2080
C2279
L2081
DRX_SW_IN_B3_4RX
L2082
C2280
L2083
DRX_SW_IN_B1_4RX
RFFE3_CLK
RFFE3_DATA
L7M_VDDQ_1P8
C2042
L24M_VDD_FR1_FEM_1P8
1026
C2043
U2009
C2026
LPWR
DRX_SW_IN_B1_4RX
DRX_SW_IN_B3_4RX
L2014
DRX_TR_IN_MB_4RX
PDRX_MB_4RX_SEL
C2037
DRX_SW_IN_MHB4
DRX_SW_IN_MHB2_4RX
L2020
DRX_TR_IN_B32_MHB2_4RX
PDRX_B32_4RX_SEL
U2015
ground

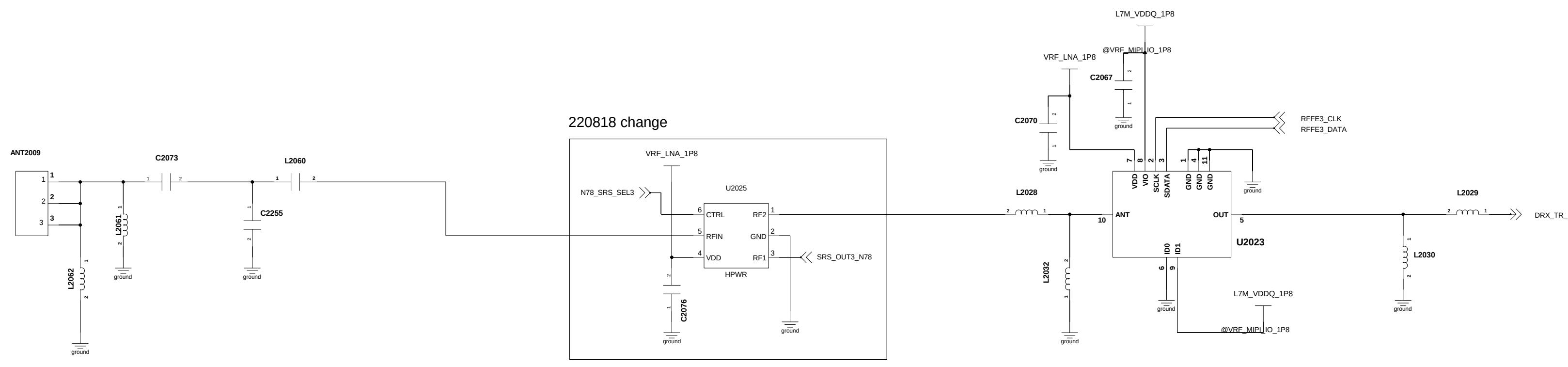
SUB ANT7

MB DRX MIMO


ANT2002
C2049
C2050
L2022
L2023
DRX_MB_4RX
ground

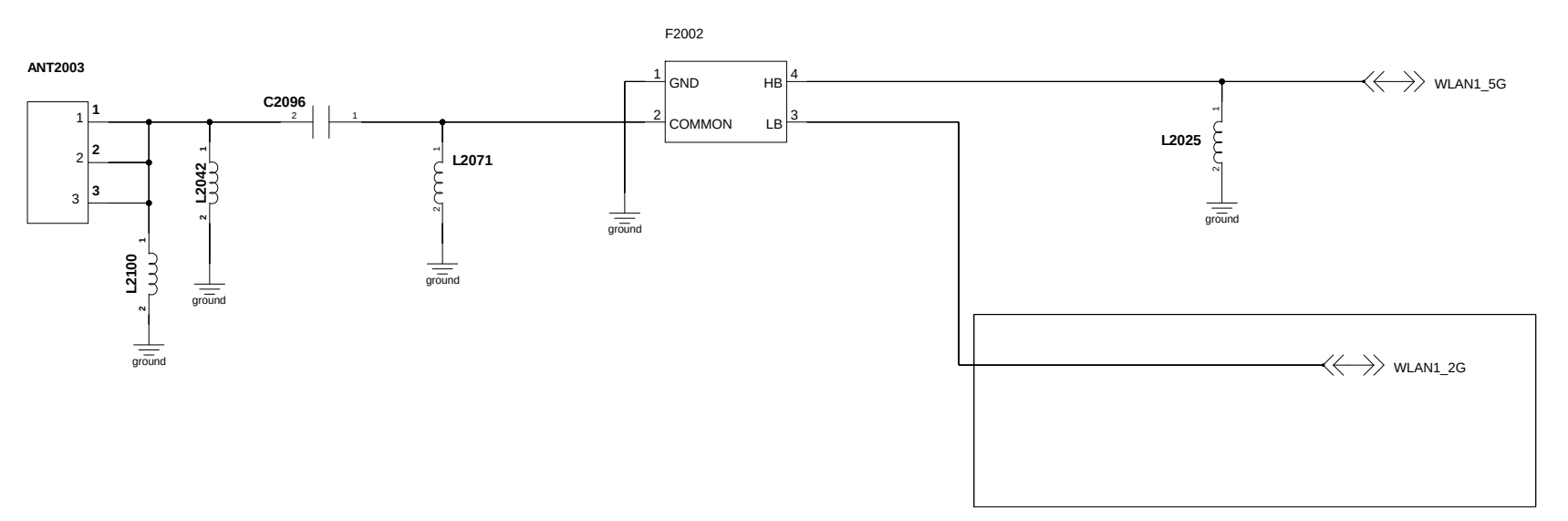
SUB ANT8

N78 DRX MIMO


ANT2009
L2061
L2062
C2073
L2060
C2255
220818 change
VRF_LNA_1P8
N78_SRS_SEL3
U2025
HPWR
C2076
SRS_OUT3_N78
L2028
L2032
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2067
C2070
RFFE3_CLK
RFFE3_DATA
U2023
L2029
L2030
DRX_TR_IN_N78_4RX
ground

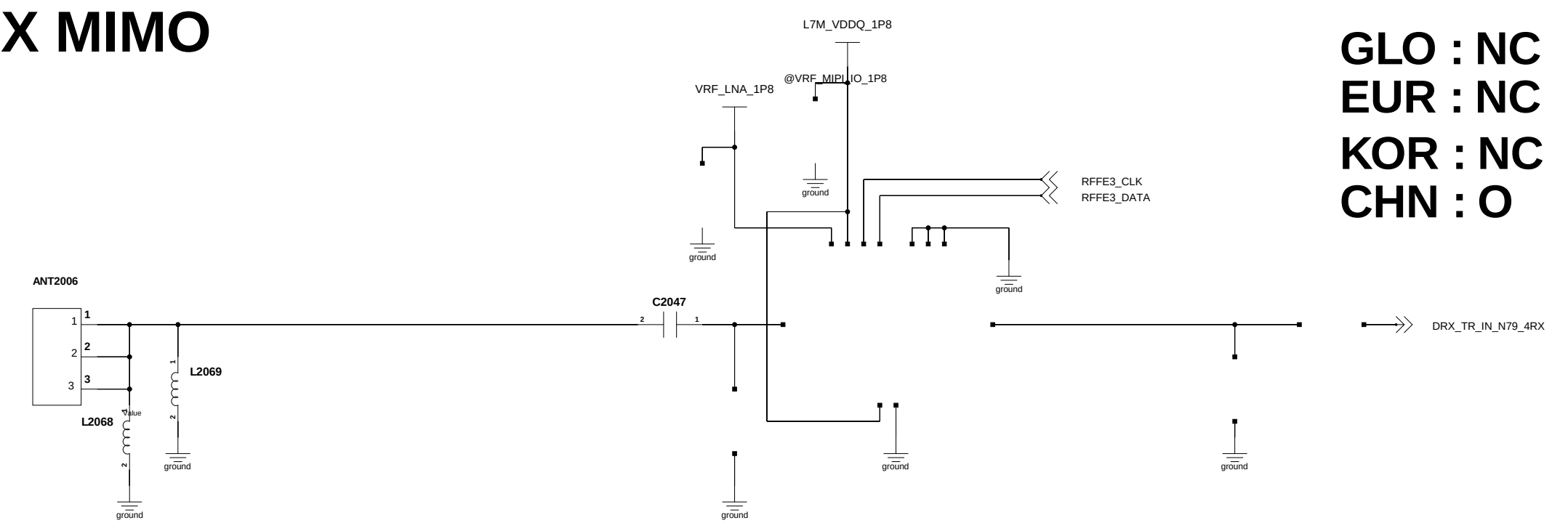
SUB ANT9

WIFI_2.4G+WIFI_5G


ANT2003
L2100
L2042
C2096
L2071
F2002
HB
COMMON
LB
L2025
WLAN1_5G
WLAN1_2G
ground

SUB ANT11

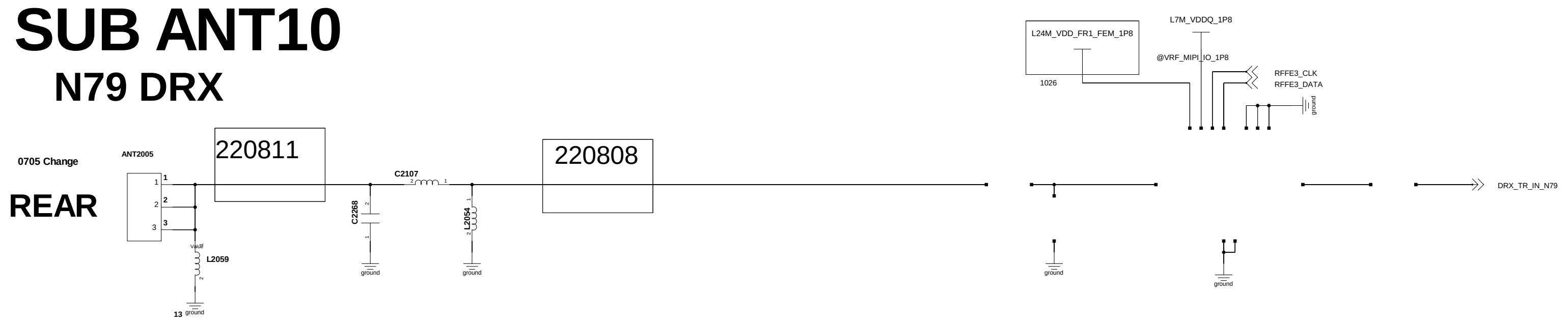
n79 DRX MIMO


ANT2006
L2068
L2069
C2047
L7M_VDDQ_1P8
VRF_LNA_1P8
@VRF_MIPI_IO_1P8
RFFE3_CLK
RFFE3_DATA
DRX_TR_IN_N79_4RX
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

SUB ANT10

N79 DRX


0705 Change
REAR
ANT2005
L2059
220811
C2268
C2107
L2054
220808
L24M_VDD_FR1_FEM_1P8
1026
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
RFFE3_CLK
RFFE3_DATA
DRX_TR_IN_N79
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

Engineer:	A54	
Drawn by:		
R&D CHK:	TITLE:	Size:
DOC CTRL CHK:	RF TRCV/PAM/APT	
MFG ENGR CHK:		
QA CHK:	REV:	Drawing Number:

Changed by: Date Changed: Time Changed:

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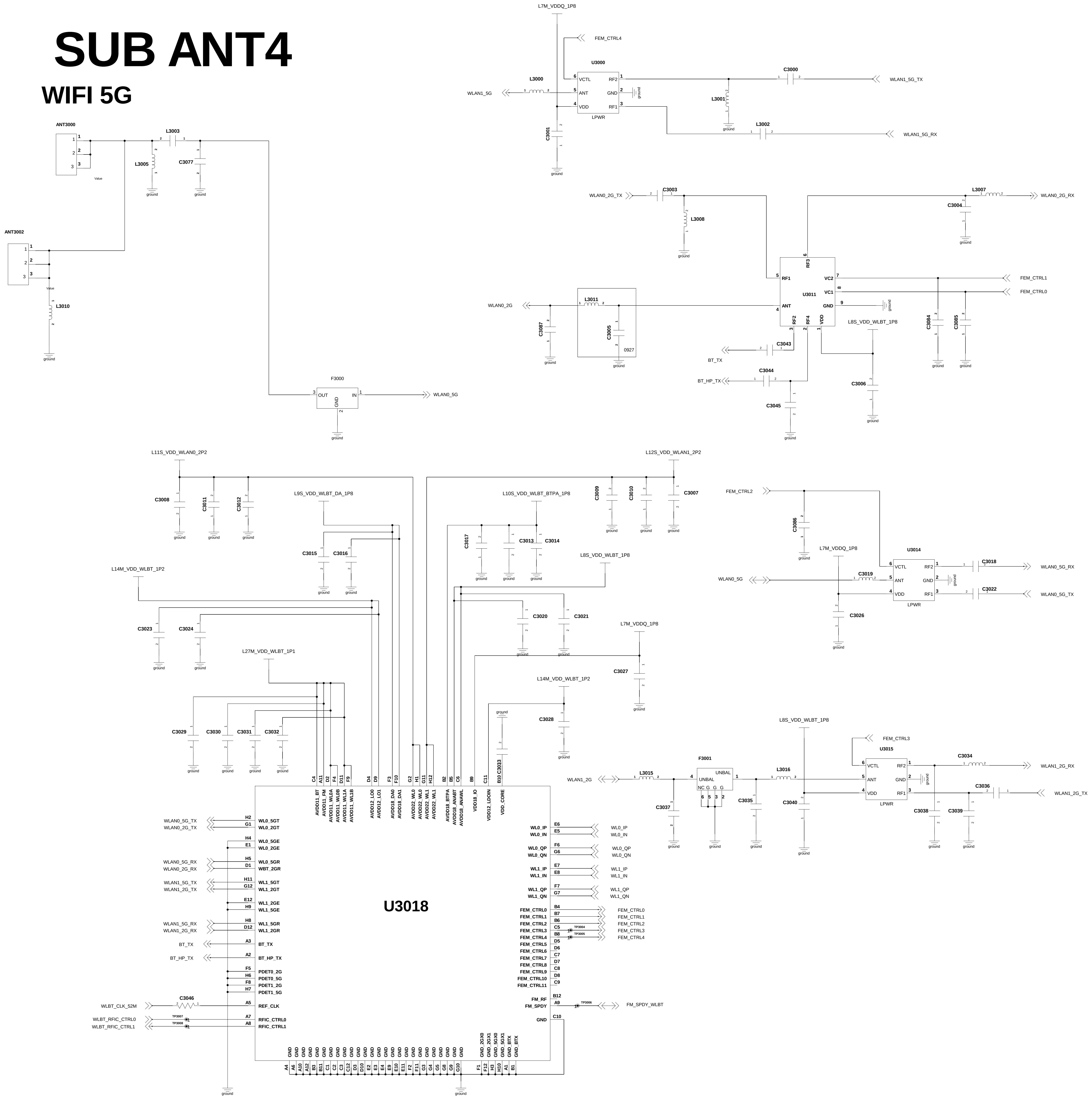
WIFI/BT/FM

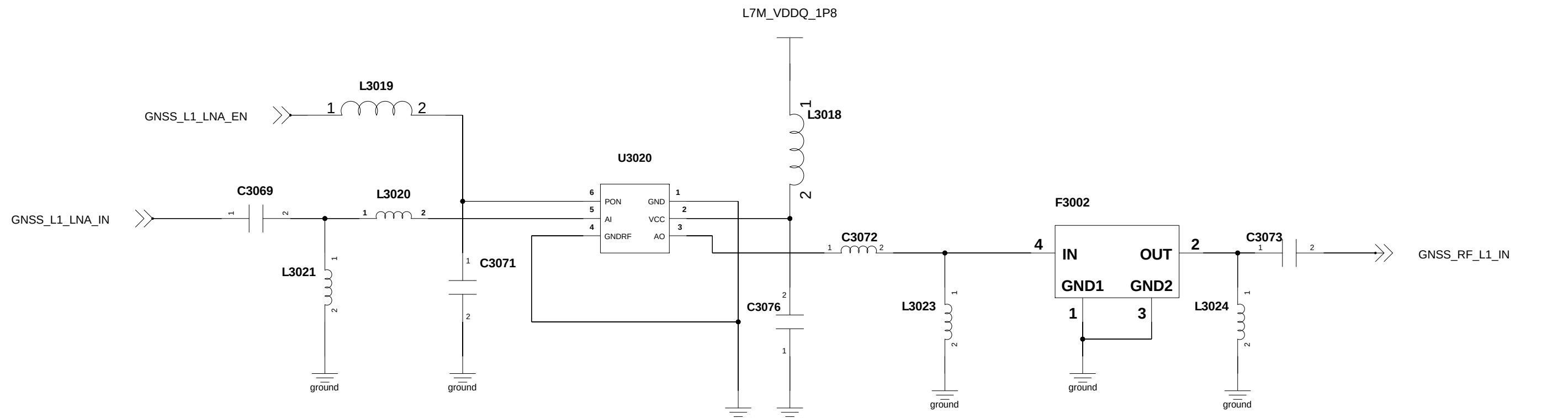
WIFI 2.4G - sub5/sub9
WIFI 5G - sub4/sub9

SUB ANT4

WIFI 5G

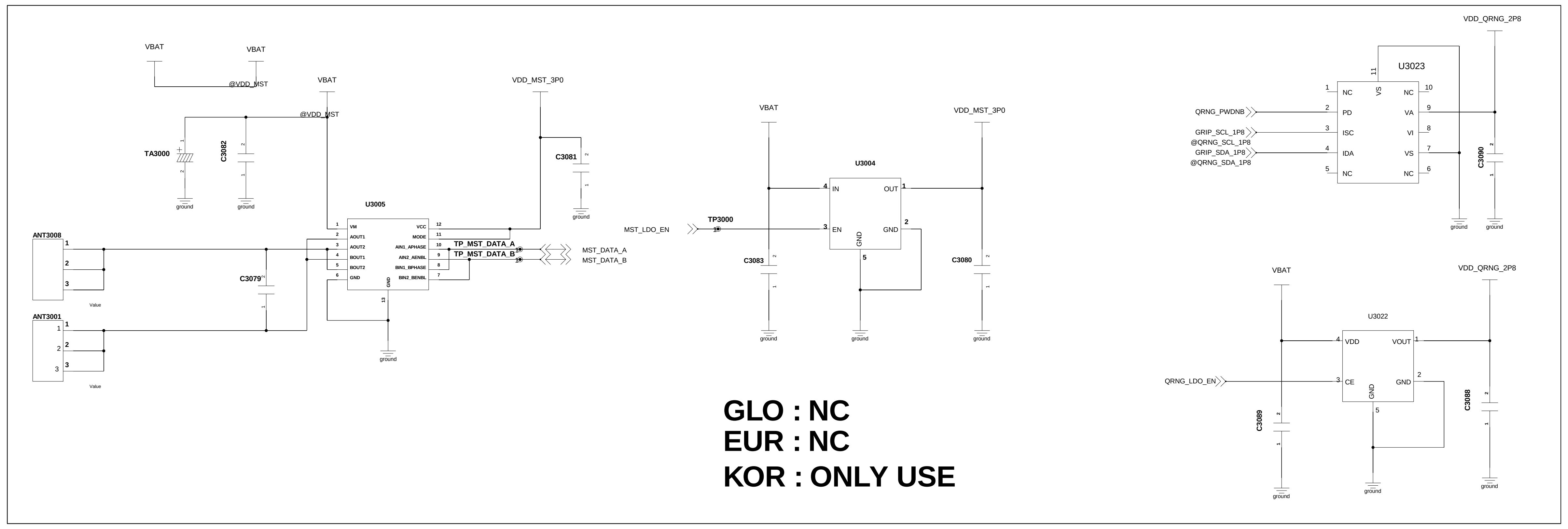

ANT3000
ANT3002
L3003
L3005
C3077
L3010
F3000
OUT
IN
GND
WLAN0_5G
L7M_VDDQ_1P8
FEM_CTRL4
U3000
L3000
WLAN1_5G
VCTL
ANT
VDD
RF2
GND
RF1
LPWR
C3001
C3000
WLAN1_5G_TX
L3001
L3002
WLAN1_5G_RX
WLAN0_2G_TX
C3003
L3008
L3007
WLAN0_2G_RX
C3004
U3011
RF1
RF3
VC2
VC1
ANT
GND
RF2
RF4
VDD
FEM_CTRL1
FEM_CTRL0
WLAN0_2G
L3011
C3087
C3005
0927
BT_TX
C3043
BT_HP_TX
C3044
C3045
L8S_VDD_WLBT_1P8
C3006
C3084
C3085
L11S_VDD_WLAN0_2P2
C3008
C3011
C3012
L9S_VDD_WLBT_DA_1P8
C3015
C3016
L10S_VDD_WLBT_BTPA_1P8
C3017
C3013
C3014
L12S_VDD_WLAN1_2P2
C3009
C3010
C3007
FEM_CTRL2
C3086
L7M_VDDQ_1P8
U3014
C3019
WLAN0_5G
C3018
WLAN0_5G_RX
C3022
WLAN0_5G_TX
C3026
L14M_VDD_WLBT_1P2
C3023
C3024
L8S_VDD_WLBT_1P8
C3020
C3021
L7M_VDDQ_1P8
C3027
L27M_VDD_WLBT_1P1
L14M_VDD_WLBT_1P2
C3028
C3029
C3030
C3031
C3032
C3033
U3018
AVDD11_BT
AVDD11_FM
AVDD11_WL0A
AVDD11_WL0B
AVDD11_WL1A
AVDD11_WL1B
AVDD12_LO0
AVDD12_LO1
AVDD18_DA0
AVDD18_DA1
AVDD22_WL0
AVDD22_WL0
AVDD22_WL1
AVDD22_WL1
AVDD18_BTPA
AVDD18_ANABT
AVDD18_ANAWL
VDD18_IO
VDD12_LDOIN
VDD_CORE
WLAN0_5G_TX
WLAN0_2G_TX
WL0_5GT
WL0_2GT
WL0_5GE
WL0_2GE
WLAN0_5G_RX
WLAN0_2G_RX
WL0_5GR
WBT_2GR
WLAN1_5G_TX
WLAN1_2G_TX
WL1_5GT
WL1_2GT
WL1_2GE
WL1_5GE
WLAN1_5G_RX
WLAN1_2G_RX
WL1_5GR
WL1_2GR
BT_TX
BT_HP_TX
PDET0_2G
PDET0_5G
PDET1_2G
PDET1_5G
WLBT_CLK_52M
C3046
REF_CLK
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
TP3007
TP3008
RFIC_CTRL0
RFIC_CTRL1
WL0_IP
WL0_IN
WL0_QP
WL0_QN
WL1_IP
WL1_IN
WL1_QP
WL1_QN
FEM_CTRL0
FEM_CTRL1
FEM_CTRL2
FEM_CTRL3
FEM_CTRL4
FEM_CTRL5
FEM_CTRL6
FEM_CTRL7
FEM_CTRL8
FEM_CTRL9
FEM_CTRL10
FEM_CTRL11
TP3004
TP3005
FM_RF
FM_SPDY
TP3006
FM_SPDY_WLBT
GND
GND_2GX0
GND_2GX1
GND_5GX0
GND_5GX1
GND_BTX
GND_BTX
WLAN1_2G
L3015
F3001
UNBAL
C3037
C3035
L3016
C3040
FEM_CTRL3
U3015
VCTL
ANT
VDD
RF2
GND
RF1
LPWR
C3034
WLAN1_2G_RX
C3036
WLAN1_2G_TX
C3038
C3039
ground

GPS L1 / GNSS/BEIDOU


L7M_VDDQ_1P8
GNSS_L1_LNA_EN
L3019
L3018
U3020
PON
AI
GNDRF
GND
VCC
AO
GNSS_L1_LNA_IN
C3069
L3020
L3021
C3071
C3076
C3072
L3023
F3002
IN
OUT
GND1
GND2
L3024
C3073
GNSS_RF_L1_IN
ground

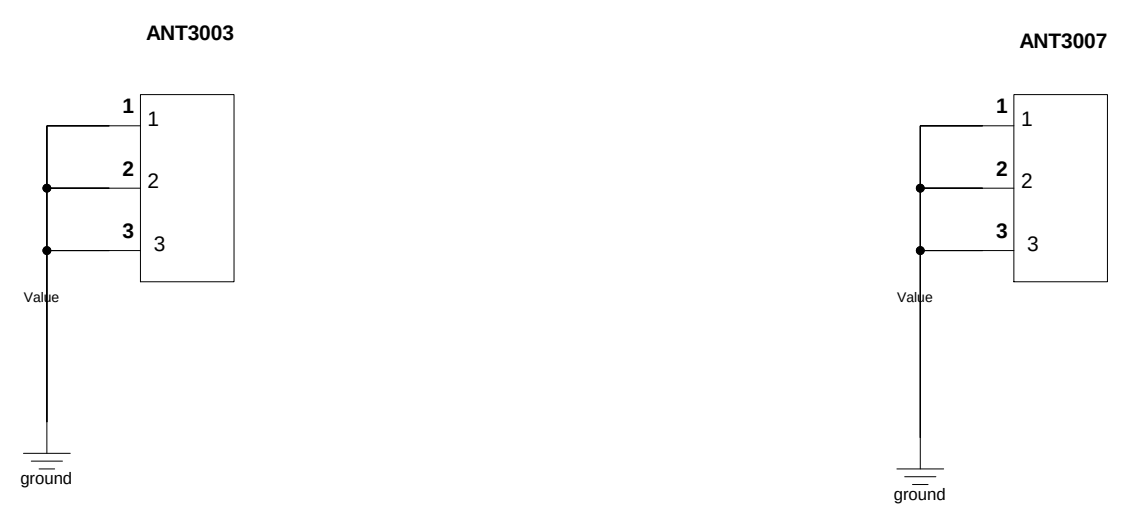
MST KOR : ONLY USE

QRNG KOR : ONLY USE


VBAT
VBAT
@VDD_MST
VBAT
@VDD_MST
VDD_MST_3P0
TA3000
C3082
C3081
U3005
VM
AOUT1
AOUT2
BOUT1
BOUT2
GND
VCC
MODE
AIN1_APHASE
AIN2_AENBL
BIN1_BPHASE
BIN2_BENBL
TP_MST_DATA_A
TP_MST_DATA_B
MST_DATA_A
MST_DATA_B
ANT3008
ANT3001
C3079
VBAT
VDD_MST_3P0
U3004
IN
OUT
EN
GND
MST_LDO_EN
TP3000
C3083
C3080
VDD_QRNG_2P8
U3023
NC
PD
ISC
IDA
VS
VA
VI
QRNG_PWDNB
GRIP_SCL_1P8
@QRNG_SCL_1P8
GRIP_SDA_1P8
@QRNG_SDA_1P8
C3090
VBAT
VDD_QRNG_2P8
U3022
VDD
VOUT
CE
GND
QRNG_LDO_EN
C3089
C3088
ground

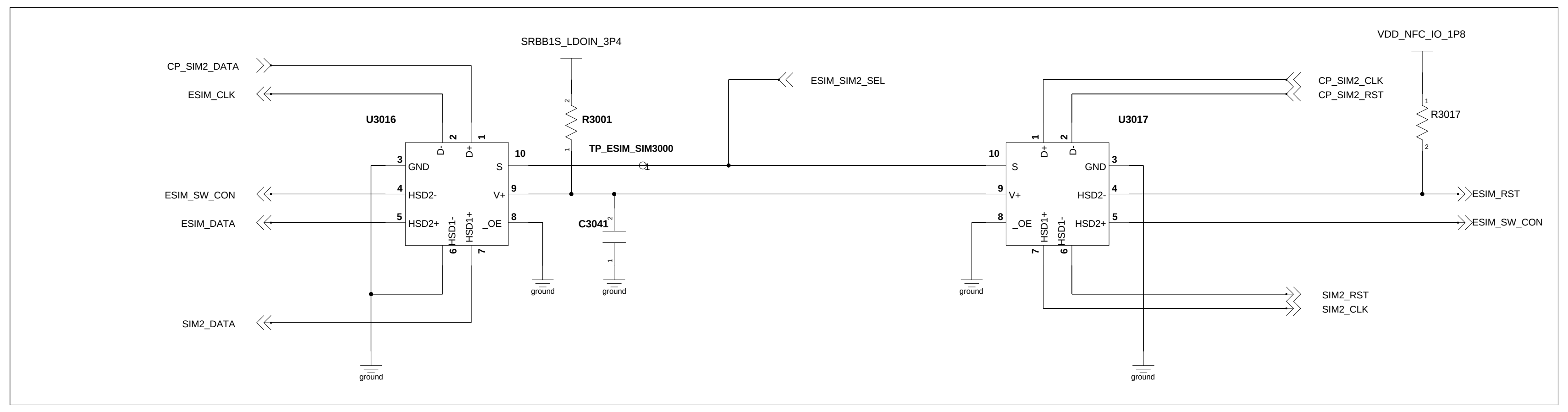
GLO : NC
EUR : NC
KOR : ONLY USE

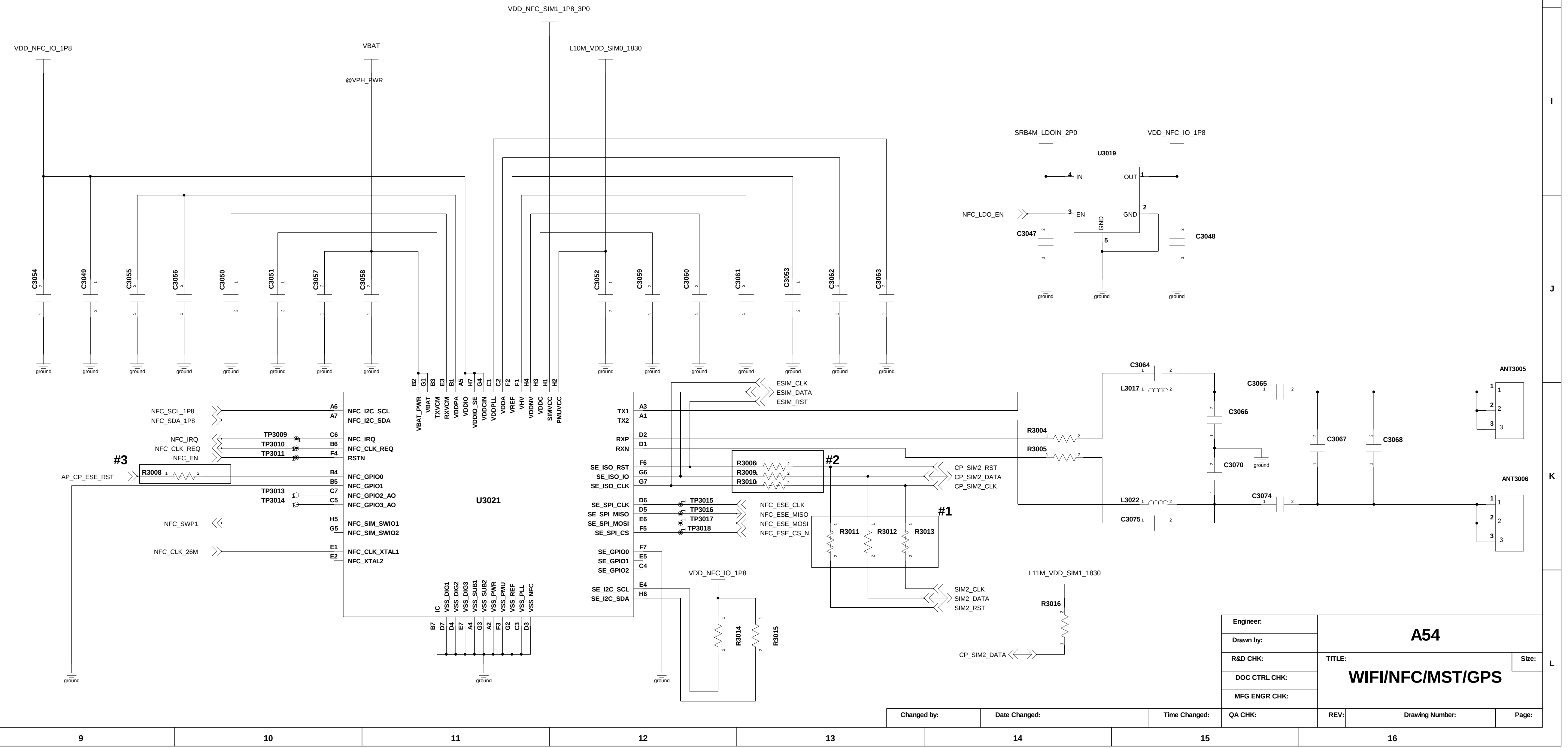
GND CCP


ANT3003
ANT3007
Value
ground

NFC

		NFC IC	#1	#2	#3		Switch IC	ESIM_SIM2_SEL
EUR	2-Nano + eSIM	SN220UV2	OPEN	OPEN	SHORT		SMD	HSD1(H)_SIM2 HSD2(L)_eSIM
KOR/NA	Nano + eSIM	SN220UV2	OPEN	SHORT	SHORT		NC	
JPN(Felica)	Nano + eSIM	SN220PV2	OPEN	SHORT	SHORT		NC	
Other	eSIM X	PN560	SHORT	OPEN	OPEN		NC	
CHN	eSIM X(eSE O)	SN220T	SHORT	OPEN	OPEN		NC	


CP_SIM2_DATA
ESIM_CLK
U3016
GND
HSD2-
HSD2+
HSD1-
HSD1+
S
V+
_OE
ESIM_SW_CON
ESIM_DATA
SIM2_DATA
SRBB1S_LDOIN_3P4
R3001
TP_ESIM_SIM3000
C3041
ESIM_SIM2_SEL
U3017
CP_SIM2_CLK
CP_SIM2_RST
VDD_NFC_IO_1P8
R3017
ESIM_RST
ESIM_SW_CON
SIM2_RST
SIM2_CLK
ground


VDD_NFC_SIM1_1P8_3P0
L10M_VDD_SIM0_1830
VDD_NFC_IO_1P8
VBAT
@VPH_PWR
C3054
C3049
C3055
C3056
C3050
C3051
C3057
C3058
C3052
C3059
C3060
C3061
C3053
C3062
C3063
SRB4M_LDOIN_2P0
VDD_NFC_IO_1P8
U3019
IN
OUT
EN
GND
NFC_LDO_EN
C3047
C3048
U3021
VBAT_PWR
VBAT
TXVCM
RXVCM
VDDPA
VDDIO
VDDIO_SE
VDDCIN
VDDPLL
VDDA
VREF
VHV
VDDNV
VDDC
SIMVCC
PMUVCC
NFC_SCL_1P8
NFC_SDA_1P8
NFC_I2C_SCL
NFC_I2C_SDA
NFC_IRQ
NFC_CLK_REQ
NFC_EN
TP3009
TP3010
TP3011
RSTN
#3
R3008
AP_CP_ESE_RST
NFC_GPIO0
NFC_GPIO1
NFC_GPIO2_AO
NFC_GPIO3_AO
TP3013
TP3014
NFC_SWP1
NFC_SIM_SWIO1
NFC_SIM_SWIO2
NFC_CLK_26M
NFC_CLK_XTAL1
NFC_XTAL2
IC
VSS_DIG1
VSS_DIG2
VSS_DIG3
VSS_SUB1
VSS_SUB2
VSS_PWR
VSS_PMU
VSS_REF
VSS_PLL
VSS_NFC
TX1
TX2
RXP
RXN
SE_ISO_RST
SE_ISO_IO
SE_ISO_CLK
SE_SPI_CLK
SE_SPI_MISO
SE_SPI_MOSI
SE_SPI_CS
SE_GPIO0
SE_GPIO1
SE_GPIO2
SE_I2C_SCL
SE_I2C_SDA
ESIM_CLK
ESIM_DATA
ESIM_RST
R3006
R3009
R3010
#2
CP_SIM2_RST
CP_SIM2_DATA
CP_SIM2_CLK
TP3015
TP3016
TP3017
TP3018
NFC_ESE_CLK
NFC_ESE_MISO
NFC_ESE_MOSI
NFC_ESE_CS_N
#1
R3011
R3012
R3013
SIM2_CLK
SIM2_DATA
SIM2_RST
VDD_NFC_IO_1P8
R3014
R3015
L11M_VDD_SIM1_1830
R3016
CP_SIM2_DATA
C3064
L3017
R3004
R3005
L3022
C3075
C3066
C3070
C3065
C3074
C3067
C3068
ANT3005
ANT3006
ground

Engineer:	A54	
Drawn by:		
R&D CHK:	TITLE: WIFI/NFC/MST/GPS	Size:
DOC CTRL CHK:		
MFG ENGR CHK:		

Changed by:	Date Changed:	Time Changed:	QA CHK:	REV:	Drawing Number:	Page:

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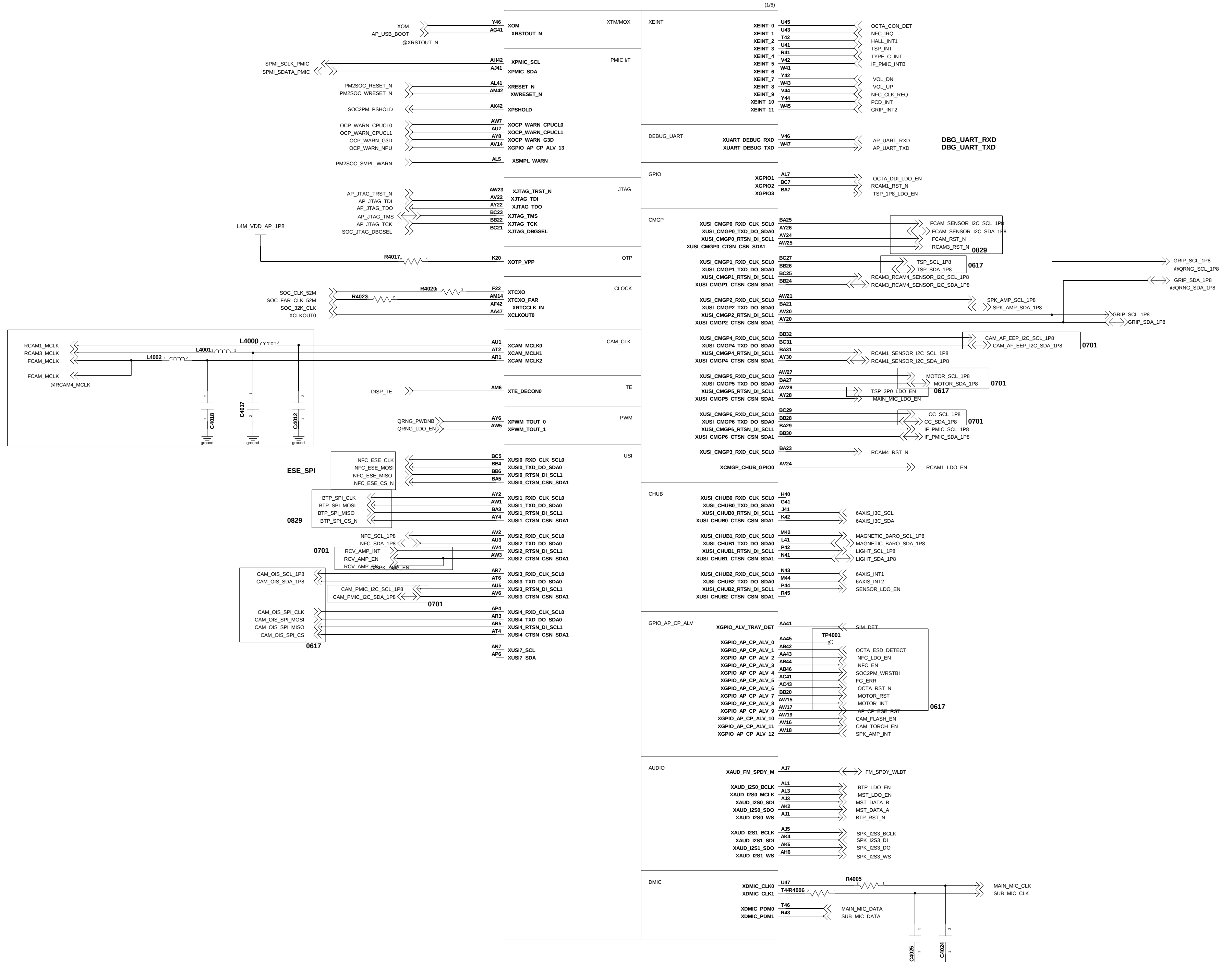
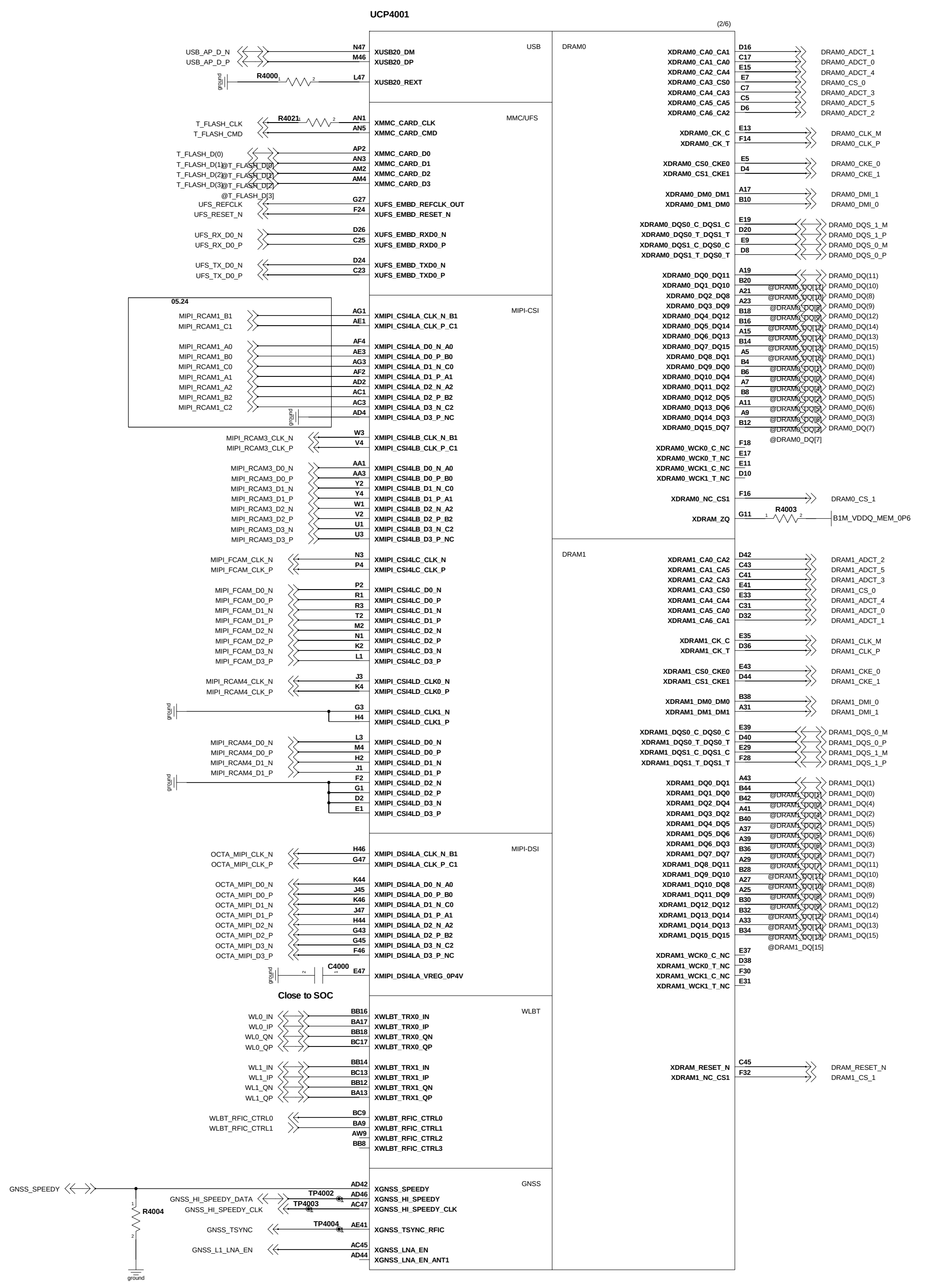
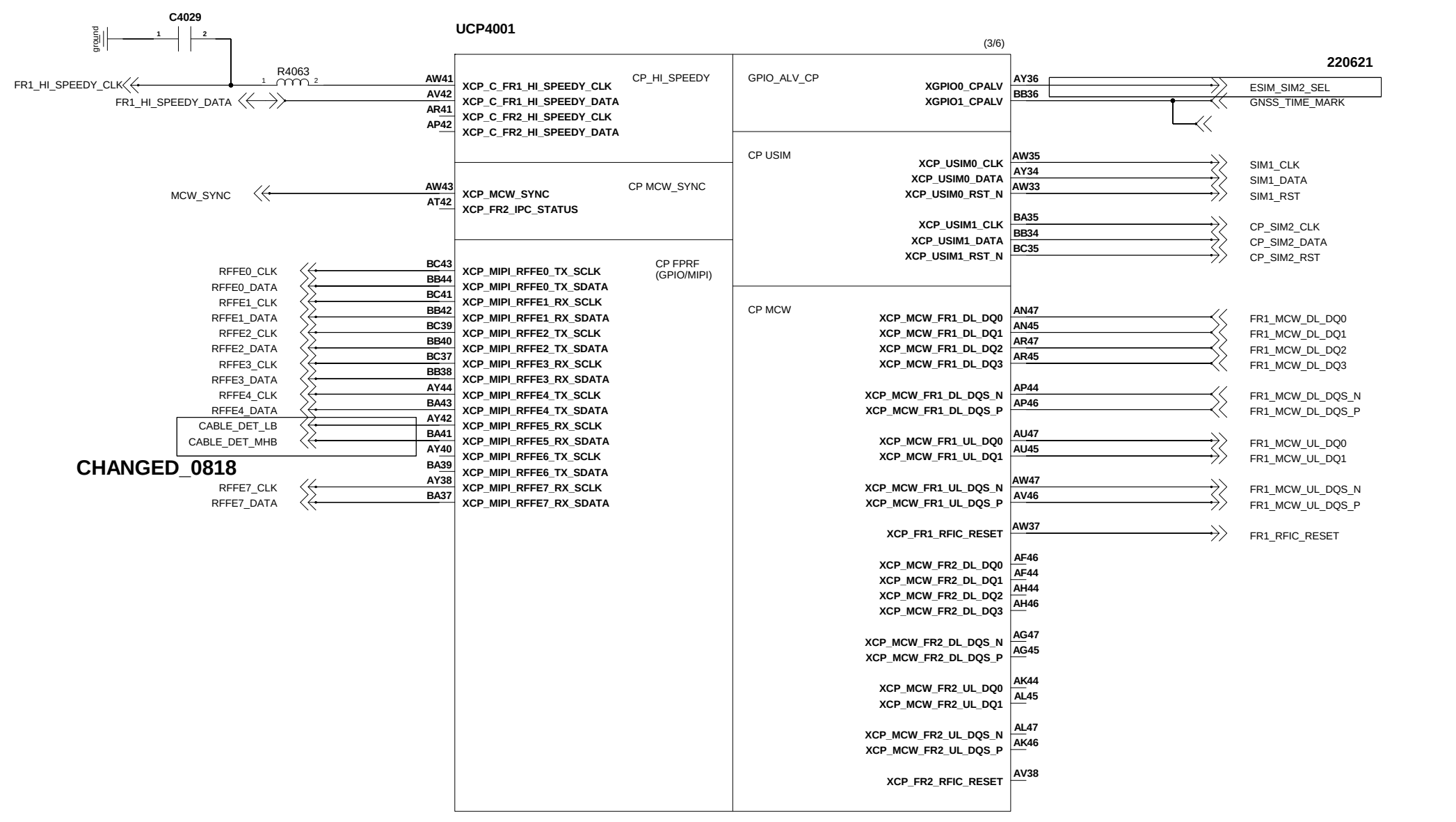
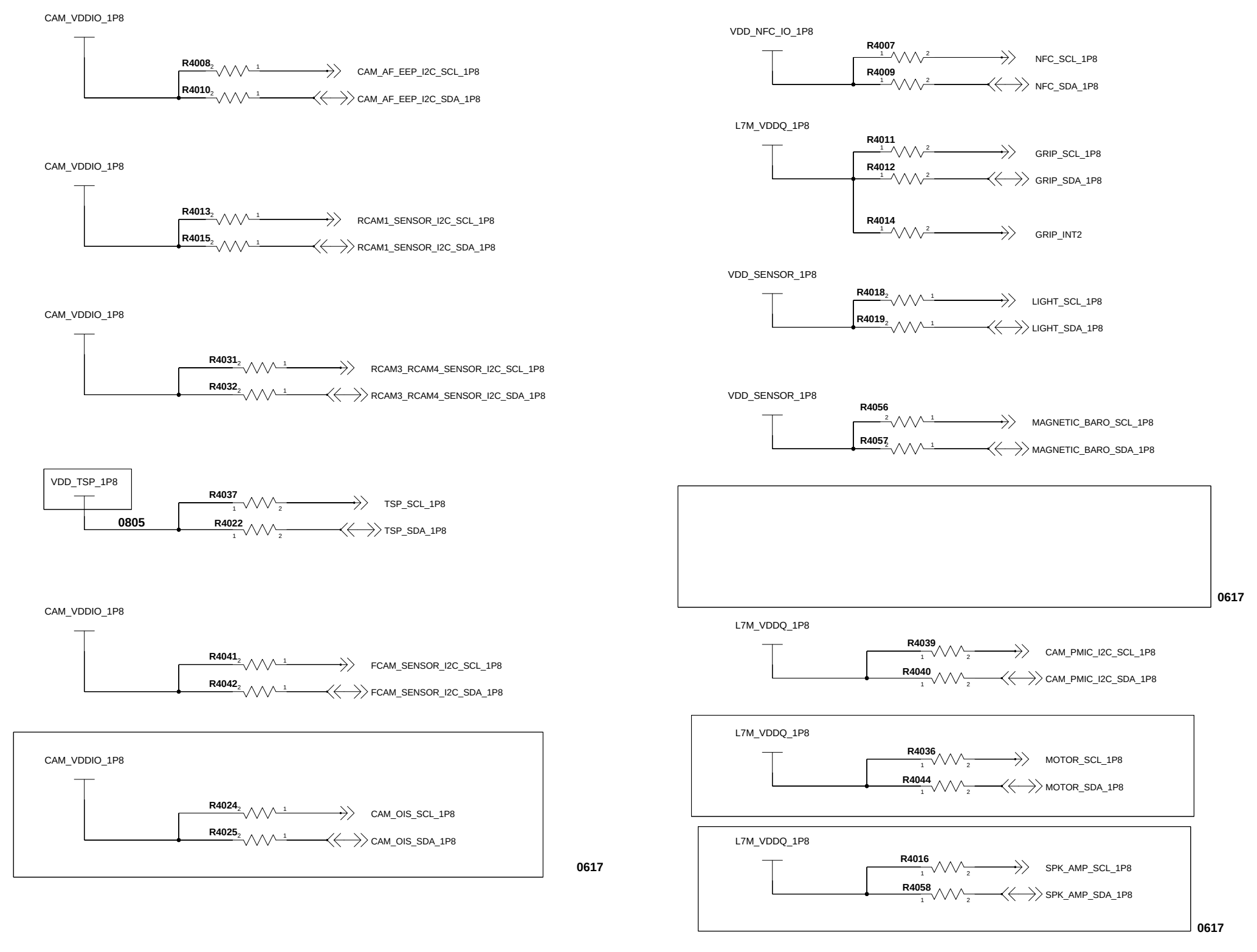
Quartz

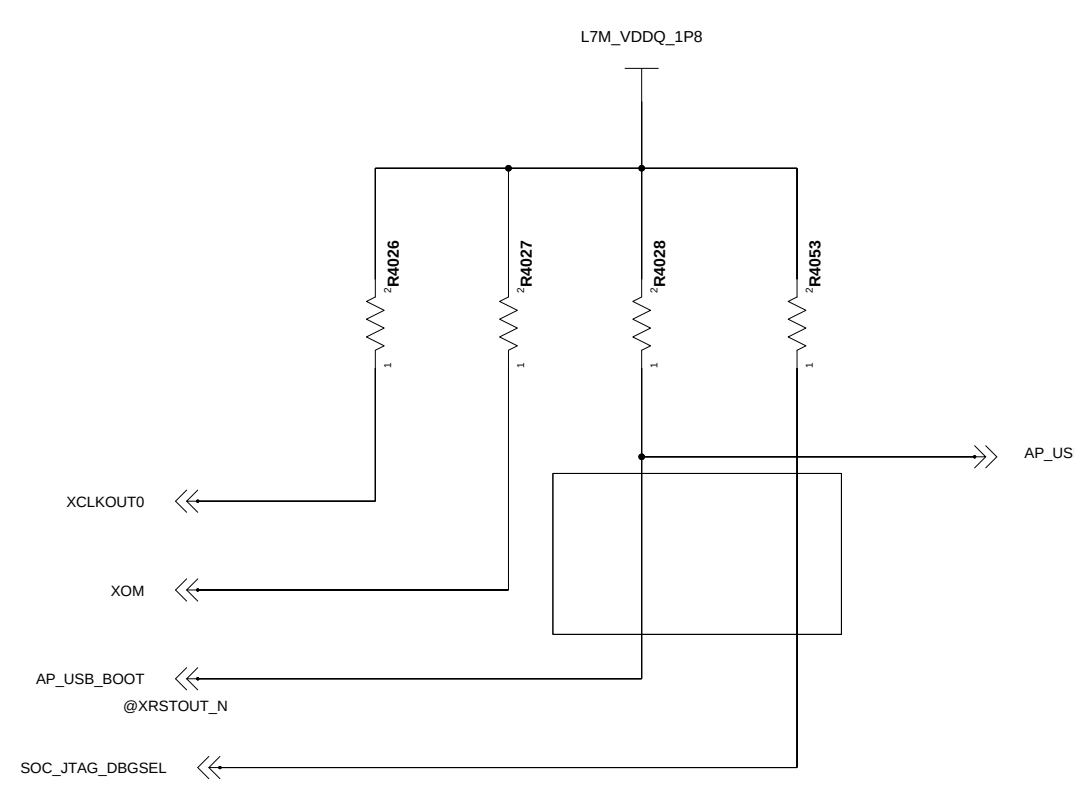

UCP4001
(1/6)
XTM/MOX
XOM
XRSTOUT_N
AP_USB_BOOT
@XRSTOUT_N
PMIC I/F
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
PM2SOC_RESET_N
PM2SOC_WRESET_N
SOC2PM_PSHOLD
OCP_WARN_CPUCL0
OCP_WARN_CPUCL1
OCP_WARN_G3D
OCP_WARN_NPU
PM2SOC_SMPL_WARN
JTAG
AP_JTAG_TRST_N
AP_JTAG_TDI
AP_JTAG_TDO
AP_JTAG_TMS
AP_JTAG_TCK
SOC_JTAG_DBGSEL
L4M_VDD_AP_1P8
R4017
OTP
XOTP_VPP
CLOCK
SOC_CLK_52M
SOC_FAR_CLK_52M
SOC_32K_CLK
XCLKOUT0
R4020
R4023
CAM_CLK
RCAM1_MCLK
RCAM3_MCLK
FCAM_MCLK
@RCAM4_MCLK
L4000
L4001
L4002
C4018
C4017
C4012
TE
DISP_TE
PWM
QRNG_PWDNB
QRNG_LDO_EN
USI
ESE_SPI
NFC_ESE_CLK
NFC_ESE_MOSI
NFC_ESE_MISO
NFC_ESE_CS_N
0829
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
BTP_SPI_CS_N
0701
NFC_SCL_1P8
NFC_SDA_1P8
RCV_AMP_INT
RCV_AMP_EN
SPK_AMP_EN
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
CAM_OIS_SPI_CLK
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
0617
XEINT
OCTA_CON_DET
NFC_IRQ
HALL_INT1
TSP_INT
TYPE_C_INT
IF_PMIC_INTB
VOL_DN
VOL_UP
NFC_CLK_REQ
PCD_INT
GRIP_INT2
DEBUG_UART
AP_UART_RXD
AP_UART_TXD
DBG_UART_RXD
DBG_UART_TXD
GPIO
OCTA_DDI_LDO_EN
RCAM1_RST_N
TSP_1P8_LDO_EN
CMGP
FCAM_SENSOR_I2C_SCL_1P8
FCAM_SENSOR_I2C_SDA_1P8
FCAM_RST_N
RCAM3_RST_N
TSP_SCL_1P8
TSP_SDA_1P8
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
GRIP_SCL_1P8
GRIP_SDA_1P8
@QRNG_SCL_1P8
@QRNG_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
RCAM1_SENSOR_I2C_SCL_1P8
RCAM1_SENSOR_I2C_SDA_1P8
MOTOR_SCL_1P8
MOTOR_SDA_1P8
TSP_3P0_LDO_EN
MAIN_MIC_LDO_EN
CC_SCL_1P8
CC_SDA_1P8
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
RCAM4_RST_N
RCAM1_LDO_EN
CHUB
6AXIS_I3C_SCL
6AXIS_I3C_SDA
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
LIGHT_SCL_1P8
LIGHT_SDA_1P8
6AXIS_INT1
6AXIS_INT2
SENSOR_LDO_EN
GPIO_AP_CP_ALV
SIM_DET
TP4001
OCTA_ESD_DETECT
NFC_LDO_EN
NFC_EN
SOC2PM_WRSTBI
FG_ERR
OCTA_RST_N
MOTOR_RST
MOTOR_INT
AP_CP_ESE_RST
CAM_FLASH_EN
CAM_TORCH_EN
SPK_AMP_INT
AUDIO
FM_SPDY_WLBT
BTP_LDO_EN
MST_LDO_EN
MST_DATA_B
MST_DATA_A
BTP_RST_N
SPK_I2S3_BCLK
SPK_I2S3_DI
SPK_I2S3_DO
SPK_I2S3_WS
DMIC
R4005
R4006
MAIN_MIC_CLK
SUB_MIC_CLK
MAIN_MIC_DATA
SUB_MIC_DATA
C4025
C4024
ground


UCP4001
(2/6)
USB
USB_AP_D_N
USB_AP_D_P
R4000
XUSB20_REXT
MMC/UFS
R4021
T_FLASH_CLK
T_FLASH_CMD
T_FLASH_D(0)
T_FLASH_D(1)
T_FLASH_D(2)
T_FLASH_D(3)
UFS_REFCLK
UFS_RESET_N
UFS_RX_D0_N
UFS_RX_D0_P
UFS_TX_D0_N
UFS_TX_D0_P
MIPI-CSI
05.24
MIPI_RCAM1_B1
MIPI_RCAM1_C1
MIPI_RCAM3_CLK_N
MIPI_RCAM3_CLK_P
MIPI_FCAM_CLK_N
MIPI_FCAM_CLK_P
MIPI_RCAM4_CLK_N
MIPI_RCAM4_CLK_P
MIPI-DSI
OCTA_MIPI_CLK_N
OCTA_MIPI_CLK_P
C4000
Close to SOC
WLBT
WL0_IN
WL0_IP
WL0_QN
WL0_QP
WL1_IN
WL1_IP
WL1_QN
WL1_QP
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
GNSS
GNSS_SPEEDY
GNSS_HI_SPEEDY_DATA
GNSS_HI_SPEEDY_CLK
TP4002
TP4003
TP4004
R4004
GNSS_TSYNC
GNSS_L1_LNA_EN
DRAM0
DRAM0_CLK_M
DRAM0_CLK_P
DRAM0_CKE_0
DRAM0_CKE_1
R4003
B1M_VDDQ_MEM_0P6
DRAM1
DRAM1_CLK_M
DRAM1_CLK_P
DRAM_RESET_N
DRAM1_CS_1


UCP4001
(3/6)
C4029
R4063
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
CP_HI_SPEEDY
MCW_SYNC
CP MCW_SYNC
CP FPRF (GPIO/MIPI)
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
RFFE4_CLK
RFFE4_DATA
CABLE_DET_LB
CABLE_DET_MHB
CHANGED_0818
RFFE7_CLK
RFFE7_DATA
GPIO_ALV_CP
220621
ESIM_SIM2_SEL
GNSS_TIME_MARK
CP USIM
SIM1_CLK
SIM1_DATA
SIM1_RST
CP_SIM2_CLK
CP_SIM2_DATA
CP_SIM2_RST
CP MCW
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
FR1_RFIC_RESET

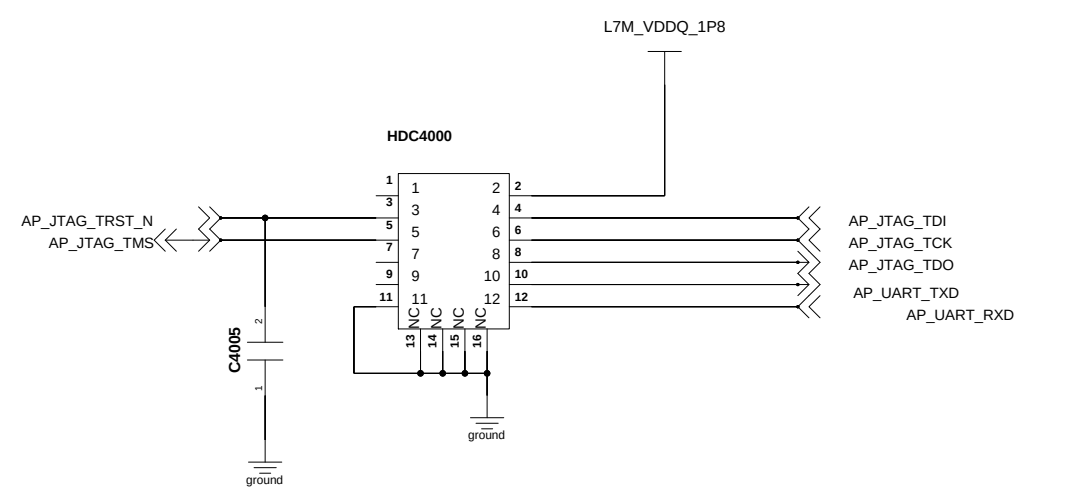
I2C PU RESISTOR


CAM_VDDIO_1P8
R4008
R4010
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
R4013
R4015
RCAM1_SENSOR_I2C_SCL_1P8
RCAM1_SENSOR_I2C_SDA_1P8
R4031
R4032
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
VDD_TSP_1P8
0805
R4037
R4022
TSP_SCL_1P8
TSP_SDA_1P8
R4041
R4042
FCAM_SENSOR_I2C_SCL_1P8
FCAM_SENSOR_I2C_SDA_1P8
R4024
R4025
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
0617
VDD_NFC_IO_1P8
R4007
R4009
NFC_SCL_1P8
NFC_SDA_1P8
L7M_VDDQ_1P8
R4011
R4012
R4014
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
VDD_SENSOR_1P8
R4018
R4019
LIGHT_SCL_1P8
LIGHT_SDA_1P8
R4056
R4057
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
0617
R4039
R4040
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
R4036
R4044
MOTOR_SCL_1P8
MOTOR_SDA_1P8
R4016
R4058
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
0617


L7M_VDDQ_1P8
R4026
R4027
R4028
R4053
AP_USB_BOOT
XCLKOUT0
XOM
AP_USB_BOOT
@XRSTOUT_N
SOC_JTAG_DBGSEL

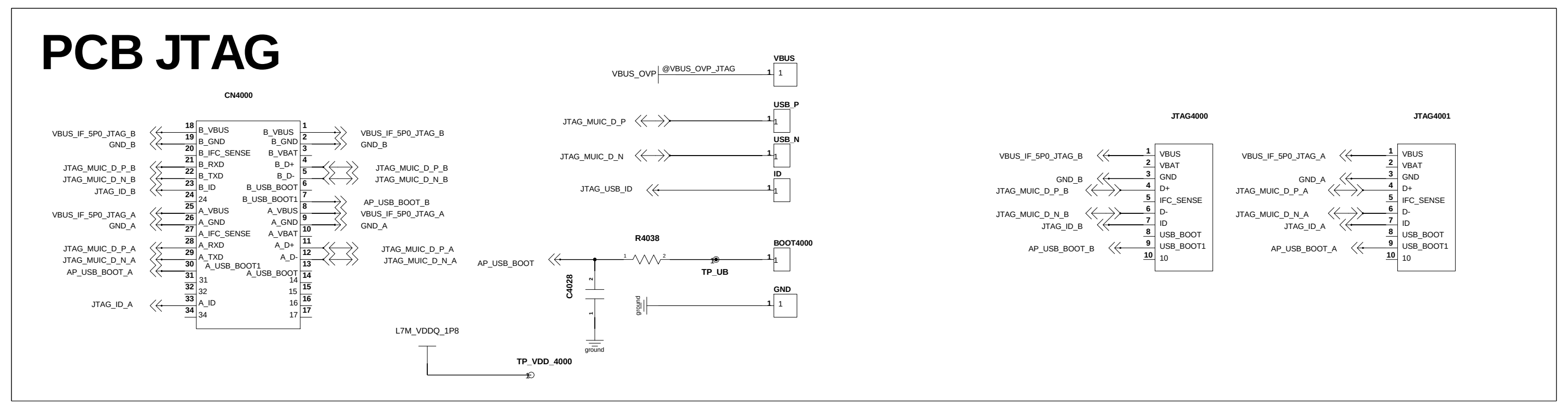
PIN			Function	
XCLKOUT0	XRSTOUT_N	XOM	1st Boot	2nd Boot
0	0	0	UFS	USB
0	1	X	USB	----
1	0	0	SD	USB

BOOT CONFIGURATIONS

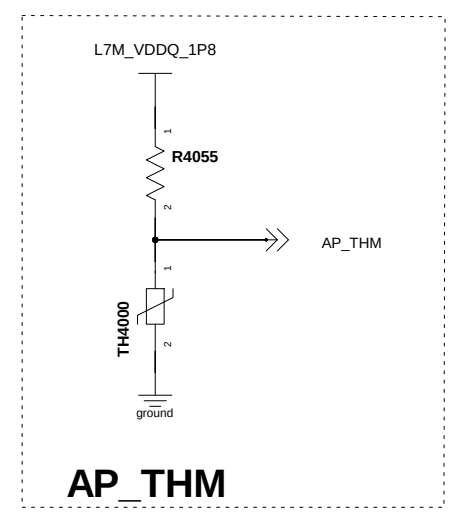

L7M_VDDQ_1P8
HDC4000
AP_JTAG_TRST_N
AP_JTAG_TMS
C4005
AP_JTAG_TDI
AP_JTAG_TCK
AP_JTAG_TDO
AP_UART_TXD
AP_UART_RXD
ground

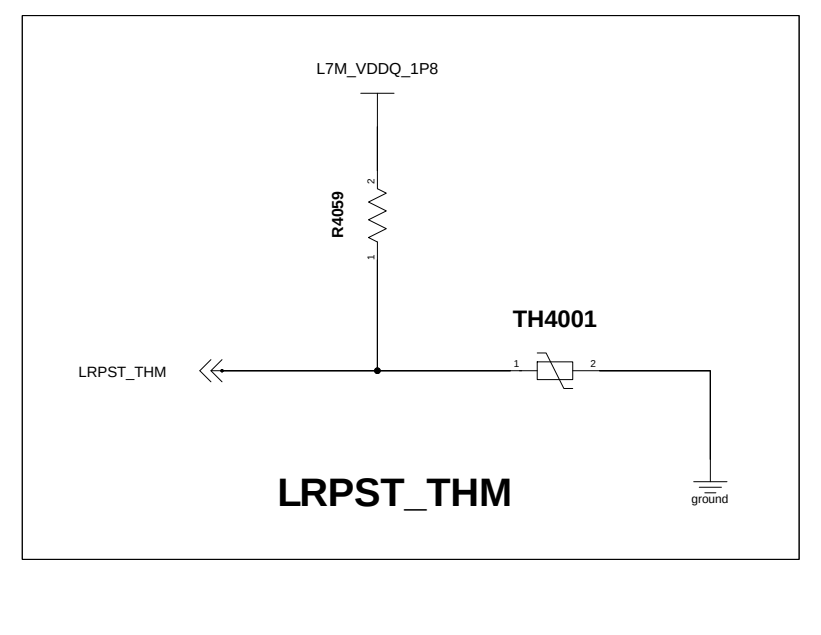
AP JTAG

PCB JTAG

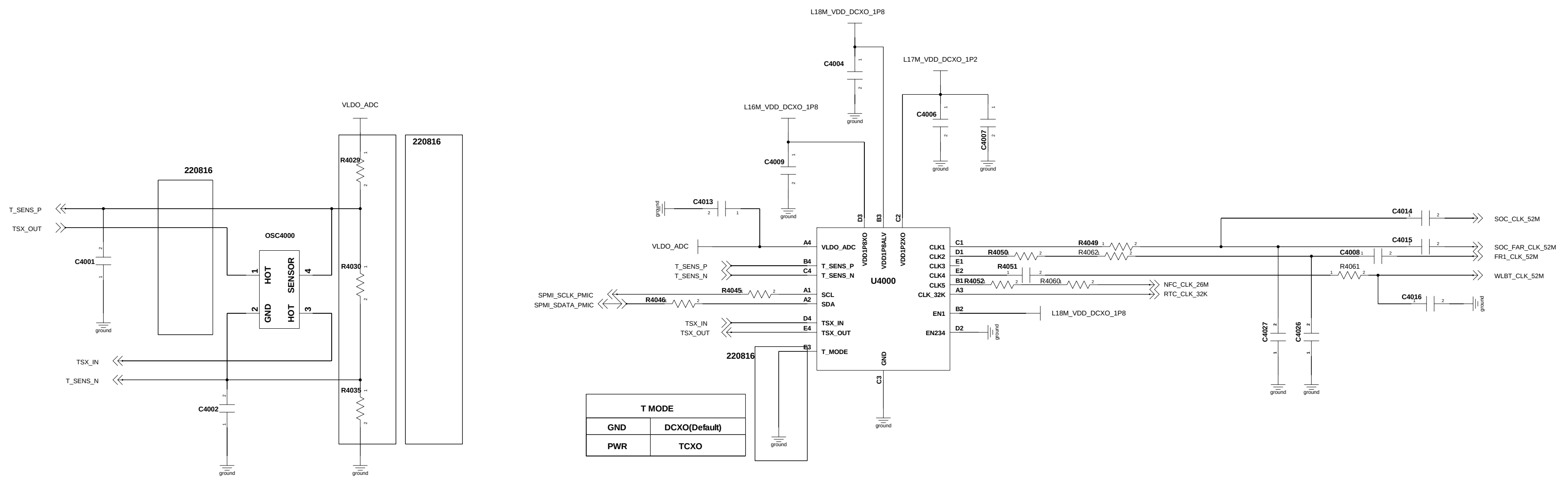

CN4000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
AP_USB_BOOT_A
JTAG_ID_A
AP_USB_BOOT_B
VBUS_OVP
@VBUS_OVP_JTAG
VBUS
USB_P
USB_N
ID
JTAG_MUIC_D_P
JTAG_MUIC_D_N
JTAG_USB_ID
R4038
AP_USB_BOOT
C4028
TP_UB
BOOT4000
GND
L7M_VDDQ_1P8
TP_VDD_4000
JTAG4000
JTAG4001
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT
USB_BOOT1

SOC THERMISTOR


L7M_VDDQ_1P8
R4055
AP_THM
TH4000
ground
AP_THM


L7M_VDDQ_1P8
R4059
TH4001
LRPST_THM
ground
LRPST_THM

SYSTEM CLOCK


VLDO_ADC
220816
T_SENS_P
TSX_OUT
C4001
OSC4000
HOT
SENSOR
GND
HOT
TSX_IN
T_SENS_N
C4002
R4029
R4030
R4035
220816
ground
L18M_VDD_DCXO_1P8
C4004
L17M_VDD_DCXO_1P2
C4006
C4007
L16M_VDD_DCXO_1P8
C4009
C4013
VLDO_ADC
T_SENS_P
T_SENS_N
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
R4045
R4046
TSX_IN
TSX_OUT
220816
U4000
VLDO_ADC
SCL
SDA
T_MODE
VDD1P8XO
VDD1P8ALV
VDD1P2XO
CLK1
CLK2
CLK3
CLK4
CLK5
CLK_32K
EN1
EN234
GND
R4049
R4050
R4051
R4052
R4060
R4061
R4062
C4014
C4015
C4008
C4016
C4027
C4026
SOC_CLK_52M
SOC_FAR_CLK_52M
FR1_CLK_52M
WLBT_CLK_52M
NFC_CLK_26M
RTC_CLK_32K
L18M_VDD_DCXO_1P8
ground

T MODE	
GND	DCXO(Default)
PWR	TCXO

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:

A54

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Drawing Number:
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UCP4001
(4/6)
(5/6)
(6/6)
UME5000
B1M_VDDQ_MEM_0P6
@RSP1M
@RSN1M
C5007
C5000
C5125
C5126
VDDQ06_DRAM0
VDDQ06_DRAM1
220608
B2_3M_VDD_CAM_1P3
@RSP2_3M
@RSN2_3M
C5132
C5016
C5012
C5001
VDD_CAM
B4_5_6M_VDD_CPUCL1_BIG_1P3
@RSP4_5_6M
@RSN4_5_6M
C5229
C5228
C5138
C5139
C5140
C5219
C5220
C5227
C5221
C5222
C5223
C5224
C5225
C5226
VDD_CPUCL1_BIG
B7M_VDD_CPUCL0_LIT_1P3
@RSP7M
@RSN7M
C5043
C5035
VDD_CPUCL0_LIT
B8M_VDD2_MEM_1P1
C5213
C5053
VDDQ11_DRAM0_CKE
VDDQ11_DRAM1_CKE
L1M_VDD_ALIVE_0P75
C5215
C5216
VDD_ALIVE
VDD_ALIVE_JSC
VDD_CHUBVTS
AVDD075_USB20
VDD075_DLL_GNSS
VDD_GNSS_ALV
VDD_ALV_CP
C5145
C5144
C5143
C5142
L2M_VDD_AP_1833
VDDQ1833_MMC_CARD
C5146
ground
C5006
AVDD085_UFS_EMBD
AVDD085_MIPI_DSI
AVDD085_MIPI_CSI
VDD_ICPU_CSILINK
L3M_VDD_AP_0P85
C5147
C5148
C5149
C5150
AVDD18_UFS_EMBD
AVDD18_MIPI_DSI
AVDD18_MIPI_CSI
AVDD18_OTP
AVDD18_TS
L4M_VDD_AP_1P8
C5151
C5152
C5153
C5154
AVDD12_PLL_AUD
AVDD12_PLL_CPU
AVDD12_PLL_SHARED
AVDD12_PLL_MIF
AVDD12_MIPI_DSI
VDDQ12_UFS_EMBD
L5M_VDD_AP_PLL_1P2
C5155
C5156
C5157
AVDD33_USB20
L6M_VDD_USB20_3P0
C5004
AVDD18_TCXO_FAR
AVDD18_TCXO
AVDD18_USB20
VDDQ18_GPIO_ALV_CP
VDDQ18_MCW_BIG
VDDQ18_MCW_CP
VDDQ1218_GNSS
VDDQ18_WLBT
VDDQ18_GNSS
VDDQ18_GPIO_B
VDDQ18_GPIO_T
VDDQ18_CMGP
L7M_VDDQ_1P8
C5158
C5159
C5160
C5161
VDD_MCW_CP
L8M_VDD_MCW_0P75
C5046
C5162
VDD06_MCW_FR12_CP
L9M_VDD_MCW_0P6
C5164
C5165
VDDQ1830_USIM0_CP
L10M_VDD_SIM0_1830
C5163
VDDQ1830_USIM1_CP
L11M_VDD_SIM1_1830
C5166
VDD_GNSS
L12M_VDD_GNSS_0P75
C5167
C5168
AVDD12_CDPLL_CP
AVDD12_IMPLL_CP
L13M_VDD_CP_PLL_1P2
C5169
C5170
AVDD12_WLBT
L14M_VDD_WLBT_1P2
C5171
VDD_WLBT_M
L20M_VDD_WLBT_M_0P75
C5172
VDD_SRAM_PPC_LIT
L34M_VDD_SRAM_PPC_LIT_0P75
C5173
C5174
VDD_SRAM_PPC_BIG
L35M_VDD_SRAM_PPC_BIG_0P75
C5175
C5176
C5177
VSS
B8M_VDD2_MEM_1P1
@RSP8M
@RSN8M
C5207
C5047
C5050
C5051
C5206
C5208
220801
B1S_VDD_MIF_0P75
@RSP1S
@RSN1S
C5209
C5130
C5068
VDD_MIF
B2S_VDD_WLBT_0P75
@RSP2S
@RSN2S
C5214
C5217
VDD_WLBT
B3S_VDD_NPU_0P75
@RSP3S
@RSN3S
C5087
C5210
VDD_NPU
B4_5S_VDD_G3D_0P75
@RSP4_5S
@RSN4_5S
C5178
C5088
C5089
VDD_G3D
B6S_VDD_CPU_CP_0P75
@RSP6S
@RSN6S
C5184
C5180
VDD_CPU_CP
VDD_CP
B7_8S_VDD_CP_0P75
@RSP7_8S
@RSN7_8S
C5185
C5186
C5191
C5190
VDD_INT
B9S_VDD_INT_0P75
@RSP9S
@RSN9S
C5192
C5193
C5198
C5200
VDD_SRAM
B10S_VDD_SRAM_0P75
@RSP10S
@RSN10S
C5201
C5202
C5203
AVDD18_WLBT
L8S_VDD_WLBT_1P8
C5204
VDD_SRAM_PPC_NPU
L15S_VDD_SRAM_PPC_NPU_0P75
C5205
L29M_VDD_UFS_2P95
C5076
C5077
L30M_VDD_UFS_1P8
C5060
C5061
C5062
L7M_VDDQ_1P8
C5063
C5064
C5065
C5066
C5078
C5212
C5211
B1M_VDDQ_MEM_0P6
C5069
C5070
C5071
C5072
C5073
C5074
C5075
R5000
R5001
VCC
VCCQ
VCCQ2
VDD1
VDD2
VDDI
VDDIQ
VDDIQ2
VDDQ
ZQ0_A
ZQ1_A
DRAM0_ADCT_0
DRAM0_ADCT_1
DRAM0_ADCT_2
DRAM0_ADCT_3
DRAM0_ADCT_4
DRAM0_ADCT_5
DRAM0_CLK_M
DRAM0_CLK_P
DRAM0_CKE_0
DRAM0_CKE_1
DRAM0_CS_0
DRAM0_CS_1
DRAM0_DMI_0
DRAM0_DMI_1
DRAM0_DQ(0)
DRAM0_DQS_0_M
DRAM0_DQS_0_P
DRAM0_DQS_1_M
DRAM0_DQS_1_P
UFS_TX_D0_N
UFS_TX_D0_P
UFS_RX_D0_N
UFS_RX_D0_P
UFS_REFCLK
TP5003
TP5004
TP5005
TP5006
TP5007
DIN0_C
DIN0_T
DIN1_C
DIN1_T
DOUT0_C
DOUT0_T
DOUT1_C
DOUT1_T
REF_CLK
RFU
NC
VSS_G
DRAM1_ADCT_0
DRAM1_ADCT_1
DRAM1_ADCT_2
DRAM1_ADCT_3
DRAM1_ADCT_4
DRAM1_ADCT_5
DRAM1_CLK_M
DRAM1_CLK_P
DRAM1_CKE_0
DRAM1_CKE_1
DRAM1_CS_0
DRAM1_CS_1
DRAM1_DMI_0
DRAM1_DMI_1
DRAM1_DQ(0)
DRAM1_DQS_0_M
DRAM1_DQS_0_P
DRAM1_DQS_1_M
DRAM1_DQS_1_P
ODT_B
RESET_N
RST_N
TP5001
TP5002
DRAM_RESET_N
UFS_RESET_N
DNU
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
TP5008
TP5009

6GB + 128GB uMCP : **1108-000714**
8GB + 128GB uMCP : **1108-000721**
8GB + 256GB uMCP : **1108-000722**

Engineer:	A54		
Drawn by:			
R&D CHK:	TITLE:		Size:
DOC CTRL CHK:	AP/CP/uMCP		
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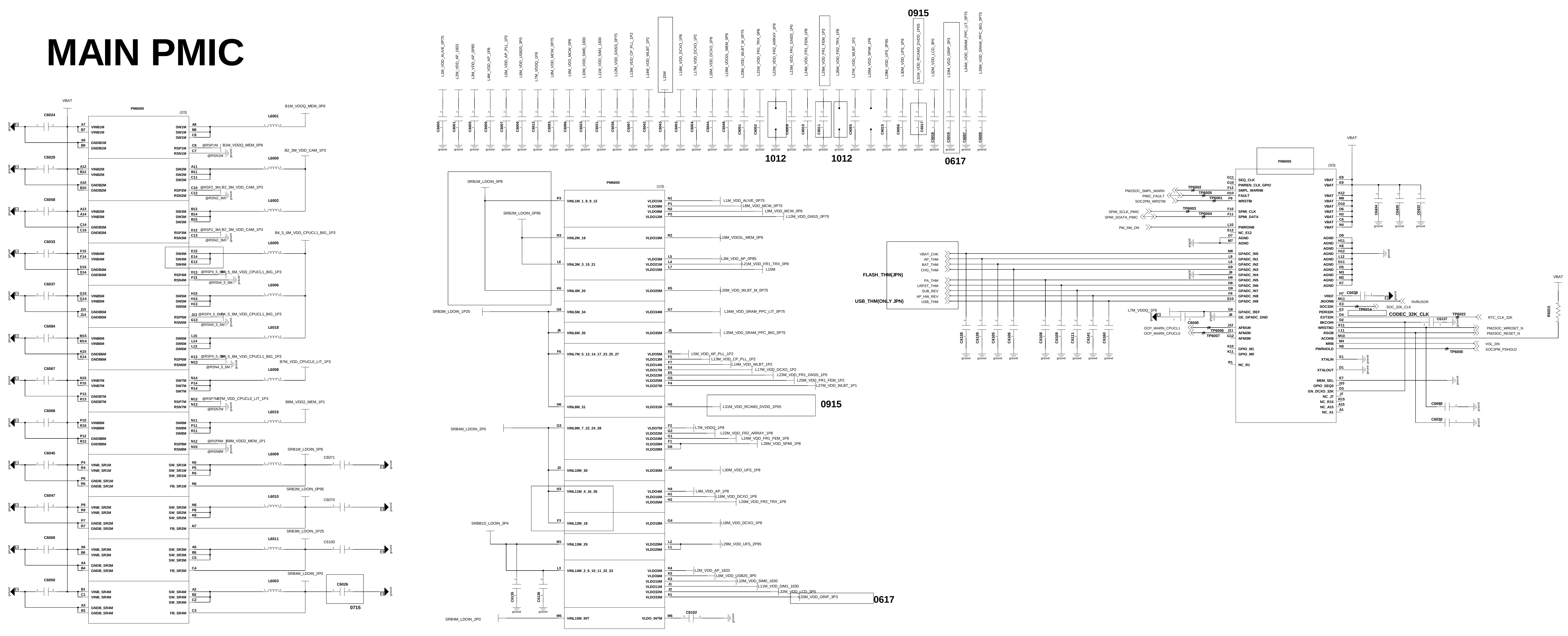
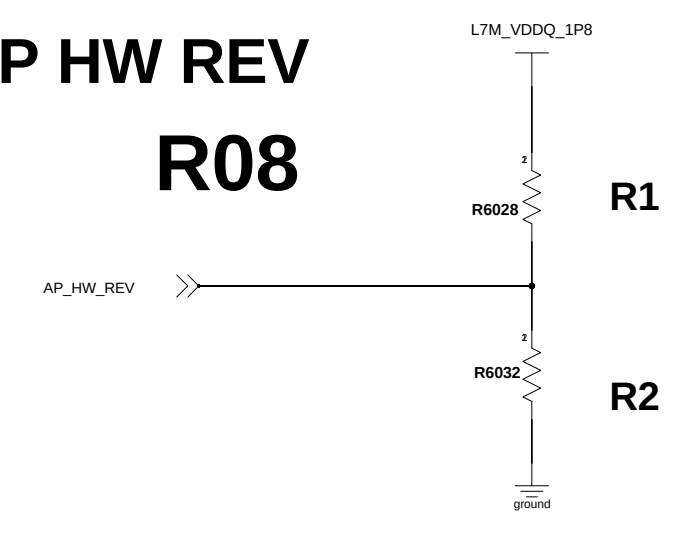
MAIN PMIC


PM6000
(2/3)
(1/3)
(3/3)
VBAT
B1M_VDDQ_MEM_0P6
B2_3M_VDD_CAM_1P3
B4_5_6M_VDD_CPUCL1_BIG_1P3
B7M_VDD_CPUCL0_LIT_1P3
B8M_VDD2_MEM_1P1
SRB1M_LDOIN_0P8
SRB2M_LDOIN_0P95
SRB3M_LDOIN_1P25
SRB4M_LDOIN_2P0
SRBB1S_LDOIN_3P4
L1M_VDD_ALIVE_0P75
L2M_VDD_AP_1833
L3M_VDD_AP_0P85
L4M_VDD_AP_1P8
L5M_VDD_AP_PLL_1P2
L6M_VDD_USB20_3P0
L7M_VDDQ_1P8
L8M_VDD_MCW_0P75
L9M_VDD_MCW_0P6
L10M_VDD_SIM0_1830
L11M_VDD_SIM1_1830
L12M_VDD_GNSS_0P75
L13M_VDD_CP_PLL_1P2
L14M_VDD_WLBT_1P2
L15M
L16M_VDD_DCXO_1P8
L17M_VDD_DCXO_1P2
L18M_VDD_DCXO_1P8
L19M_VDD2L_MEM_0P9
L20M_VDD_WLBT_M_0P75
L21M_VDD_FR1_TRX_0P8
L22M_VDD_FR2_ARRAY_1P8
L23M_VDD_FR1_GNSS_1P0
L24M_VDD_FR1_FEM_1P8
L25M_VDD_FR1_FEM_1P2
L26M_VDD_FR2_TRX_1P8
L27M_VDD_WLBT_1P1
L28M_VDD_SPMI_1P8
L29M_VDD_UFS_2P95
L30M_VDD_UFS_1P8
L31M_VDD_RCAM3_DVDD_1P05
L32M_VDD_LCD_3P0
L33M_VDD_GRIP_3P3
L34M_VDD_SRAM_PPC_LIT_0P75
L35M_VDD_SRAM_PPC_BIG_0P75
1012
1012
0915
0617
0715
FLASH_THM(JPN)
USB_THM(ONLY JPN)
VBAT_CHK
AP_THM
BAT_THM
CHG_THM
PA_THM
LRPST_THM
SUB_REV
AP_HW_REV
USB_THM
PM2SOC_SMPL_WARN
PMIC_FAULT
SOC2PM_WRSTBI
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
PM_SW_ON
OCP_WARN_CPUCL1
OCP_WARN_CPUCL0
NVBUSOK
SOC_32K_CLK
CODEC_32K_CLK
RTC_CLK_32K
PM2SOC_WRESET_N
PM2SOC_RESET_N
VOL_DN
SOC2PM_PSHOLD

REV	R1(Kohm)	R2(Kohm)	AP	
0	NC	0	R00	Bringup00
1	430	30	R01	Bringup01
2	390	62	R02	Real REV00
3	360	91	R03	Real REV01
4	330	120	R04	Real REV02
5	300	150	R05	Real REV03
6	270	180	R06	Real REV04
7	270	240	R07	Real REV04B
8	240	270	R08	Real REV04C
9	180	270		
10	150	300		

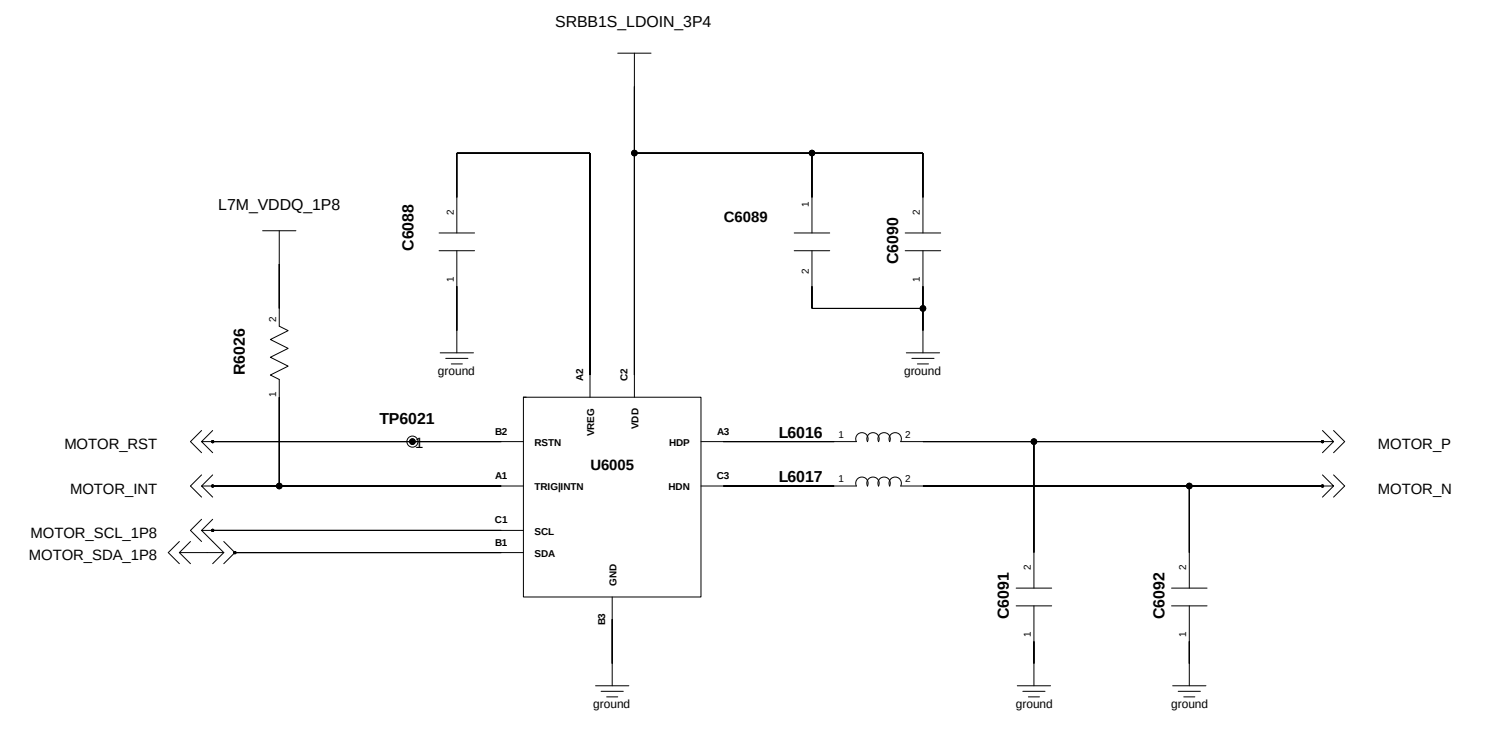
AP HW REV

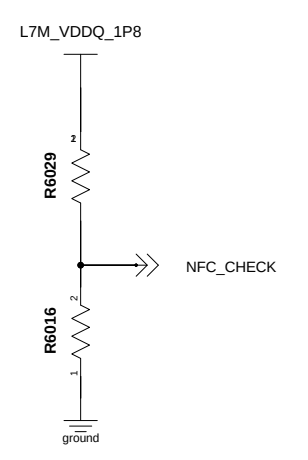
R08


L7M_VDDQ_1P8
R6028
R1
AP_HW_REV
R6032
R2
ground

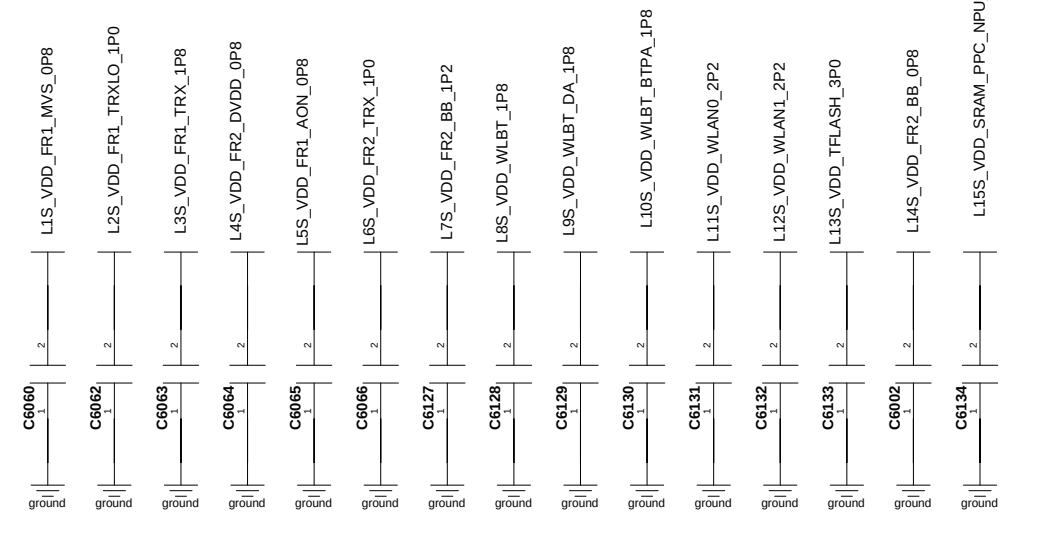
220620

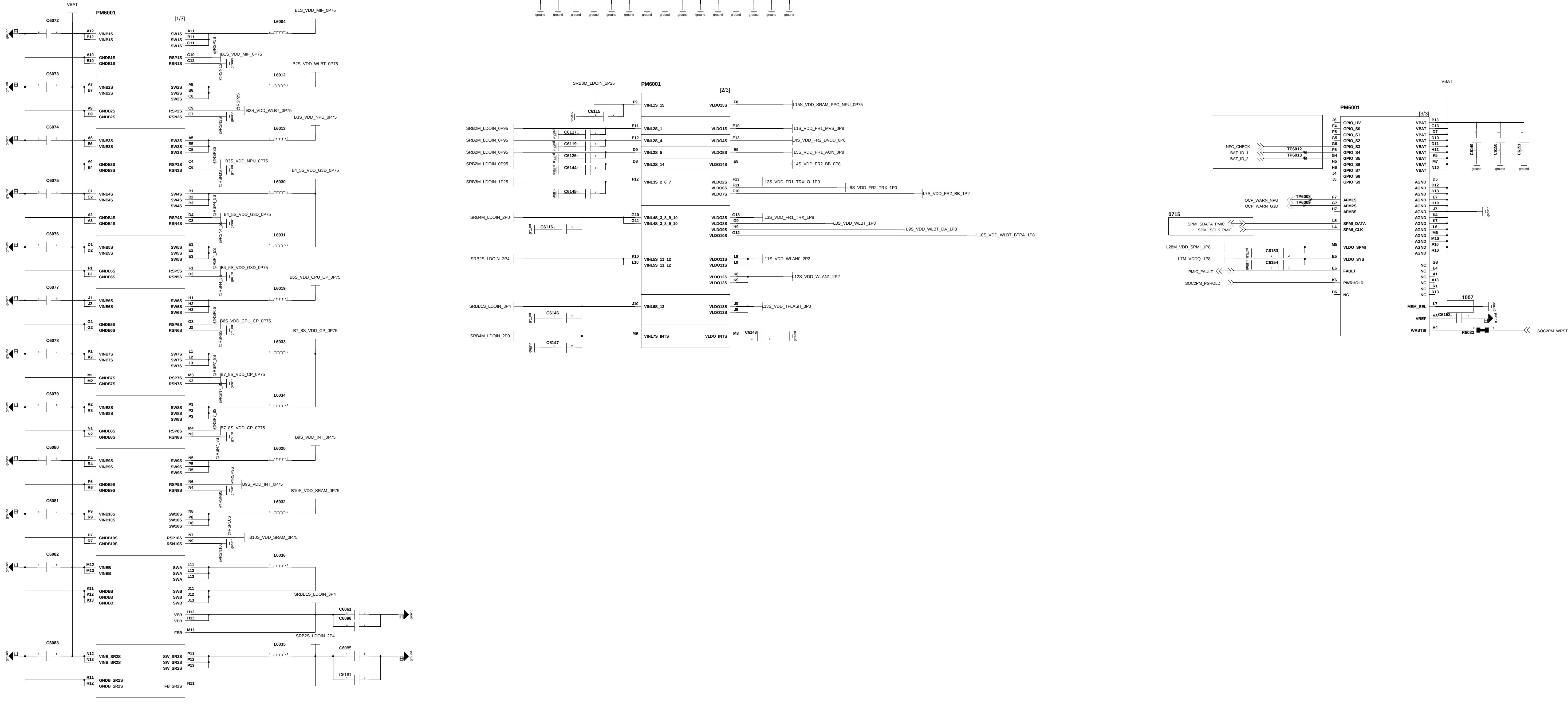
MOT DRV


SRBB1S_LDOIN_3P4
L7M_VDDQ_1P8
R6026
TP6021
U6005
L6016
L6017
MOTOR_RST
MOTOR_INT
MOTOR_SCL_1P8
MOTOR_SDA_1P8
MOTOR_P
MOTOR_N
C6088
C6089
C6090
C6091
C6092

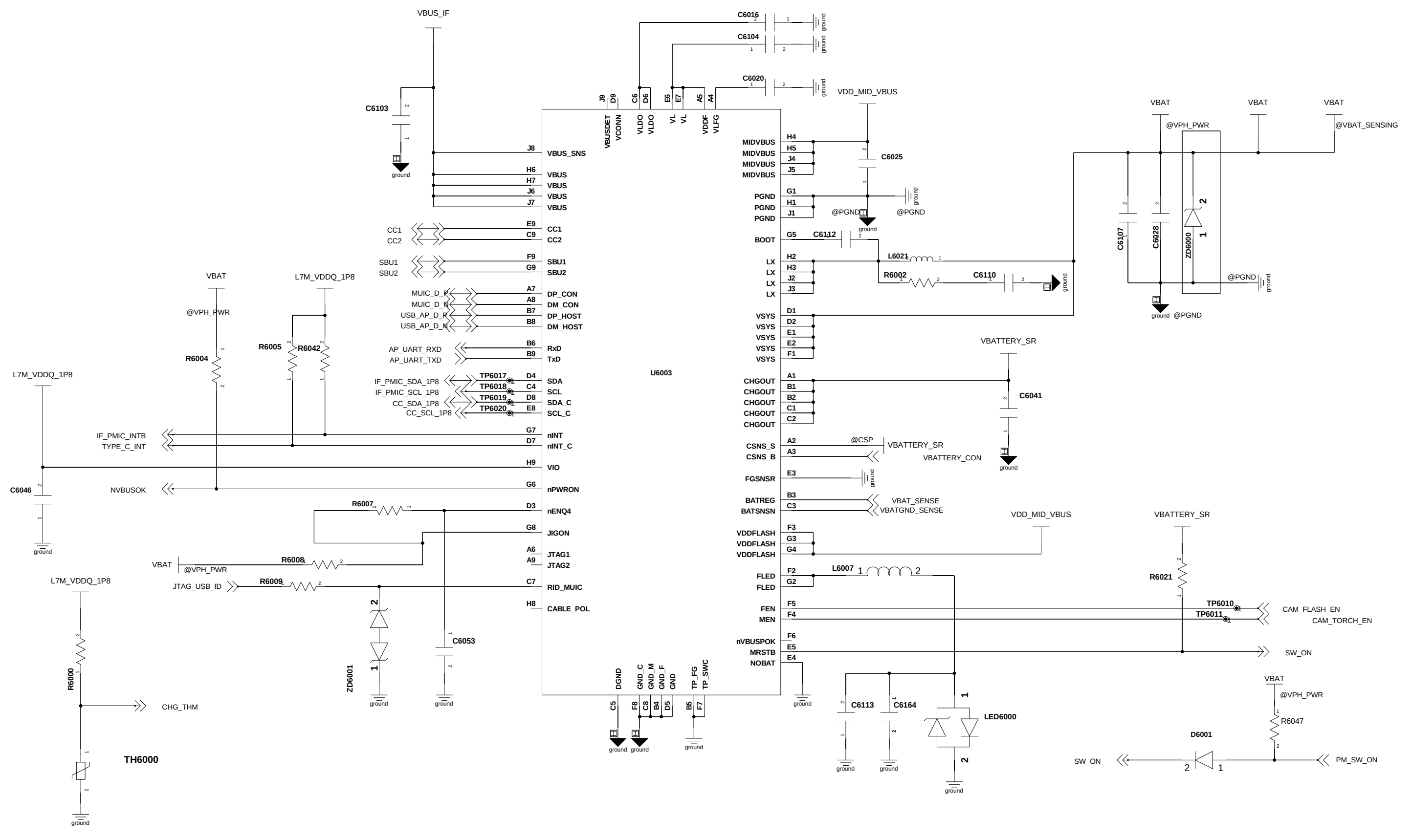

L7M_VDDQ_1P8
R6029
R6016
NFC_CHECK

SUB PMIC


L1S_VDD_FR1_MVS_0P8
L2S_VDD_FR1_TRXLO_1P0
L3S_VDD_FR1_TRX_1P8
L4S_VDD_FR2_DVDD_0P8
L5S_VDD_FR1_AON_0P8
L6S_VDD_FR2_TRX_1P0
L7S_VDD_FR2_BB_1P2
L8S_VDD_WLBT_1P8
L9S_VDD_WLBT_DA_1P8
L10S_VDD_WLBT_BTPA_1P8
L11S_VDD_WLAN0_2P2
L12S_VDD_WLAN1_2P2
L13S_VDD_TFLASH_3P0
L14S_VDD_FR2_BB_0P8
L15S_VDD_SRAM_PPC_NPU_0P75

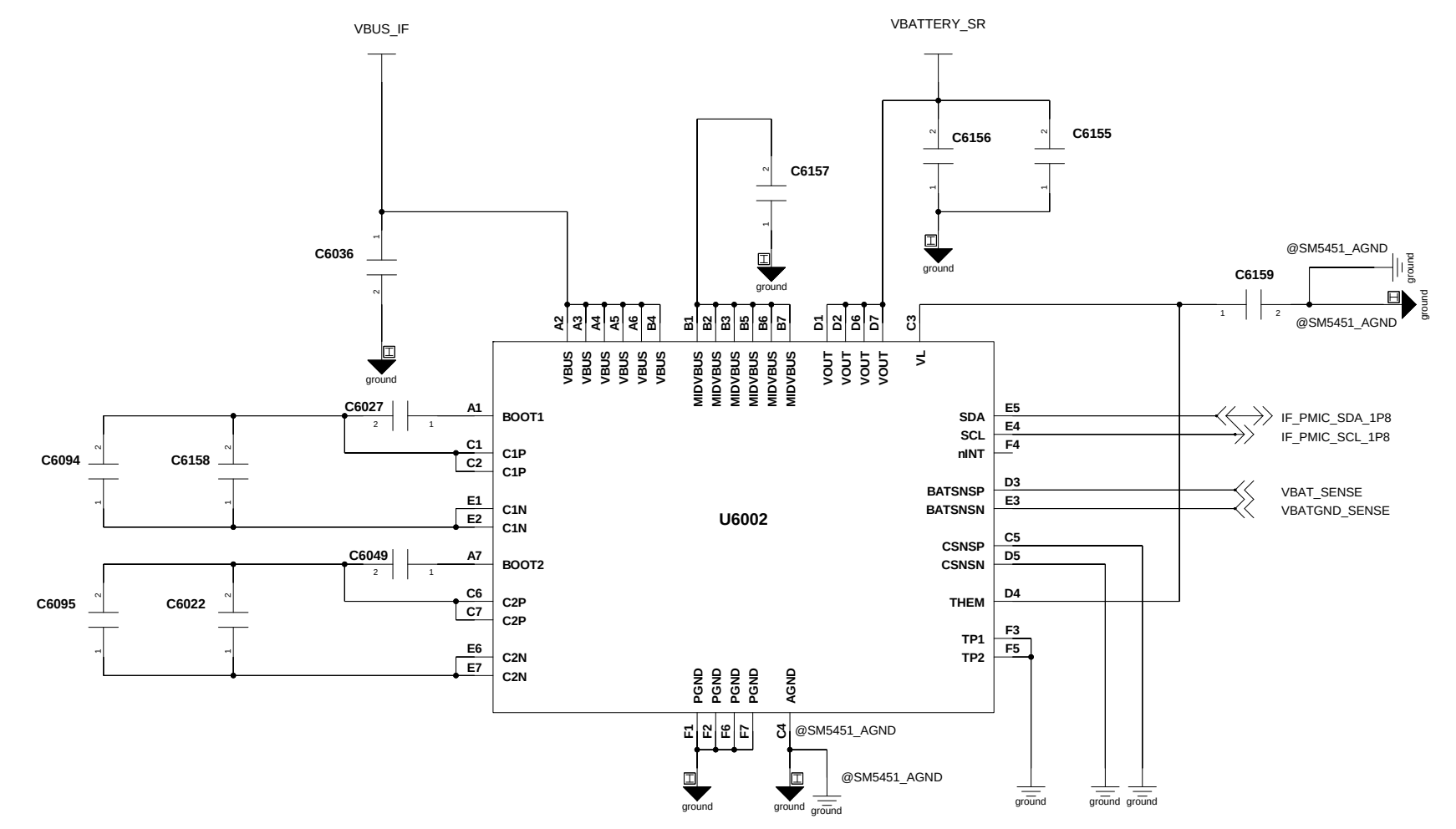

PM6001
[1/3]
[2/3]
[3/3]
VBAT
B1S_VDD_MIF_0P75
B2S_VDD_WLBT_0P75
B3S_VDD_NPU_0P75
B4_5S_VDD_G3D_0P75
B6S_VDD_CPU_CP_0P75
B7_8S_VDD_CP_0P75
B9S_VDD_INT_0P75
B10S_VDD_SRAM_0P75
SRBB1S_LDOIN_3P4
SRB2S_LDOIN_2P4
SRB3M_LDOIN_1P25
SRB2M_LDOIN_0P95
SRB4M_LDOIN_2P0
NFC_CHECK
BAT_ID_1
BAT_ID_2
OCP_WARN_NPU
OCP_WARN_G3D
0715
SPMI_SDATA_PMIC
SPMI_SCLK_PMIC
L28M_VDD_SPMI_1P8
L7M_VDDQ_1P8
PMIC_FAULT
SOC2PM_PSHOLD
1007
SOC2PM_WRSTBI

IF PMIC(SM5714)

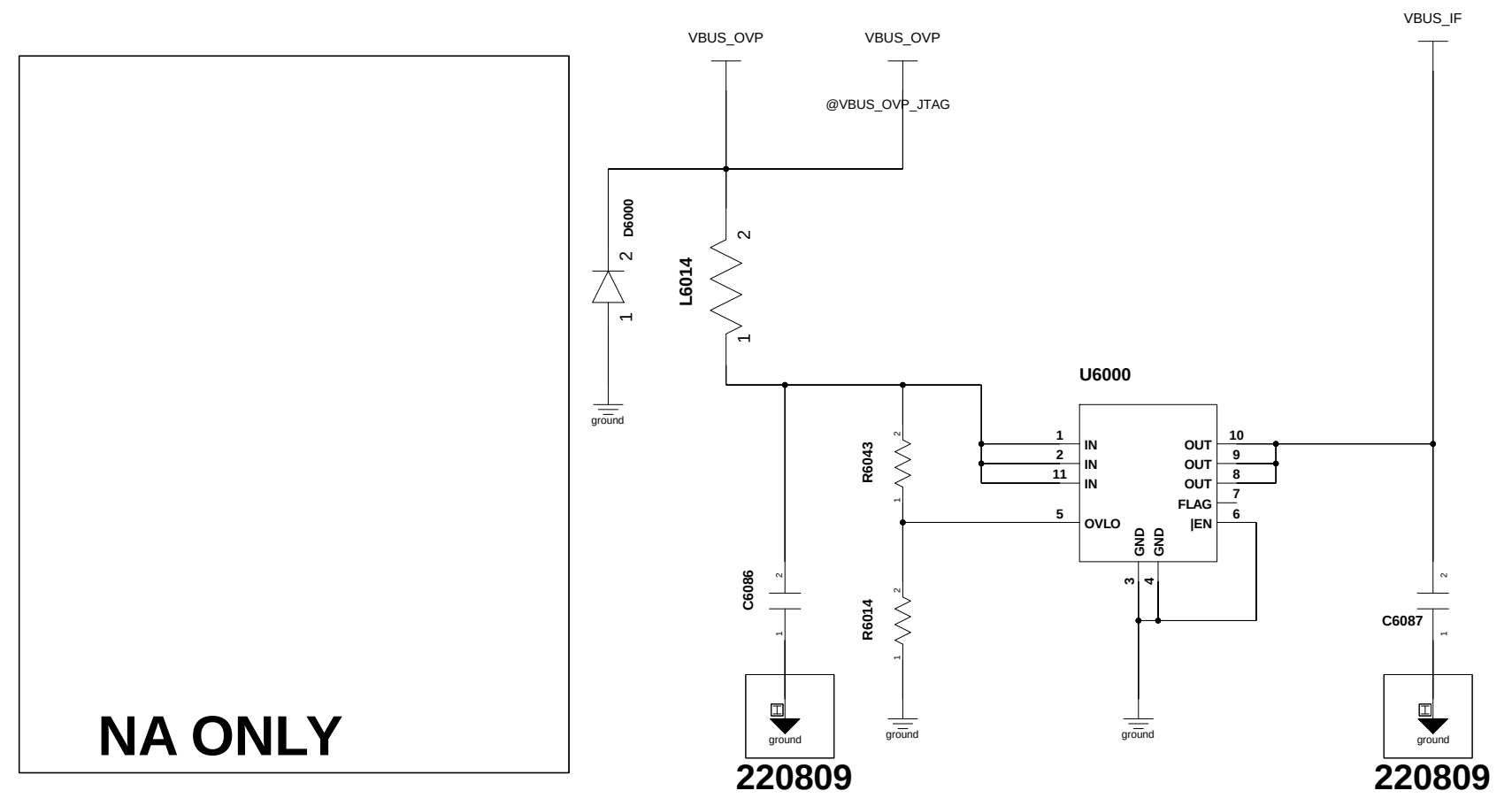

VBUS_IF
U6003
VDD_MID_VBUS
VBAT
@VPH_PWR
@VBAT_SENSING
CC1
CC2
SBU1
SBU2
MUIC_D_P
MUIC_D_N
USB_AP_D_P
USB_AP_D_N
AP_UART_RXD
AP_UART_TXD
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
CC_SDA_1P8
CC_SCL_1P8
IF_PMIC_INTB
TYPE_C_INT
NVBUSOK
JTAG_USB_ID
L7M_VDDQ_1P8
CHG_THM
TH6000
VBATTERY_SR
VBATTERY_CON
VBAT_SENSE
VBATGND_SENSE
CAM_FLASH_EN
CAM_TORCH_EN
SW_ON
PM_SW_ON
LED6000
D6001
ZD6000
ZD6001

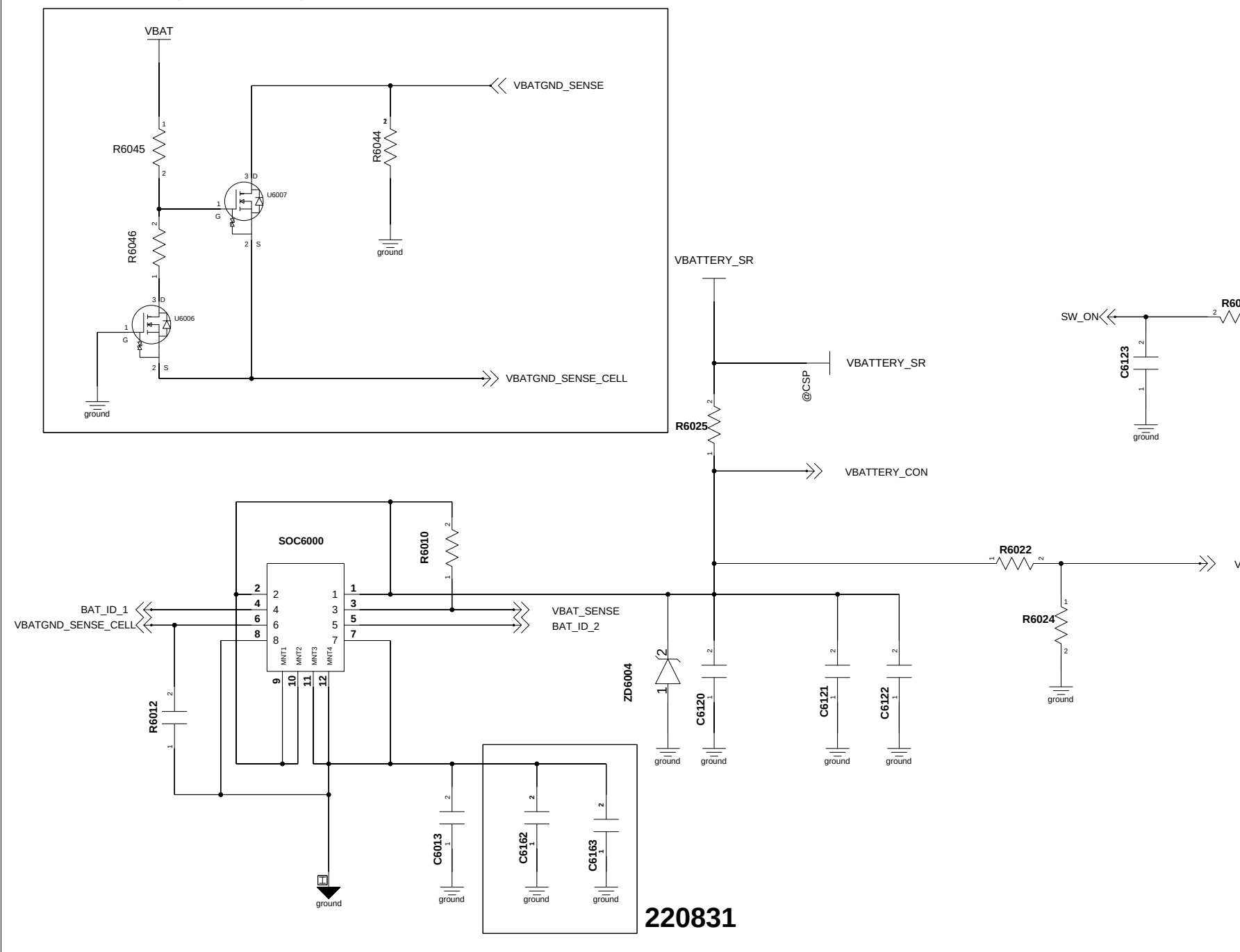
DIRECT CHARGER

(SM5451)

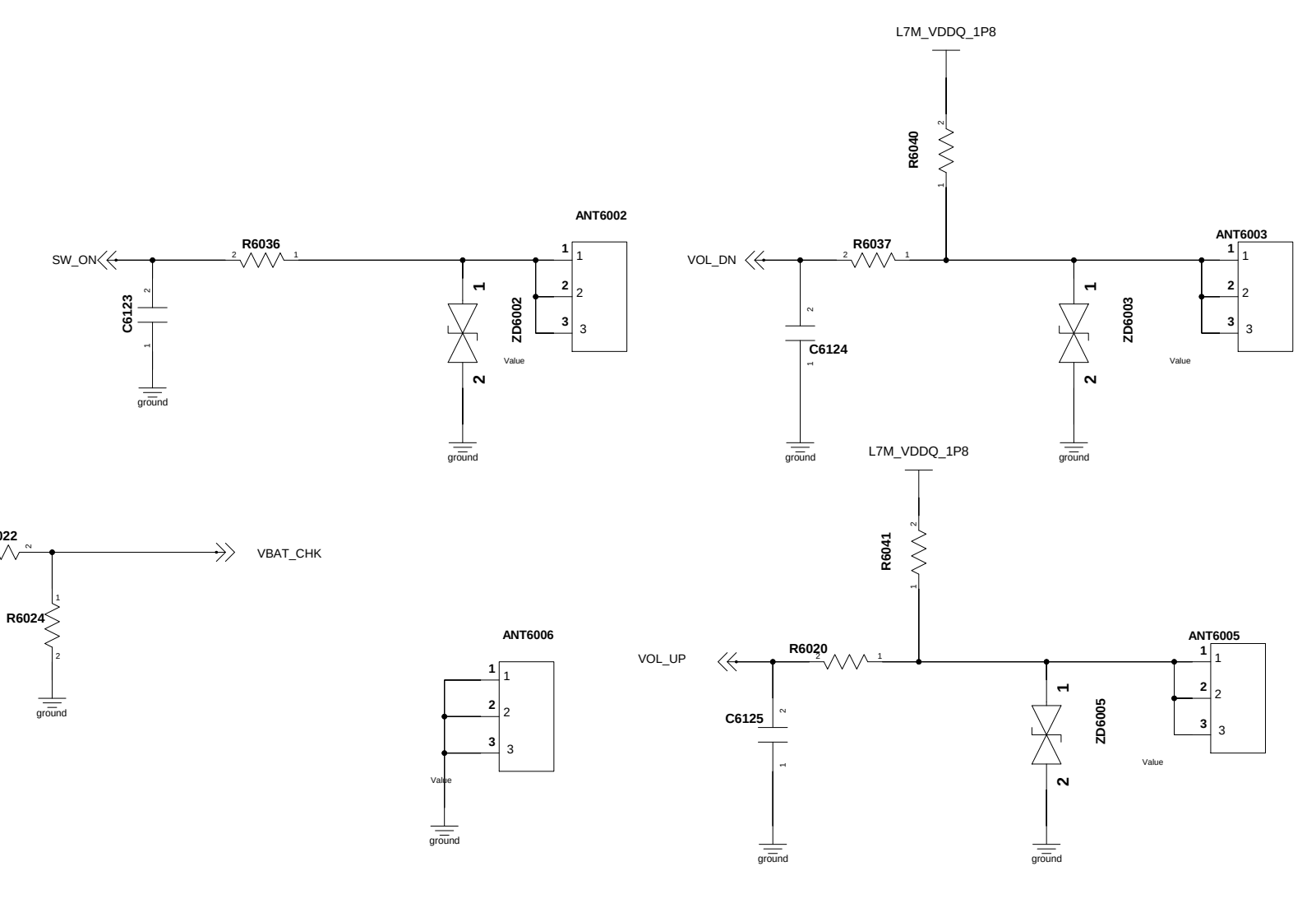

VBUS_IF
VBATTERY_SR
U6002
@SM5451_AGND
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
VBAT_SENSE
VBATGND_SENSE

OVP


NA ONLY
VBUS_OVP
@VBUS_OVP_JTAG
VBUS_IF
D6000
L6014
U6000
R6043
R6014
C6086
C6087
220809
220809


Cell leakage Sensing 1006
VBAT
VBATGND_SENSE
VBATGND_SENSE_CELL
U6007
U6006
VBATTERY_SR
@CSP
VBATTERY_CON
VBAT_CHK
SOC6000
BAT_ID_1
VBAT_SENSE
BAT_ID_2
220831

BATTERY CON


L7M_VDDQ_1P8
SW_ON
VOL_DN
VOL_UP
ANT6002
ANT6003
ANT6005
ANT6006
ZD6002
ZD6003
ZD6005

POWER & VOL KEY

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:
A54
TITLE: POWER/MOT/KEY
Size:
Changed by:
Date Changed:
Time Changed:
REV:
Drawing Number:

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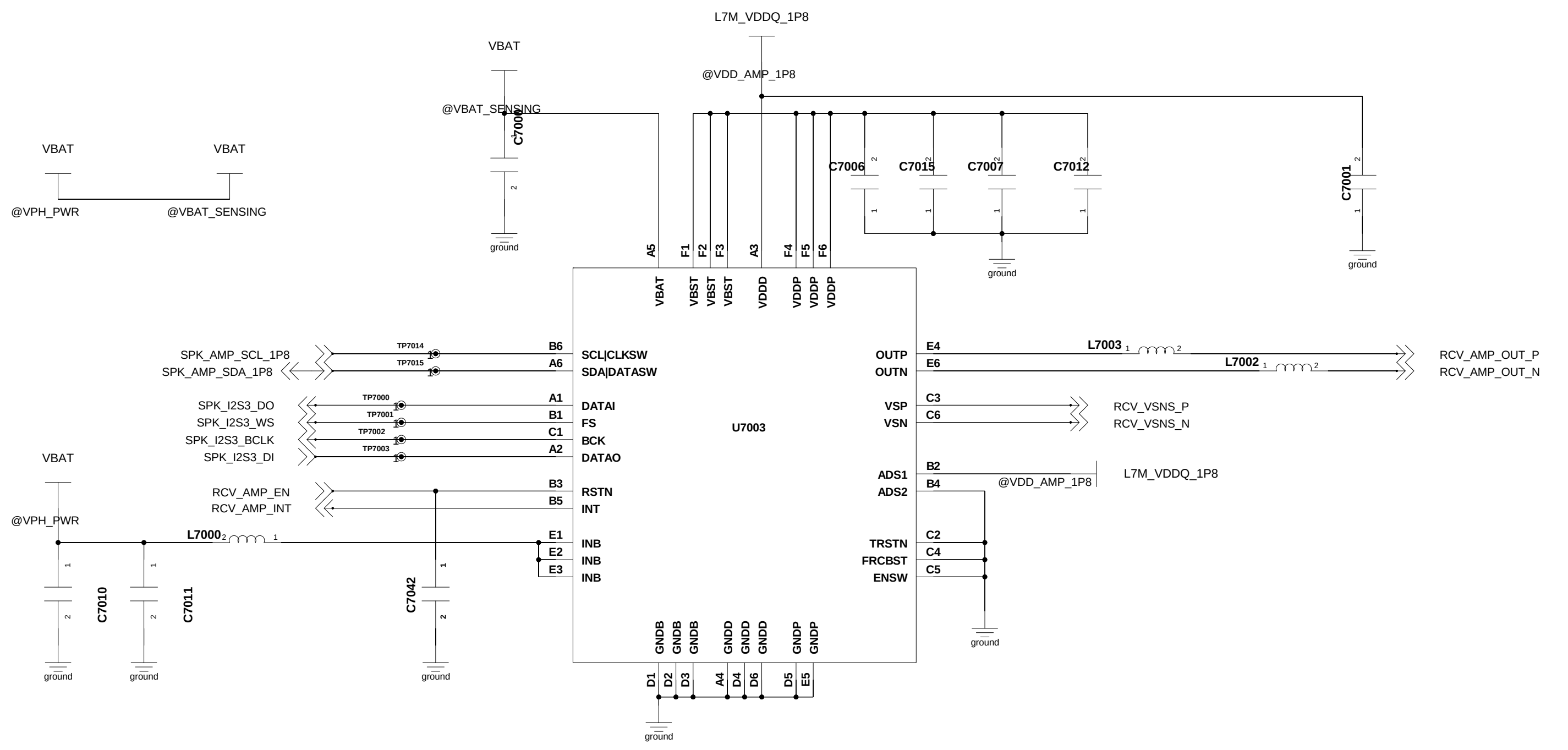
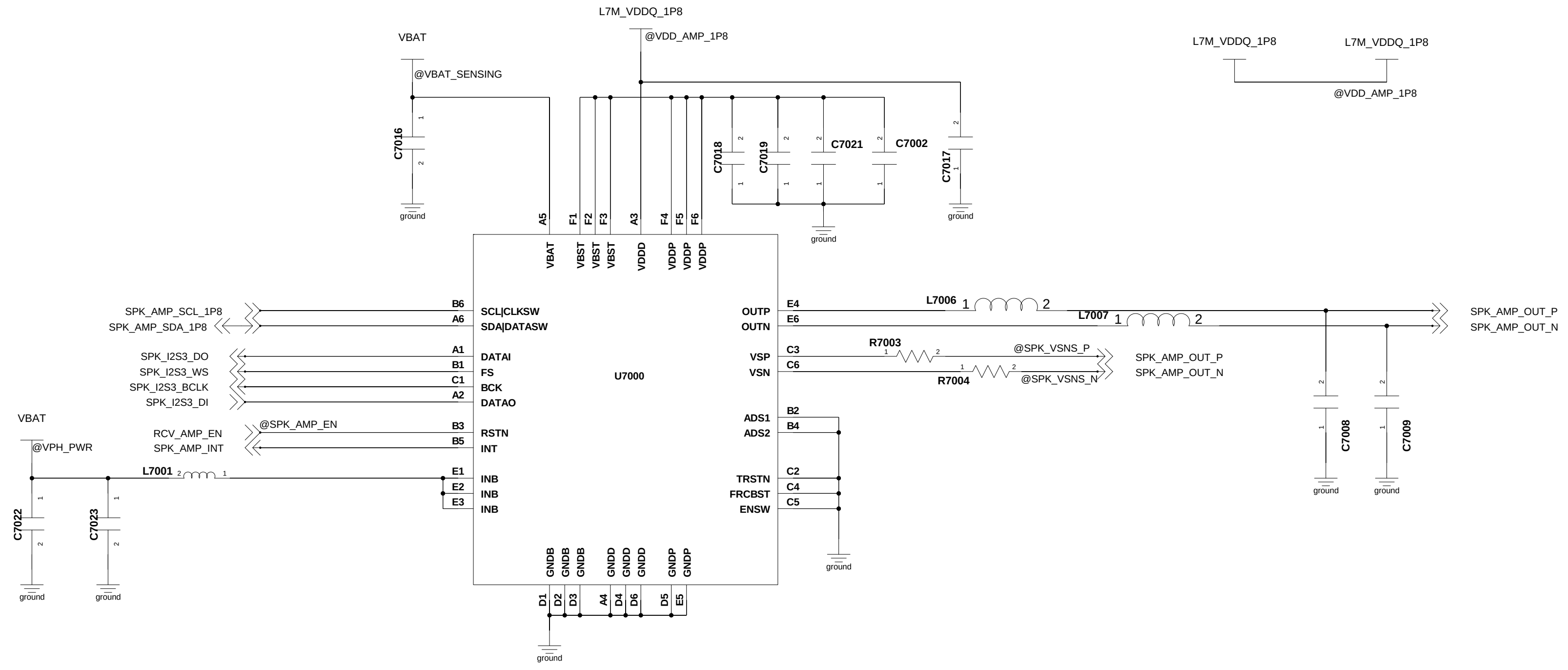
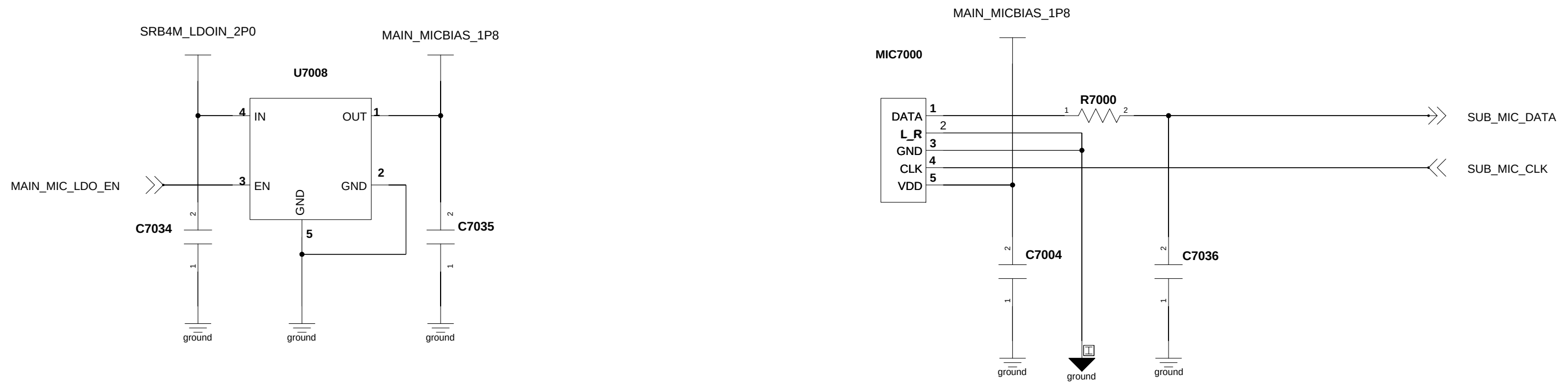
AUDIO & SENSOR PART


VBAT
@VPH_PWR
@VBAT_SENSING
L7M_VDDQ_1P8
@VDD_AMP_1P8
C7000
C7006
C7015
C7007
C7012
C7001
U7003
VBAT
VBST
VDDD
VDDP
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
TP7014
TP7015
SCL|CLKSW
SDA|DATASW
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
TP7000
TP7001
TP7002
TP7003
DATAI
FS
BCK
DATAO
RCV_AMP_EN
RCV_AMP_INT
RSTN
INT
L7000
INB
C7010
C7011
C7042
OUTP
OUTN
L7003
L7002
RCV_AMP_OUT_P
RCV_AMP_OUT_N
VSP
VSN
RCV_VSNS_P
RCV_VSNS_N
ADS1
ADS2
L7M_VDDQ_1P8
TRSTN
FRCBST
ENSW
GNDB
GNDD
GNDP
ground

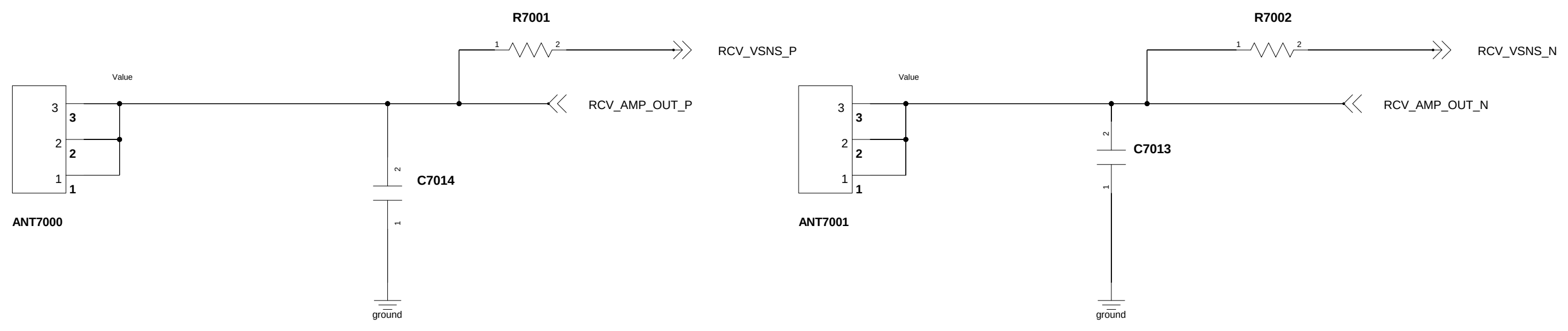
RCV AMP


VBAT
@VBAT_SENSING
L7M_VDDQ_1P8
@VDD_AMP_1P8
L7M_VDDQ_1P8
L7M_VDDQ_1P8
@VDD_AMP_1P8
C7016
C7018
C7019
C7021
C7002
C7017
U7000
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
SCL|CLKSW
SDA|DATASW
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
DATAI
FS
BCK
DATAO
@SPK_AMP_EN
RCV_AMP_EN
SPK_AMP_INT
RSTN
INT
VBAT
@VPH_PWR
L7001
INB
C7022
C7023
OUTP
OUTN
L7006
L7007
R7003
R7004
@SPK_VSNS_P
@SPK_VSNS_N
SPK_AMP_OUT_P
SPK_AMP_OUT_N
VSP
VSN
ADS1
ADS2
TRSTN
FRCBST
ENSW
C7008
C7009
GNDB
GNDD
GNDP
ground

SPK AMP

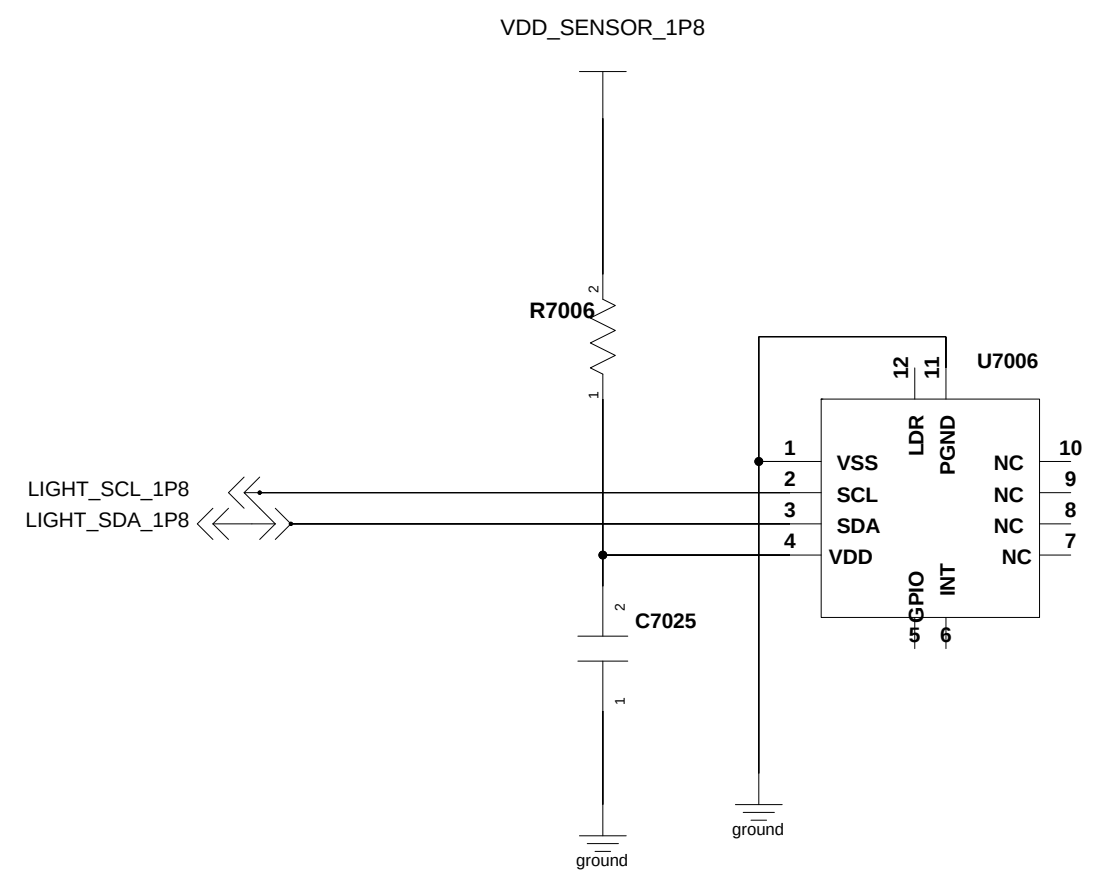

SRB4M_LDOIN_2P0
MAIN_MICBIAS_1P8
U7008
IN
OUT
EN
GND
MAIN_MIC_LDO_EN
C7034
C7035
ground
MAIN_MICBIAS_1P8
MIC7000
DATA
L_R
GND
CLK
VDD
R7000
SUB_MIC_DATA
SUB_MIC_CLK
C7004
C7036
ground

2nd MIC


Value
ANT7000
R7001
RCV_VSNS_P
RCV_AMP_OUT_P
C7014
Value
ANT7001
R7002
RCV_VSNS_N
RCV_AMP_OUT_N
C7013
ground

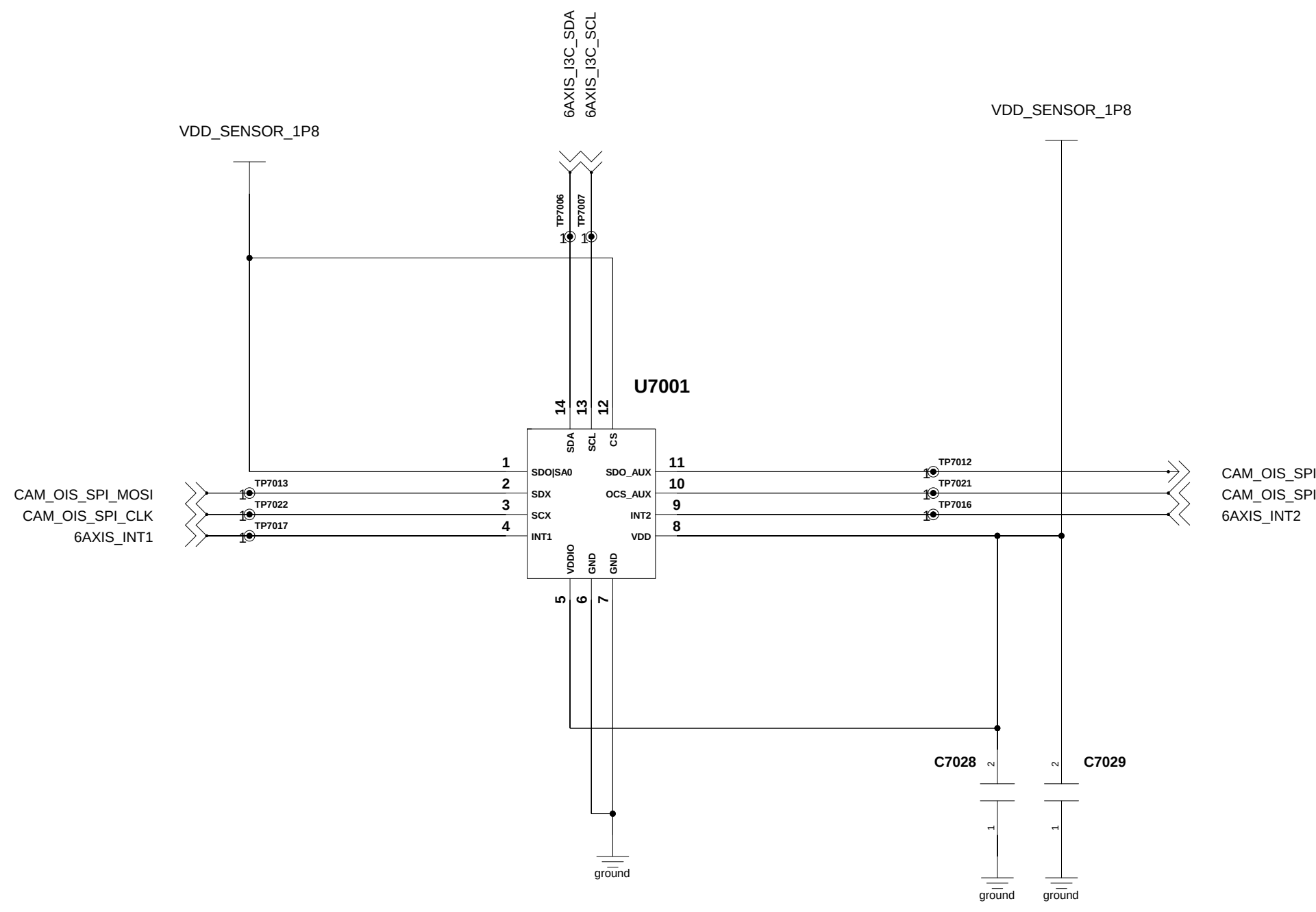
RCV CONTACT

SENSOR

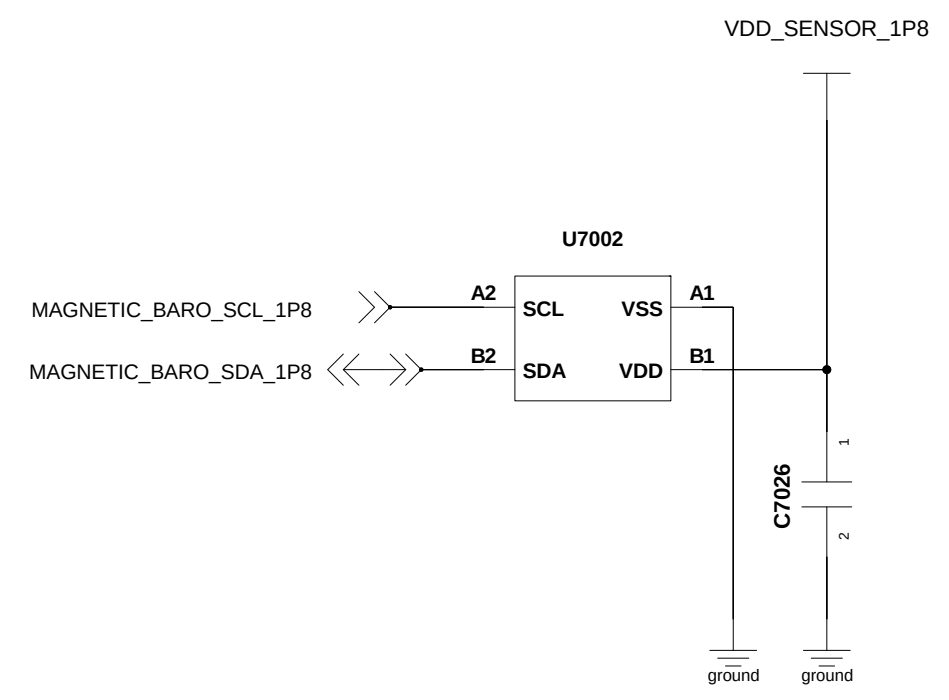

VDD_SENSOR_1P8
R7006
U7006
LDR
PGND
VSS
SCL
SDA
VDD
NC
GPIO
INT
LIGHT_SCL_1P8
LIGHT_SDA_1P8
C7025
ground

AMBIENT SENSOR

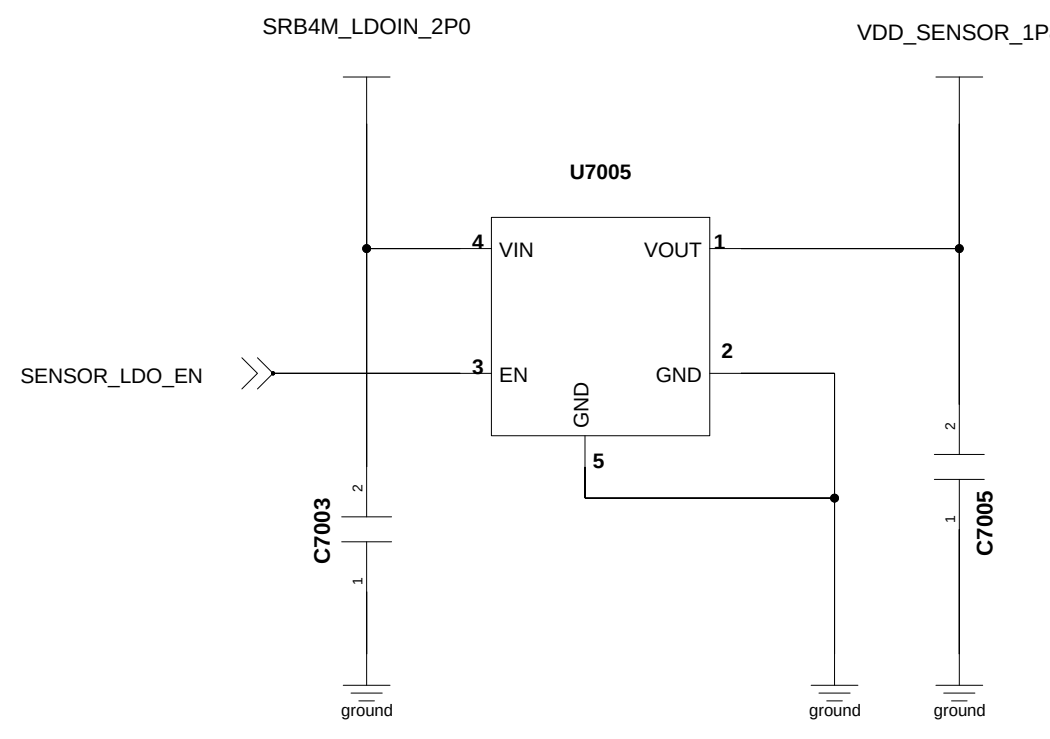
HALL IC


6AXIS_I3C_SDA
6AXIS_I3C_SCL
TP7006
TP7007
VDD_SENSOR_1P8
VDD_SENSOR_1P8
U7001
SDA
SCL
CS
SDO|SA0
SDX
SCX
INT1
SDO_AUX
OCS_AUX
INT2
VDD
VDDIO
GND
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_CLK
6AXIS_INT1
TP7013
TP7022
TP7017
TP7012
TP7021
TP7016
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
6AXIS_INT2
C7028
C7029
ground

ACC&GYRO SENSOR

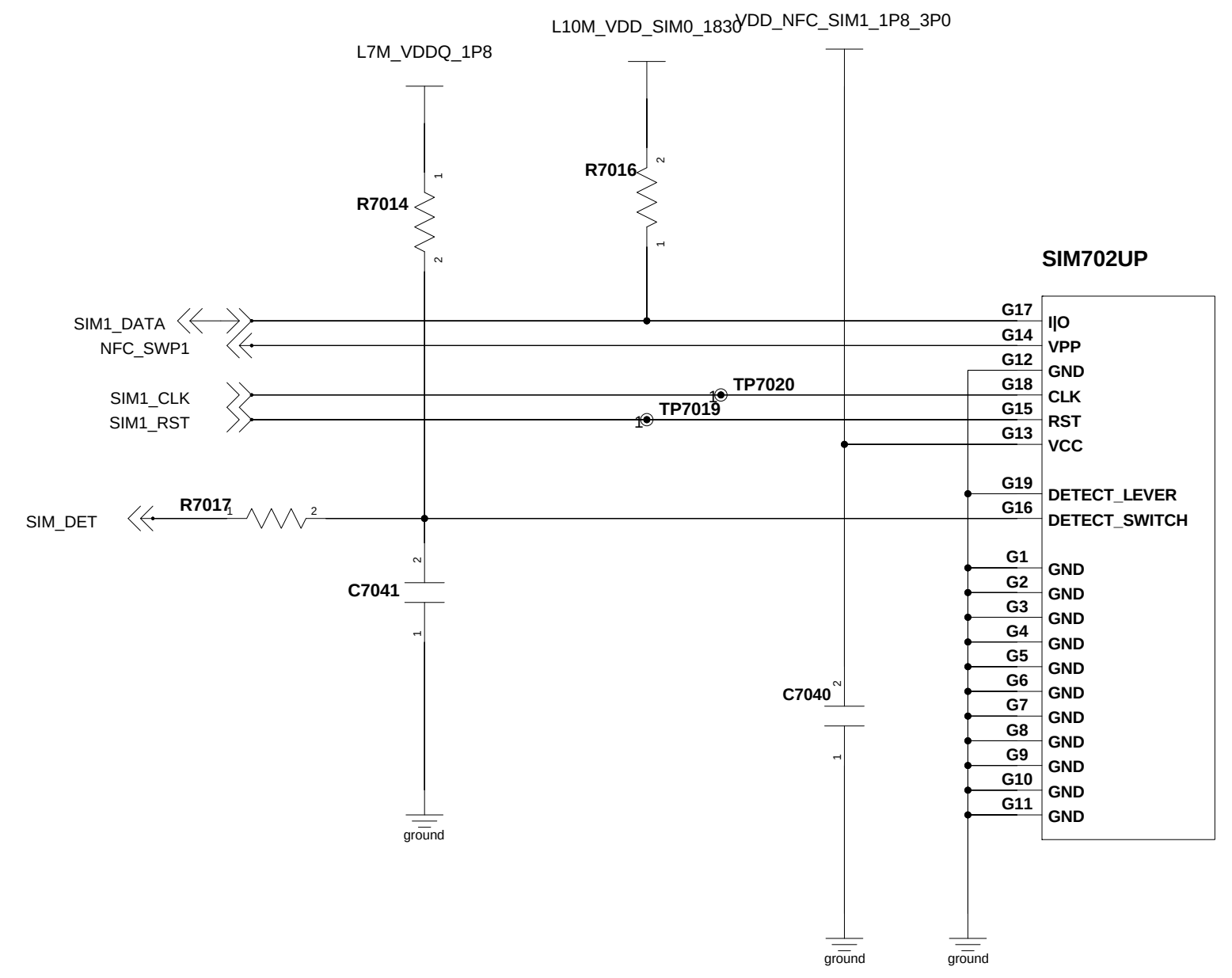

VDD_SENSOR_1P8
U7002
SCL
SDA
VSS
VDD
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
C7026
ground

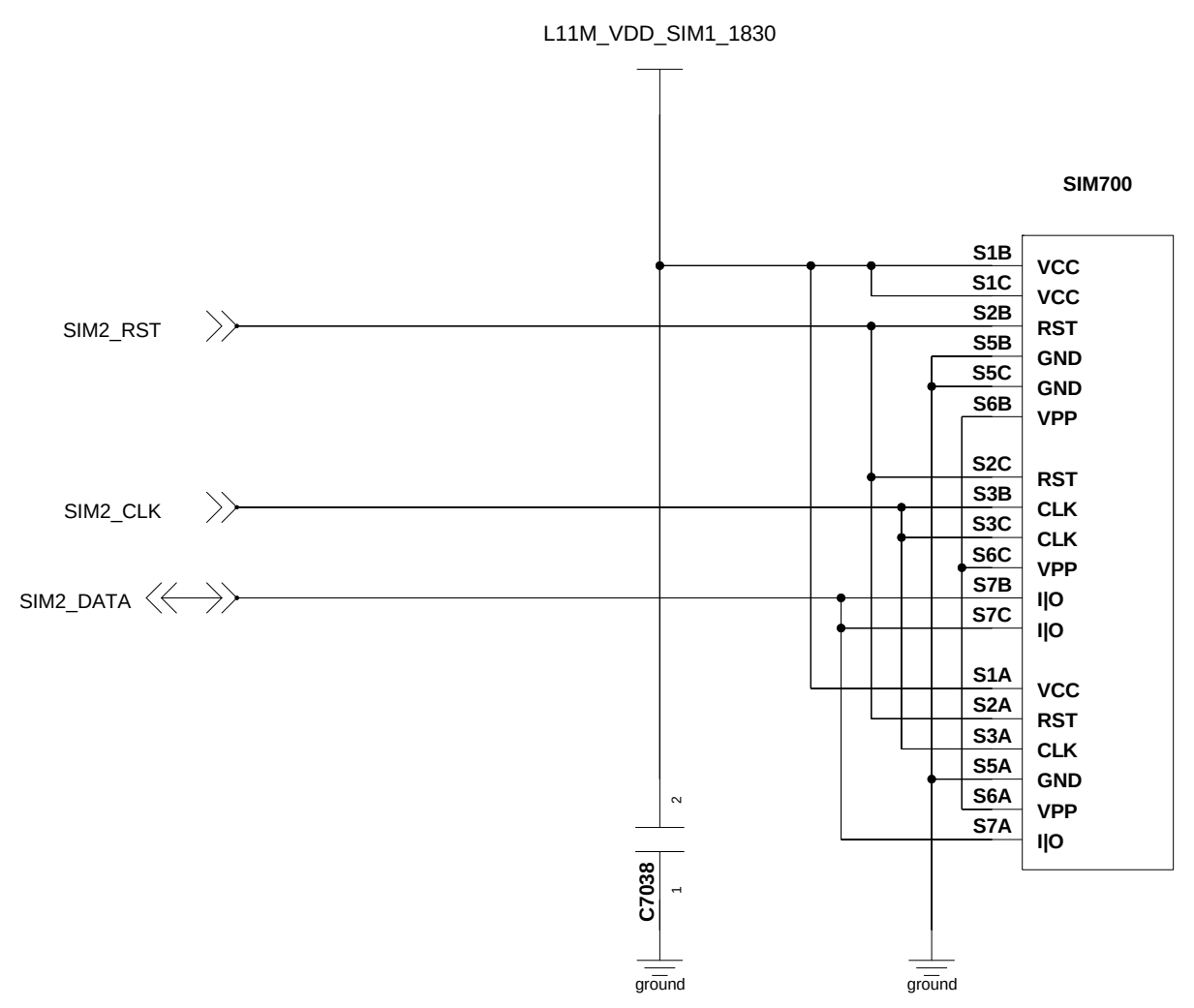
MAGNETIC SENSOR

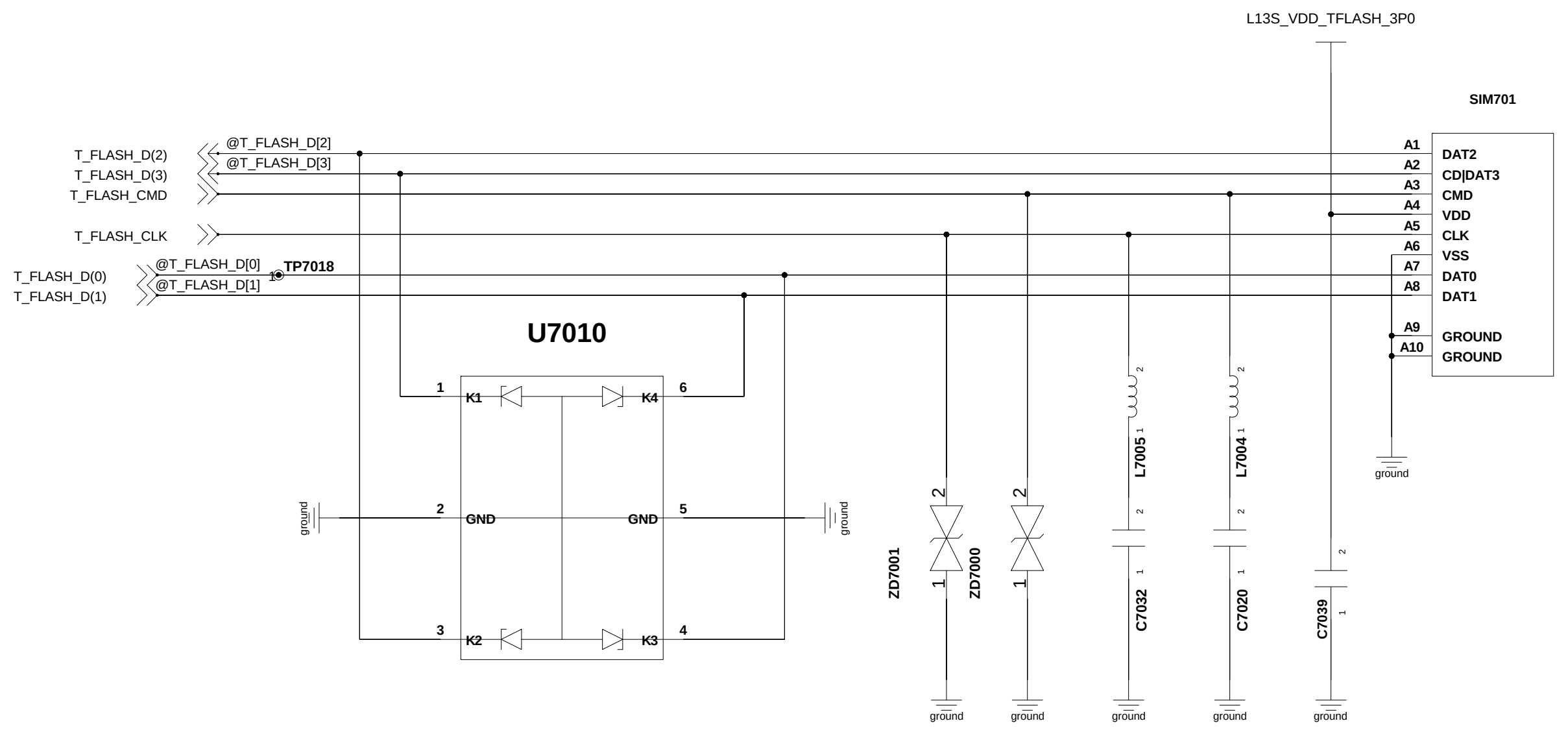

SRB4M_LDOIN_2P0
VDD_SENSOR_1P8
U7005
VIN
VOUT
EN
GND
SENSOR_LDO_EN
C7003
C7005
ground

NA ONLY

DUAL SIM / T-FLASH (3in2)


L7M_VDDQ_1P8
L10M_VDD_SIM0_1830
VDD_NFC_SIM1_1P8_3P0
R7014
R7016
SIM702UP
SIM1_DATA
NFC_SWP1
SIM1_CLK
SIM1_RST
TP7020
TP7019
I|O
VPP
GND
CLK
RST
VCC
DETECT_LEVER
DETECT_SWITCH
SIM_DET
R7017
C7041
C7040
ground


L11M_VDD_SIM1_1830
SIM700
SIM2_RST
SIM2_CLK
SIM2_DATA
VCC
RST
GND
VPP
CLK
I|O
C7038
ground


L13S_VDD_TFLASH_3P0
SIM701
T_FLASH_D(2)
T_FLASH_D(3)
T_FLASH_CMD
T_FLASH_CLK
T_FLASH_D(0)
T_FLASH_D(1)
@T_FLASH_D[2]
@T_FLASH_D[3]
@T_FLASH_D[0]
@T_FLASH_D[1]
TP7018
DAT2
CD|DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
GROUND
U7010
K1
K2
K3
K4
GND
ZD7001
ZD7000
L7005
L7004
C7032
C7020
C7039
ground

Stacked Dual SIM SOCKET (Reverse)

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
A54
TITLE:
Size:
Changed by:
Date Changed:
Time Changed:
QA CHK:
REV:
Drawing Number:
Page:

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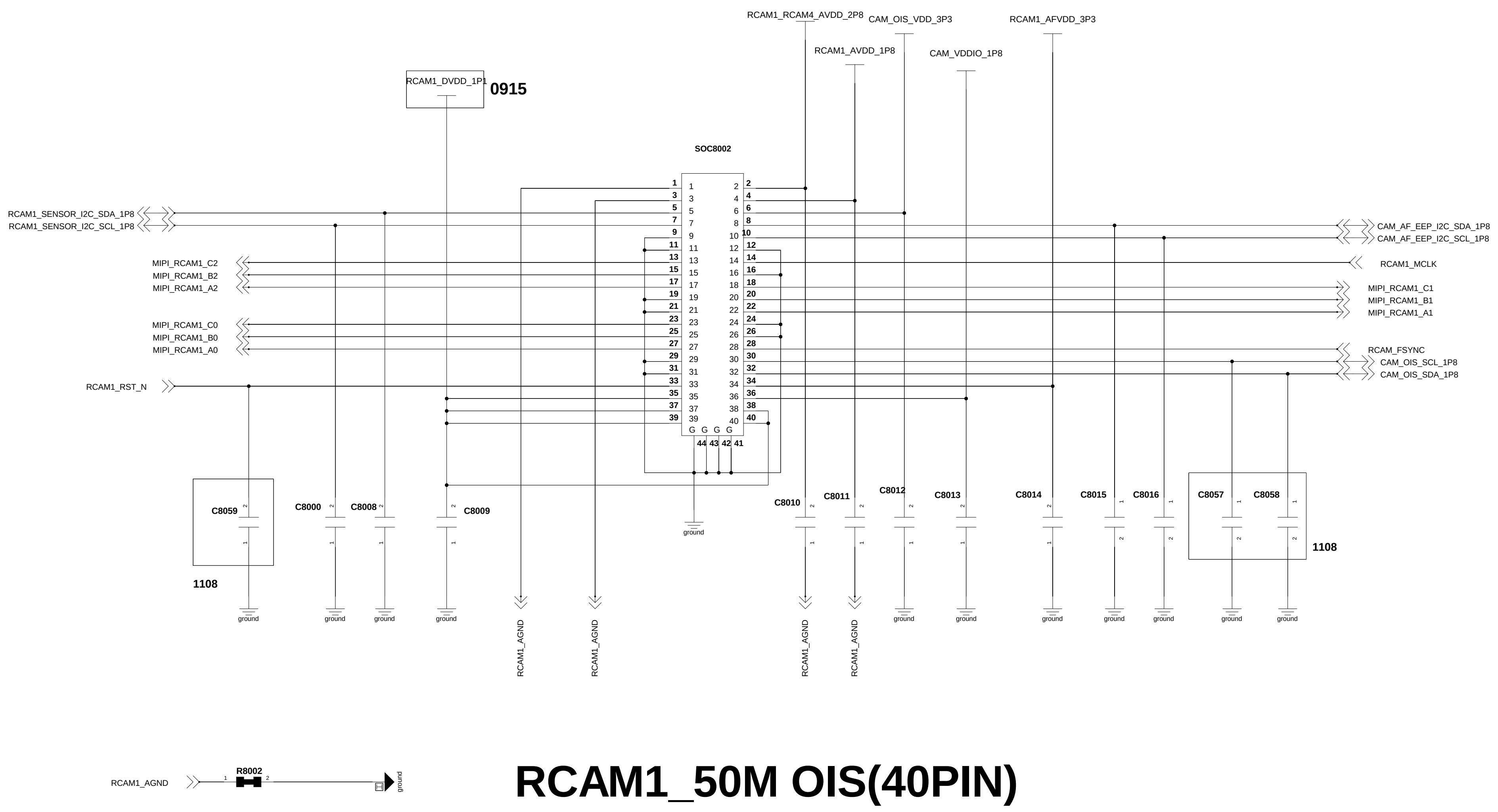
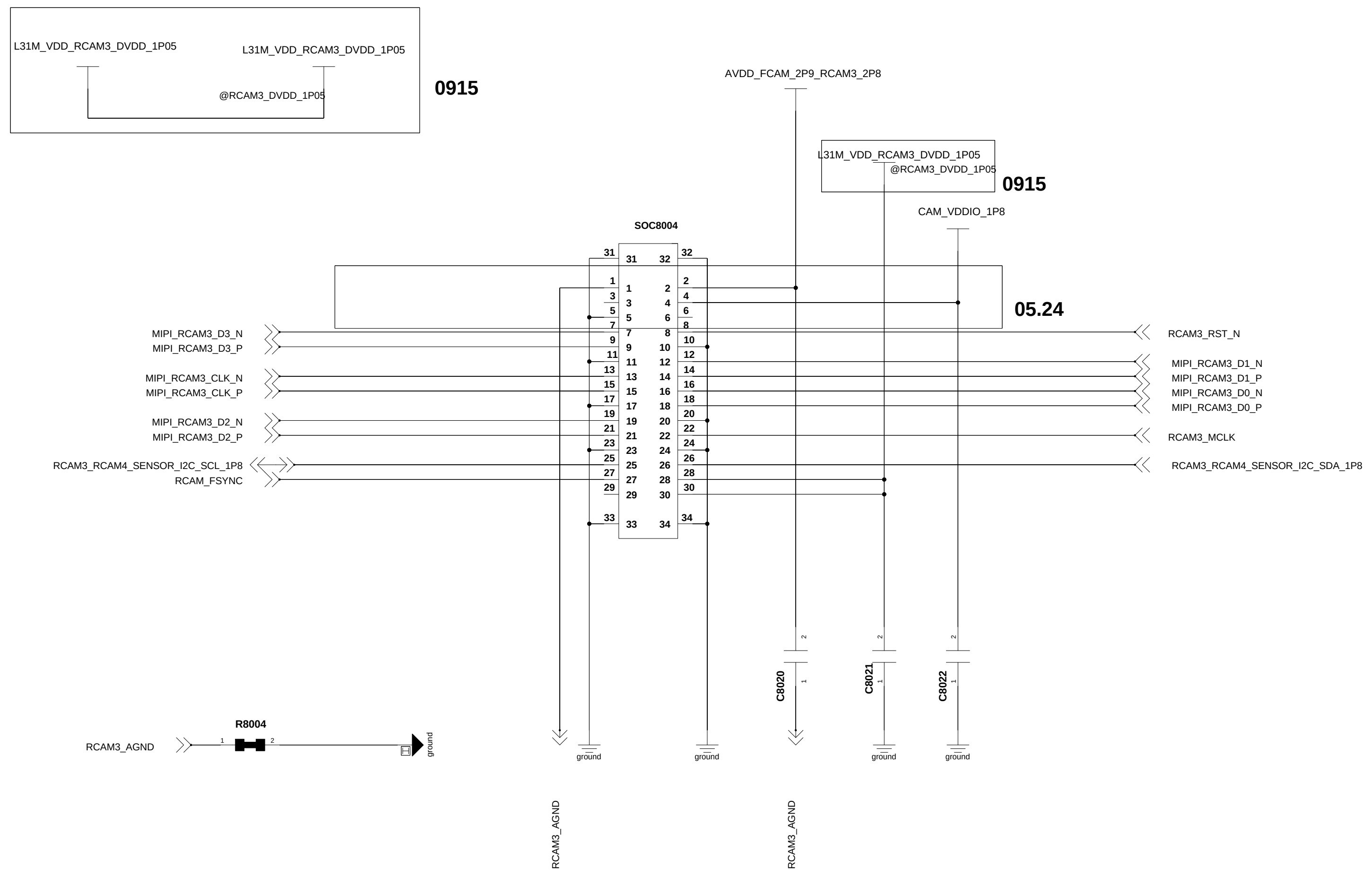
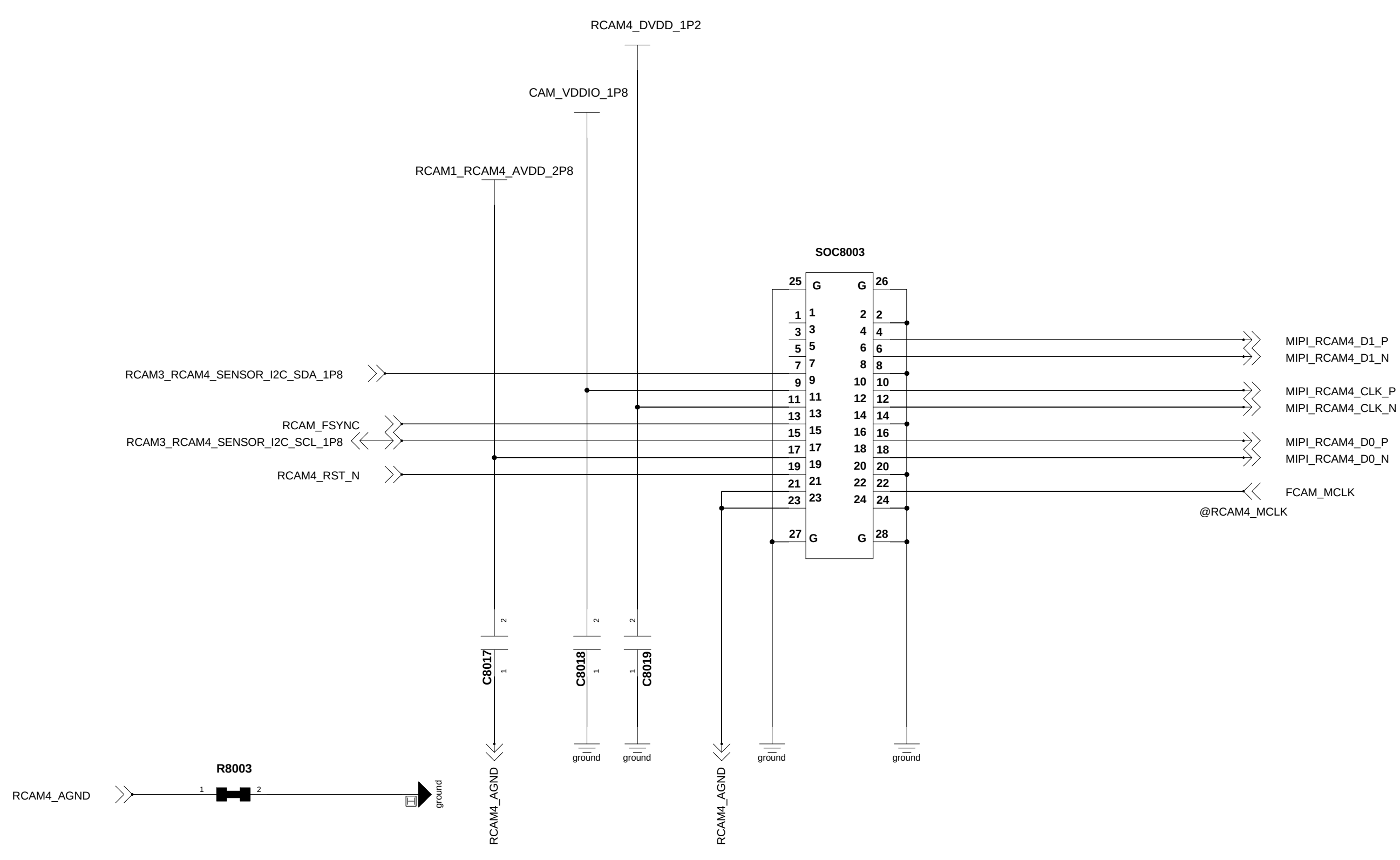
CAMERA


RCAM1_DVDD_1P1
0915
RCAM1_RCAM4_AVDD_2P8
CAM_OIS_VDD_3P3
RCAM1_AVDD_1P8
CAM_VDDIO_1P8
RCAM1_AFVDD_3P3
SOC8002
RCAM1_SENSOR_I2C_SDA_1P8
RCAM1_SENSOR_I2C_SCL_1P8
MIPI_RCAM1_C2
MIPI_RCAM1_B2
MIPI_RCAM1_A2
MIPI_RCAM1_C0
MIPI_RCAM1_B0
MIPI_RCAM1_A0
RCAM1_RST_N
CAM_AF_EEP_I2C_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
RCAM1_MCLK
MIPI_RCAM1_C1
MIPI_RCAM1_B1
MIPI_RCAM1_A1
RCAM_FSYNC
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
C8059
1108
C8000
C8008
C8009
C8010
C8011
C8012
C8013
C8014
C8015
C8016
C8057
C8058
1108
ground
RCAM1_AGND
R8002

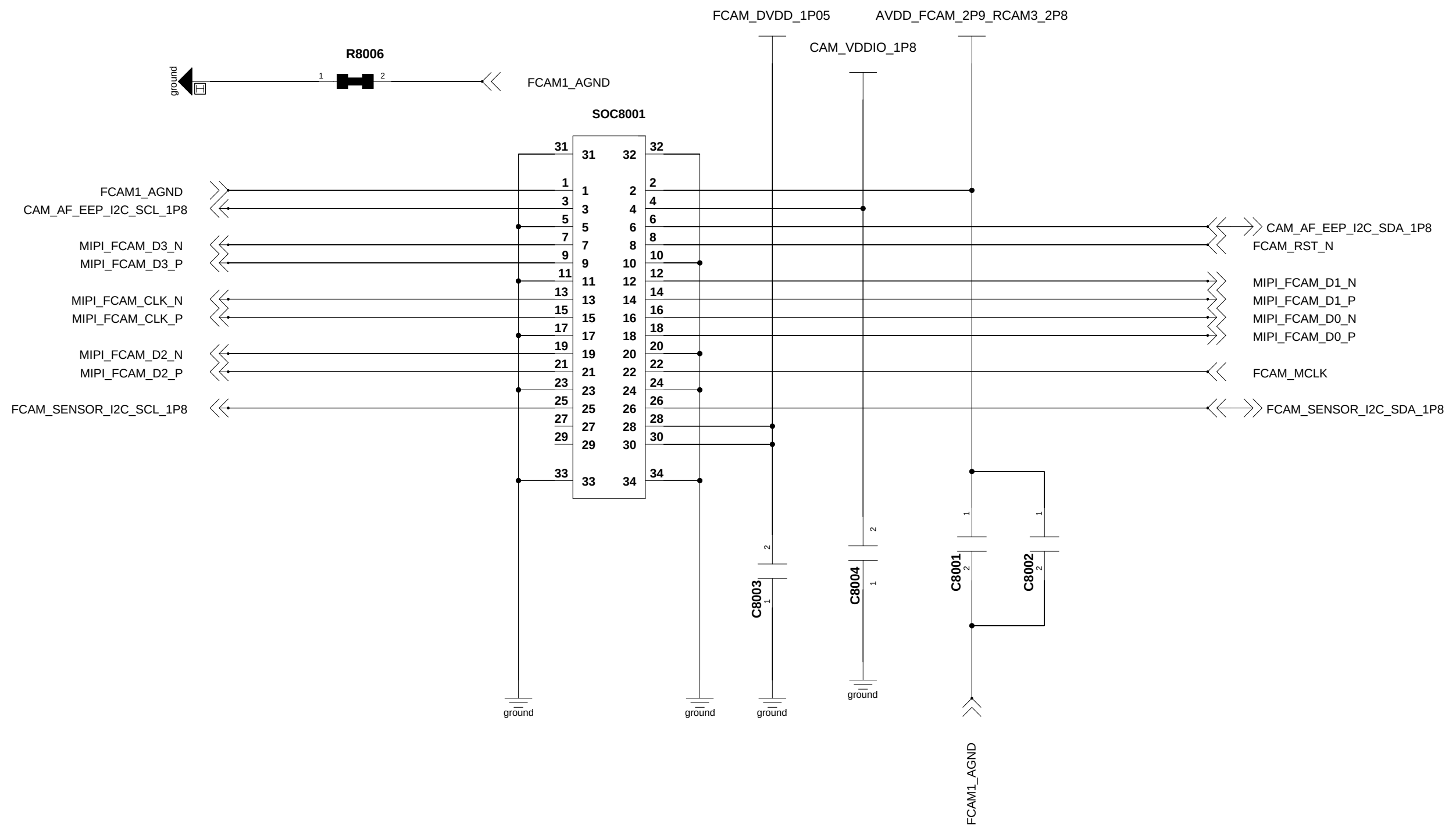
RCAM1_50M OIS(40PIN)


L31M_VDD_RCAM3_DVDD_1P05
L31M_VDD_RCAM3_DVDD_1P05
@RCAM3_DVDD_1P05
0915
AVDD_FCAM_2P9_RCAM3_2P8
L31M_VDD_RCAM3_DVDD_1P05
@RCAM3_DVDD_1P05
0915
CAM_VDDIO_1P8
SOC8004
05.24
MIPI_RCAM3_D3_N
MIPI_RCAM3_D3_P
MIPI_RCAM3_CLK_N
MIPI_RCAM3_CLK_P
MIPI_RCAM3_D2_N
MIPI_RCAM3_D2_P
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM_FSYNC
RCAM3_RST_N
MIPI_RCAM3_D1_N
MIPI_RCAM3_D1_P
MIPI_RCAM3_D0_N
MIPI_RCAM3_D0_P
RCAM3_MCLK
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
RCAM3_AGND
R8004
C8020
C8021
C8022
ground
RCAM3_AGND

RCAM3_SUPER WIDE_12M CAMERA(30pin)

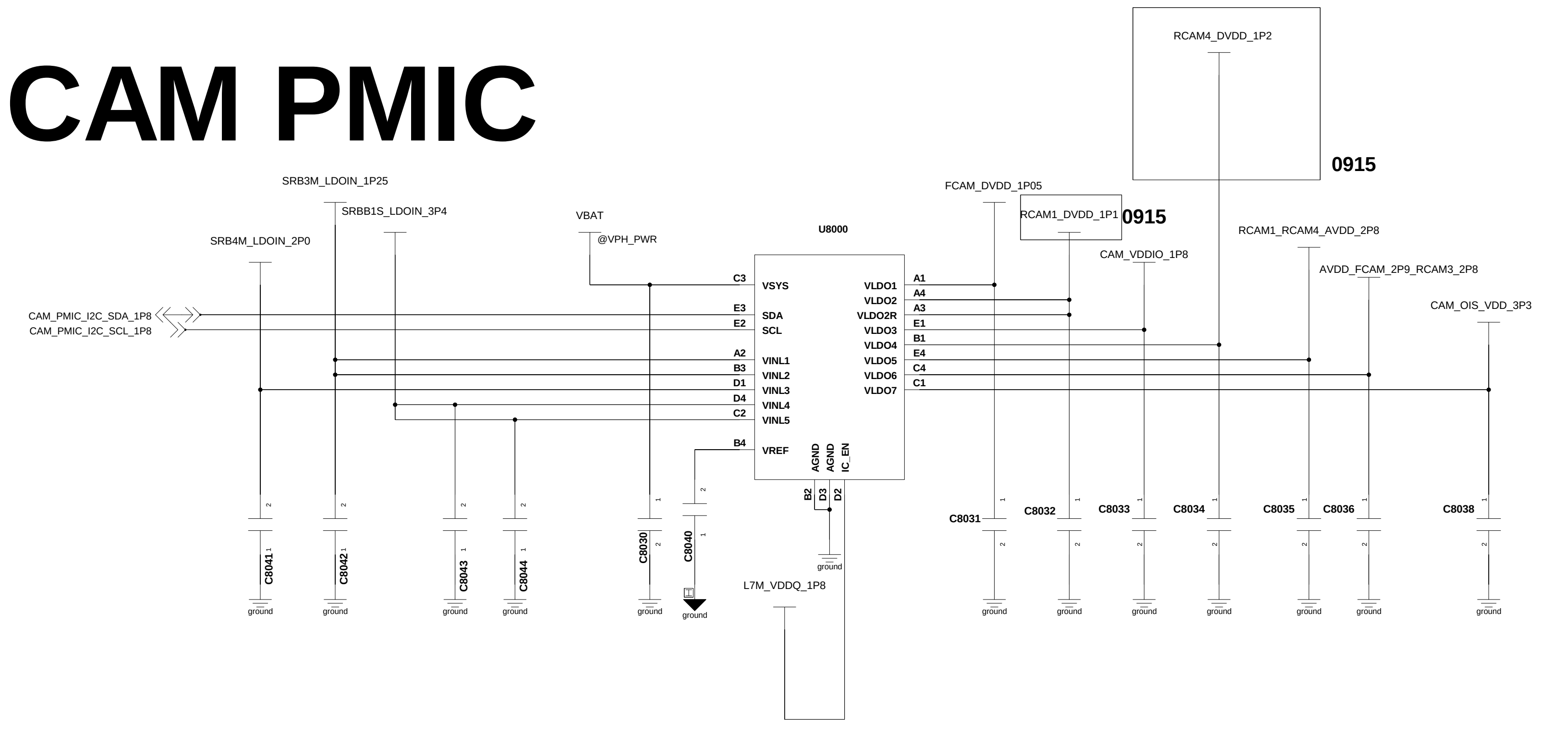

RCAM4_DVDD_1P2
CAM_VDDIO_1P8
RCAM1_RCAM4_AVDD_2P8
SOC8003
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
RCAM_FSYNC
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM4_RST_N
MIPI_RCAM4_D1_P
MIPI_RCAM4_D1_N
MIPI_RCAM4_CLK_P
MIPI_RCAM4_CLK_N
MIPI_RCAM4_D0_P
MIPI_RCAM4_D0_N
FCAM_MCLK
@RCAM4_MCLK
C8017
C8018
C8019
ground
RCAM4_AGND
R8003

RCAM4_5M_MACRO_CAMERA(24PIN)

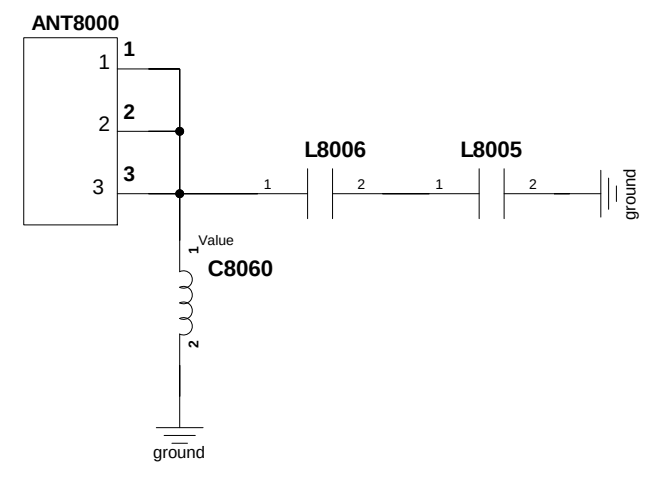

FCAM_DVDD_1P05
AVDD_FCAM_2P9_RCAM3_2P8
CAM_VDDIO_1P8
R8006
FCAM1_AGND
SOC8001
FCAM1_AGND
CAM_AF_EEP_I2C_SCL_1P8
MIPI_FCAM_D3_N
MIPI_FCAM_D3_P
MIPI_FCAM_CLK_N
MIPI_FCAM_CLK_P
MIPI_FCAM_D2_N
MIPI_FCAM_D2_P
FCAM_SENSOR_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
FCAM_RST_N
MIPI_FCAM_D1_N
MIPI_FCAM_D1_P
MIPI_FCAM_D0_N
MIPI_FCAM_D0_P
FCAM_MCLK
FCAM_SENSOR_I2C_SDA_1P8
C8003
C8004
C8001
C8002
ground
FCAM1_AGND

FRONT_32M CAMERA (30PIN)

CAM PMIC


RCAM4_DVDD_1P2
0915
SRB3M_LDOIN_1P25
SRBB1S_LDOIN_3P4
SRB4M_LDOIN_2P0
VBAT
@VPH_PWR
U8000
FCAM_DVDD_1P05
RCAM1_DVDD_1P1
0915
CAM_VDDIO_1P8
RCAM1_RCAM4_AVDD_2P8
AVDD_FCAM_2P9_RCAM3_2P8
CAM_OIS_VDD_3P3
CAM_PMIC_I2C_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
VSYS
SDA
SCL
VINL1
VINL2
VINL3
VINL4
VINL5
VREF
VLDO1
VLDO2
VLDO2R
VLDO3
VLDO4
VLDO5
VLDO6
VLDO7
AGND
AGND
IC_EN
C8041
C8042
C8043
C8044
C8030
C8040
C8031
C8032
C8033
C8034
C8035
C8036
C8038
L7M_VDDQ_1P8
ground

DECO CONTACT


ANT8000
L8006
L8005
C8060
Value
ground

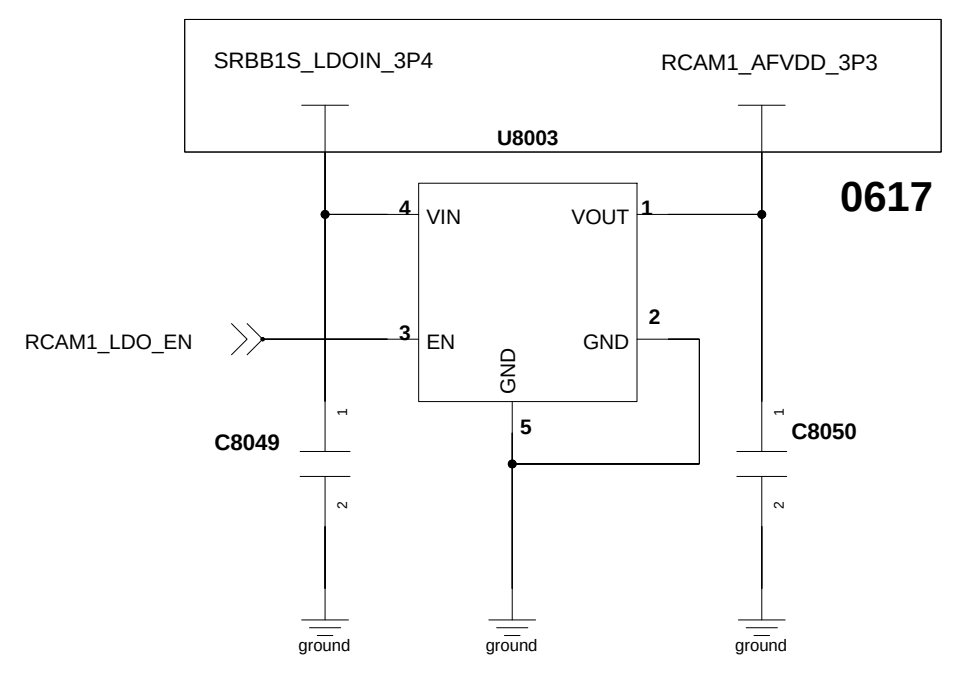
220915

RCAM4_AVDD

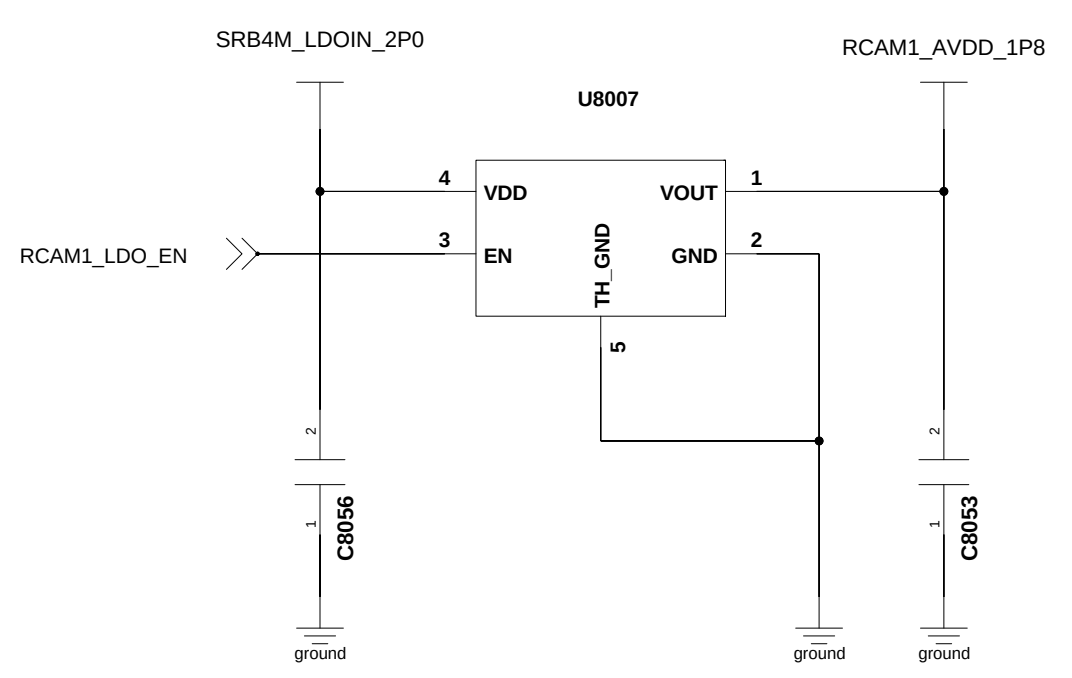
RCAM3_AVDD

CAM MCU_DVDD

0617


SRBB1S_LDOIN_3P4
RCAM1_AFVDD_3P3
U8003
0617
VIN
VOUT
EN
GND
GND
RCAM1_LDO_EN
C8049
C8050
ground

RCAM1_AF


SRB4M_LDOIN_2P0
RCAM1_AVDD_1P8
U8007
VDD
VOUT
EN
GND
TH_GND
RCAM1_LDO_EN
C8056
C8053
ground

RCAM1_AVDD

CAM MCU

RCAM4_DVDD

220915

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: CAMERA
DOC CTRL CHK:	Size:
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

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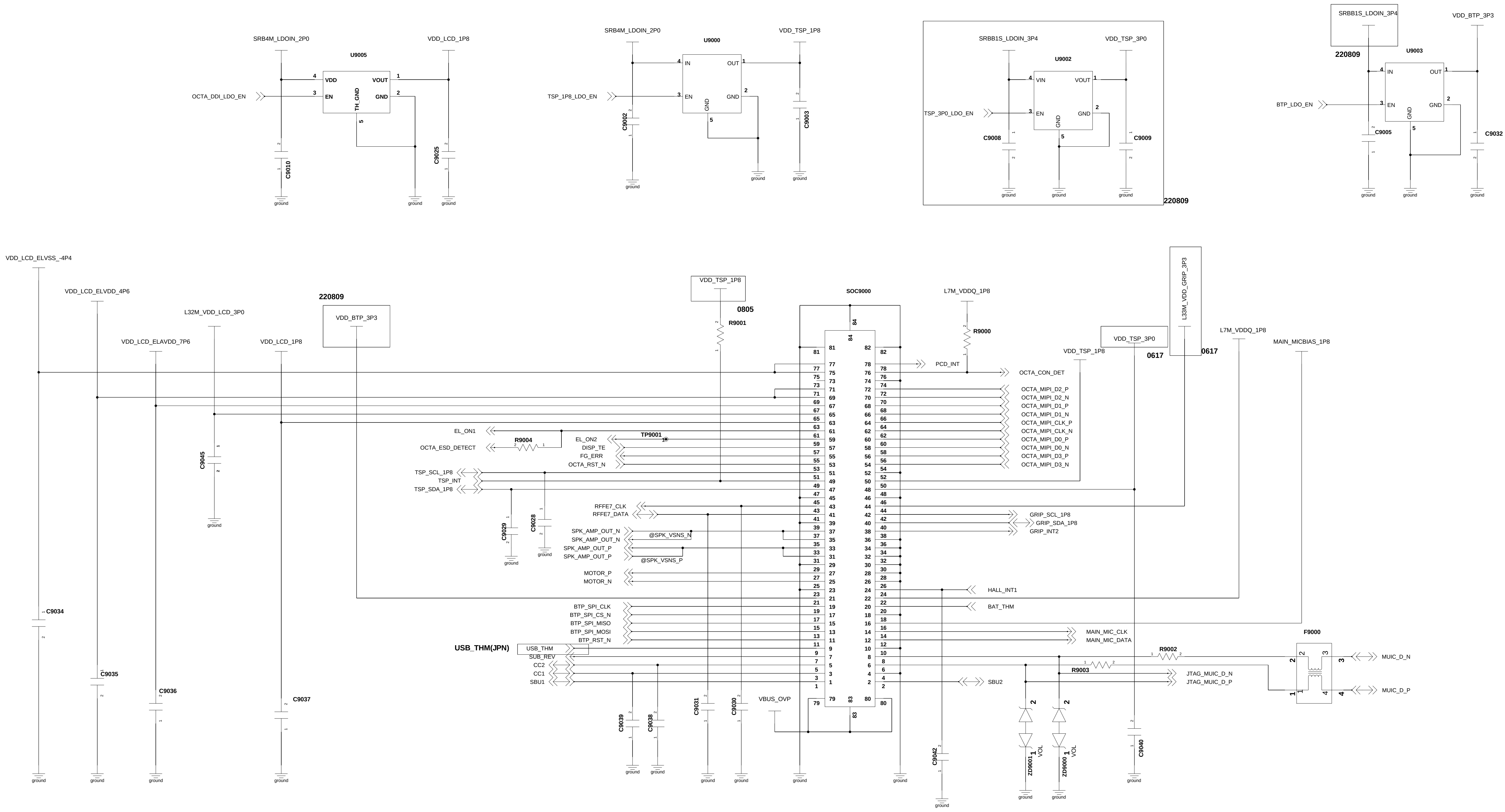
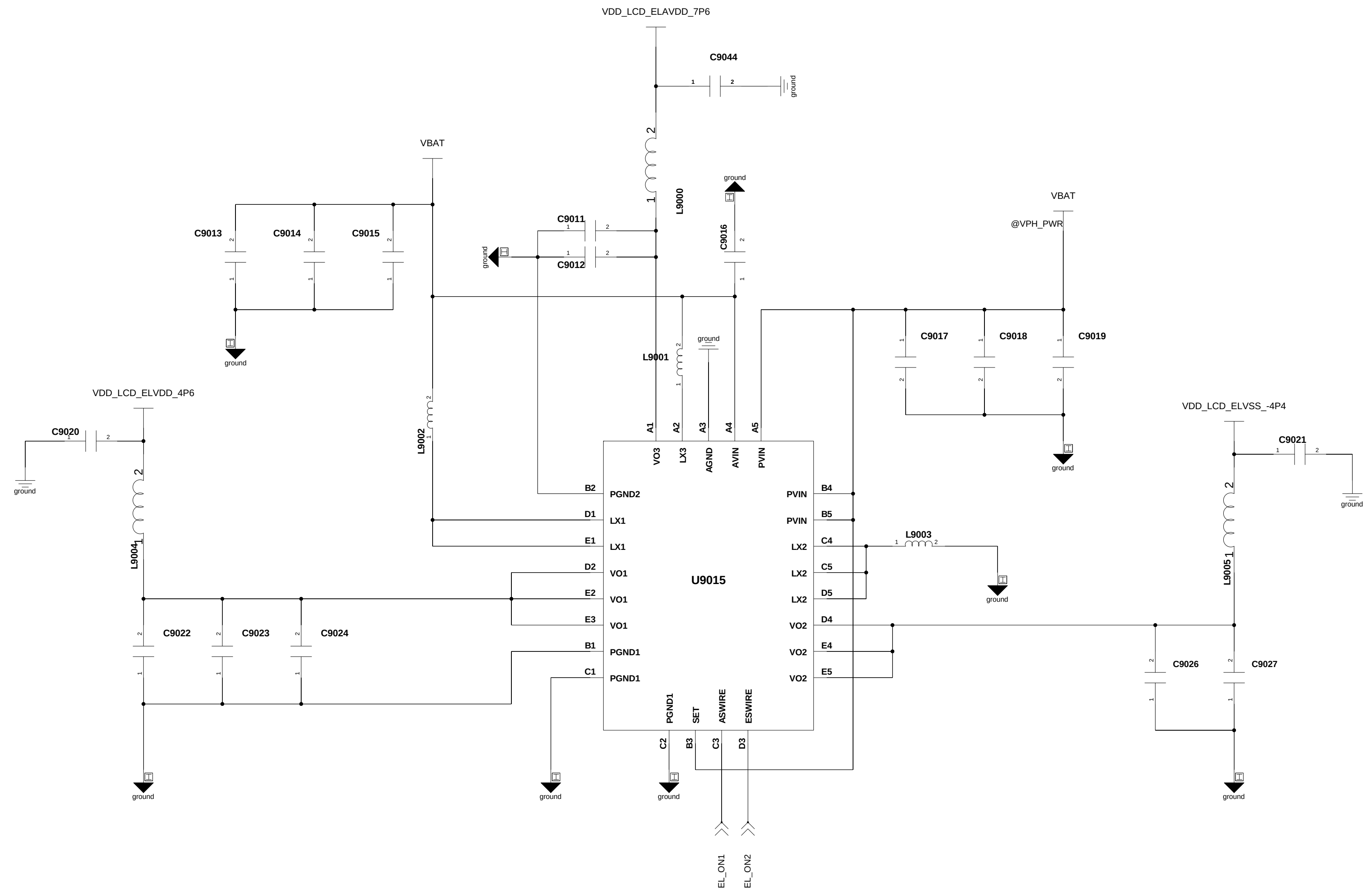
C2C (Conn to Conn FPCB)


SRB4M_LDOIN_2P0
VDD_LCD_1P8
U9005
VDD
VOUT
EN
GND
TH_GND
OCTA_DDI_LDO_EN
C9010
C9025
SRB4M_LDOIN_2P0
U9000
IN
OUT
EN
GND
VDD_TSP_1P8
TSP_1P8_LDO_EN
C9002
C9003
SRBB1S_LDOIN_3P4
VDD_TSP_3P0
U9002
VIN
VOUT
EN
GND
TSP_3P0_LDO_EN
C9008
C9009
220809
SRBB1S_LDOIN_3P4
220809
U9003
VDD_BTP_3P3
IN
OUT
EN
GND
BTP_LDO_EN
C9005
C9032
ground
VDD_LCD_ELVSS_-4P4
VDD_LCD_ELVDD_4P6
L32M_VDD_LCD_3P0
VDD_LCD_ELAVDD_7P6
VDD_LCD_1P8
220809
VDD_BTP_3P3
VDD_TSP_1P8
0805
R9001
SOC9000
L7M_VDDQ_1P8
R9000
C9045
EL_ON1
R9004
OCTA_ESD_DETECT
EL_ON2
TP9001
DISP_TE
FG_ERR
OCTA_RST_N
TSP_SCL_1P8
TSP_INT
TSP_SDA_1P8
C9029
C9028
RFFE7_CLK
RFFE7_DATA
SPK_AMP_OUT_N
SPK_AMP_OUT_N
@SPK_VSNS_N
SPK_AMP_OUT_P
SPK_AMP_OUT_P
@SPK_VSNS_P
MOTOR_P
MOTOR_N
BTP_SPI_CLK
BTP_SPI_CS_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_RST_N
USB_THM(JPN)
USB_THM
SUB_REV
CC2
CC1
SBU1
C9034
C9035
C9036
C9037
C9039
C9038
C9031
C9030
VBUS_OVP
PCD_INT
OCTA_CON_DET
OCTA_MIPI_D2_P
OCTA_MIPI_D2_N
OCTA_MIPI_D1_P
OCTA_MIPI_D1_N
OCTA_MIPI_CLK_P
OCTA_MIPI_CLK_N
OCTA_MIPI_D0_P
OCTA_MIPI_D0_N
OCTA_MIPI_D3_P
OCTA_MIPI_D3_N
VDD_TSP_1P8
VDD_TSP_3P0
0617
L33M_VDD_GRIP_3P3
0617
L7M_VDDQ_1P8
MAIN_MICBIAS_1P8
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
HALL_INT1
BAT_THM
MAIN_MIC_CLK
MAIN_MIC_DATA
F9000
R9002
R9003
MUIC_D_N
MUIC_D_P
JTAG_MUIC_D_N
JTAG_MUIC_D_P
SBU2
C9042
ZD9001
ZD9000
VOL
VOL
C9040
ground

DISPLAY

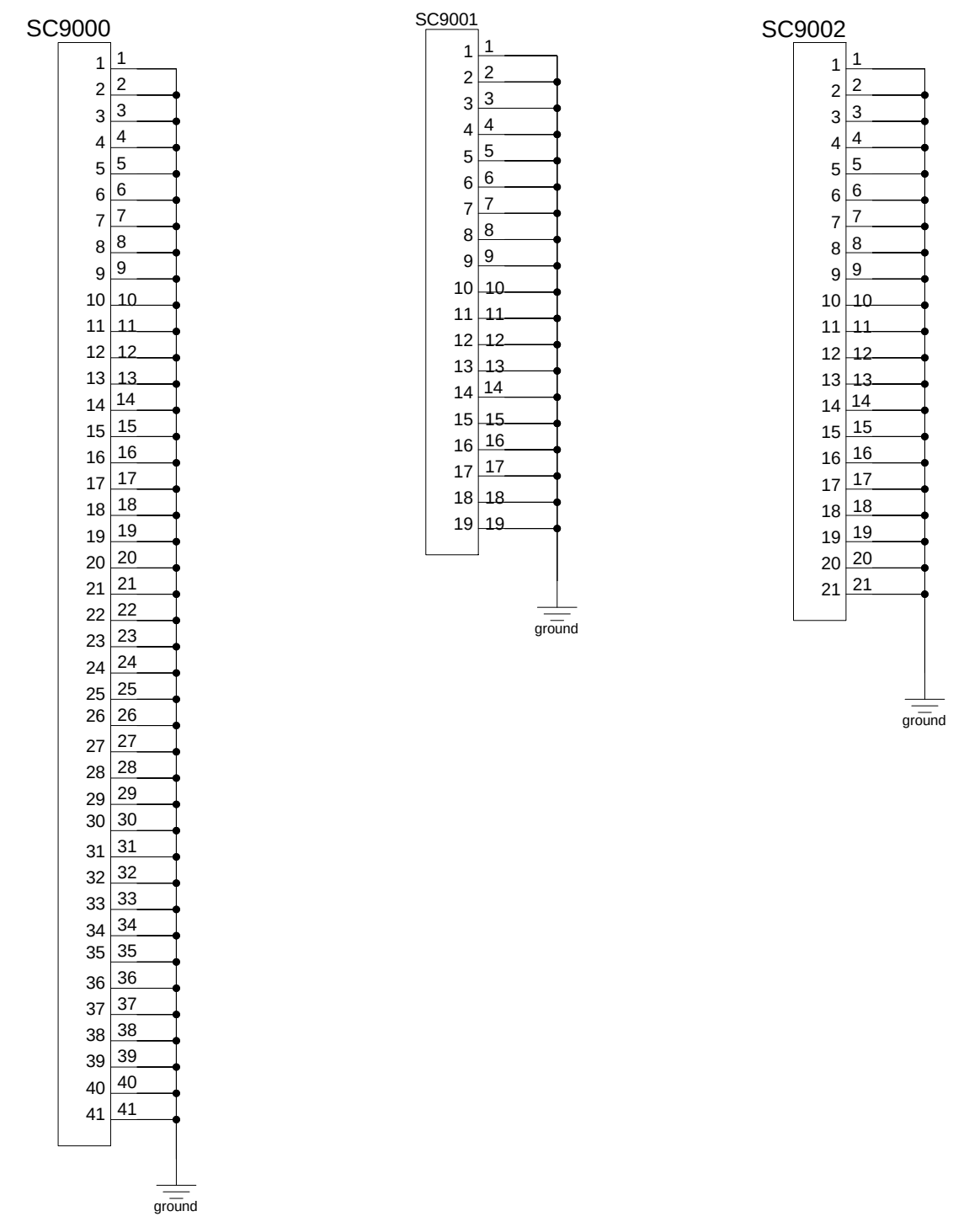

VDD_LCD_ELAVDD_7P6
C9044
ground
L9000
VBAT
C9013
C9014
C9015
C9011
C9012
C9016
ground
L9001
L9002
VBAT
@VPH_PWR
C9017
C9018
C9019
VDD_LCD_ELVDD_4P6
C9020
L9004
C9022
C9023
C9024
U9015
A1
A2
A3
A4
A5
VO3
LX3
AGND
AVIN
PVIN
B2
D1
E1
D2
E2
E3
B1
C1
PGND2
LX1
LX1
VO1
VO1
VO1
PGND1
PGND1
B4
B5
C4
C5
D5
D4
E4
E5
PVIN
PVIN
LX2
LX2
LX2
VO2
VO2
VO2
PGND1
SET
ASWIRE
ESWIRE
C2
B3
C3
D3
L9003
VDD_LCD_ELVSS_-4P4
C9021
L9005
C9026
C9027
EL_ON1
EL_ON2
ground

DISPLAY PMIC

2D Barcode

CN9000
CN9001
221101
SUS9000
SUS9002
1
2
3
4
5
BOTH
ground

SHIELD CAN


SC9000
SC9001
SC9002
ground

REGION		MEMORY		
R9013	B : EUR	R9019	6/128	
R9014	0 : HK CHN	R9015	8/128	
R9018	S : KOR SKT	R9017	8/256	
R9021	E : SEA, MEA, SWA E : AFR, LA, CIS			

mmW_ANT_CONN (Right)

NA ONLY

ES2.0 : OPEN
ES3.0 : 0 ohm

5G IFIC

NA ONLY

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:

A54

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Page:

SM-A5460
Service Schematic

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A54 5G EUR_REV04C

2022.12

Sheet01 :	GSM / WCDMA / LTE / NR(N41) RF
Sheet02 :	TRANSCEIVER, SUB ANT
Sheet03 :	CONNECTIVITY(BT/WIFI/FM), NFC
Sheet04 :	AP(Quartz-INTERFACE,GPIO)
Sheet05 :	AP(POWER), MEM(uMCP)
Sheet06 :	PMIC/SUB PMIC, IF PMIC, DC IC
Sheet07 :	AUDIO AMP, SENSOR, SIM, SD
Sheet08 :	CAMERA
Sheet09 :	DISPLAY PMIC, C2C

Engineer:				
Drawn by:				
R&D CHK:	TITLE:			Size:
DOC CTRL CHK:	A54			
MFG ENGR CHK:				
QA CHK:	REV:	Drawing Number:		Page:

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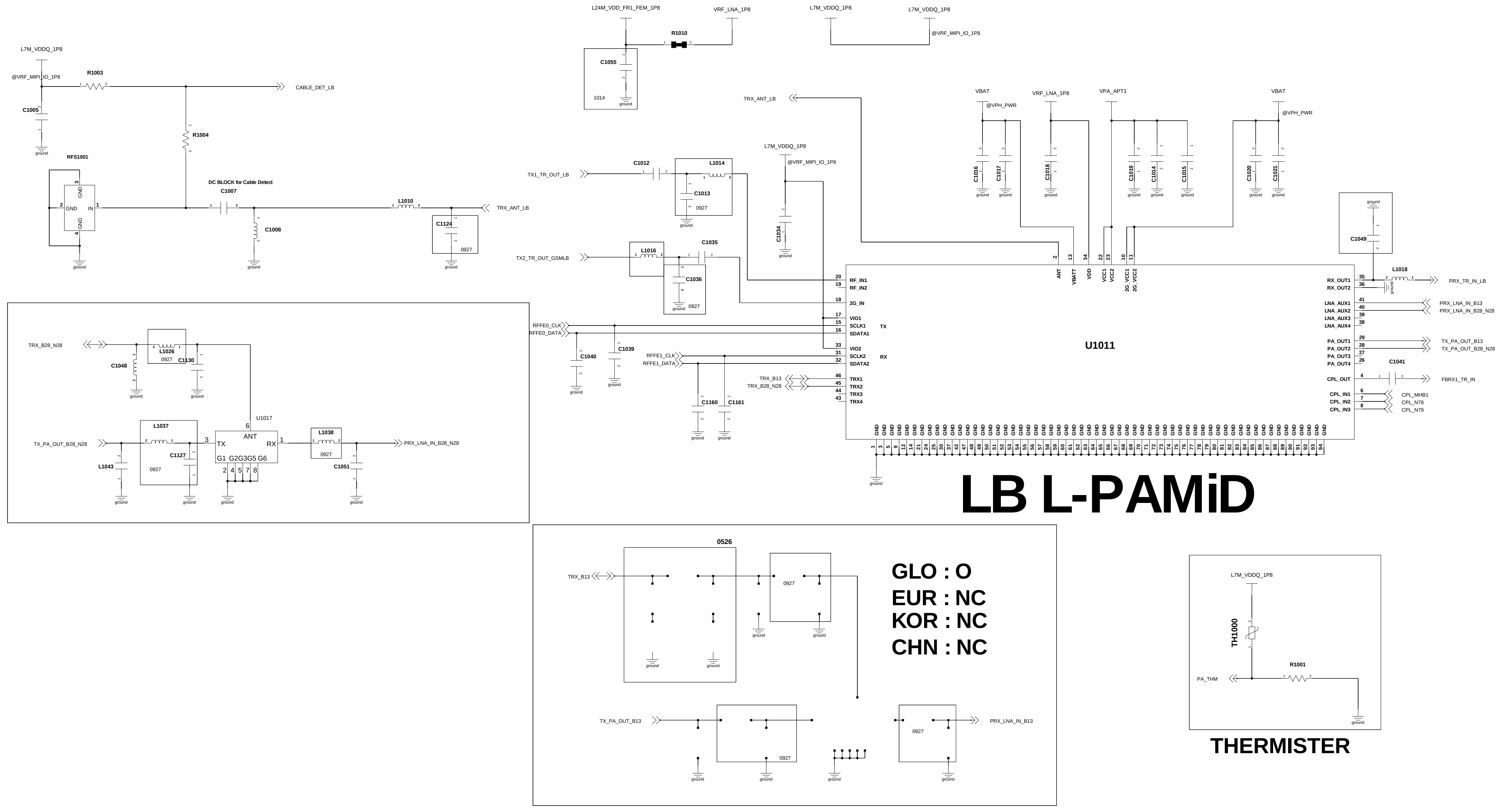
RF PART 1

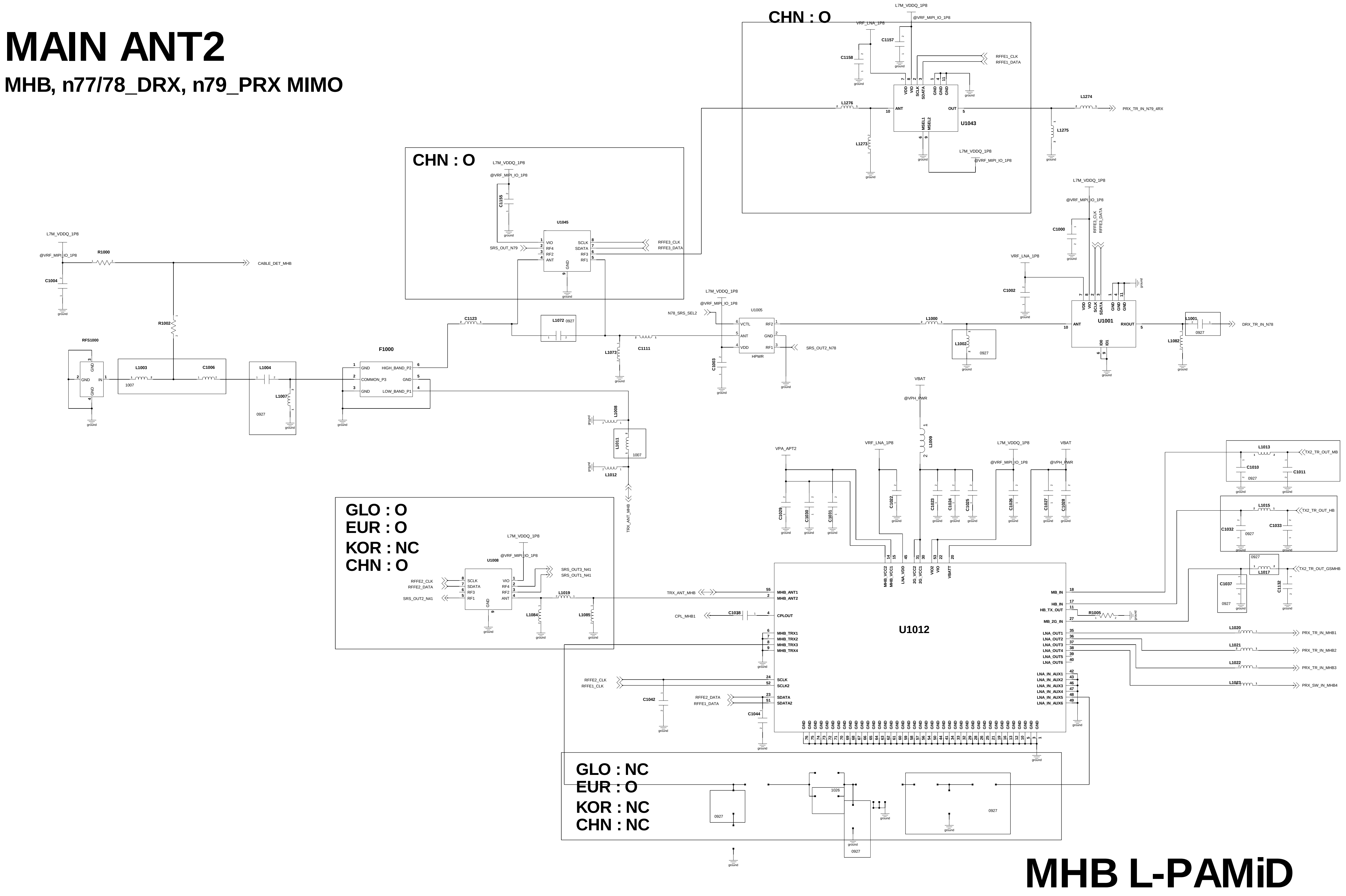
MAIN ANT1

LB


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1003
CABLE_DET_LB
C1005
R1004
RFS1001
DC BLOCK for Cable Detect
C1007
C1008
L1010
TRX_ANT_LB
C1124
L24M_VDD_FR1_FEM_1P8
VRF_LNA_1P8
R1010
C1055
1014
L7M_VDDQ_1P8
TRX_ANT_LB
TX1_TR_OUT_LB
C1012
L1014
C1013
TX2_TR_OUT_GSMLB
L1016
C1035
C1036
C1034
RFFE0_CLK
RFFE0_DATA
C1040
C1039
RFFE1_CLK
RFFE1_DATA
TRX_B13
TRX_B28_N28
C1160
C1161
VBAT
@VPH_PWR
VRF_LNA_1P8
VPA_APT1
C1016
C1017
C1018
C1019
C1014
C1015
C1020
C1021
C1049
U1011
RF_IN1
RF_IN2
2G_IN
VIO1
SCLK1
SDATA1
TX
VIO2
SCLK2
SDATA2
RX
TRX1
TRX2
TRX3
TRX4
ANT
VBATT
VDD
VCC1
VCC2
2G_VCC1
2G_VCC2
RX_OUT1
RX_OUT2
L1018
PRX_TR_IN_LB
LNA_AUX1
LNA_AUX2
LNA_AUX3
LNA_AUX4
PRX_LNA_IN_B13
PRX_LNA_IN_B28_N28
PA_OUT1
PA_OUT2
PA_OUT3
PA_OUT4
TX_PA_OUT_B13
TX_PA_OUT_B28_N28
CPL_OUT
C1041
FBRX1_TR_IN
CPL_IN1
CPL_IN2
CPL_IN3
CPL_MHB1
CPL_N78
CPL_N79
LB L-PAMiD
TRX_B28_N28
L1026
C1130
C1048
U1017
TX_PA_OUT_B28_N28
L1037
C1127
L1043
TX
ANT
RX
G1 G2G3G5 G6
L1038
C1051
PRX_LNA_IN_B28_N28
0526
TRX_B13
GLO : O
EUR : NC
KOR : NC
CHN : NC
TX_PA_OUT_B13
PRX_LNA_IN_B13
L7M_VDDQ_1P8
TH1000
R1001
PA_THM
THERMISTER

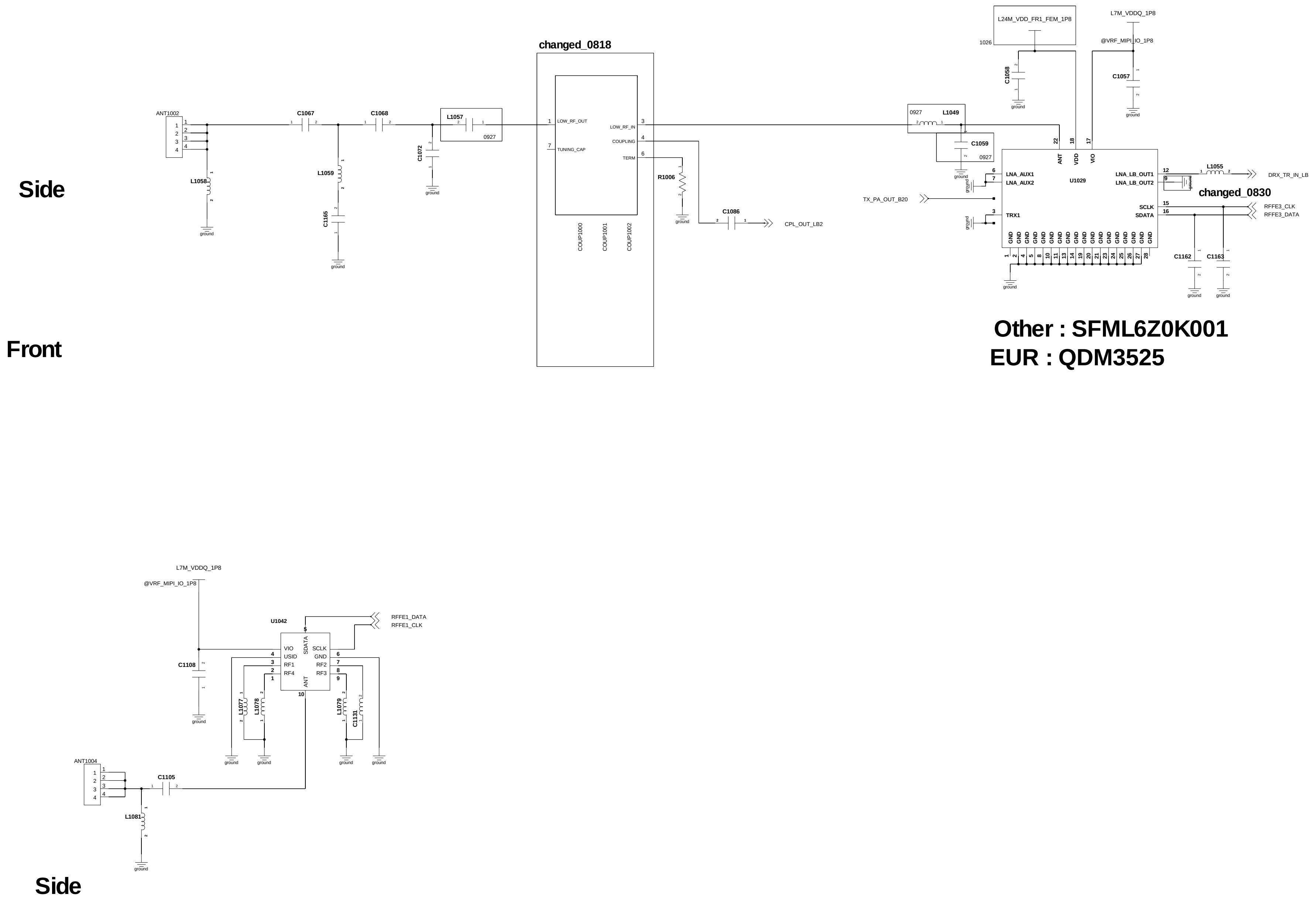
MAIN ANT2

MHB, n77/78_DRX, n79_PRX MIMO


CHN : O
VRF_LNA_1P8
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1158
C1157
RFFE1_CLK
RFFE1_DATA
L1276
L1273
U1043
ANT
OUT
MSEL1
MSEL2
L1274
L1275
PRX_TR_IN_N79_4RX
CHN : O
C1155
U1045
VIO
RF4
RF2
ANT
SCLK
SDATA
RF3
RF1
SRS_OUT_N79
RFFE3_CLK
RFFE3_DATA
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
R1000
CABLE_DET_MHB
C1004
R1002
RFS1000
L1003
C1006
1007
L1004
L1007
F1000
HIGH_BAND_P2
COMMON_P3
LOW_BAND_P1
C1123
L1072
L1073
C1111
N78_SRS_SEL2
U1005
VCTL
RF2
RF1
HPWR
C1003
SRS_OUT2_N78
L1000
L1002
VRF_LNA_1P8
C1002
C1000
RFFE3_CLK
RFFE3_DATA
U1001
RXOUT
ID0
ID1
L1001
L1082
DRX_TR_IN_N78
L1008
L1011
L1012
TRX_ANT_MHB
GLO : O
EUR : O
KOR : NC
CHN : O
U1008
RFFE2_CLK
RFFE2_DATA
SRS_OUT2_N41
SRS_OUT3_N41
SRS_OUT1_N41
L1019
L1084
L1085
VBAT
@VPH_PWR
L1009
VPA_APT2
C1029
C1030
C1031
C1022
C1023
C1024
C1025
C1026
C1027
C1028
U1012
MHB_ANT1
MHB_ANT2
CPLOUT
C1038
CPL_MHB1
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
SCLK
SCLK2
SDATA
SDATA2
RFFE2_CLK
RFFE1_CLK
RFFE2_DATA
RFFE1_DATA
C1042
C1044
MB_IN
HB_IN
HB_TX_OUT
MB_2G_IN
R1005
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
L1020
L1021
L1022
L1023
PRX_TR_IN_MHB1
PRX_TR_IN_MHB2
PRX_TR_IN_MHB3
PRX_SW_IN_MHB4
L1013
C1010
C1011
TX2_TR_OUT_MB
L1015
C1032
C1033
TX2_TR_OUT_HB
L1017
C1037
C1132
TX2_TR_OUT_GSMHB
GLO : NC
EUR : O
KOR : NC
CHN : NC
1026
MHB L-PAMiD

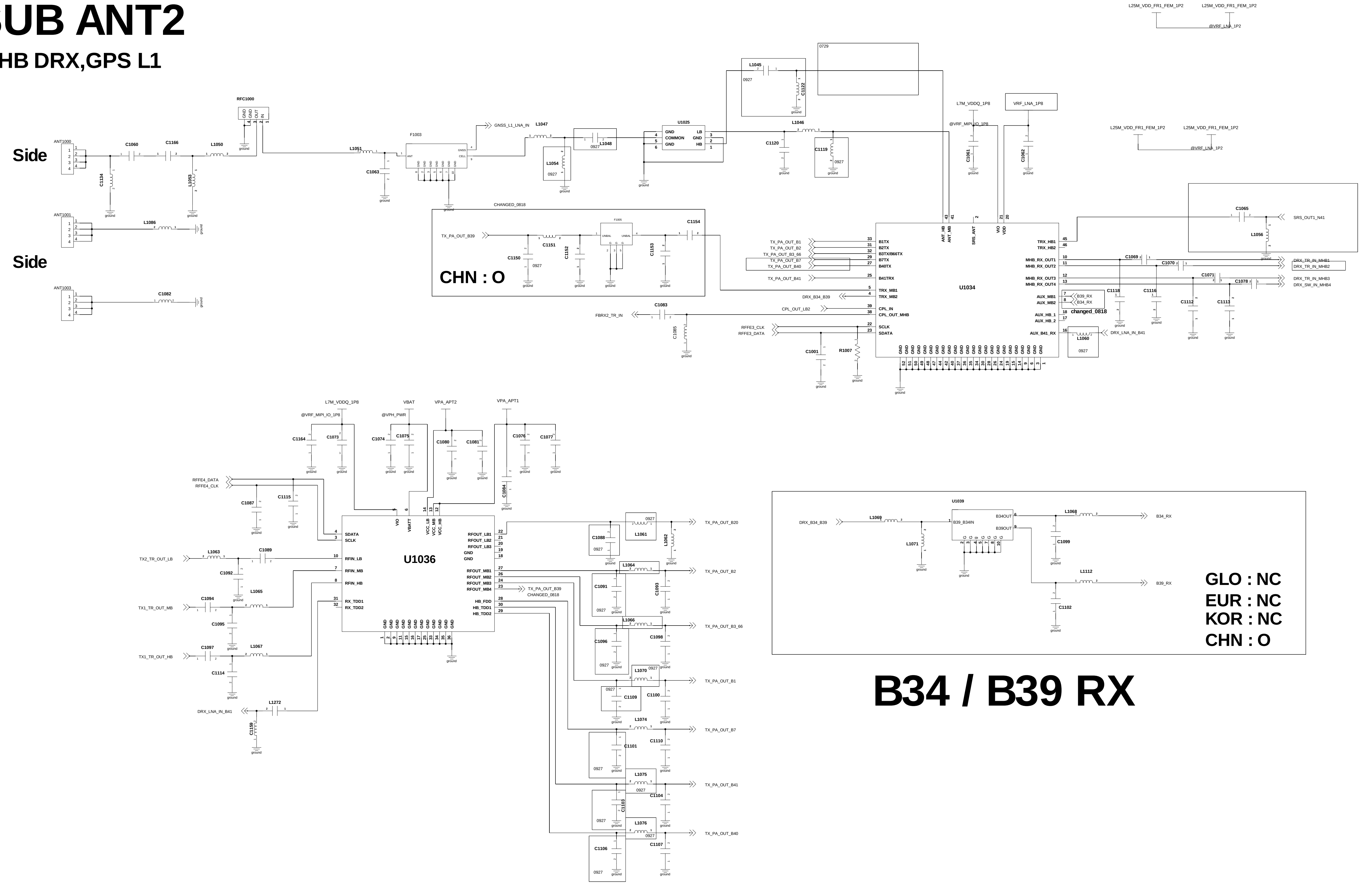
SUB ANT1

LB DRX


changed_0818
ANT1002
C1067
C1068
L1057
C1072
L1058
L1059
C1165
LOW_RF_OUT
TUNING_CAP
LOW_RF_IN
COUPLING
TERM
COUP1000
COUP1001
COUP1002
R1006
C1086
CPL_OUT_LB2
L24M_VDD_FR1_FEM_1P8
1026
C1058
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C1057
L1049
C1059
U1029
LNA_AUX1
LNA_AUX2
TRX1
TX_PA_OUT_B20
LNA_LB_OUT1
LNA_LB_OUT2
SCLK
SDATA
L1055
DRX_TR_IN_LB
changed_0830
RFFE3_CLK
RFFE3_DATA
C1162
C1163
Side
Front
Other : SFML6Z0K001
EUR : QDM3525
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
U1042
VIO
USID
RF1
RF4
SDATA
SCLK
GND
RF2
RF3
ANT
RFFE1_DATA
RFFE1_CLK
C1108
L1077
L1078
L1079
C1131
ANT1004
C1105
L1081
Side

SUB ANT2

MHB DRX,GPS L1


L25M_VDD_FR1_FEM_1P2
@VRF_LNA_1P2
RFC1000
Side
ANT1000
C1060
C1166
L1050
C1134
L1053
L1051
C1063
F1003
GNSS_L1_LNA_IN
L1047
L1048
L1054
U1025
COMMON
LB
HB
L1045
C1122
0729
L1046
C1120
C1119
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
C1061
C1062
ANT1001
L1086
Side
ANT1003
C1082
CHANGED_0818
TX_PA_OUT_B39
C1150
C1151
C1152
F1005
UNBAL
C1153
C1154
CHN : O
FBRX2_TR_IN
C1083
C1085
U1034
TX_PA_OUT_B1
TX_PA_OUT_B2
TX_PA_OUT_B3_66
TX_PA_OUT_B7
TX_PA_OUT_B40
TX_PA_OUT_B41
B1TX
B2TX
B3TX/B66TX
B7TX
B40TX
B41TRX
TRX_MB1
TRX_MB2
DRX_B34_B39
CPL_IN
CPL_OUT_MHB
CPL_OUT_LB2
SCLK
SDATA
RFFE3_CLK
RFFE3_DATA
C1001
R1007
ANT_HB
ANT_MB
SRS_ANT
VIO
VDD
TRX_HB1
TRX_HB2
MHB_RX_OUT1
MHB_RX_OUT2
MHB_RX_OUT3
MHB_RX_OUT4
AUX_MB1
AUX_MB2
AUX_HB_1
AUX_HB_2
AUX_B41_RX
B39_RX
B34_RX
changed_0818
L1060
DRX_LNA_IN_B41
C1065
L1056
SRS_OUT1_N41
C1069
C1070
C1071
C1078
DRX_TR_IN_MHB1
DRX_TR_IN_MHB2
DRX_TR_IN_MHB3
DRX_SW_IN_MHB4
C1118
C1116
C1112
C1113
L7M_VDDQ_1P8
VBAT
VPA_APT2
VPA_APT1
C1164
C1073
C1074
C1075
C1080
C1081
C1076
C1077
C1084
RFFE4_DATA
RFFE4_CLK
C1087
C1115
U1036
SDATA
SCLK
RFIN_LB
RFIN_MB
RFIN_HB
RX_TDD1
RX_TDD2
VBATT
VCC_LB
VCC_MB
VCC_HB
RFOUT_LB1
RFOUT_LB2
RFOUT_LB3
RFOUT_MB1
RFOUT_MB2
RFOUT_MB3
RFOUT_MB4
HB_FDD
HB_TDD1
HB_TDD2
TX2_TR_OUT_LB
L1063
C1089
C1092
TX1_TR_OUT_MB
C1094
C1095
L1065
TX1_TR_OUT_HB
C1097
C1114
L1067
DRX_LNA_IN_B41
L1272
C1159
C1088
L1061
L1062
TX_PA_OUT_B20
L1064
C1091
C1093
TX_PA_OUT_B2
L1066
C1096
C1098
TX_PA_OUT_B3_66
L1070
C1109
C1100
TX_PA_OUT_B1
L1074
C1101
C1110
TX_PA_OUT_B7
L1075
C1103
C1104
TX_PA_OUT_B41
L1076
C1106
C1107
TX_PA_OUT_B40
U1039
DRX_B34_B39
L1069
L1071
B39_B34IN
B34OUT
B39OUT
L1068
C1099
B34_RX
L1112
C1102
B39_RX
GLO : NC
EUR : NC
KOR : NC
CHN : O
B34 / B39 RX

Engineer:	A54	
Drawn by:		
R&D CHK:	TITLE:	Size:
DOC CTRL CHK:	RF PAM/FEM	
MFG ENGR CHK:		
QA CHK:	REV:	Drawing Number:

Changed by: | Date Changed: | Time Changed:

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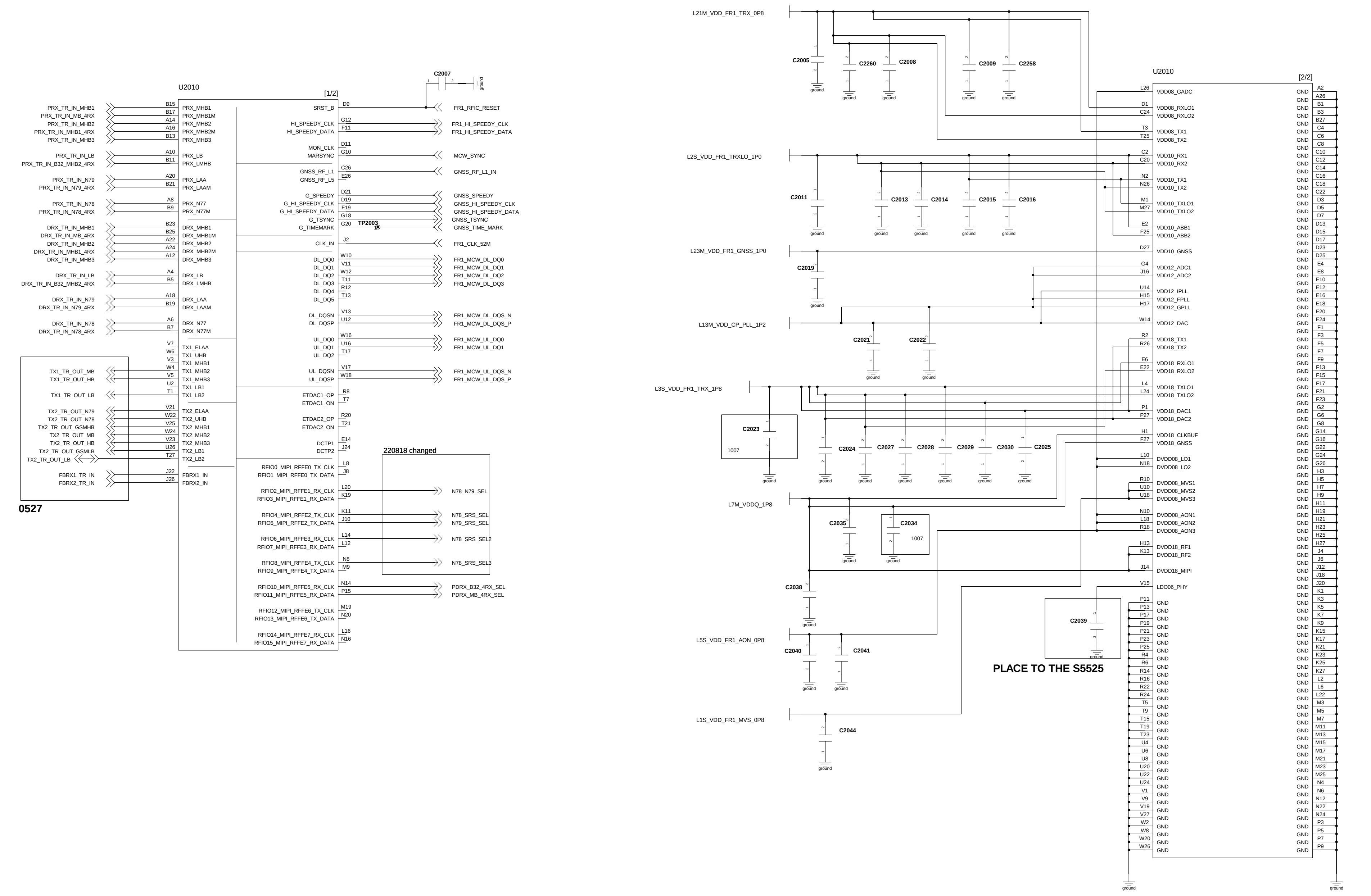
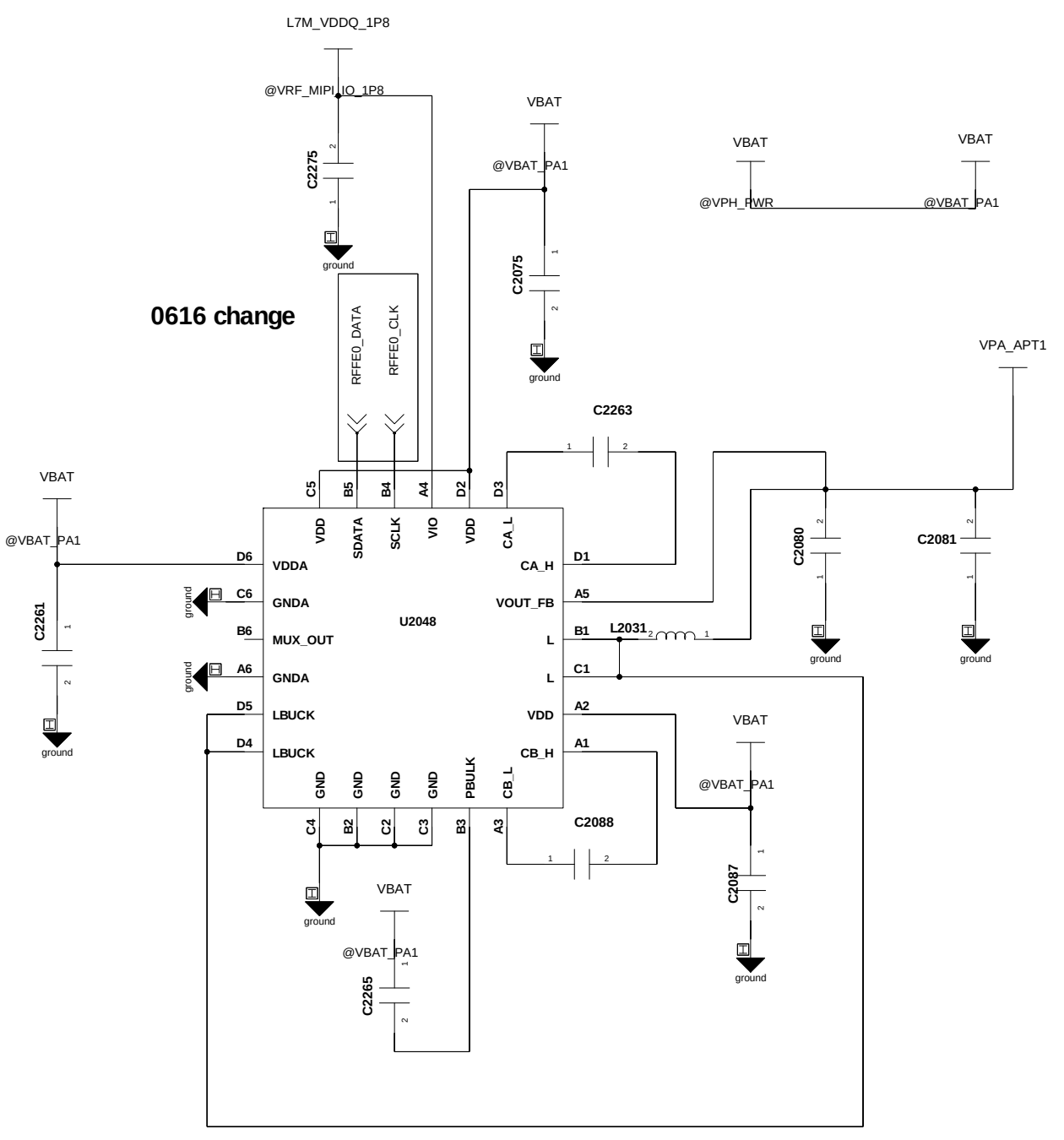
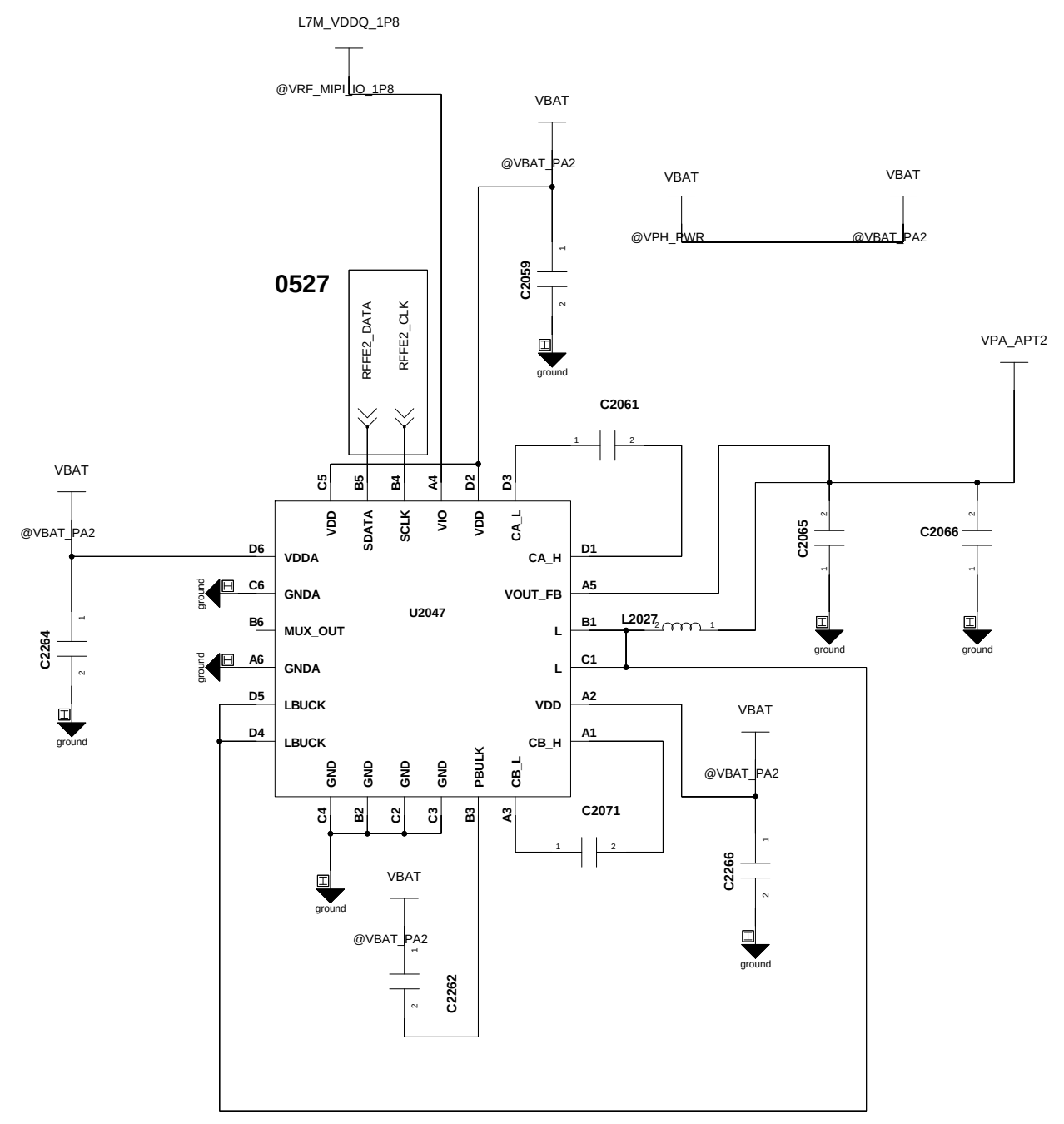
TRANSCEIVER (S5525)


U2010
[1/2]
PRX_TR_IN_MHB1
PRX_TR_IN_MB_4RX
PRX_TR_IN_MHB2
PRX_TR_IN_MHB1_4RX
PRX_TR_IN_MHB3
PRX_TR_IN_LB
PRX_TR_IN_B32_MHB2_4RX
PRX_TR_IN_N79
PRX_TR_IN_N79_4RX
PRX_TR_IN_N78
PRX_TR_IN_N78_4RX
DRX_TR_IN_MHB1
DRX_TR_IN_MB_4RX
DRX_TR_IN_MHB2
DRX_TR_IN_MHB1_4RX
DRX_TR_IN_MHB3
DRX_TR_IN_LB
DRX_TR_IN_B32_MHB2_4RX
DRX_TR_IN_N79
DRX_TR_IN_N79_4RX
DRX_TR_IN_N78
DRX_TR_IN_N78_4RX
TX1_TR_OUT_MB
TX1_TR_OUT_HB
TX1_TR_OUT_LB
TX2_TR_OUT_N79
TX2_TR_OUT_N78
TX2_TR_OUT_GSMHB
TX2_TR_OUT_MB
TX2_TR_OUT_HB
TX2_TR_OUT_GSMLB
TX2_TR_OUT_LB
FBRX1_TR_IN
FBRX2_TR_IN
0527
PRX_MHB1 PRX_MHB1M PRX_MHB2 PRX_MHB2M PRX_MHB3 PRX_LB PRX_LMHB PRX_LAA PRX_LAAM PRX_N77 PRX_N77M
DRX_MHB1 DRX_MHB1M DRX_MHB2 DRX_MHB2M DRX_MHB3 DRX_LB DRX_LMHB DRX_LAA DRX_LAAM DRX_N77 DRX_N77M
TX1_ELAA TX1_UHB TX1_MHB1 TX1_MHB2 TX1_MHB3 TX1_LB1 TX1_LB2
TX2_ELAA TX2_UHB TX2_MHB1 TX2_MHB2 TX2_MHB3 TX2_LB1 TX2_LB2
FBRX1_IN FBRX2_IN
SRST_B
HI_SPEEDY_CLK
HI_SPEEDY_DATA
MON_CLK
MARSYNC
GNSS_RF_L1
GNSS_RF_L5
G_SPEEDY
G_HI_SPEEDY_CLK
G_HI_SPEEDY_DATA
G_TSYNC
G_TIMEMARK
TP2003
CLK_IN
DL_DQ0 DL_DQ1 DL_DQ2 DL_DQ3 DL_DQ4 DL_DQ5
DL_DQSN DL_DQSP
UL_DQ0 UL_DQ1 UL_DQ2
UL_DQSN UL_DQSP
ETDAC1_OP ETDAC1_ON
ETDAC2_OP ETDAC2_ON
DCTP1 DCTP2
RFIO0_MIPI_RFFE0_TX_CLK
RFIO1_MIPI_RFFE0_TX_DATA
RFIO2_MIPI_RFFE1_RX_CLK
RFIO3_MIPI_RFFE1_RX_DATA
RFIO4_MIPI_RFFE2_TX_CLK
RFIO5_MIPI_RFFE2_TX_DATA
RFIO6_MIPI_RFFE3_RX_CLK
RFIO7_MIPI_RFFE3_RX_DATA
RFIO8_MIPI_RFFE4_TX_CLK
RFIO9_MIPI_RFFE4_TX_DATA
RFIO10_MIPI_RFFE5_RX_CLK
RFIO11_MIPI_RFFE5_RX_DATA
RFIO12_MIPI_RFFE6_TX_CLK
RFIO13_MIPI_RFFE6_TX_DATA
RFIO14_MIPI_RFFE7_RX_CLK
RFIO15_MIPI_RFFE7_RX_DATA
C2007
FR1_RFIC_RESET
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
MCW_SYNC
GNSS_RF_L1_IN
GNSS_SPEEDY
GNSS_HI_SPEEDY_CLK
GNSS_HI_SPEEDY_DATA
GNSS_TSYNC
GNSS_TIME_MARK
FR1_CLK_52M
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
220818 changed
N78_N79_SEL
N78_SRS_SEL
N79_SRS_SEL
N78_SRS_SEL2
N78_SRS_SEL3
PDRX_B32_4RX_SEL
PDRX_MB_4RX_SEL
L21M_VDD_FR1_TRX_0P8
C2005 C2260 C2008 C2009 C2258
L2S_VDD_FR1_TRXLO_1P0
C2011 C2013 C2014 C2015 C2016
L23M_VDD_FR1_GNSS_1P0
C2019
L13M_VDD_CP_PLL_1P2
C2021 C2022
L3S_VDD_FR1_TRX_1P8
C2023 1007
C2024 C2027 C2028 C2029 C2030 C2025
L7M_VDDQ_1P8
C2035 C2034 1007
C2038
L5S_VDD_FR1_AON_0P8
C2040 C2041
C2039
PLACE TO THE S5525
L1S_VDD_FR1_MVS_0P8
C2044
U2010
[2/2]
VDD08_GADC
VDD08_RXLO1
VDD08_RXLO2
VDD08_TX1
VDD08_TX2
VDD10_RX1
VDD10_RX2
VDD10_TX1
VDD10_TX2
VDD10_TXLO1
VDD10_TXLO2
VDD10_ABB1
VDD10_ABB2
VDD10_GNSS
VDD12_ADC1
VDD12_ADC2
VDD12_IPLL
VDD12_FPLL
VDD12_GPLL
VDD12_DAC
VDD18_TX1
VDD18_TX2
VDD18_RXLO1
VDD18_RXLO2
VDD18_TXLO1
VDD18_TXLO2
VDD18_DAC1
VDD18_DAC2
VDD18_CLKBUF
VDD18_GNSS
DVDD08_LO1
DVDD08_LO2
DVDD08_MVS1
DVDD08_MVS2
DVDD08_MVS3
DVDD08_AON1
DVDD08_AON2
DVDD08_AON3
DVDD18_RF1
DVDD18_RF2
DVDD18_MIPI
LDO06_PHY
GND
ground

APT1


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
C2275
VBAT
@VBAT_PA1
C2075
VBAT
@VPH_PWR
@VBAT_PA1
0616 change
RFFE0_DATA
RFFE0_CLK
VPA_APT1
C2263
C2080
C2081
C2261
U2048
VDD SDATA SCLK VIO VDD CA_L
VDDA GNDA MUX_OUT GNDA LBUCK LBUCK
CA_H VOUT_FB L L VDD CB_H
GND GND GND GND PBULK CB_L
L2031
C2088
C2087
C2265
ground

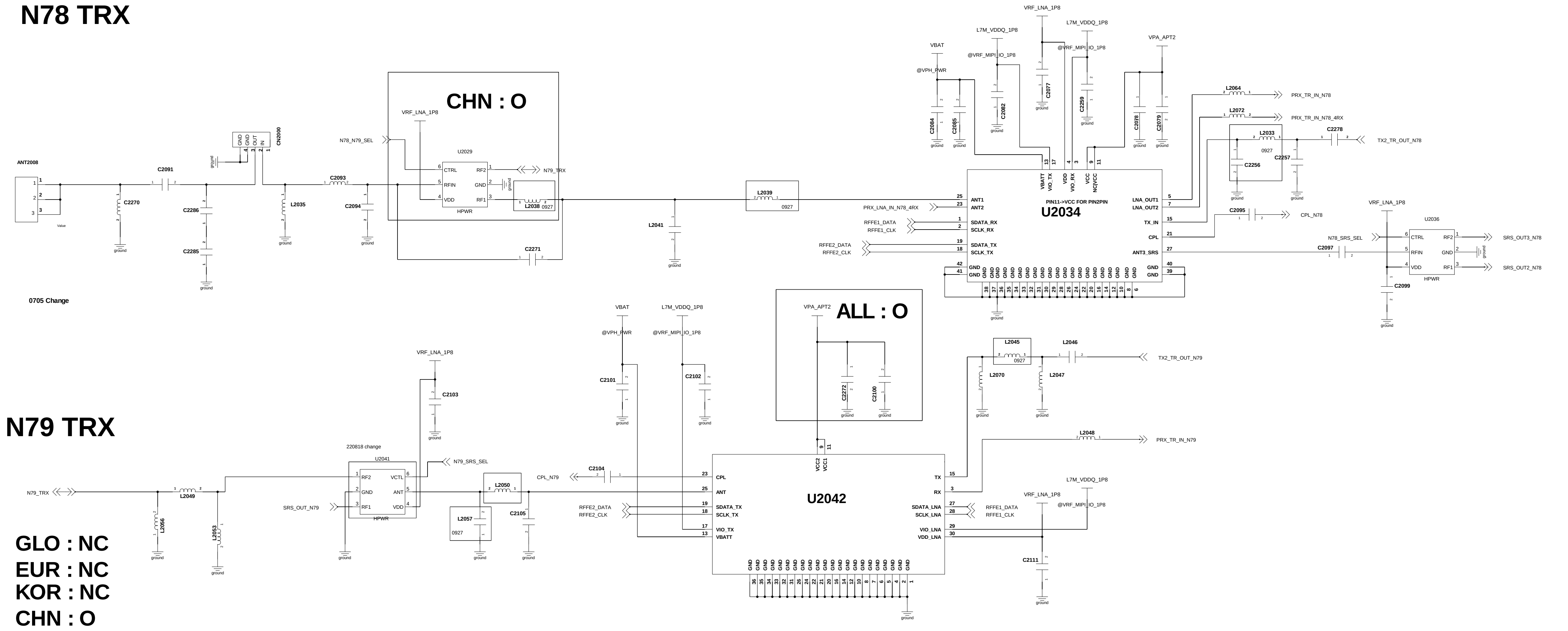
APT2


L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VBAT
@VBAT_PA2
C2059
VBAT
@VPH_PWR
@VBAT_PA2
0527
RFFE2_DATA
RFFE2_CLK
VPA_APT2
C2061
C2065
C2066
C2264
U2047
VDDA GNDA MUX_OUT GNDA LBUCK LBUCK
CA_H VOUT_FB L L VDD CB_H
L2027
C2071
C2266
C2262
ground

SUB ANT3

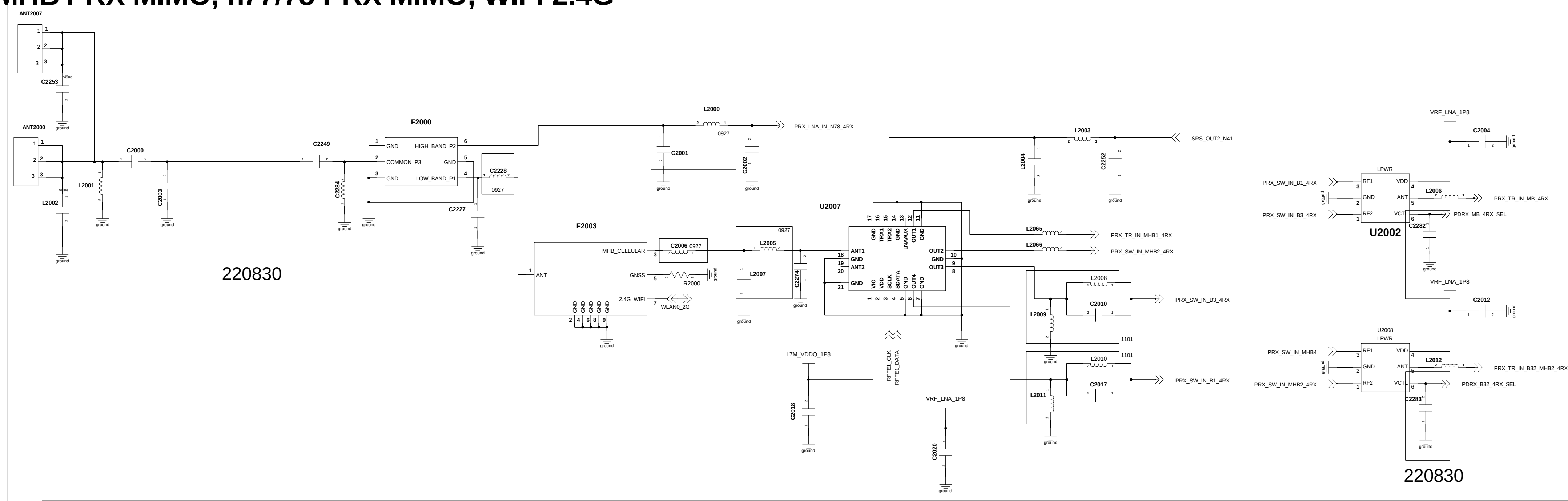
N78 TRX

N79 TRX


ANT2008
C2270
C2091
C2286
C2285
CN2000
GND GND OUT IN
L2035
C2093
C2094
CHN : O
VRF_LNA_1P8
N78_N79_SEL
U2029
CTRL RFIN VDD RF2 GND RF1
HPWR
N79_TRX
L2038 0927
C2271
L2041
L2039 0927
0705 Change
PRX_LNA_IN_N78_4RX
RFFE1_DATA
RFFE1_CLK
RFFE2_DATA
RFFE2_CLK
VBAT
@VPH_PWR
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
VRF_LNA_1P8
C2084 C2085 C2082 C2077 C2259
VPA_APT2
C2078 C2079
U2034
PIN11->VCC FOR PIN2PIN
VBATT VIO_TX VDD VIO_RX VCC NC/VCC
ANT1 ANT2 SDATA_RX SCLK_RX SDATA_TX SCLK_TX
LNA_OUT1 LNA_OUT2 TX_IN CPL ANT3_SRS
L2064 PRX_TR_IN_N78
L2072 PRX_TR_IN_N78_4RX
L2033 0927
C2256 C2257
C2278 TX2_TR_OUT_N78
C2095 CPL_N78
C2097
N78_SRS_SEL
U2036
CTRL RFIN VDD RF2 GND RF1
HPWR
SRS_OUT3_N78
SRS_OUT2_N78
C2099
VPA_APT2
ALL : O
C2272 C2100
N79_TRX
L2049
L2056
L2053
VRF_LNA_1P8
C2103
220818 change
U2041
RF2 GND RF1 VCTL ANT VDD
HPWR
N79_SRS_SEL
SRS_OUT_N79
L2050
L2057 0927
C2105
CPL_N79
C2104
C2101
C2102
U2042
CPL ANT SDATA_TX SCLK_TX VIO_TX VBATT
VCC2 VCC1
TX RX SDATA_LNA SCLK_LNA VIO_LNA VDD_LNA
L2045 0927
L2046 TX2_TR_OUT_N79
L2070
L2047
L2048 PRX_TR_IN_N79
C2111
GLO : NC
EUR : NC
KOR : NC
CHN : O
ground

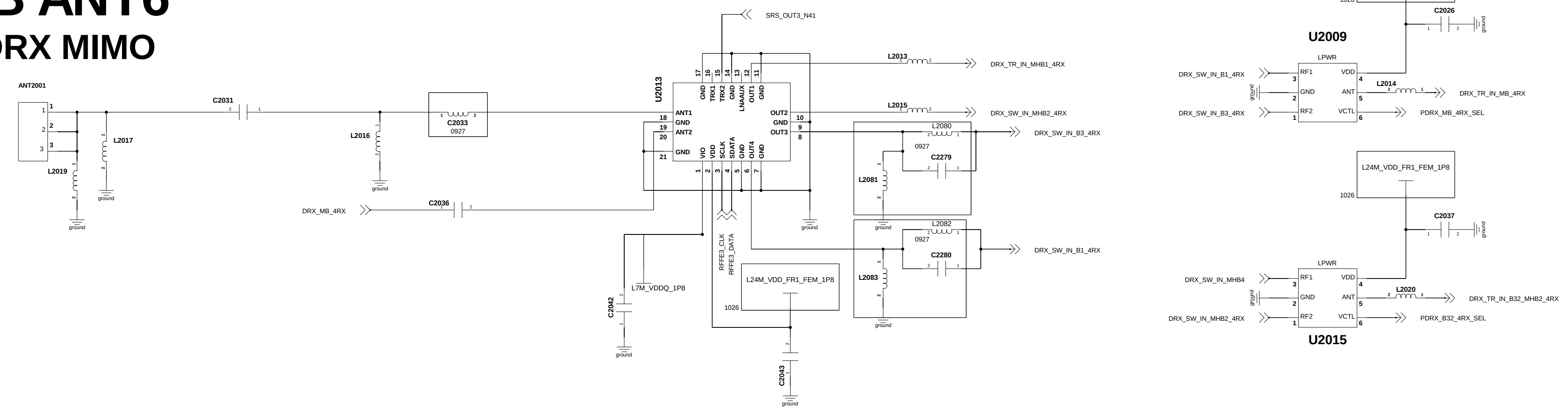
SUB ANT5

MHB PRX MIMO, n77/78 PRX MIMO, WIFI 2.4G


ANT2007
C2253
ANT2000
L2001
L2002
C2000
C2003
220830
C2249
C2284
F2000
GND COMMON_P3 GND HIGH_BAND_P2 GND LOW_BAND_P1
C2228 0927
C2227
L2000 0927
C2001
C2002
PRX_LNA_IN_N78_4RX
F2003
ANT MHB_CELLULAR GNSS 2.4G_WIFI
C2006 0927
R2000
WLAN0_2G
L2005 0927
L2007
C2274
U2007
ANT1 GND ANT2 GND
OUT2 GND OUT3
VIO VDD SCLK SDATA GND OUT4 GND
GND TRX1 TRX2 GND LNAAUX OUT1 GND
RFFE1_CLK
RFFE1_DATA
L7M_VDDQ_1P8
C2018
VRF_LNA_1P8
C2020
L2003
L2004
C2252
SRS_OUT2_N41
L2065 PRX_TR_IN_MHB1_4RX
L2066 PRX_SW_IN_MHB2_4RX
L2008 C2010 L2009 PRX_SW_IN_B3_4RX 1101
L2010 C2017 L2011 PRX_SW_IN_B1_4RX
VRF_LNA_1P8
C2004
LPWR
PRX_SW_IN_B1_4RX
PRX_SW_IN_B3_4RX
RF1 GND RF2 VDD ANT VCTL
L2006 PRX_TR_IN_MB_4RX
PDRX_MB_4RX_SEL
C2282
U2002
VRF_LNA_1P8
C2012
U2008
LPWR
PRX_SW_IN_MHB4
PRX_SW_IN_MHB2_4RX
L2012 PRX_TR_IN_B32_MHB2_4RX
PDRX_B32_4RX_SEL
C2283
220830
ground

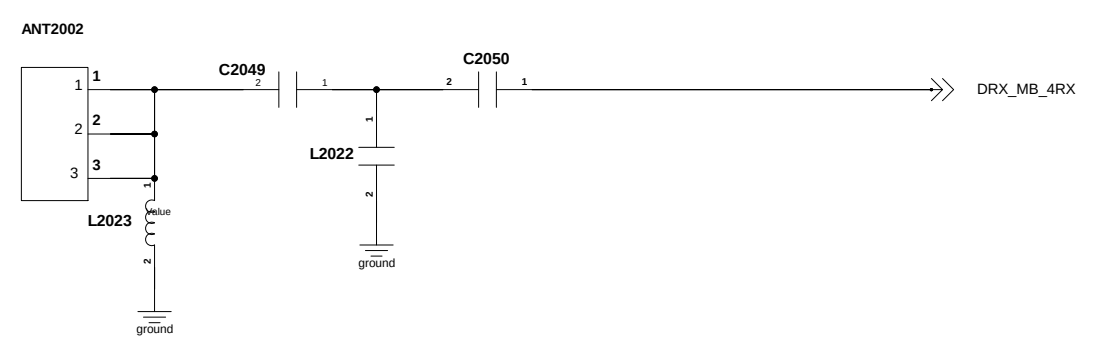
SUB ANT6

HB DRX MIMO


ANT2001
L2017
L2019
C2031
L2016
C2033 0927
DRX_MB_4RX
C2036
U2013
ANT1 GND ANT2 GND
GND TRX1 TRX2 GND LNAAUX OUT1 GND
VIO VDD SCLK SDATA GND OUT4 GND
OUT2 GND OUT3
SRS_OUT3_N41
L2013 DRX_TR_IN_MHB1_4RX
L2015 DRX_SW_IN_MHB2_4RX
L2080 0927 C2279 L2081 DRX_SW_IN_B3_4RX
L2082 0927 C2280 L2083 DRX_SW_IN_B1_4RX
RFFE3_CLK
RFFE3_DATA
L7M_VDDQ_1P8
C2042
L24M_VDD_FR1_FEM_1P8
1026
C2043
L24M_VDD_FR1_FEM_1P8
1026
C2026
U2009
LPWR
DRX_SW_IN_B1_4RX
DRX_SW_IN_B3_4RX
RF1 GND RF2 VDD ANT VCTL
L2014 DRX_TR_IN_MB_4RX
PDRX_MB_4RX_SEL
L24M_VDD_FR1_FEM_1P8
1026
C2037
LPWR
DRX_SW_IN_MHB4
DRX_SW_IN_MHB2_4RX
L2020 DRX_TR_IN_B32_MHB2_4RX
PDRX_B32_4RX_SEL
U2015
ground

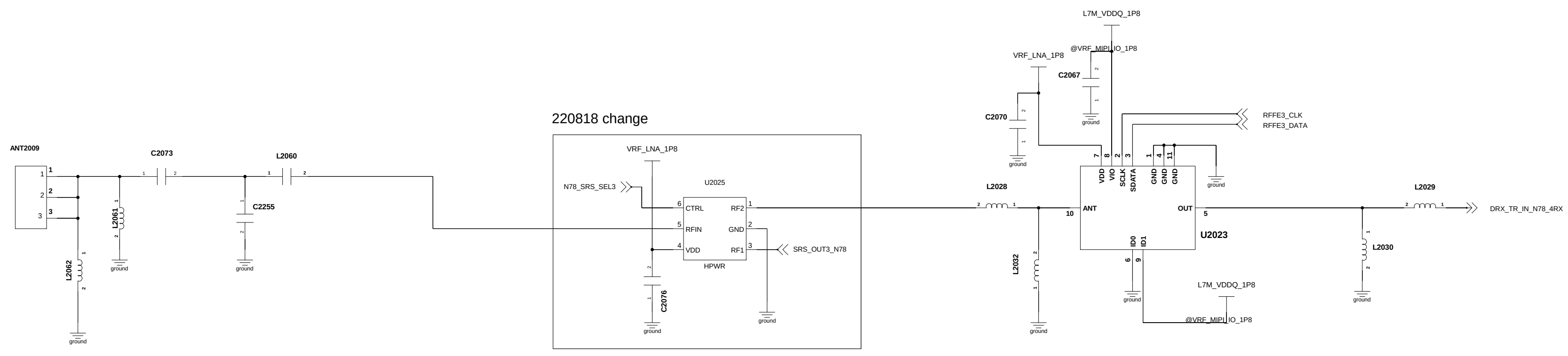
SUB ANT7

MB DRX MIMO


ANT2002
C2049
C2050
L2022
L2023
DRX_MB_4RX
ground

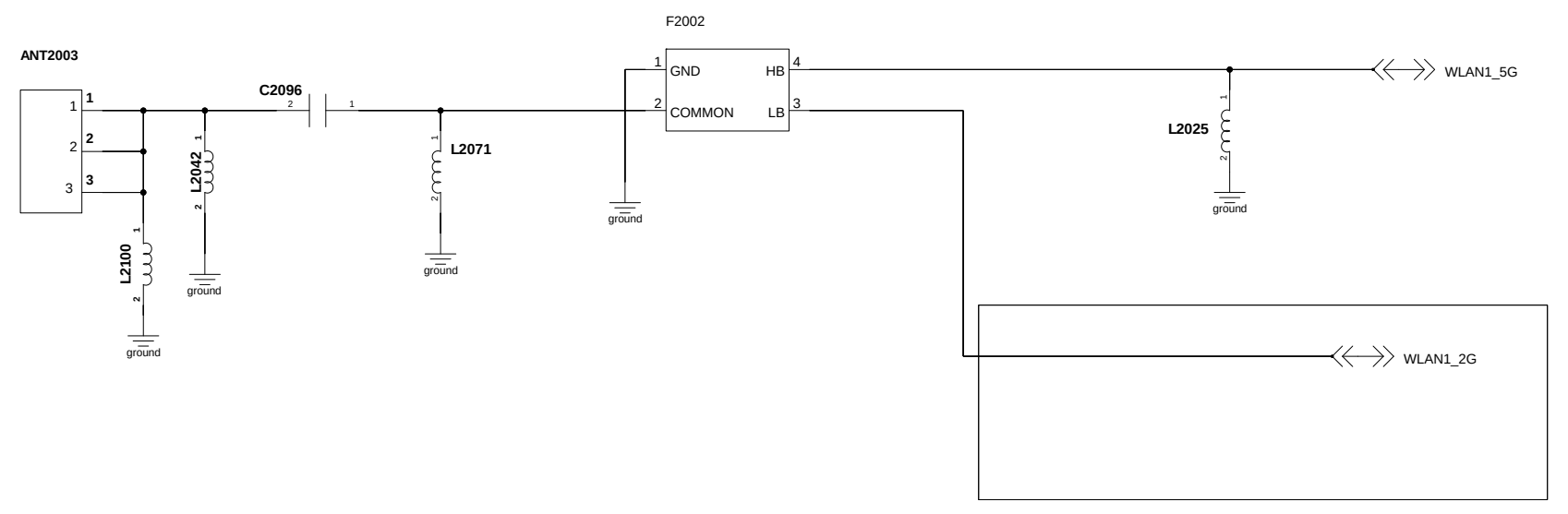
SUB ANT8

N78 DRX MIMO


ANT2009
L2061
L2062
C2073
L2060
C2255
220818 change
VRF_LNA_1P8
N78_SRS_SEL3
U2025
CTRL RFIN VDD RF2 GND RF1
HPWR
SRS_OUT3_N78
C2076
L7M_VDDQ_1P8
VRF_LNA_1P8
@VRF_MIPI_IO_1P8
C2067
C2070
RFFE3_CLK
RFFE3_DATA
L2028
L2032
U2023
VDD VIO SCLK SDATA GND GND GND
ANT OUT ID0 ID1
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
L2029 DRX_TR_IN_N78_4RX
L2030
ground

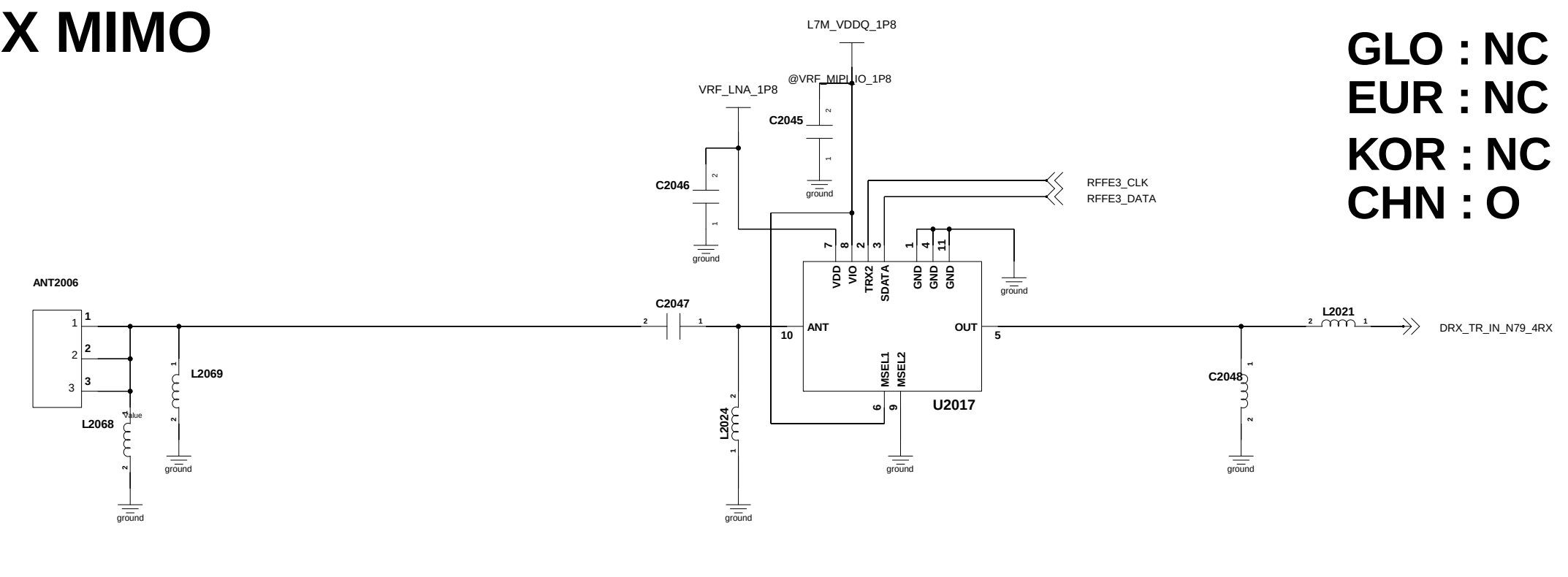
SUB ANT9

WIFI_2.4G+WIFI_5G


ANT2003
L2100
L2042
C2096
L2071
F2002
GND COMMON HB LB
WLAN1_5G
L2025
WLAN1_2G
ground

SUB ANT11

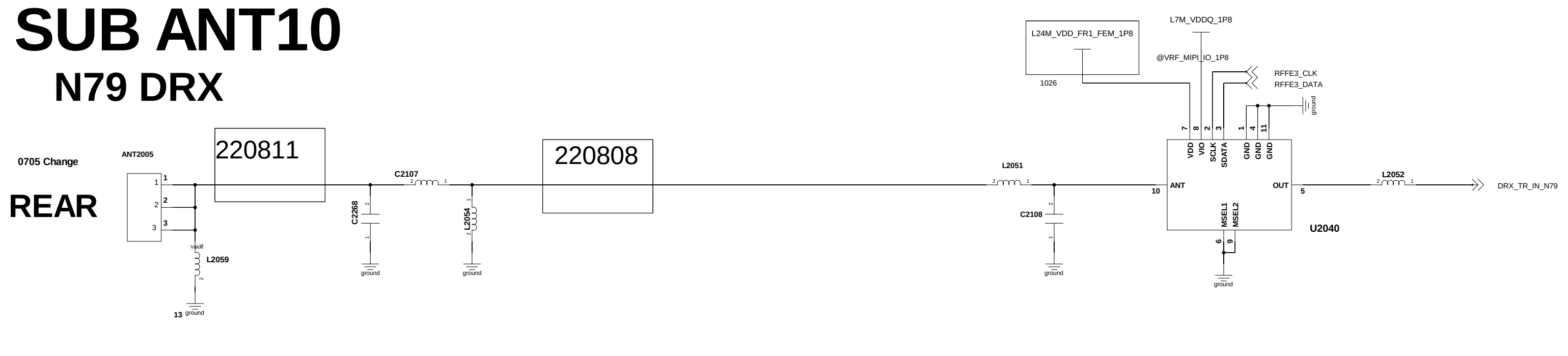
n79 DRX MIMO


ANT2006
L2068
L2069
C2047
L2024
L7M_VDDQ_1P8
VRF_LNA_1P8
@VRF_MIPI_IO_1P8
C2045
C2046
RFFE3_CLK
RFFE3_DATA
U2017
VDD VIO TRX2 SDATA GND GND GND
ANT OUT MSEL1 MSEL2
L2021 DRX_TR_IN_N79_4RX
C2048
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

SUB ANT10

N79 DRX


0705 Change
REAR
ANT2005
220811
220808
L2059
C2268
C2107
L2054
L2051
C2108
L24M_VDD_FR1_FEM_1P8
1026
L7M_VDDQ_1P8
@VRF_MIPI_IO_1P8
RFFE3_CLK
RFFE3_DATA
U2040
VDD VIO SCLK SDATA GND GND GND
ANT OUT MSEL1 MSEL2
L2052 DRX_TR_IN_N79
ground

GLO : NC
EUR : NC
KOR : NC
CHN : O

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:
A54
TITLE:
RF TRCV/PAM/APT
Size:
REV:
Drawing Number:
Changed by:
Date Changed:
Time Changed:

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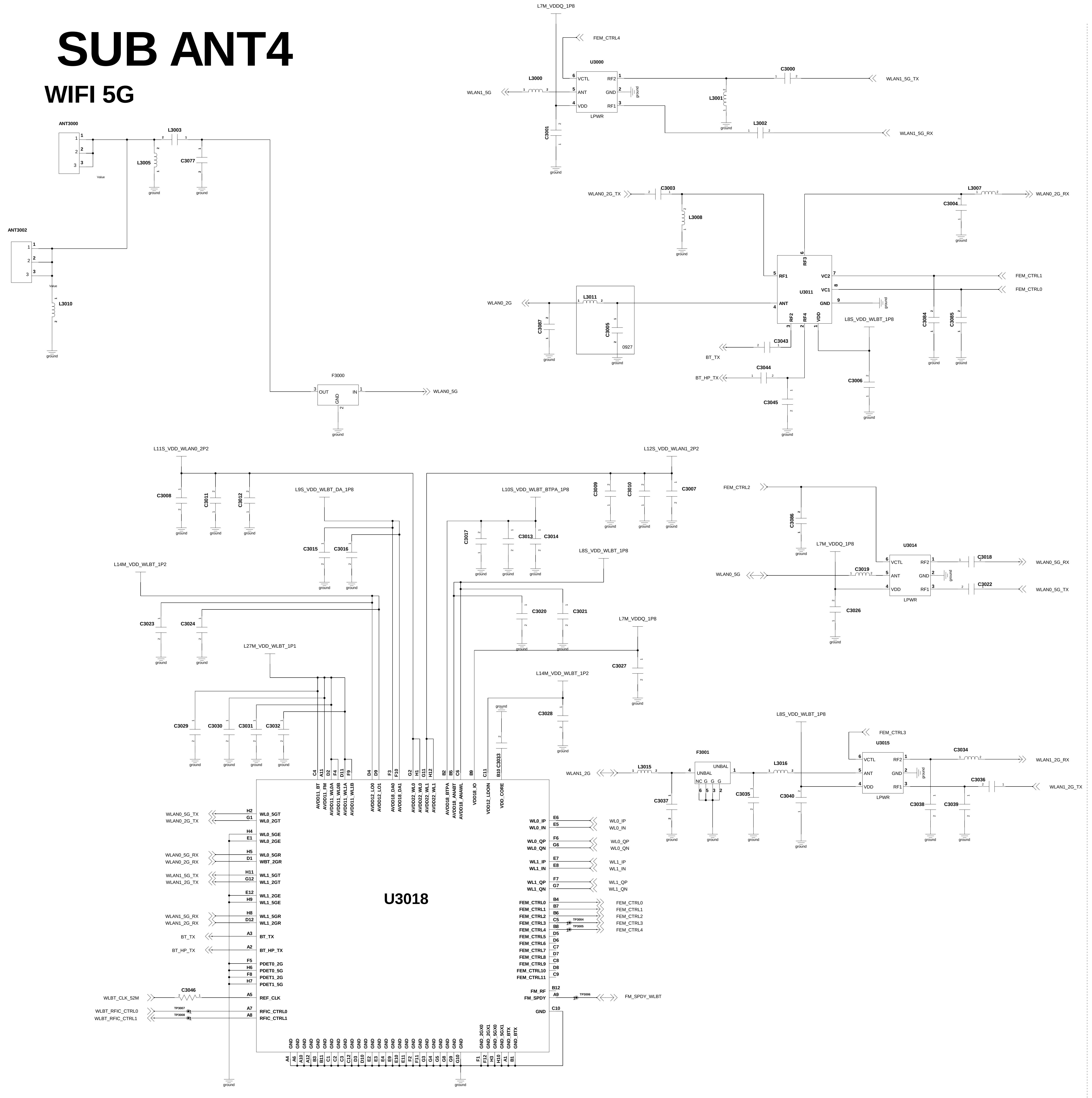
WIFI/BT/FM

WIFI 2.4G - sub5/sub9
WIFI 5G - sub4/sub9

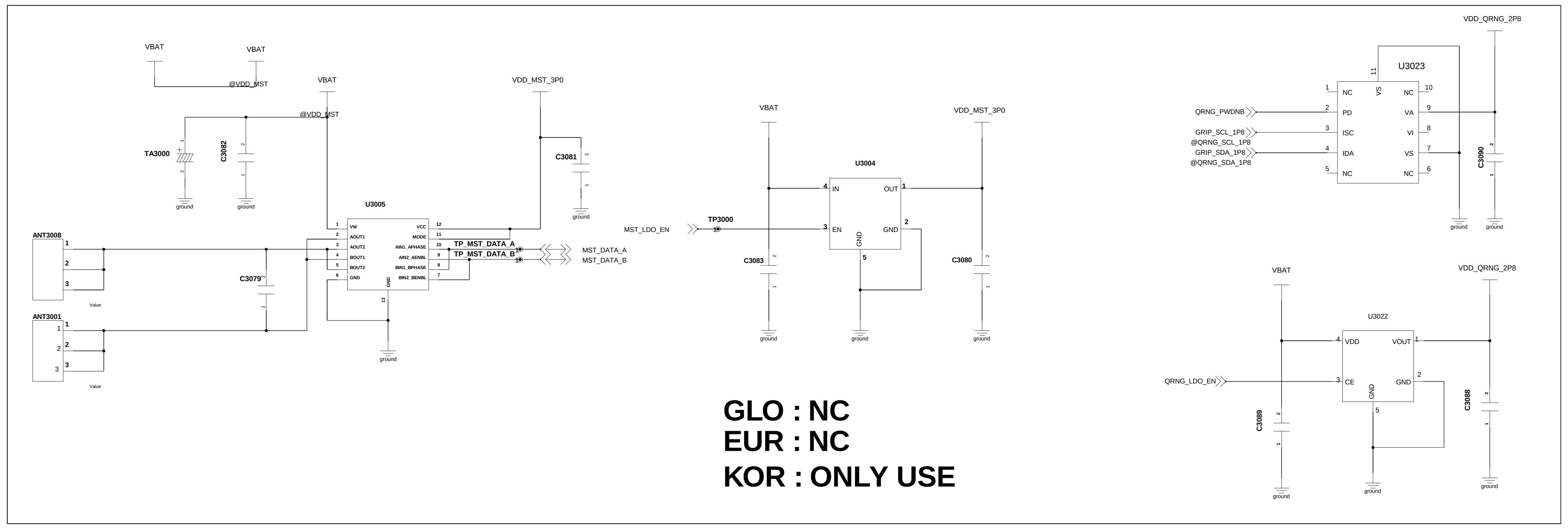
SUB ANT4

WIFI 5G


ANT3000
Value
L3003
L3005
C3077
ANT3002
L3010
F3000
OUT
IN
GND
WLAN0_5G
L7M_VDDQ_1P8
FEM_CTRL4
U3000
VCTL
ANT
VDD
RF2
GND
RF1
LPWR
L3000
WLAN1_5G
C3001
L3001
C3000
WLAN1_5G_TX
L3002
WLAN1_5G_RX
C3003
WLAN0_2G_TX
L3008
L3007
WLAN0_2G_RX
C3004
U3011
RF1
RF3
VC2
VC1
GND
ANT
RF2
RF4
VDD
FEM_CTRL1
FEM_CTRL0
WLAN0_2G
L3011
C3087
C3005
0927
C3043
BT_TX
C3044
BT_HP_TX
C3045
L8S_VDD_WLBT_1P8
C3006
C3084
C3085
L11S_VDD_WLAN0_2P2
C3008
C3011
C3012
L9S_VDD_WLBT_DA_1P8
C3015
C3016
L12S_VDD_WLAN1_2P2
C3009
C3010
C3007
L10S_VDD_WLBT_BTPA_1P8
C3017
C3013
C3014
L8S_VDD_WLBT_1P8
C3020
C3021
L7M_VDDQ_1P8
C3027
L14M_VDD_WLBT_1P2
C3023
C3024
L27M_VDD_WLBT_1P1
C3029
C3030
C3031
C3032
L14M_VDD_WLBT_1P2
C3028
C3033
FEM_CTRL2
C3086
L7M_VDDQ_1P8
U3014
C3019
WLAN0_5G
C3026
C3018
WLAN0_5G_RX
C3022
WLAN0_5G_TX
LPWR
L8S_VDD_WLBT_1P8
FEM_CTRL3
U3015
F3001
UNBAL
NC G G G
L3015
WLAN1_2G
C3037
C3035
L3016
C3040
C3034
WLAN1_2G_RX
C3036
WLAN1_2G_TX
C3038
C3039
U3018
AVDD11_BT
AVDD11_FM
AVDD11_WL0A
AVDD11_WL0B
AVDD11_WL1A
AVDD11_WL1B
AVDD12_LO0
AVDD12_LO1
AVDD18_DA0
AVDD18_DA1
AVDD22_WL0
AVDD22_WL0
AVDD22_WL1
AVDD22_WL1
AVDD18_BTPA
AVDD18_ANABT
AVDD18_ANAWL
VDD18_IO
VDD12_LDOIN
VDD_CORE
WLAN0_5G_TX
WLAN0_2G_TX
WL0_5GT
WL0_2GT
WL0_5GE
WL0_2GE
WLAN0_5G_RX
WLAN0_2G_RX
WL0_5GR
WBT_2GR
WLAN1_5G_TX
WLAN1_2G_TX
WL1_5GT
WL1_2GT
WL1_2GE
WL1_5GE
WLAN1_5G_RX
WLAN1_2G_RX
WL1_5GR
WL1_2GR
BT_TX
BT_HP_TX
PDET0_2G
PDET0_5G
PDET1_2G
PDET1_5G
C3046
WLBT_CLK_52M
REF_CLK
TP3007
TP3008
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
RFIC_CTRL0
RFIC_CTRL1
WL0_IP
WL0_IN
WL0_QP
WL0_QN
WL1_IP
WL1_IN
WL1_QP
WL1_QN
FEM_CTRL0
FEM_CTRL1
FEM_CTRL2
FEM_CTRL3
FEM_CTRL4
FEM_CTRL5
FEM_CTRL6
FEM_CTRL7
FEM_CTRL8
FEM_CTRL9
FEM_CTRL10
FEM_CTRL11
TP3004
TP3005
FM_RF
FM_SPDY
TP3006
FM_SPDY_WLBT
GND
GND_2GX0
GND_2GX1
GND_5GX0
GND_5GX1
GND_BTX
GND_BTX
ground

MST KOR : ONLY USE

QRNG KOR : ONLY USE


VBAT
@VDD_MST
TA3000
C3082
VDD_MST_3P0
C3081
U3005
VM
AOUT1
AOUT2
BOUT1
BOUT2
GND
VCC
MODE
AIN1_APHASE
AIN2_AENBL
BIN1_BPHASE
BIN2_BENBL
ANT3008
ANT3001
Value
C3079
TP_MST_DATA_A
TP_MST_DATA_B
MST_DATA_A
MST_DATA_B
MST_LDO_EN
TP3000
U3004
IN
OUT
EN
GND
C3083
C3080
VDD_QRNG_2P8
U3023
NC
PD
ISC
IDA
VS
VA
VI
QRNG_PWDNB
GRIP_SCL_1P8
@QRNG_SCL_1P8
GRIP_SDA_1P8
@QRNG_SDA_1P8
C3090
U3022
VDD
CE
VOUT
QRNG_LDO_EN
C3089
C3088
ground

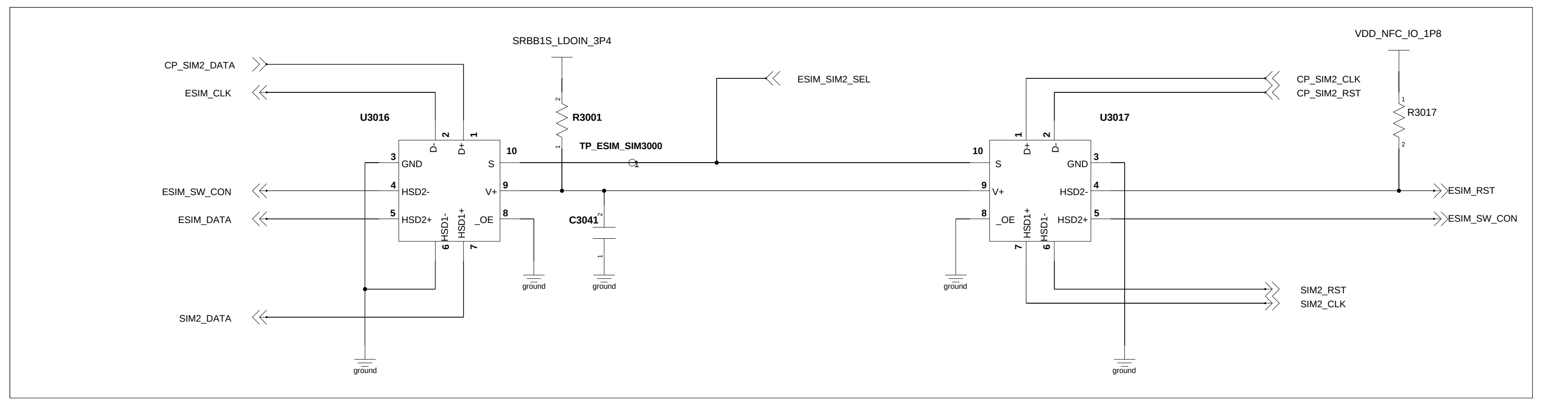
GLO : NC
EUR : NC
KOR : ONLY USE

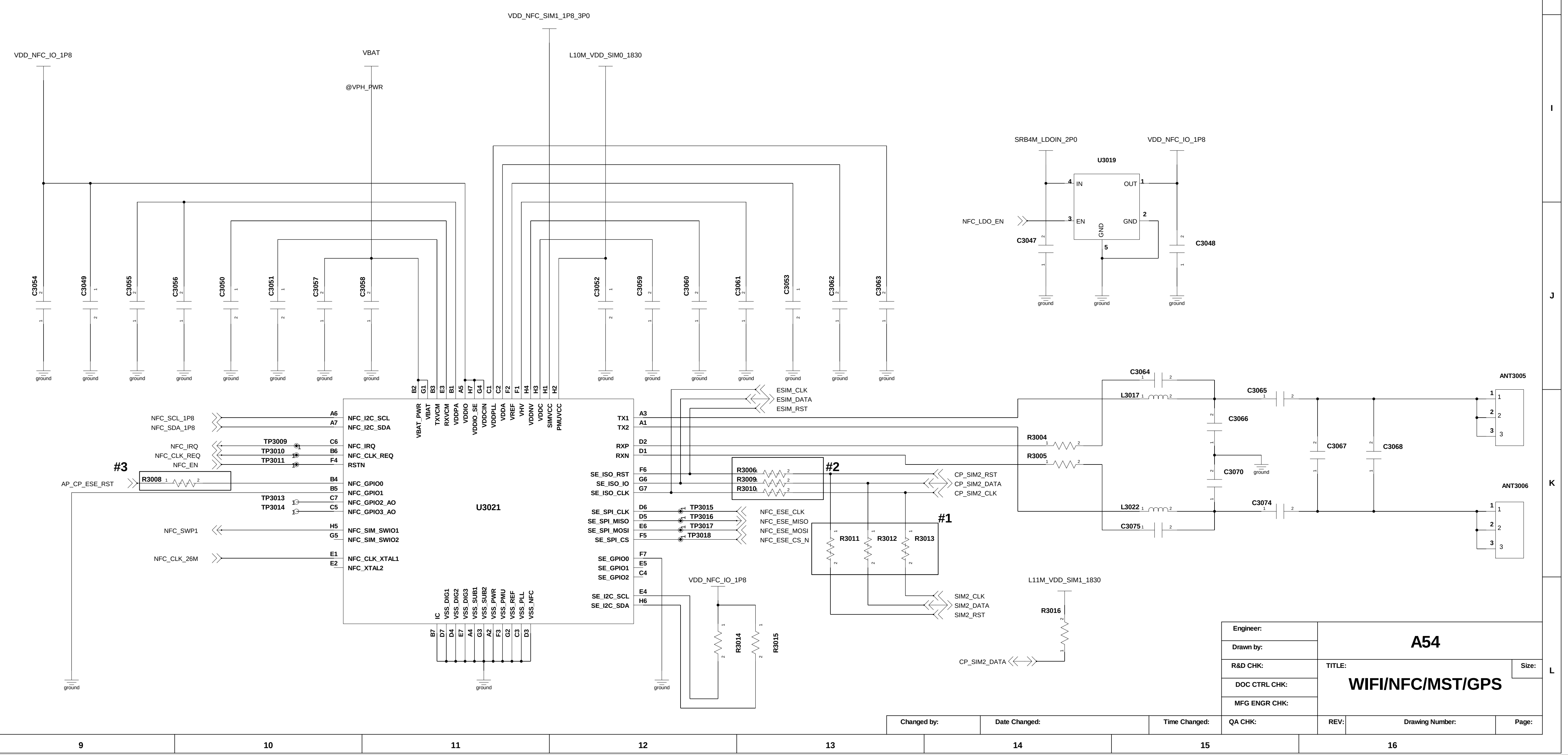
GND CCP

ANT3003
ANT3007
Value
ground

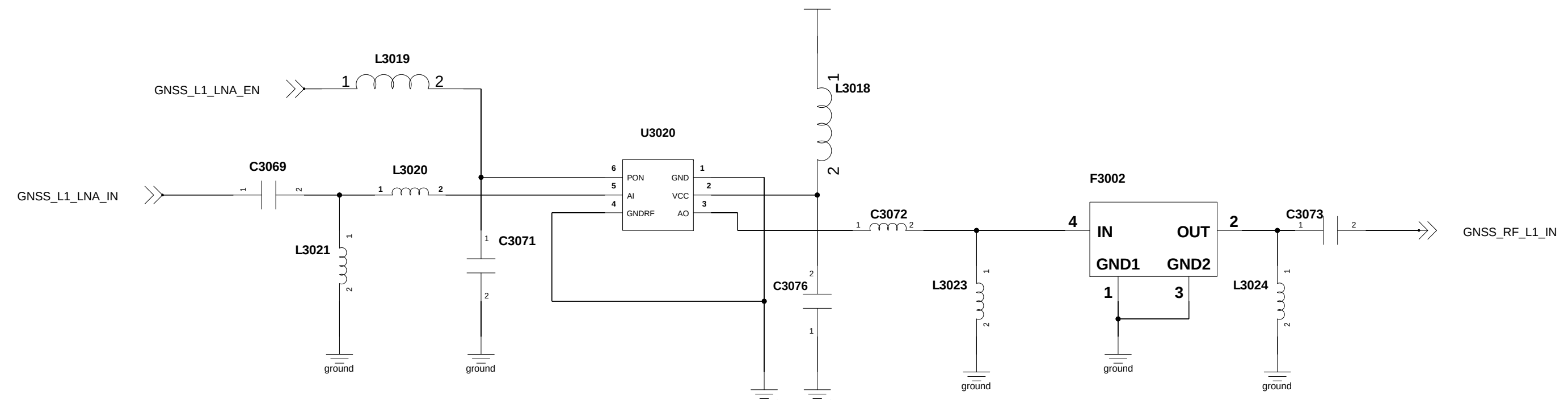
NFC

		NFC IC	#1	#2	#3		Switch IC	ESIM_SIM2_SEL
EUR	2-Nano + eSIM	SN220UV2	OPEN	OPEN	SHORT		SMD	HSD1(H)_SIM2 HSD2(L)_eSIM
KOR/NA	Nano + eSIM	SN220UV2	OPEN	SHORT	SHORT		NC	
JPN(Felica)	Nano + eSIM	SN220PV2	OPEN	SHORT	SHORT		NC	
Other	eSIM X	PN560	SHORT	OPEN	OPEN		NC	
CHN	eSIM X(eSE O)	SN220T	SHORT	OPEN	OPEN		NC	


CP_SIM2_DATA
ESIM_CLK
U3016
GND
HSD2-
HSD2+
D-
D+
S
V+
_OE
HSD1-
HSD1+
SRBB1S_LDOIN_3P4
R3001
TP_ESIM_SIM3000
ESIM_SIM2_SEL
ESIM_SW_CON
ESIM_DATA
C3041
SIM2_DATA
U3017
CP_SIM2_CLK
CP_SIM2_RST
VDD_NFC_IO_1P8
R3017
ESIM_RST
ESIM_SW_CON
SIM2_RST
SIM2_CLK
ground


VDD_NFC_IO_1P8
VBAT
@VPH_PWR
VDD_NFC_SIM1_1P8_3P0
L10M_VDD_SIM0_1830
C3054
C3049
C3055
C3056
C3050
C3051
C3057
C3058
C3052
C3059
C3060
C3061
C3053
C3062
C3063
U3021
VBAT_PWR
VBAT
TXVCM
RXVCM
VDDPA
VDDIO
VDDIO_SE
VDDCIN
VDDPLL
VDDA
VREF
VHV
VDDNV
VDDC
SIMVCC
PMUVCC
NFC_SCL_1P8
NFC_SDA_1P8
NFC_I2C_SCL
NFC_I2C_SDA
NFC_IRQ
NFC_CLK_REQ
NFC_EN
TP3009
TP3010
TP3011
RSTN
#3
R3008
AP_CP_ESE_RST
NFC_GPIO0
NFC_GPIO1
NFC_GPIO2_AO
NFC_GPIO3_AO
TP3013
TP3014
NFC_SWP1
NFC_SIM_SWIO1
NFC_SIM_SWIO2
NFC_CLK_26M
NFC_CLK_XTAL1
NFC_XTAL2
IC
VSS_DIG1
VSS_DIG2
VSS_DIG3
VSS_SUB1
VSS_SUB2
VSS_PWR
VSS_PMU
VSS_REF
VSS_PLL
VSS_NFC
TX1
TX2
RXP
RXN
SE_ISO_RST
SE_ISO_IO
SE_ISO_CLK
SE_SPI_CLK
SE_SPI_MISO
SE_SPI_MOSI
SE_SPI_CS
TP3015
TP3016
TP3017
TP3018
NFC_ESE_CLK
NFC_ESE_MISO
NFC_ESE_MOSI
NFC_ESE_CS_N
SE_GPIO0
SE_GPIO1
SE_GPIO2
SE_I2C_SCL
SE_I2C_SDA
ESIM_CLK
ESIM_DATA
ESIM_RST
R3006
R3009
R3010
#2
CP_SIM2_RST
CP_SIM2_DATA
CP_SIM2_CLK
#1
R3011
R3012
R3013
SIM2_CLK
SIM2_DATA
SIM2_RST
VDD_NFC_IO_1P8
R3014
R3015
L11M_VDD_SIM1_1830
R3016
CP_SIM2_DATA
SRB4M_LDOIN_2P0
VDD_NFC_IO_1P8
U3019
IN
OUT
EN
GND
NFC_LDO_EN
C3047
C3048
R3004
R3005
C3064
L3017
C3065
C3066
C3070
C3067
C3068
L3022
C3074
C3075
ANT3005
ANT3006
ground

GPS L1 / GNSS/BEIDOU


GNSS_L1_LNA_EN
L3019
GNSS_L1_LNA_IN
C3069
L3020
L3021
C3071
U3020
PON
AI
GNDRF
GND
VCC
AO
L7M_VDDQ_1P8
L3018
C3076
C3072
L3023
F3002
IN
OUT
GND1
GND2
C3073
L3024
GNSS_RF_L1_IN
ground

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
QA CHK:

A54

TITLE:
Size:
WIFI/NFC/MST/GPS

Changed by:
Date Changed:
Time Changed:
REV:
Drawing Number:
Page:

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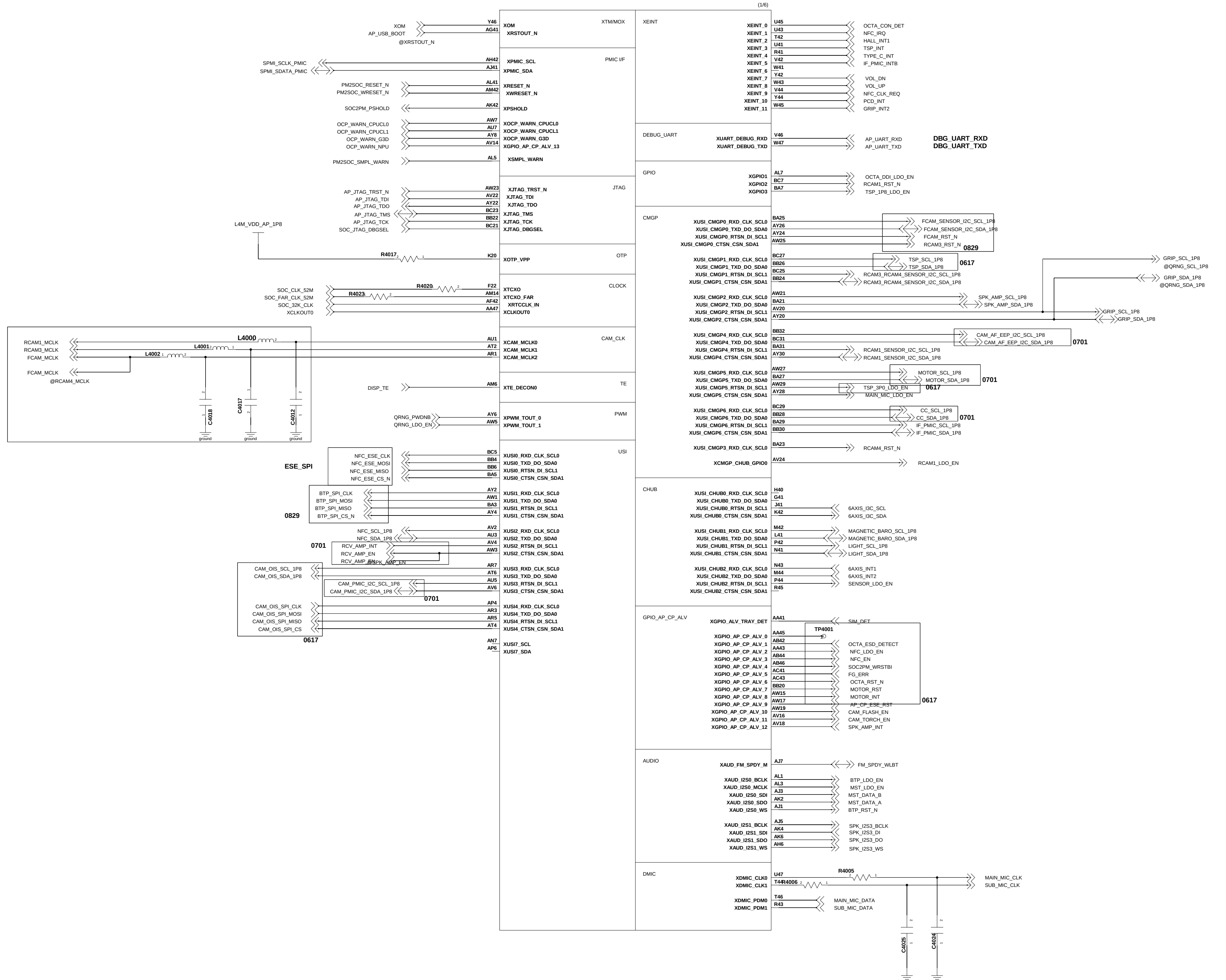
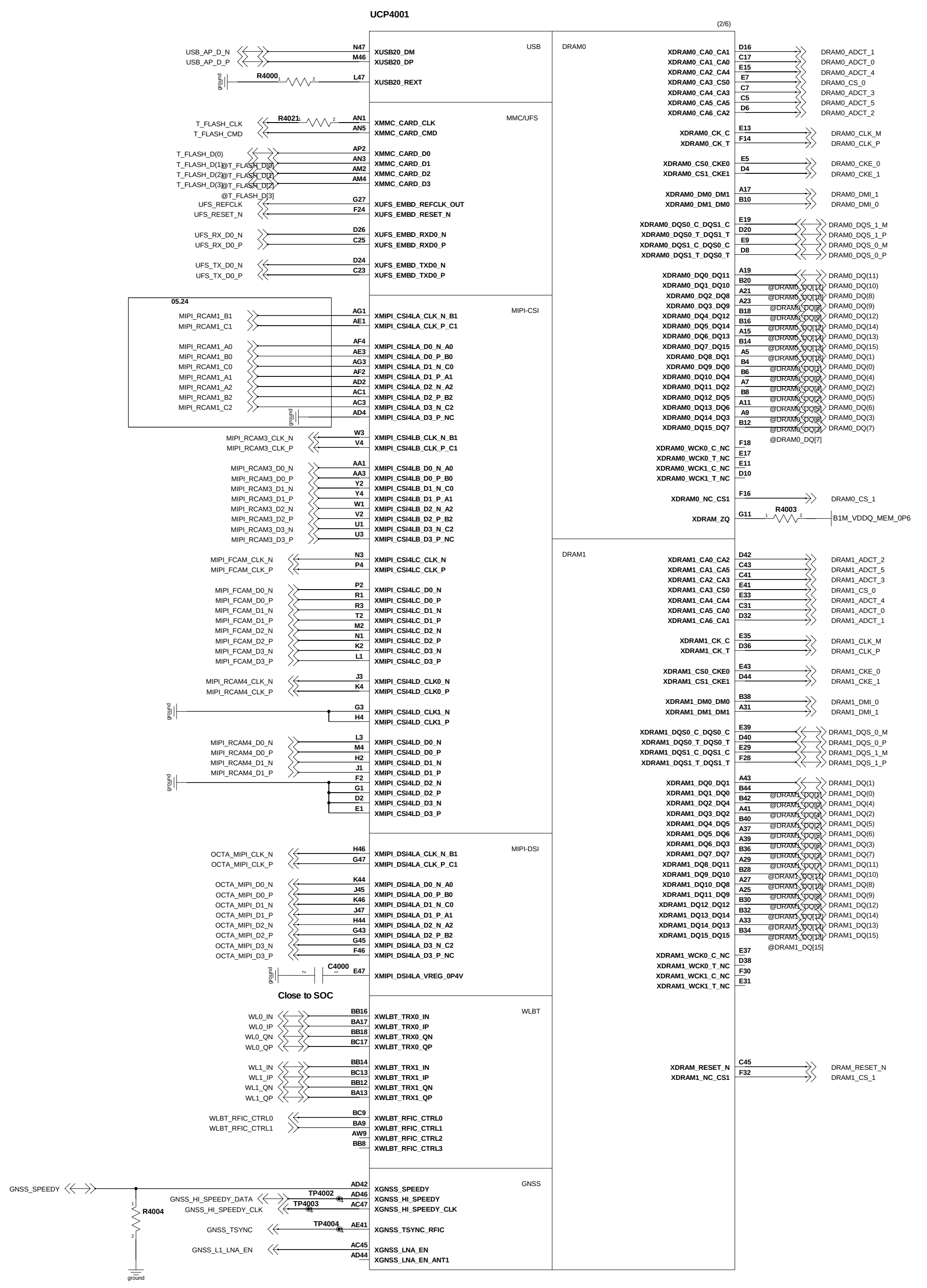
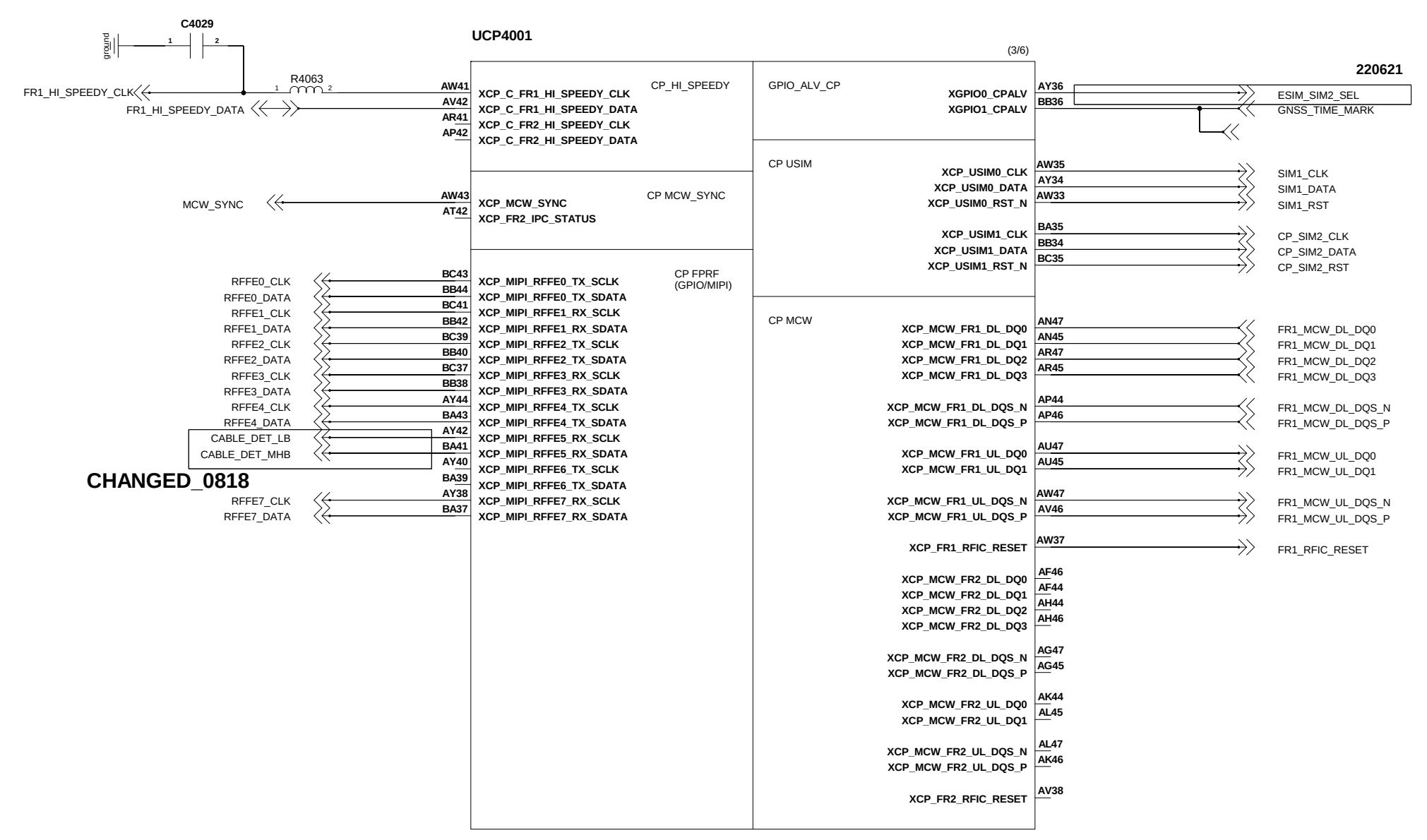
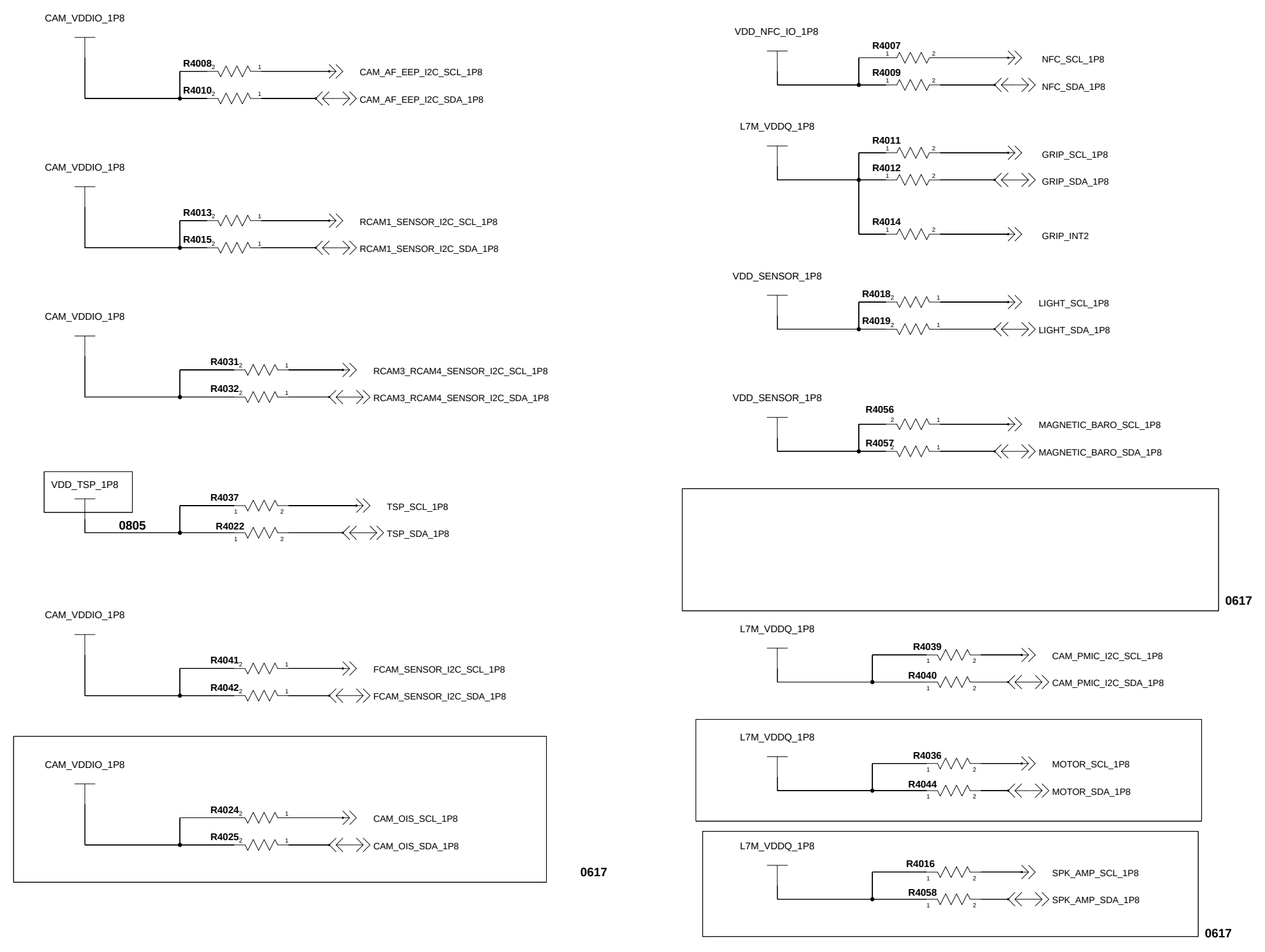
Quartz

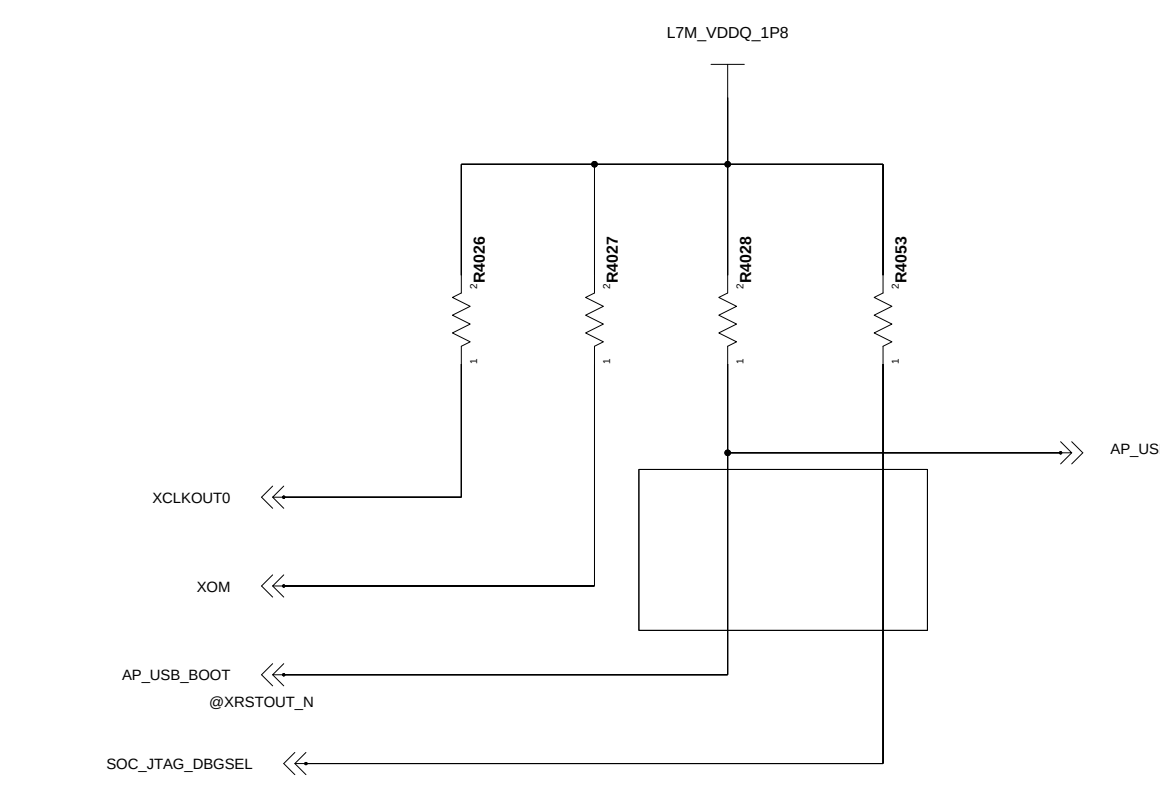

UCP4001
(1/6)
XTM/MOX
XOM
XRSTOUT_N
AP_USB_BOOT
@XRSTOUT_N
PMIC I/F
XPMIC_SCL
XPMIC_SDA
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
PM2SOC_RESET_N
PM2SOC_WRESET_N
SOC2PM_PSHOLD
OCP_WARN_CPUCL0
OCP_WARN_CPUCL1
OCP_WARN_G3D
OCP_WARN_NPU
PM2SOC_SMPL_WARN
JTAG
AP_JTAG_TRST_N
AP_JTAG_TDI
AP_JTAG_TDO
AP_JTAG_TMS
AP_JTAG_TCK
SOC_JTAG_DBGSEL
OTP
L4M_VDD_AP_1P8
R4017
XOTP_VPP
CLOCK
SOC_CLK_52M
SOC_FAR_CLK_52M
SOC_32K_CLK
XCLKOUT0
R4020
R4023
CAM_CLK
RCAM1_MCLK
RCAM3_MCLK
FCAM_MCLK
@RCAM4_MCLK
L4000
L4001
L4002
C4018
C4017
C4012
TE
DISP_TE
PWM
QRNG_PWDNB
QRNG_LDO_EN
USI
NFC_ESE_CLK
NFC_ESE_MOSI
NFC_ESE_MISO
NFC_ESE_CS_N
ESE_SPI
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
BTP_SPI_CS_N
0829
NFC_SCL_1P8
NFC_SDA_1P8
RCV_AMP_INT
RCV_AMP_EN
SPK_AMP_EN
0701
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
CAM_OIS_SPI_CLK
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
0617
XEINT
OCTA_CON_DET
NFC_IRQ
HALL_INT1
TSP_INT
TYPE_C_INT
IF_PMIC_INTB
VOL_DN
VOL_UP
NFC_CLK_REQ
PCD_INT
GRIP_INT2
DEBUG_UART
AP_UART_RXD
AP_UART_TXD
DBG_UART_RXD
DBG_UART_TXD
GPIO
OCTA_DDI_LDO_EN
RCAM1_RST_N
TSP_1P8_LDO_EN
CMGP
FCAM_SENSOR_I2C_SCL_1P8
FCAM_SENSOR_I2C_SDA_1P8
FCAM_RST_N
RCAM3_RST_N
TSP_SCL_1P8
TSP_SDA_1P8
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
GRIP_SCL_1P8
GRIP_SDA_1P8
@QRNG_SCL_1P8
@QRNG_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
RCAM1_SENSOR_I2C_SCL_1P8
RCAM1_SENSOR_I2C_SDA_1P8
MOTOR_SCL_1P8
MOTOR_SDA_1P8
TSP_3P0_LDO_EN
MAIN_MIC_LDO_EN
CC_SCL_1P8
CC_SDA_1P8
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
RCAM4_RST_N
RCAM1_LDO_EN
CHUB
6AXIS_I3C_SCL
6AXIS_I3C_SDA
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
LIGHT_SCL_1P8
LIGHT_SDA_1P8
6AXIS_INT1
6AXIS_INT2
SENSOR_LDO_EN
GPIO_AP_CP_ALV
SIM_DET
TP4001
OCTA_ESD_DETECT
NFC_LDO_EN
NFC_EN
SOC2PM_WRSTBI
FG_ERR
OCTA_RST_N
MOTOR_RST
MOTOR_INT
AP_CP_ESE_RST
CAM_FLASH_EN
CAM_TORCH_EN
SPK_AMP_INT
AUDIO
FM_SPDY_WLBT
BTP_LDO_EN
MST_LDO_EN
MST_DATA_B
MST_DATA_A
BTP_RST_N
SPK_I2S3_BCLK
SPK_I2S3_DI
SPK_I2S3_DO
SPK_I2S3_WS
DMIC
R4005
R4006
MAIN_MIC_CLK
SUB_MIC_CLK
MAIN_MIC_DATA
SUB_MIC_DATA
C4025
C4024
ground


UCP4001
(2/6)
USB
USB_AP_D_N
USB_AP_D_P
R4000
XUSB20_REXT
MMC/UFS
R4021
T_FLASH_CLK
T_FLASH_CMD
T_FLASH_D(0)
T_FLASH_D(1)
T_FLASH_D(2)
T_FLASH_D(3)
UFS_REFCLK
UFS_RESET_N
UFS_RX_D0_N
UFS_RX_D0_P
UFS_TX_D0_N
UFS_TX_D0_P
MIPI-CSI
05.24
MIPI_RCAM1_B1
MIPI_RCAM1_C1
MIPI_RCAM1_A0
MIPI_RCAM1_B0
MIPI_RCAM1_C0
MIPI_RCAM1_A1
MIPI_RCAM1_A2
MIPI_RCAM1_B2
MIPI_RCAM1_C2
MIPI_RCAM3_CLK_N
MIPI_RCAM3_CLK_P
MIPI_RCAM3_D0_N
MIPI_RCAM3_D0_P
MIPI_RCAM3_D1_N
MIPI_RCAM3_D1_P
MIPI_RCAM3_D2_N
MIPI_RCAM3_D2_P
MIPI_RCAM3_D3_N
MIPI_RCAM3_D3_P
MIPI_FCAM_CLK_N
MIPI_FCAM_CLK_P
MIPI_FCAM_D0_N
MIPI_FCAM_D0_P
MIPI_FCAM_D1_N
MIPI_FCAM_D1_P
MIPI_FCAM_D2_N
MIPI_FCAM_D2_P
MIPI_FCAM_D3_N
MIPI_FCAM_D3_P
MIPI_RCAM4_CLK_N
MIPI_RCAM4_CLK_P
MIPI_RCAM4_D0_N
MIPI_RCAM4_D0_P
MIPI_RCAM4_D1_N
MIPI_RCAM4_D1_P
MIPI-DSI
OCTA_MIPI_CLK_N
OCTA_MIPI_CLK_P
OCTA_MIPI_D0_N
OCTA_MIPI_D0_P
OCTA_MIPI_D1_N
OCTA_MIPI_D1_P
OCTA_MIPI_D2_N
OCTA_MIPI_D2_P
OCTA_MIPI_D3_N
OCTA_MIPI_D3_P
C4000
Close to SOC
WLBT
WL0_IN
WL0_IP
WL0_QN
WL0_QP
WL1_IN
WL1_IP
WL1_QN
WL1_QP
WLBT_RFIC_CTRL0
WLBT_RFIC_CTRL1
GNSS
GNSS_SPEEDY
GNSS_HI_SPEEDY_DATA
GNSS_HI_SPEEDY_CLK
TP4002
TP4003
TP4004
R4004
GNSS_TSYNC
GNSS_L1_LNA_EN
DRAM0
DRAM0_ADCT_1
DRAM0_ADCT_0
DRAM0_ADCT_4
DRAM0_CS_0
DRAM0_ADCT_3
DRAM0_ADCT_5
DRAM0_ADCT_2
DRAM0_CLK_M
DRAM0_CLK_P
DRAM0_CKE_0
DRAM0_CKE_1
DRAM0_DMI_1
DRAM0_DMI_0
DRAM0_DQS_1_M
DRAM0_DQS_1_P
DRAM0_DQS_0_M
DRAM0_DQS_0_P
DRAM0_DQ(0)–DRAM0_DQ(15)
DRAM0_CS_1
R4003
B1M_VDDQ_MEM_0P6
DRAM1
DRAM1_ADCT_2
DRAM1_ADCT_5
DRAM1_ADCT_3
DRAM1_CS_0
DRAM1_ADCT_4
DRAM1_ADCT_0
DRAM1_ADCT_1
DRAM1_CLK_M
DRAM1_CLK_P
DRAM1_CKE_0
DRAM1_CKE_1
DRAM1_DMI_0
DRAM1_DMI_1
DRAM1_DQS_0_M
DRAM1_DQS_0_P
DRAM1_DQS_1_M
DRAM1_DQS_1_P
DRAM1_DQ(0)–DRAM1_DQ(15)
DRAM_RESET_N
DRAM1_CS_1


C4029
R4063
UCP4001
(3/6)
CP_HI_SPEEDY
FR1_HI_SPEEDY_CLK
FR1_HI_SPEEDY_DATA
CP MCW_SYNC
MCW_SYNC
CP FPRF (GPIO/MIPI)
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
RFFE4_CLK
RFFE4_DATA
CABLE_DET_LB
CABLE_DET_MHB
CHANGED_0818
RFFE7_CLK
RFFE7_DATA
GPIO_ALV_CP
220621
ESIM_SIM2_SEL
GNSS_TIME_MARK
CP USIM
SIM1_CLK
SIM1_DATA
SIM1_RST
CP_SIM2_CLK
CP_SIM2_DATA
CP_SIM2_RST
CP MCW
FR1_MCW_DL_DQ0
FR1_MCW_DL_DQ1
FR1_MCW_DL_DQ2
FR1_MCW_DL_DQ3
FR1_MCW_DL_DQS_N
FR1_MCW_DL_DQS_P
FR1_MCW_UL_DQ0
FR1_MCW_UL_DQ1
FR1_MCW_UL_DQS_N
FR1_MCW_UL_DQS_P
FR1_RFIC_RESET

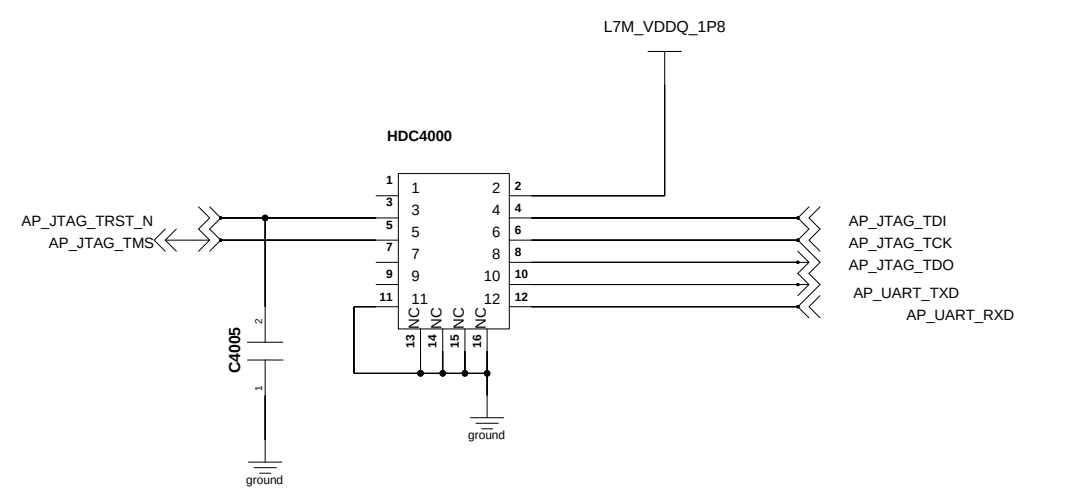
I2C PU RESISTOR


CAM_VDDIO_1P8
R4008
R4010
CAM_AF_EEP_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
R4013
R4015
RCAM1_SENSOR_I2C_SCL_1P8
RCAM1_SENSOR_I2C_SDA_1P8
R4031
R4032
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
VDD_TSP_1P8
0805
R4037
R4022
TSP_SCL_1P8
TSP_SDA_1P8
R4041
R4042
FCAM_SENSOR_I2C_SCL_1P8
FCAM_SENSOR_I2C_SDA_1P8
R4024
R4025
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
0617
VDD_NFC_IO_1P8
R4007
R4009
NFC_SCL_1P8
NFC_SDA_1P8
L7M_VDDQ_1P8
R4011
R4012
R4014
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
VDD_SENSOR_1P8
R4018
R4019
LIGHT_SCL_1P8
LIGHT_SDA_1P8
R4056
R4057
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
0617
R4039
R4040
CAM_PMIC_I2C_SCL_1P8
CAM_PMIC_I2C_SDA_1P8
R4036
R4044
MOTOR_SCL_1P8
MOTOR_SDA_1P8
R4016
R4058
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
0617


L7M_VDDQ_1P8
R4026
R4027
R4028
R4053
AP_USB_BOOT
XCLKOUT0
XOM
AP_USB_BOOT
@XRSTOUT_N
SOC_JTAG_DBGSEL

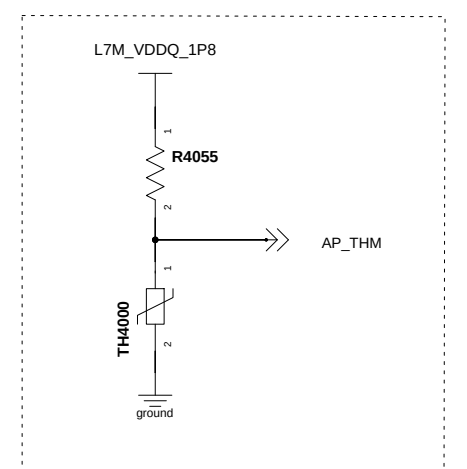
PIN			Function	
XCLKOUT0	XRSTOUT_N	XOM	1st Boot	2nd Boot
0	0	0	UFS	USB
0	1	X	USB	----
1	0	0	SD	USB

BOOT CONFIGURATIONS

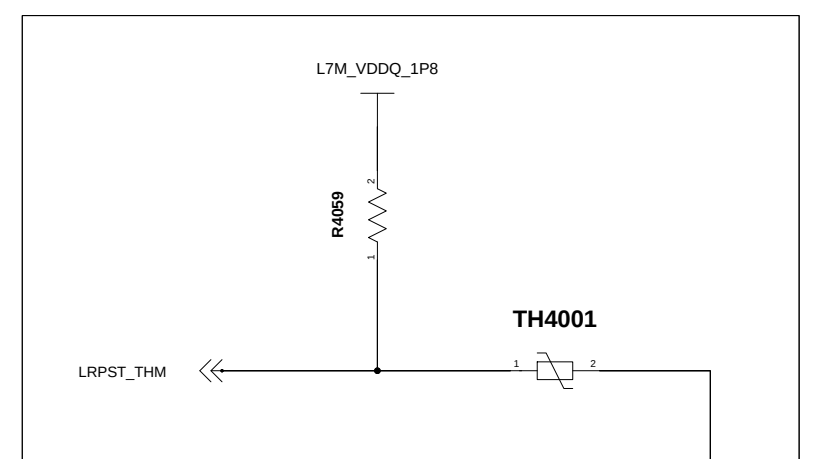

L7M_VDDQ_1P8
HDC4000
AP_JTAG_TRST_N
AP_JTAG_TMS
C4005
AP_JTAG_TDI
AP_JTAG_TCK
AP_JTAG_TDO
AP_UART_TXD
AP_UART_RXD
NC
ground

AP JTAG

SOC THERMISTOR

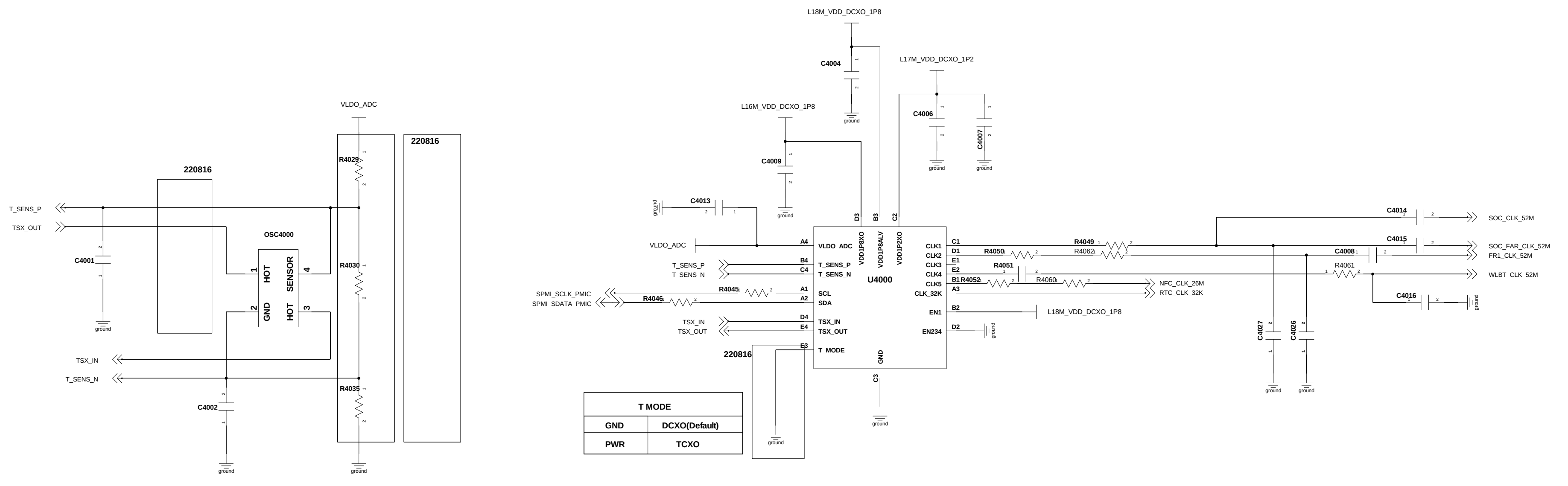

L7M_VDDQ_1P8
R4055
AP_THM
TH4000
ground

AP_THM


L7M_VDDQ_1P8
R4059
TH4001
LRPST_THM
ground

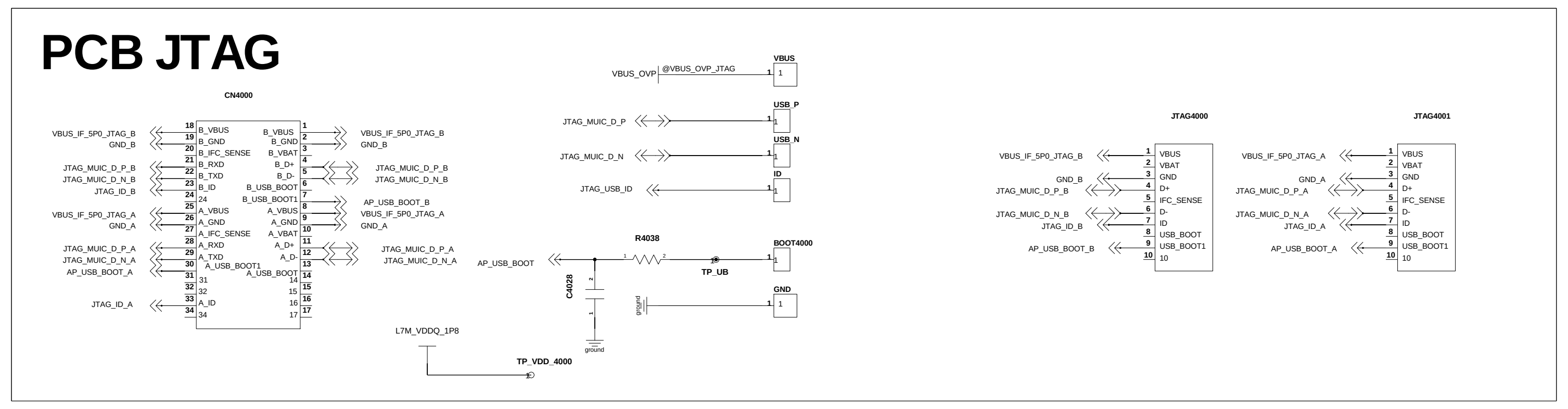
LRPST_THM

SYSTEM CLOCK


VLDO_ADC
220816
T_SENS_P
TSX_OUT
C4001
OSC4000
HOT
SENSOR
GND
HOT
TSX_IN
T_SENS_N
C4002
R4029
R4030
R4035
220816
L18M_VDD_DCXO_1P8
C4004
L17M_VDD_DCXO_1P2
C4006
C4007
L16M_VDD_DCXO_1P8
C4009
C4013
VLDO_ADC
T_SENS_P
T_SENS_N
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
R4045
R4046
TSX_IN
TSX_OUT
220816
U4000
VDD1P8XO
VDD1P8ALV
VDD1P2XO
CLK1
CLK2
CLK3
CLK4
CLK5
CLK_32K
EN1
EN234
T_MODE
GND
R4049
R4050
R4062
R4051
R4052
R4060
R4061
C4014
C4015
C4008
C4016
C4027
C4026
SOC_CLK_52M
SOC_FAR_CLK_52M
FR1_CLK_52M
WLBT_CLK_52M
NFC_CLK_26M
RTC_CLK_32K
L18M_VDD_DCXO_1P8
ground

T MODE	
GND	DCXO(Default)
PWR	TCXO

PCB JTAG


CN4000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
AP_USB_BOOT_A
JTAG_ID_A
AP_USB_BOOT_B
VBUS_OVP
@VBUS_OVP_JTAG
VBUS
JTAG_MUIC_D_P
USB_P
JTAG_MUIC_D_N
USB_N
JTAG_USB_ID
ID
R4038
AP_USB_BOOT
TP_UB
BOOT4000
C4028
GND
L7M_VDDQ_1P8
TP_VDD_4000
JTAG4000
JTAG4001
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT
USB_BOOT1
ground

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

Changed by: Date Changed: Time Changed:

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
A B C D E F G H I J K L
UCP4001
(4/6)
(5/6)
(6/6)
B1M_VDDQ_MEM_0P6
@RSP1M
@RSN1M
C5007 C5000 C5125 C5126
VDDQ06_DRAM0 G15 G17 G19
VDDQ06_DRAM1 H28 H30 H32 H34
220608
B2_3M_VDD_CAM_1P3
@RSP2_3M
@RSN2_3M
C5132 C5016 C5012 C5001
VDD_CAM J15 K14 L15 L17 N13 N15 N17 P14 R15 R17 T14 U13 U15 U17 U19 V14 V20 W13 W15 W17
B4_5_6M_VDD_CPUCL1_BIG_1P3
@RSP4_5_6M
@RSN4_5_6M
C5229 C5228 C5138 C5139 C5140 C5219 C5220 C5227 C5221 C5222 C5223 C5224 C5225 C5226
VDD_CPUCL1_BIG K22 K24 K26 K28 K32 K34 L21 L25 M34 P34 R21 R25 R31 R33 T34 U21 U23 U25 U31 V30 W29
B7M_VDD_CPUCL0_LIT_1P3
@RSP7M
@RSN7M
C5043 C5035
VDD_CPUCL0_LIT M26 P26 R29 T26 T28 V26 V28
B8M_VDD2_MEM_1P1
C5213 C5053
VDDQ11_DRAM0_CKE H18
VDDQ11_DRAM1_CKE J31 J33
L1M_VDD_ALIVE_0P75
C5215 C5216
VDD_ALIVE H26 T20 W37 AA37 AL13 AN13 AR13 AT14 AU23 AU25
VDD_ALIVE_JSC AB36 G23
VDD_CHUBVTS T36 U37
AVDD075_USB20 P36
VDD075_DLL_GNSS AE37
VDD_GNSS_ALV AC37
VDD_ALV_CP AE29 AG29 AK34 AM34 AT36 AV32 AV34 AV36
C5145 C5144 C5143 C5142
L2M_VDD_AP_1833
VDDQ1833_MMC_CARD AC11
C5146
ground
AVDD085_UFS_EMBD F26
AVDD085_MIPI_DSI L37 M36
AVDD085_MIPI_CSI M12 P12 T12 V12
VDD_ICPU_CSILINK H12 J13 L13 R13
C5006 C5147 C5148 C5149 C5150
L3M_VDD_AP_0P85
AVDD18_UFS_EMBD H22
AVDD18_MIPI_DSI K38
AVDD18_MIPI_CSI L11 N11 R11 U11
AVDD18_OTP J21
AVDD18_TS AA23 AA25
C5151 C5152 C5153 C5154
L4M_VDD_AP_1P8
AVDD12_PLL_AUD AA11
AVDD12_PLL_CPU W25 W27
AVDD12_PLL_SHARED AG23 AH24
AVDD12_PLL_MIF J23 J25
AVDD12_MIPI_DSI K36
VDDQ12_UFS_EMBD H24
C5155 C5156 C5157
L5M_VDD_AP_PLL_1P2
AVDD33_USB20 N37
C5004
L6M_VDD_USB20_3P0
AVDD18_TCXO_FAR AM12
AVDD18_TCXO E23
AVDD18_USB20 R35
VDDQ18_GPIO_ALV_CP AU37
VDDQ18_MCW_CP AP38 AR37
VDDQ1218_GNSS AC35
VDDQ18_WLBT AL11
VDDQ18_GNSS AD36
VDDQ18_GPIO_B U35 V36 Y36
VDDQ18_GPIO_T AJ11
VDDQ18_CMGP AU27 AV26
C5158 C5159 C5160 C5161
L7M_VDDQ_1P8
VDD_MCW_CP AH36 AK36 AM36 AP36
C5046 C5162
L8M_VDD_MCW_0P75
VDD06_MCW_FR12_CP AJ35 AL35
C5164 C5165
L9M_VDD_MCW_0P6
VDDQ1830_USIM0_CP AV30
C5163
L10M_VDD_SIM0_1830
VDDQ1830_USIM1_CP AV28
C5166
L11M_VDD_SIM1_1830
VDD_GNSS AD34 AE35 AF34
C5167 C5168
L12M_VDD_GNSS_0P75
AVDD12_CDPLL_CP Y28 Y30
AVDD12_IMPLL_CP AG35
C5169 C5170
L13M_VDD_CP_PLL_1P2
AVDD12_WLBT AN11 AR11 AU11
C5171
L14M_VDD_WLBT_1P2
VDD_WLBT_M AR17
C5172
L20M_VDD_WLBT_M_0P75
VDD_SRAM_PPC_LIT P28 R27
C5173 C5174
L34M_VDD_SRAM_PPC_LIT_0P75
VDD_SRAM_PPC_BIG J27 J29 N35 P20 R19
C5175 C5176 C5177
L35M_VDD_SRAM_PPC_BIG_0P75
VSS
B1S_VDD_MIF_0P75
@RSP1S
@RSN1S
C5209 C5130 C5068
VDD_MIF G29 G31 G33 H20 J17 J19
B2S_VDD_WLBT_0P75
@RSP2S
@RSN2S
C5214 C5217
VDD_WLBT AP14 AR15 AT16 AT18 AU15 AU17
B3S_VDD_NPU_0P75
@RSP3S
@RSN3S
C5087 C5210
VDD_NPU AC23 AD18 AD22 AD24 AE17 AE19 AE21 AE23
B4_5S_VDD_G3D_0P75
@RSP4_5S
@RSN4_5S
C5178 C5088 C5089
VDD_G3D AF16 AF18 AF20 AG21 AH16 AH18 AH20 AH22 AJ25 AK16 AK18 AK20 AK22 AK24 AL17 AM16 AM18 AM20 AM22 AM24 AN25 AP20 AP22 AP24 AR21 AT20 AT22
B6S_VDD_CPU_CP_0P75
@RSP6S
@RSN6S
C5184 C5180
VDD_CPU_CP AA29 AA31 AC27 AC29 AC31 AC33 AD30 AE31 AE33 AF30
VDD_CP AE27 AF26 AF28 AG27 AH34 AJ27 AJ29 AJ31 AJ33 AK26 AL27 AL29 AL31 AL33 AM26 AN27 AN29 AN31 AN33 AP26 AP34 AR27 AR29 AR31 AR33 AT24 AT26 AT28 AU31 AU33 AU35
B7_8S_VDD_CP_0P75
@RSP7_8S
@RSN7_8S
C5185 C5186 C5191 C5190
VDD_INT H36 J35 L19 L35 M18 Y20 AA13 AA15 AA17 AA19 AA21 AB16 AB18 AC13 AC15 AE13 AF14 AG13 AG25 AH14 AH26
B9S_VDD_INT_0P75
@RSP9S
@RSN9S
C5192 C5193 C5198 C5200
VDD_SRAM G37 J37 K16 K18 W23 Y12 Y22 Y24 Y26 AB12 AD12 AF12 AH12 AH28 AH30 AH32 AJ13
B10S_VDD_SRAM_0P75
@RSP10S
@RSN10S
C5201 C5202 C5203
AVDD18_WLBT AP12 AT12
C5204
L8S_VDD_WLBT_1P8
VDD_SRAM_PPC_NPU AD20 AC21
C5205
L15S_VDD_SRAM_PPC_NPU_0P75
B8M_VDD2_MEM_1P1
@RSP8M
@RSN8M
C5207 C5047 C5050 C5051 C5206 C5208
220801
L30M_VDD_UFS_1P8
C5060 C5061 C5062
L7M_VDDQ_1P8
C5063 C5064 C5065 C5066
L29M_VDD_UFS_2P95
C5076 C5077
C5078
C5212 C5211
B1M_VDDQ_MEM_0P6
C5069 C5070 C5071 C5072 C5073 C5074 C5075
R5000 R5001
UME5000
VCC VCCQ VCCQ2 VDD1 VDD2 VDDI VDDIQ VDDIQ2 VDDQ
ZQ0_A ZQ1_A
CA0_A CA1_A CA2_A CA3_A CA4_A CA5_A
DRAM0_ADCT_0 DRAM0_ADCT_1 DRAM0_ADCT_2 DRAM0_ADCT_3 DRAM0_ADCT_4 DRAM0_ADCT_5
CK_C_A CK_T_A DRAM0_CLK_M DRAM0_CLK_P
CKE0_A CKE1_A DRAM0_CKE_0 DRAM0_CKE_1
CS0_A CS1_A DRAM0_CS_0 DRAM0_CS_1
DMI0_A DMI1_A DRAM0_DMI_0 DRAM0_DMI_1
DQ0_A … DQ15_A DRAM0_DQ(0) … DRAM0_DQ(15)
DQS0_C_A DQS0_T_A DQS1_C_A DQS1_T_A
DRAM0_DQS_0_M DRAM0_DQS_0_P DRAM0_DQS_1_M DRAM0_DQS_1_P
ODT_A
DIN0_C DIN0_T DIN1_C DIN1_T
UFS_TX_D0_N UFS_TX_D0_P
TP5003 TP5004 TP5005 TP5006 TP5007
DOUT0_C DOUT0_T DOUT1_C DOUT1_T
UFS_RX_D0_N UFS_RX_D0_P
REF_CLK UFS_REFCLK
RFU NC
CA0_B CA1_B CA2_B CA3_B CA4_B CA5_B
DRAM1_ADCT_0 DRAM1_ADCT_1 DRAM1_ADCT_2 DRAM1_ADCT_3 DRAM1_ADCT_4 DRAM1_ADCT_5
CK_C_B CK_T_B DRAM1_CLK_M DRAM1_CLK_P
CKE0_B CKE1_B DRAM1_CKE_0 DRAM1_CKE_1
CS0_B CS1_B DRAM1_CS_0 DRAM1_CS_1
DMI0_B DMI1_B DRAM1_DMI_0 DRAM1_DMI_1
DQ0_B … DQ15_B DRAM1_DQ(0) … DRAM1_DQ(15)
DQS0_C_B DQS0_T_B DQS1_C_B DQS1_T_B
DRAM1_DQS_0_M DRAM1_DQS_0_P DRAM1_DQS_1_M DRAM1_DQS_1_P
ODT_B
RESET_N RST_N DRAM_RESET_N UFS_RESET_N
TP5001 TP5002
DNU
VSF1 VSF2 VSF3 VSF4 VSF5 VSF6
TP5008 TP5009
VSS_G VSS
ground

6GB + 128GB uMCP : 1108-000714

8GB + 128GB uMCP : 1108-000721

8GB + 256GB uMCP : 1108-000722

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	AP/CP/uMCP
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

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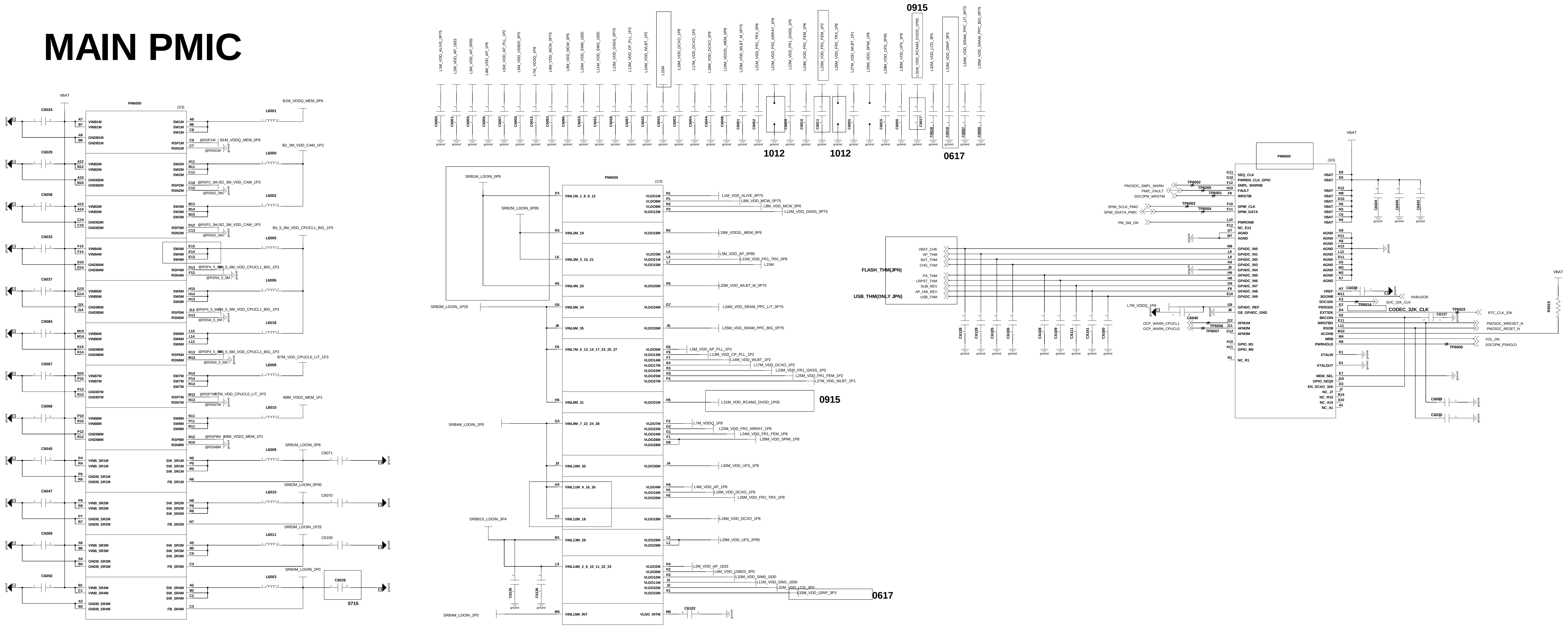
MAIN PMIC


0915
1012
1012
0617
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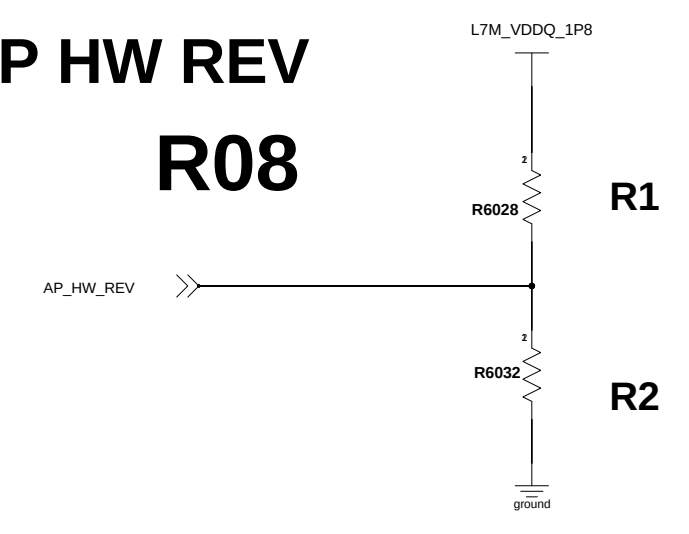
L3M_VDD_AP_0P85
L4M_VDD_AP_1P8
L5M_VDD_AP_PLL_1P2
L6M_VDD_USB20_3P0
L7M_VDDQ_1P8
L8M_VDD_MCW_0P75
L9M_VDD_MCW_0P6
L10M_VDD_SIM0_1830
L11M_VDD_SIM1_1830
L12M_VDD_GNSS_0P75
L13M_VDD_CP_PLL_1P2
L14M_VDD_WLBT_1P2
L15M
L16M_VDD_DCXO_1P8
L17M_VDD_DCXO_1P2
L18M_VDD_DCXO_1P8
L19M_VDD2L_MEM_0P9
L20M_VDD_WLBT_M_0P75
L21M_VDD_FR1_TRX_0P8
L22M_VDD_FR2_ARRAY_1P8
L23M_VDD_FR1_GNSS_1P0
L24M_VDD_FR1_FEM_1P8
L25M_VDD_FR1_FEM_1P2
L26M_VDD_FR2_TRX_1P8
L27M_VDD_WLBT_1P1
L28M_VDD_SPMI_1P8
L29M_VDD_UFS_2P95
L30M_VDD_UFS_1P8
L31M_VDD_RCAM3_DVDD_1P05
L32M_VDD_LCD_3P0
L33M_VDD_GRIP_3P3
L34M_VDD_SRAM_PPC_LIT_0P75
L35M_VDD_SRAM_PPC_BIG_0P75
C6000 C6001 C6005 C6006 C6007 C6008 C6012 C6093 C6096 C6023 C6031 C6038 C6097 C6042 C6043 C6003 C6004 C6044 C6048 C6051 C6052 C6009 C6010 C6011 C6055 C6015 C6056 C6017 C6018 C6019 C6057 C6059
VBAT
PM6000
(2/3)
C6024 C6029 C6058 C6033 C6037 C6084 C6067 C6068 C6045 C6047 C6069 C6050
VINB1M GNDB1M SW1M RSP1M RSN1M
VINB2M GNDB2M SW2M RSP2M RSN2M
VINB3M GNDB3M SW3M RSP3M RSN3M
VINB4M GNDB4M SW4M RSP4M RSN4M
VINB5M GNDB5M SW5M RSP5M RSN5M
VINB6M GNDB6M SW6M RSP6M RSN6M
VINB7M GNDB7M SW7M RSP7M RSN7M
VINB8M GNDB8M SW8M RSP8M RSN8M
VINB_SR1M GNDB_SR1M SW_SR1M FB_SR1M
VINB_SR2M GNDB_SR2M SW_SR2M FB_SR2M
VINB_SR3M GNDB_SR3M SW_SR3M FB_SR3M
VINB_SR4M GNDB_SR4M SW_SR4M FB_SR4M
L6001 L6000 L6002 L6005 L6006 L6018 L6008 L6015 L6009 L6010 L6011 L6003
B1M_VDDQ_MEM_0P6
B2_3M_VDD_CAM_1P3
B4_5_6M_VDD_CPUCL1_BIG_1P3
B7M_VDD_CPUCL0_LIT_1P3
B8M_VDD2_MEM_1P1
SRB1M_LDOIN_0P8
SRB2M_LDOIN_0P95
SRB3M_LDOIN_1P25
SRB4M_LDOIN_2P0
C6071 C6070 C6100 C6026
0715
PM6000
(1/3)
VINL1M_1_8_9_12 VINL2M_19 VINL3M_3_15_21 VINL4M_20 VINL5M_34 VINL6M_35 VINL7M_5_13_14_17_23_25_27 VINL8M_31 VINL9M_7_22_24_28 VINL10M_30 VINL11M_4_16_26 VINL12M_18 VINL13M_29 VINL14M_2_6_10_11_32_33 VINL15M_INT
VLDO_INTM
SRBB1S_LDOIN_3P4
C6135 C6136 C6102
PM6000
(3/3)
SEQ_CLK PWREN_CLK_GPIO SMPL_WARNB FAULT WRSTBI SPMI_CLK SPMI_DATA PWRONB NC_E12 AGND
GPADC_IN0 GPADC_IN1 GPADC_IN2 GPADC_IN3 GPADC_IN4 GPADC_IN5 GPADC_IN6 GPADC_IN7 GPADC_IN8 GPADC_IN9 GPADC_REF GE_GPADC_GND
AFM1M AFM2M AFM3M GPIO_M1 GPIO_M0 NC_R1
VBAT VREF JIGONB SOC32K PERI32K EXT32K BKCOIN WRSTBO RSOB ACOKB MRB PWRHOLD XTALIN XTALOUT MEM_SEL GPIO_SEQ0 EN_DCXO_32K NC_J7 NC_R15 NC_A15 NC_A1
PM2SOC_SMPL_WARN
PMIC_FAULT
SOC2PM_WRSTBI
SPMI_SCLK_PMIC
SPMI_SDATA_PMIC
PM_SW_ON
TP6002 TP6005 TP6001 TP6003 TP6004 TP6006 TP6007 TP6014 TP6022 TP6000
VBAT_CHK AP_THM BAT_THM CHG_THM PA_THM LRPST_THM SUB_REV AP_HW_REV USB_THM
FLASH_THM(JPN)
USB_THM(ONLY JPN)
C6138 C6139 C6105 C6106 C6108 C6109 C6111 C6141 C6160
L7M_VDDQ_1P8
C6040
OCP_WARN_CPUCL1
OCP_WARN_CPUCL0
C6039
NVBUSOK
SOC_32K_CLK
CODEC_32K_CLK
RTC_CLK_32K
C6137
PM2SOC_WRESET_N
PM2SOC_RESET_N
VOL_DN
SOC2PM_PSHOLD
R6015
C6034 C6035 C6032 C6099 C6030
ground

REV	R1(Kohm)	R2(Kohm)	AP	
0	NC	0	R00	Bringup00
1	430	30	R01	Bringup01
2	390	62	R02	Real REV00
3	360	91	R03	Real REV01
4	330	120	R04	Real REV02
5	300	150	R05	Real REV03
6	270	180	R06	Real REV04
7	270	240	R07	Real REV04B
8	240	270	R08	Real REV04C
9	180	270		
10	150	300		

AP HW REV

R08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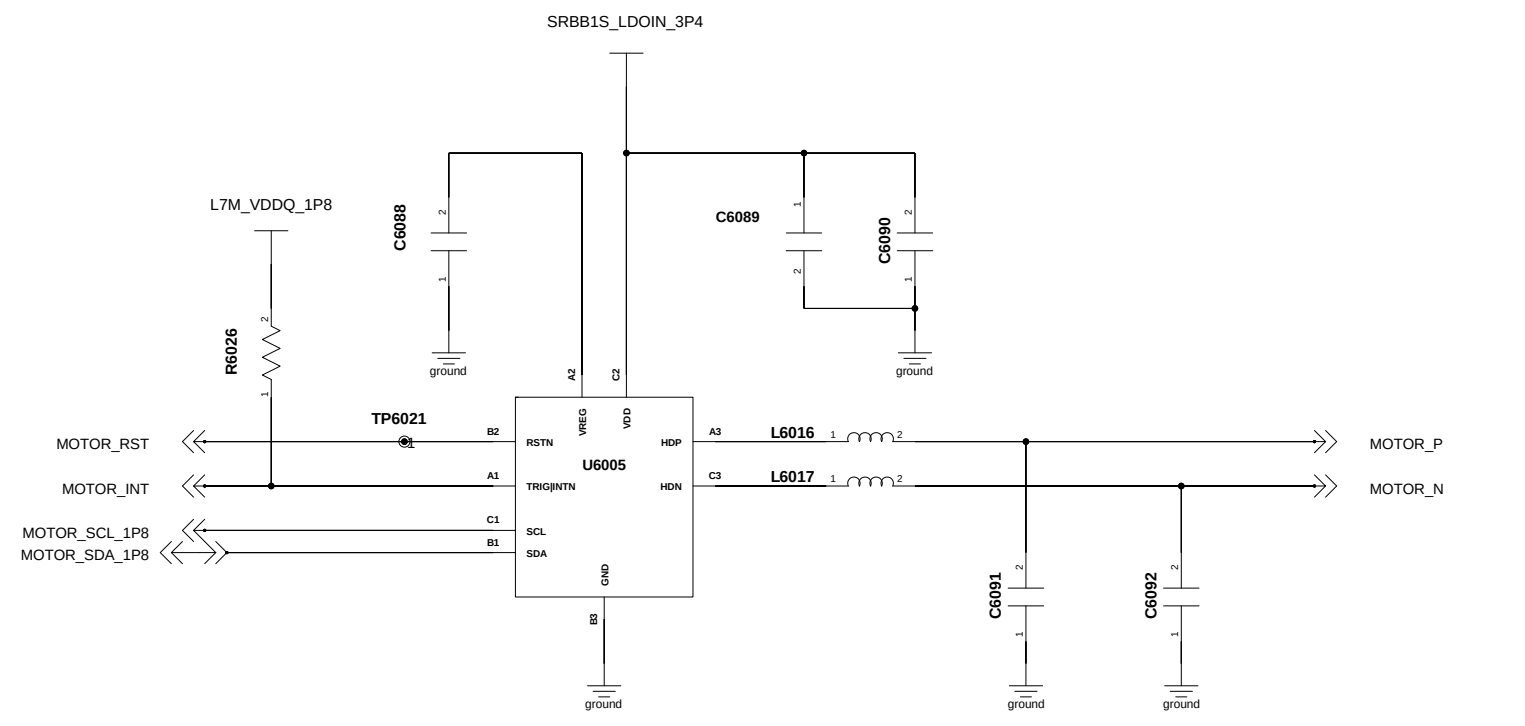
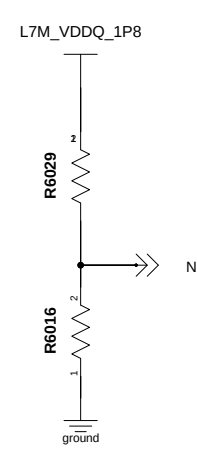
R6028
R1
AP_HW_REV
R6032
R2
ground

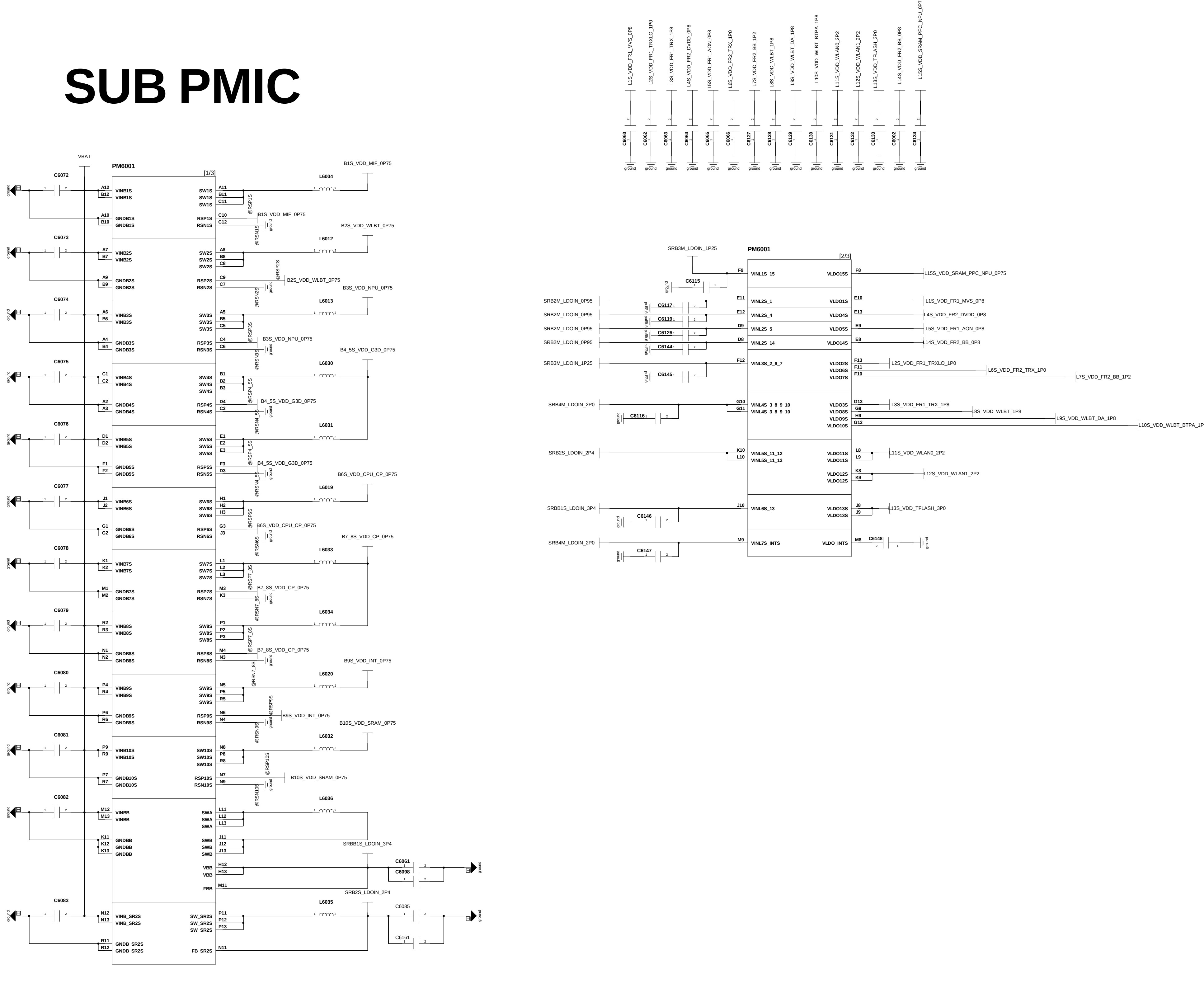
220620

MOT DRV


SRBB1S_LDOIN_3P4
L7M_VDDQ_1P8
C6088
C6089
C6090
R6026
TP6021
MOTOR_RST
MOTOR_INT
MOTOR_SCL_1P8
MOTOR_SDA_1P8
RSTN
TRIG/INTN
SCL
SDA
VREG
VDD
GND
U6005
HDP
HDN
L6016
L6017
MOTOR_P
MOTOR_N
C6091
C6092
ground


L7M_VDDQ_1P8
R6029
NFC_CHECK
R6016
ground

SUB PMIC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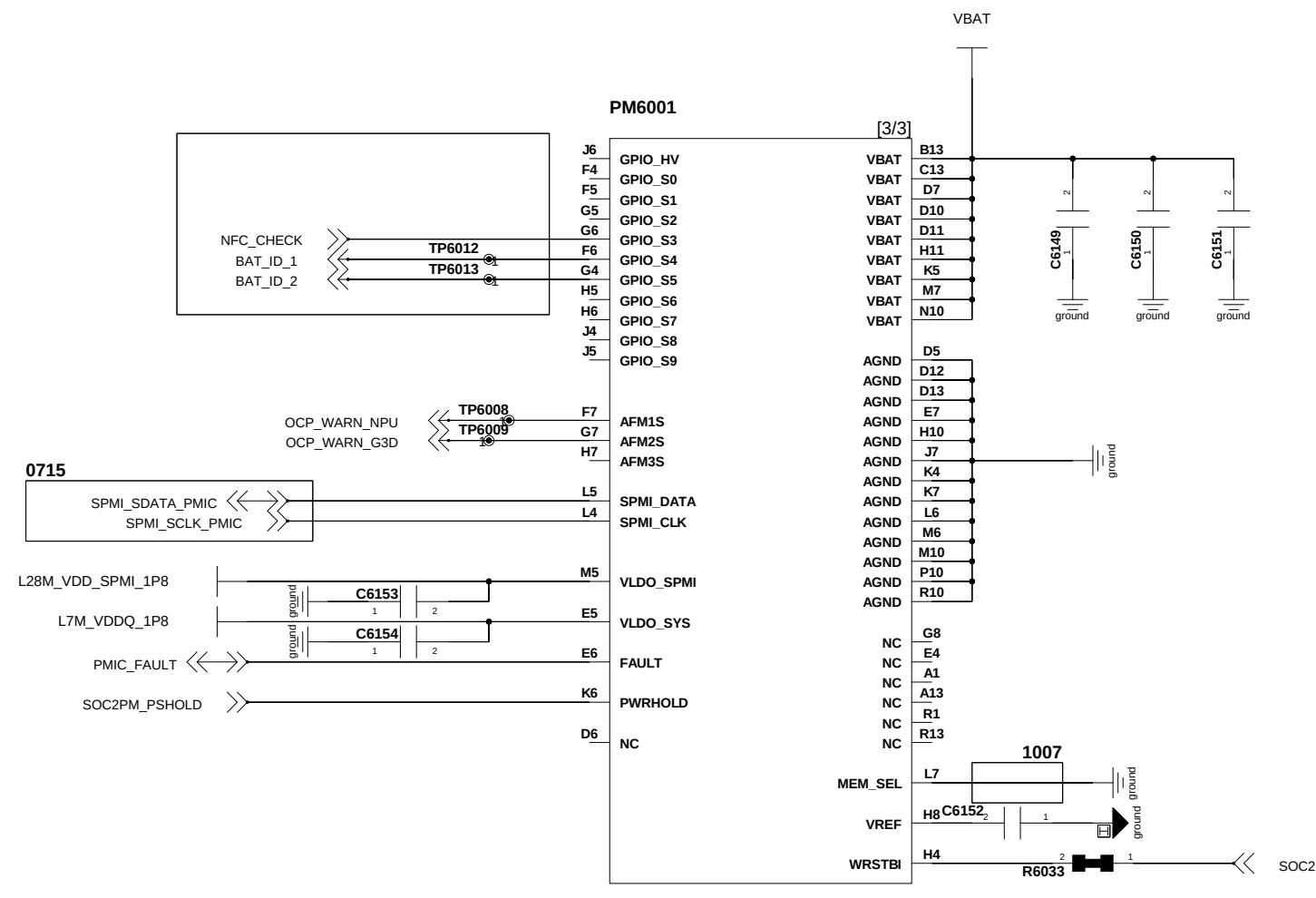
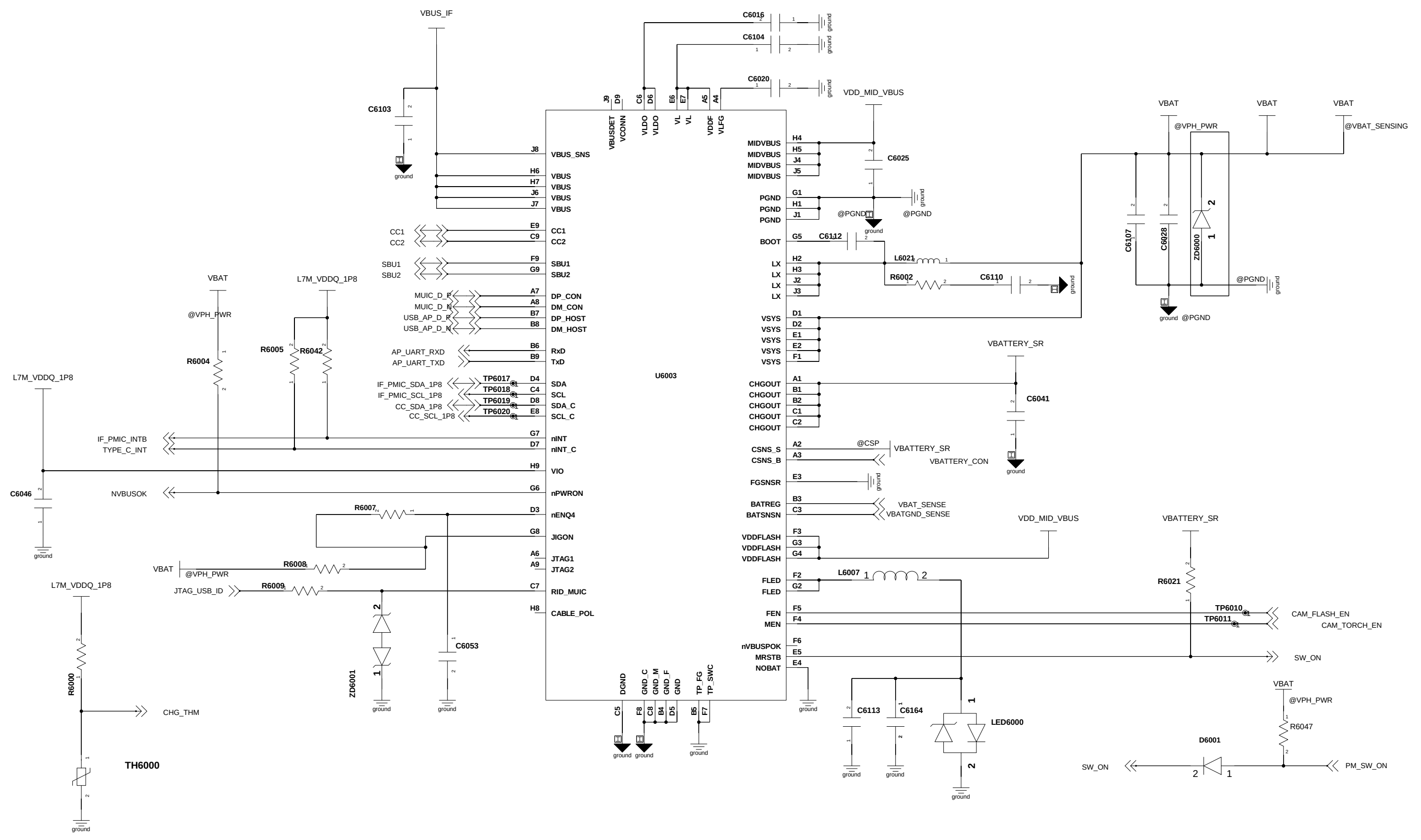
L2S_VDD_FR1_TRXLO_1P0
L3S_VDD_FR1_TRX_1P8
L4S_VDD_FR2_DVDD_0P8
L5S_VDD_FR1_AON_0P8
L6S_VDD_FR2_TRX_1P0
L7S_VDD_FR2_BB_1P2
L8S_VDD_WLBT_1P8
L9S_VDD_WLBT_DA_1P8
L10S_VDD_WLBT_BTPA_1P8
L11S_VDD_WLAN0_2P2
L12S_VDD_WLAN1_2P2
L13S_VDD_TFLASH_3P0
L14S_VDD_FR2_BB_0P8
L15S_VDD_SRAM_PPC_NPU_0P75
C6060 C6062 C6063 C6064 C6065 C6066 C6127 C6128 C6129 C6130 C6131 C6132 C6133 C6002 C6134
VBAT
PM6001
[1/3]
C6072 C6073 C6074 C6075 C6076 C6077 C6078 C6079 C6080 C6081 C6082 C6083
VINB1S GNDB1S SW1S RSP1S RSN1S
VINB2S GNDB2S SW2S RSP2S RSN2S
VINB3S GNDB3S SW3S RSP3S RSN3S
VINB4S GNDB4S SW4S RSP4S RSN4S
VINB5S GNDB5S SW5S RSP5S RSN5S
VINB6S GNDB6S SW6S RSP6S RSN6S
VINB7S GNDB7S SW7S RSP7S RSN7S
VINB8S GNDB8S SW8S RSP8S RSN8S
VINB9S GNDB9S SW9S RSP9S RSN9S
VINB10S GNDB10S SW10S RSP10S RSN10S
VINBB GNDBB SWA SWB VBB FBB
VINB_SR2S GNDB_SR2S SW_SR2S FB_SR2S
L6004 L6012 L6013 L6030 L6031 L6019 L6033 L6034 L6020 L6032 L6036 L6035
B1S_VDD_MIF_0P75
B2S_VDD_WLBT_0P75
B3S_VDD_NPU_0P75
B4_5S_VDD_G3D_0P75
B6S_VDD_CPU_CP_0P75
B7_8S_VDD_CP_0P75
B9S_VDD_INT_0P75
B10S_VDD_SRAM_0P75
SRBB1S_LDOIN_3P4
SRB2S_LDOIN_2P4
C6061 C6098 C6085 C6161
PM6001
[2/3]
SRB3M_LDOIN_1P25
SRB2M_LDOIN_0P95
SRB4M_LDOIN_2P0
SRB2S_LDOIN_2P4
SRBB1S_LDOIN_3P4
C6115 C6117 C6119 C6126 C6144 C6145 C6116 C6146 C6147 C6148
VINL1S_15 VINL2S_1 VINL2S_4 VINL2S_5 VINL2S_14 VINL3S_2_6_7 VINL4S_3_8_9_10 VINL5S_11_12 VINL6S_13 VINL7S_INTS
VLDO1S VLDO2S VLDO3S VLDO4S VLDO5S VLDO6S VLDO7S VLDO8S VLDO9S VLDO10S VLDO11S VLDO12S VLDO13S VLDO14S VLDO15S VLDO_INTS
ground

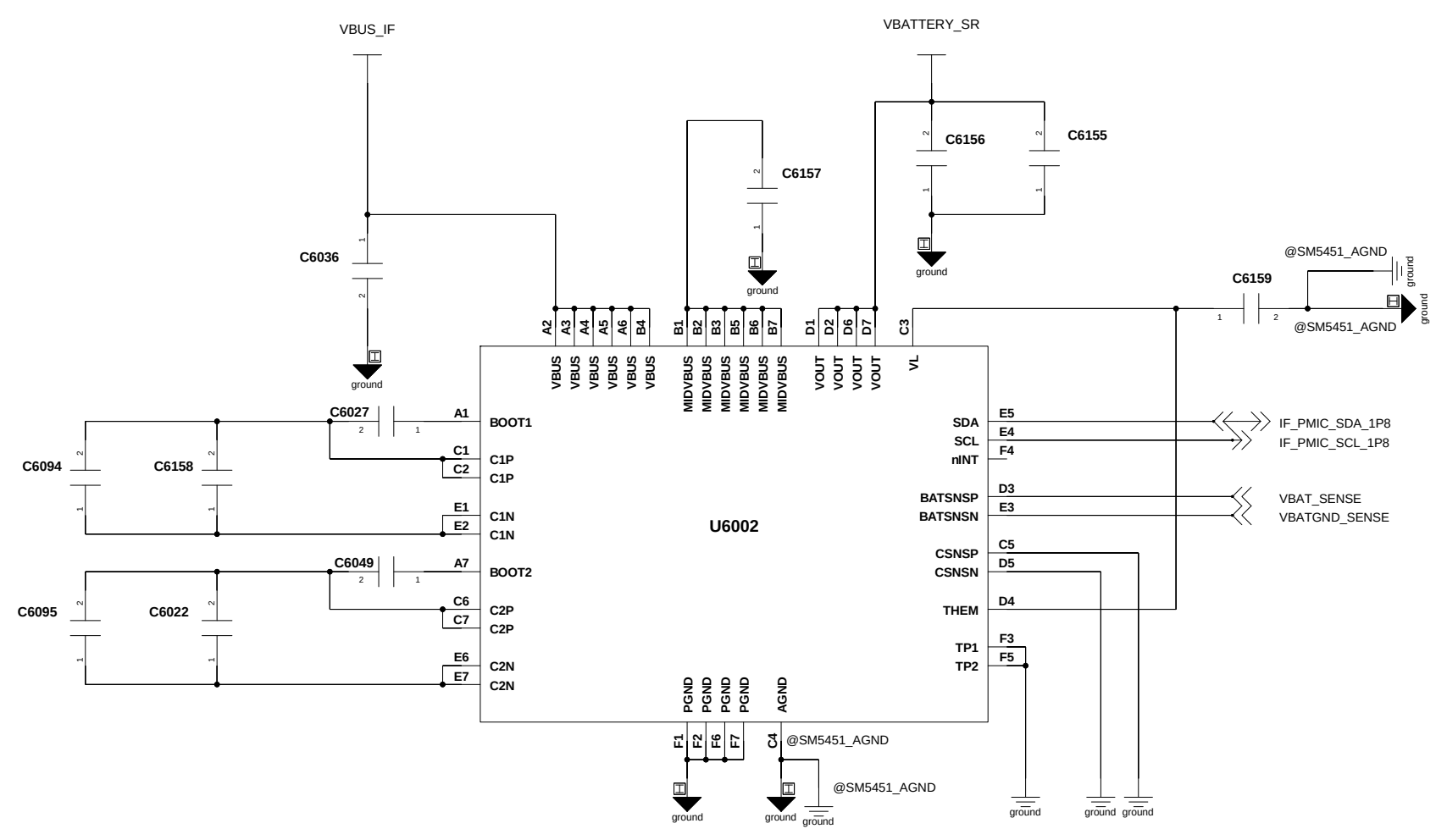

VBAT
PM6001
[3/3]
NFC_CHECK
BAT_ID_1
BAT_ID_2
TP6012
TP6013
GPIO_HV GPIO_S0 GPIO_S1 GPIO_S2 GPIO_S3 GPIO_S4 GPIO_S5 GPIO_S6 GPIO_S7 GPIO_S8 GPIO_S9
OCP_WARN_NPU
OCP_WARN_G3D
TP6008
TP6009
AFM1S AFM2S AFM3S
0715
SPMI_SDATA_PMIC
SPMI_SCLK_PMIC
SPMI_DATA
SPMI_CLK
L28M_VDD_SPMI_1P8
L7M_VDDQ_1P8
C6153
C6154
VLDO_SPMI
VLDO_SYS
PMIC_FAULT
FAULT
SOC2PM_PSHOLD
PWRHOLD
NC
VBAT
AGND
C6149 C6150 C6151
MEM_SEL
1007
VREF
C6152
WRSTBI
R6033
SOC2PM_WRSTBI
ground

IF PMIC(SM5714)


VBUS_IF
C6103
C6016
C6104
C6020
VDD_MID_VBUS
VBUS_SNS VBUS
VBUSDET VCONN VLDO VL VDDF VLFG
MIDVBUS
C6025
PGND
@PGND
BOOT
C6112
LX
L6021
R6002
C6110
VSYS
CHGOUT
VBATTERY_SR
C6041
CSNS_S
CSNS_B
@CSP
VBATTERY_CON
FGSNSR
BATREG
BATSNSN
VBAT_SENSE
VBATGND_SENSE
VDDFLASH
VDD_MID_VBUS
FLED
L6007
FEN
MEN
nVBUSPOK
MRSTB
NOBAT
VBATTERY_SR
R6021
TP6010
TP6011
CAM_FLASH_EN
CAM_TORCH_EN
SW_ON
C6113
C6164
LED6000
VBAT
@VPH_PWR
R6047
D6001
PM_SW_ON
VBAT
@VBAT_SENSING
C6107
C6028
ZD6000
CC1 CC2
SBU1 SBU2
MUIC_D_P
MUIC_D_N
USB_AP_D_P
USB_AP_D_N
DP_CON DM_CON DP_HOST DM_HOST
AP_UART_RXD
AP_UART_TXD
RxD TxD
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
CC_SDA_1P8
CC_SCL_1P8
TP6017 TP6018 TP6019 TP6020
SDA SCL SDA_C SCL_C
U6003
VBAT
L7M_VDDQ_1P8
@VPH_PWR
R6004
R6005
R6042
IF_PMIC_INTB
TYPE_C_INT
nINT nINT_C
L7M_VDDQ_1P8
VIO
C6046
NVBUSOK
nPWRON
R6007
nENQ4
JIGON
JTAG1
JTAG2
VBAT
R6008
JTAG_USB_ID
R6009
RID_MUIC
CABLE_POL
ZD6001
C6053
DGND GND_C GND_M GND_F GND TP_FG TP_SWC
R6000
CHG_THM
TH6000
ground

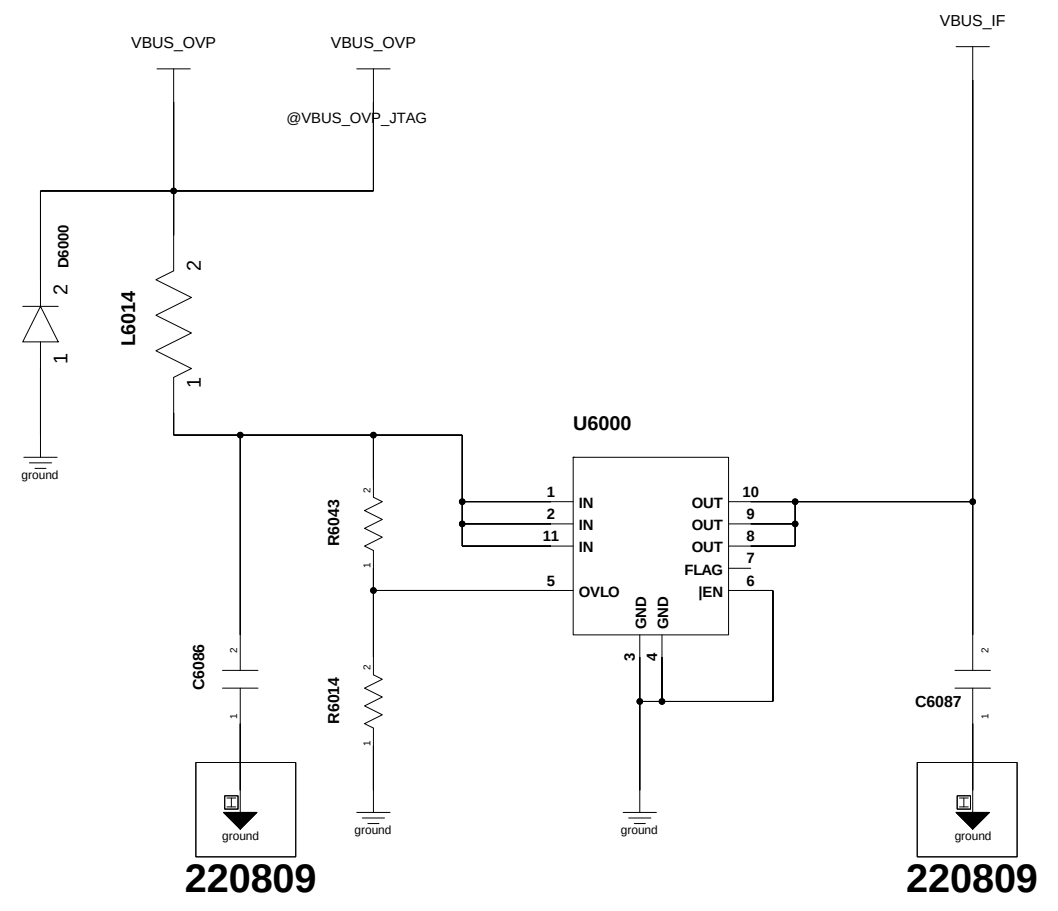
DIRECT CHARGER

(SM5451)

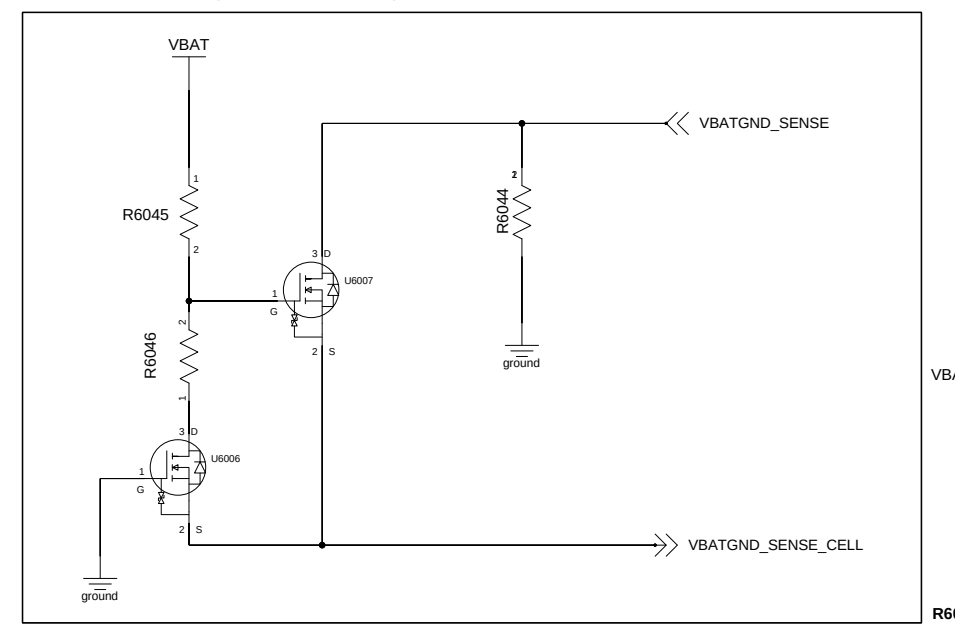

VBUS_IF
VBATTERY_SR
C6036
C6157
C6156
C6155
VBUS MIDVBUS VOUT VL
C6159
@SM5451_AGND
C6027
BOOT1
C1P C1N
C6094
C6158
C6049
BOOT2
C2P C2N
C6095
C6022
U6002
SDA SCL nINT
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
BATSNSP
BATSNSN
VBAT_SENSE
VBATGND_SENSE
CSNSP CSNSN
THEM
TP1 TP2
PGND AGND
ground

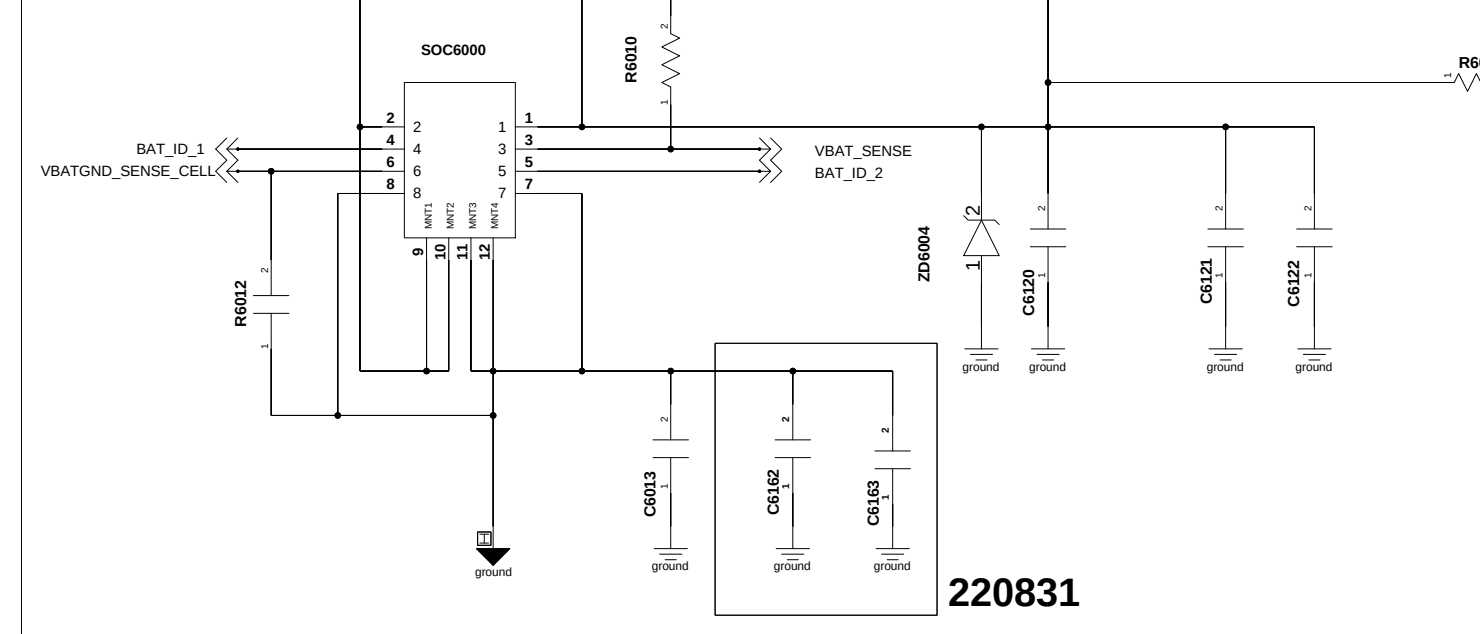
OVP

NA ONLY

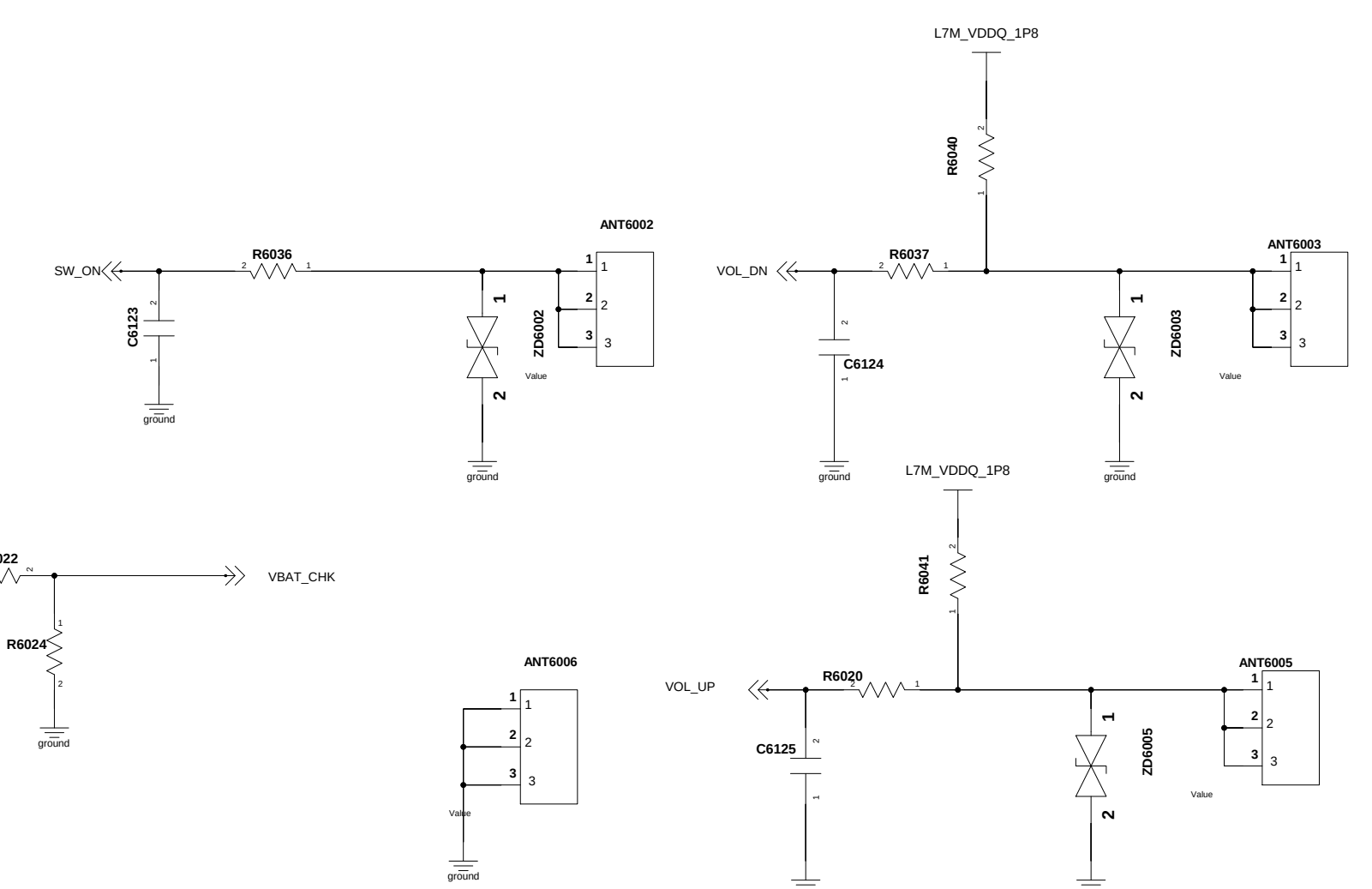

VBUS_OVP
VBUS_OVP
@VBUS_OVP_JTAG
VBUS_IF
D6000
L6014
U6000
IN
OUT
FLAG
/EN
OVLO
GND
R6043
R6014
C6086
C6087
220809
220809
ground

Cell leakage Sensing 1006


VBAT
VBATGND_SENSE
R6045
R6044
U6007
R6046
U6006
VBATGND_SENSE_CELL
ground


VBATTERY_SR
@CSP
VBATTERY_SR
R6025
VBATTERY_CON
SOC6000
BAT_ID_1
VBATGND_SENSE_CELL
VBAT_SENSE
BAT_ID_2
R6010
R6012
ZD6004
C6120
C6121
C6122
R6022
VBAT_CHK
R6024
C6013
C6162
C6163
220831
ground

BATTERY CON


L7M_VDDQ_1P8
R6040
SW_ON
R6036
C6123
ZD6002
ANT6002
VOL_DN
R6037
C6124
ZD6003
ANT6003
L7M_VDDQ_1P8
R6041
ANT6006
VOL_UP
R6020
C6125
ZD6005
ANT6005
Value
ground

POWER & VOL KEY

Engineer:	A54	
Drawn by:		
R&D CHK:	TITLE: POWER/MOT/KEY	Size:
DOC CTRL CHK:		
MFG ENGR CHK:		
QA CHK:	REV:	Drawing Number:

Changed by:	Date Changed:	Time Changed:

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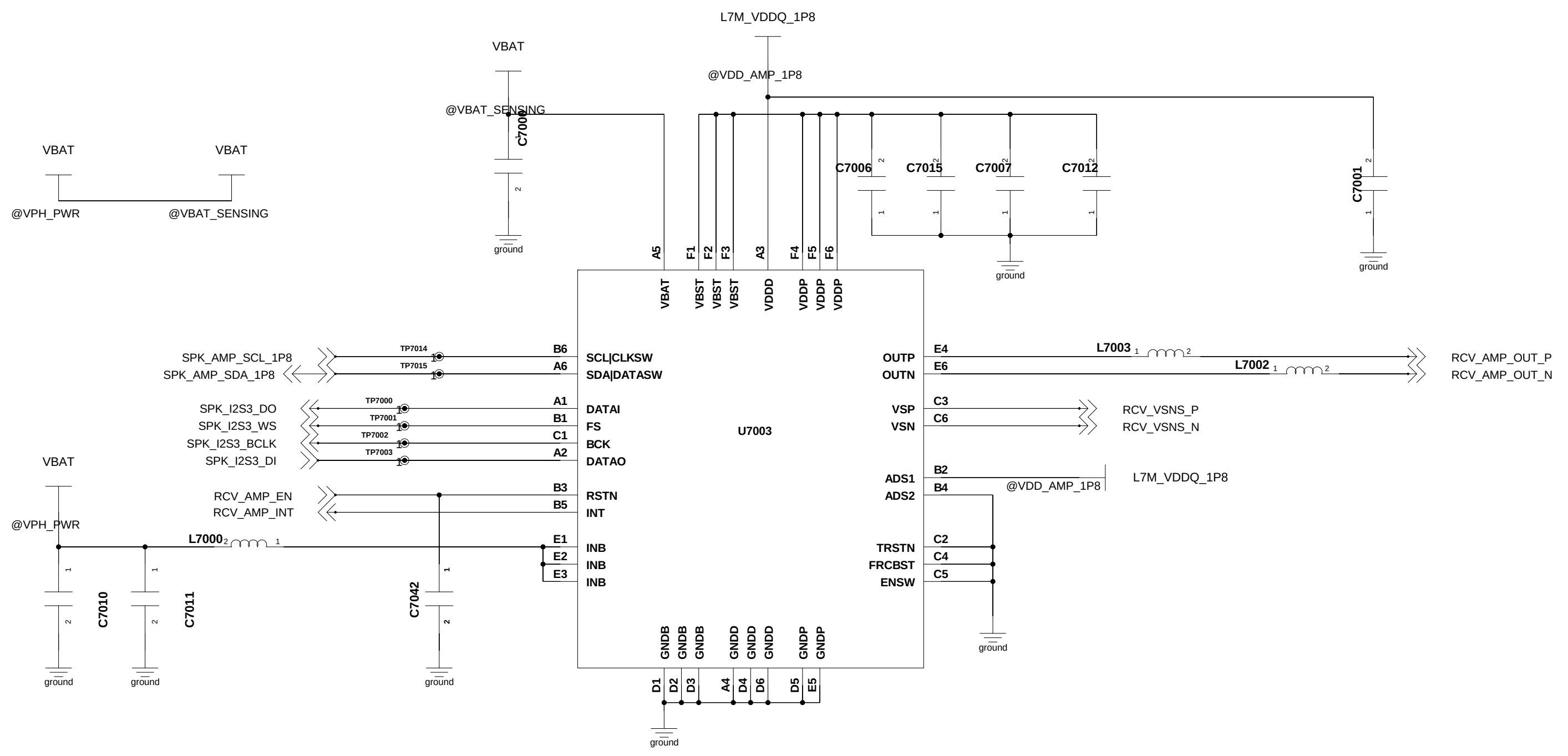
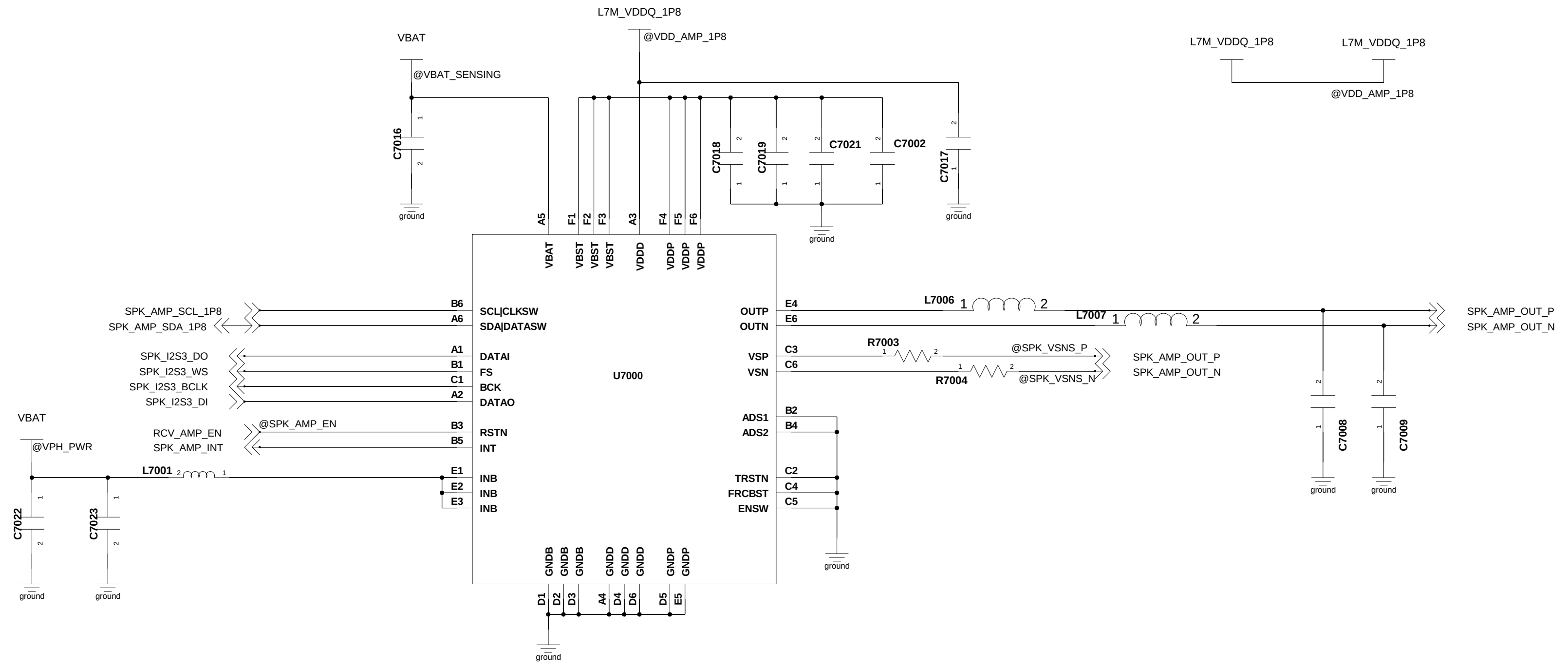
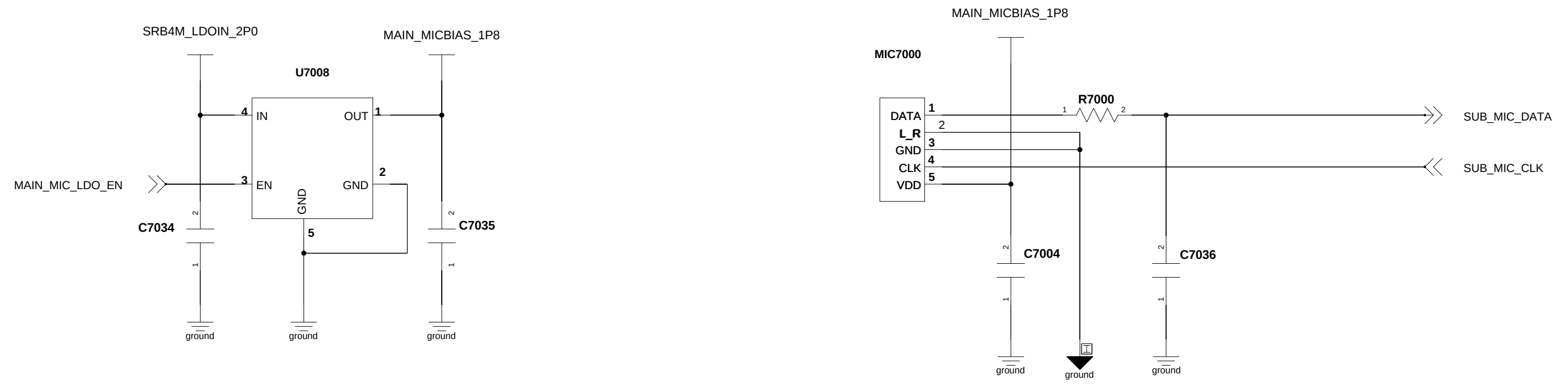
AUDIO & SENSOR PART


L7M_VDDQ_1P8
VBAT
@VDD_AMP_1P8
@VBAT_SENSING
C7000
VBAT
VBAT
@VPH_PWR
@VBAT_SENSING
C7006
C7015
C7007
C7012
C7001
U7003
VBAT VBST VBST VBST VDDD VDDP VDDP VDDP
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
TP7014
TP7015
SCL|CLKSW
SDA|DATASW
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
TP7000
TP7001
TP7002
TP7003
DATAI
FS
BCK
DATAO
RCV_AMP_EN
RCV_AMP_INT
RSTN
INT
VBAT
@VPH_PWR
L7000
INB
INB
INB
C7010
C7011
C7042
OUTP
OUTN
L7003
L7002
RCV_AMP_OUT_P
RCV_AMP_OUT_N
VSP
VSN
RCV_VSNS_P
RCV_VSNS_N
ADS1
ADS2
@VDD_AMP_1P8
L7M_VDDQ_1P8
TRSTN
FRCBST
ENSW
GNDB GNDB GNDB GNDD GNDD GNDD GNDP GNDP
ground

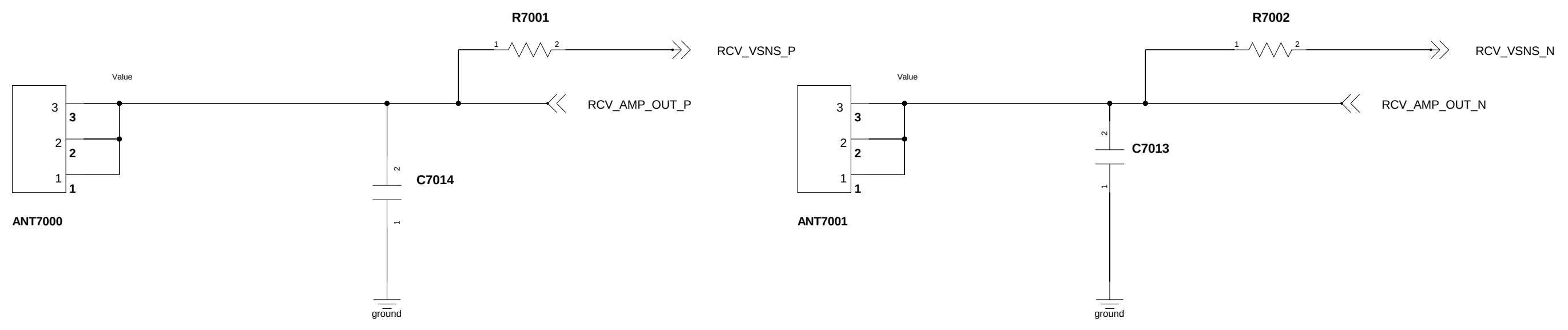
RCV AMP


L7M_VDDQ_1P8
VBAT
@VDD_AMP_1P8
@VBAT_SENSING
L7M_VDDQ_1P8
L7M_VDDQ_1P8
@VDD_AMP_1P8
C7016
C7018
C7019
C7021
C7002
C7017
U7000
SPK_AMP_SCL_1P8
SPK_AMP_SDA_1P8
SCL|CLKSW
SDA|DATASW
SPK_I2S3_DO
SPK_I2S3_WS
SPK_I2S3_BCLK
SPK_I2S3_DI
DATAI
FS
BCK
DATAO
@SPK_AMP_EN
RCV_AMP_EN
SPK_AMP_INT
RSTN
INT
VBAT
@VPH_PWR
L7001
INB
INB
INB
C7022
C7023
OUTP
OUTN
L7006
L7007
R7003
R7004
@SPK_VSNS_P
@SPK_VSNS_N
SPK_AMP_OUT_P
SPK_AMP_OUT_N
SPK_AMP_OUT_P
SPK_AMP_OUT_N
VSP
VSN
ADS1
ADS2
TRSTN
FRCBST
ENSW
C7008
C7009
ground

SPK AMP

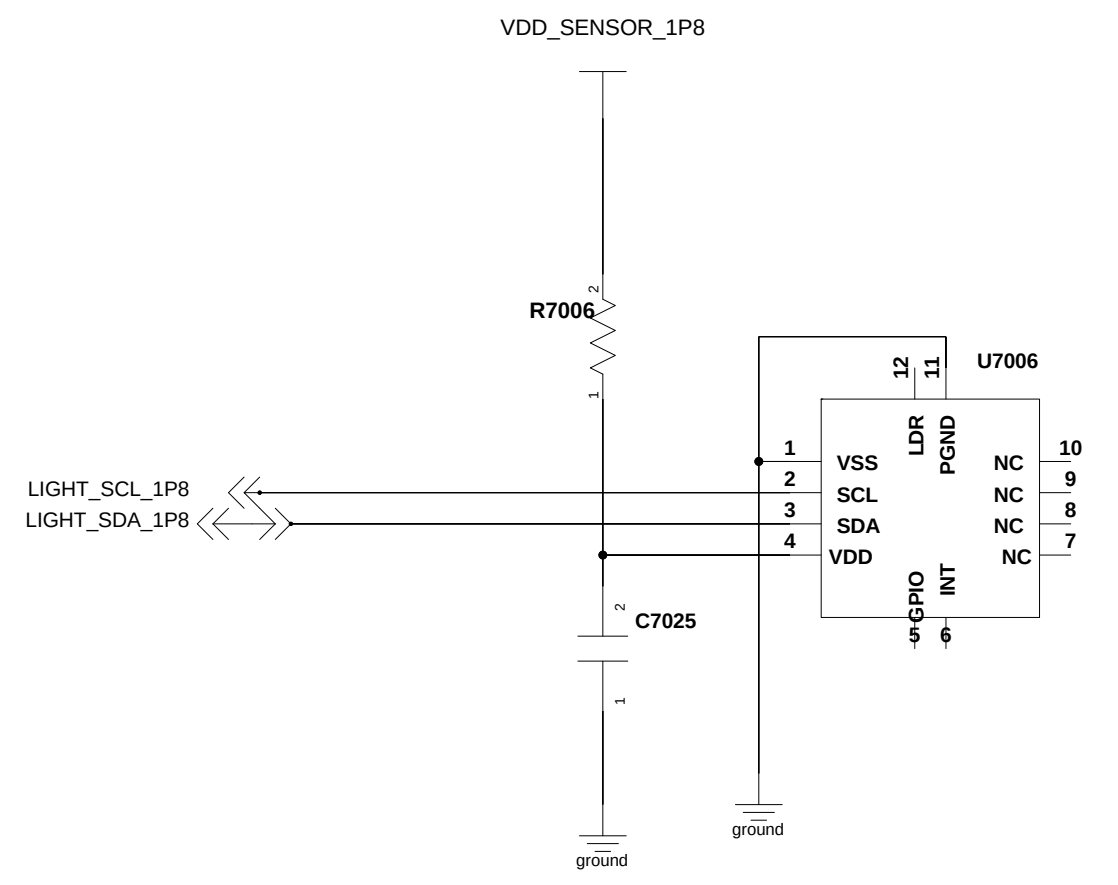

SRB4M_LDOIN_2P0
MAIN_MICBIAS_1P8
U7008
IN
OUT
MAIN_MIC_LDO_EN
EN
GND
GND
C7034
C7035
MAIN_MICBIAS_1P8
MIC7000
DATA
L_R
GND
CLK
VDD
R7000
SUB_MIC_DATA
SUB_MIC_CLK
C7004
C7036
ground

2nd MIC


R7001
RCV_VSNS_P
RCV_AMP_OUT_P
ANT7000
C7014
R7002
RCV_VSNS_N
RCV_AMP_OUT_N
ANT7001
C7013
Value
ground

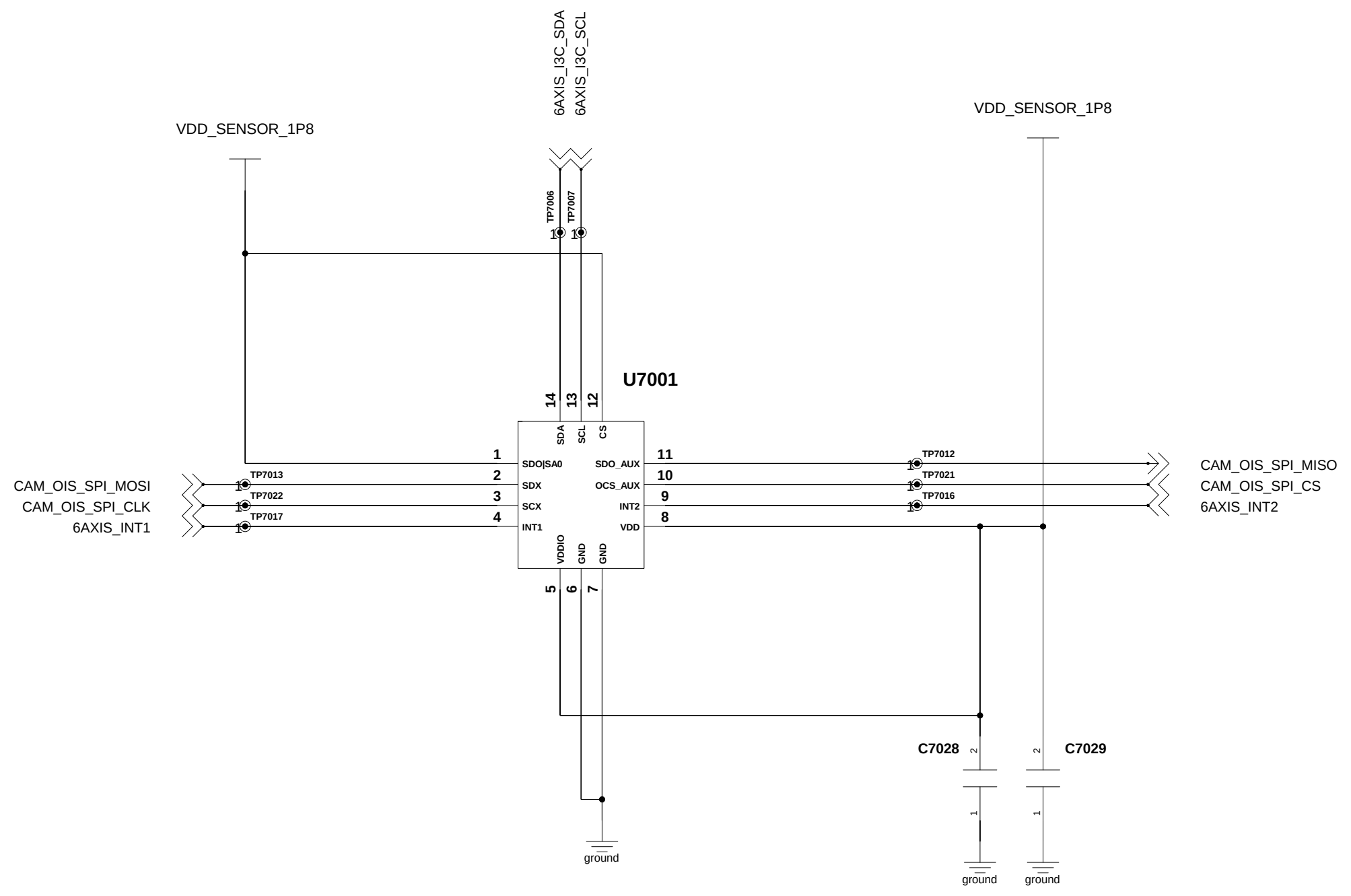
RCV CONTACT

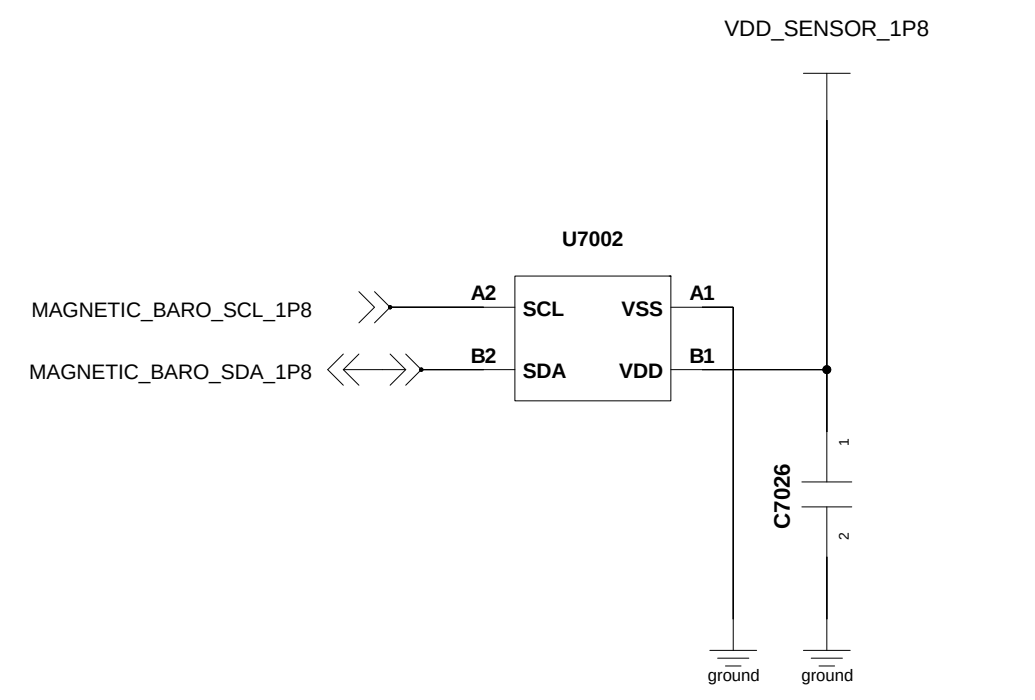
SENSOR


VDD_SENSOR_1P8
R7006
U7006
LIGHT_SCL_1P8
LIGHT_SDA_1P8
VSS
SCL
SDA
VDD
LDR
PGND
NC
NC
NC
NC
GPIO
INT
C7025
ground

AMBIENT SENSOR

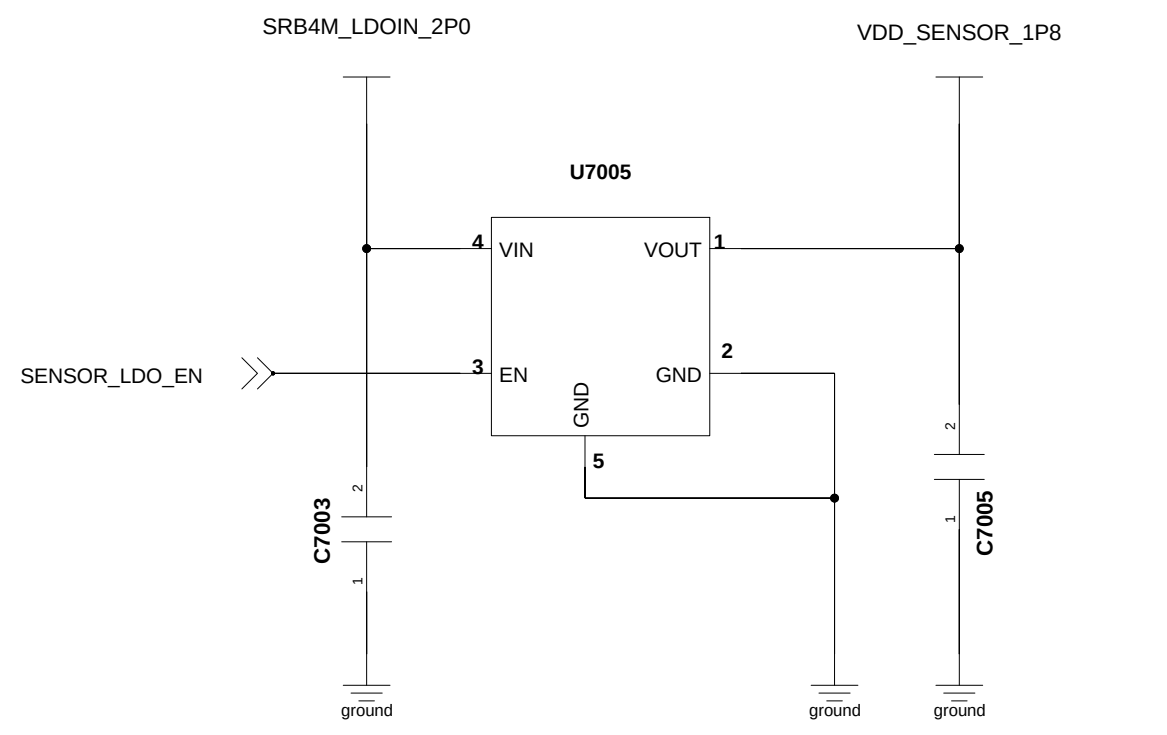
HALL IC


6AXIS_I3C_SDA
6AXIS_I3C_SCL
TP7006
TP7007
VDD_SENSOR_1P8
VDD_SENSOR_1P8
U7001
SDA SCL CS
SDO|SA0
SDX
SCX
INT1
SDO_AUX
OCS_AUX
INT2
VDD
VDDIO GND GND
CAM_OIS_SPI_MOSI
CAM_OIS_SPI_CLK
6AXIS_INT1
TP7013
TP7022
TP7017
TP7012
TP7021
TP7016
CAM_OIS_SPI_MISO
CAM_OIS_SPI_CS
6AXIS_INT2
C7028
C7029
ground


VDD_SENSOR_1P8
U7002
MAGNETIC_BARO_SCL_1P8
MAGNETIC_BARO_SDA_1P8
SCL
SDA
VSS
VDD
C7026
ground

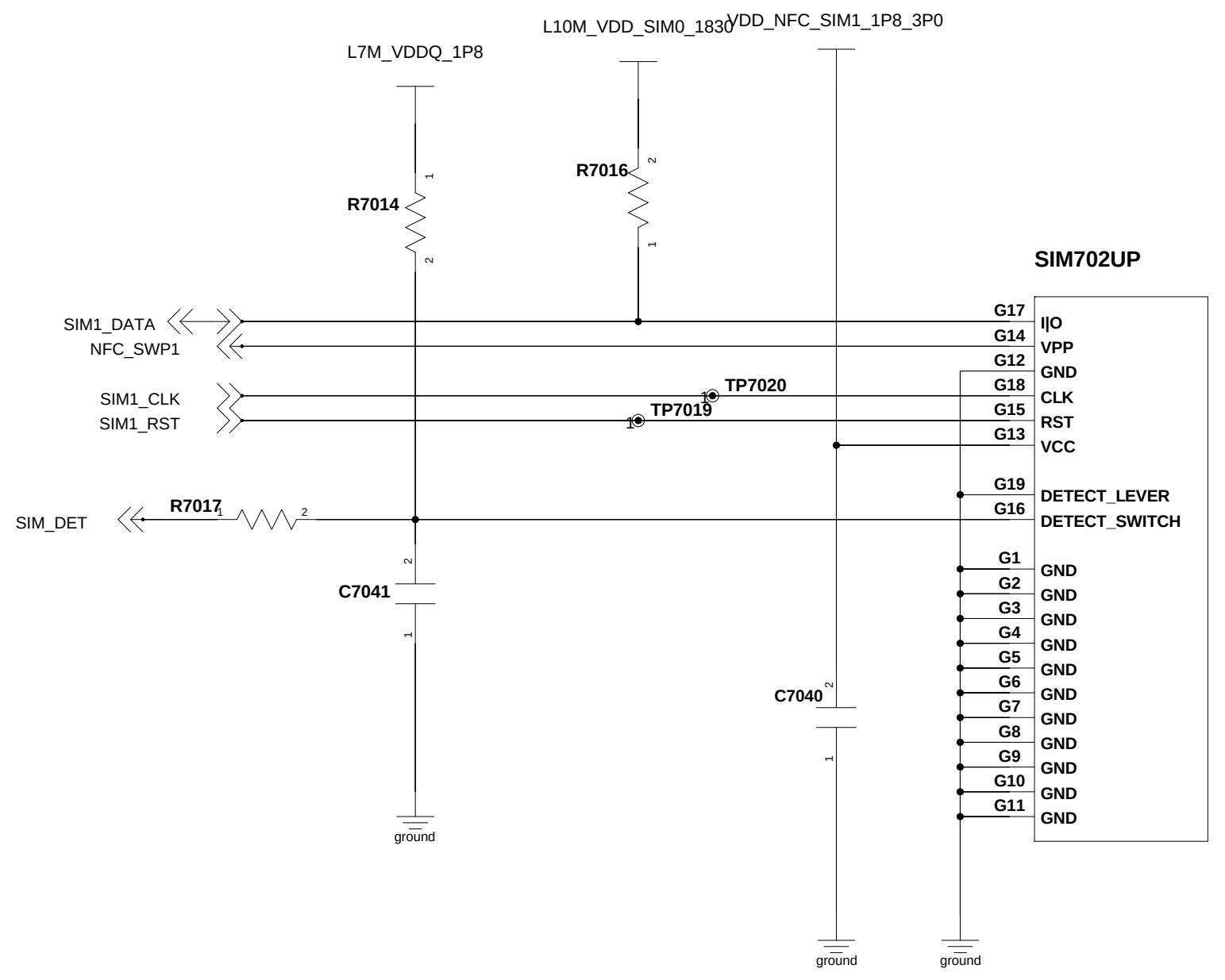
MAGNETIC SENSOR

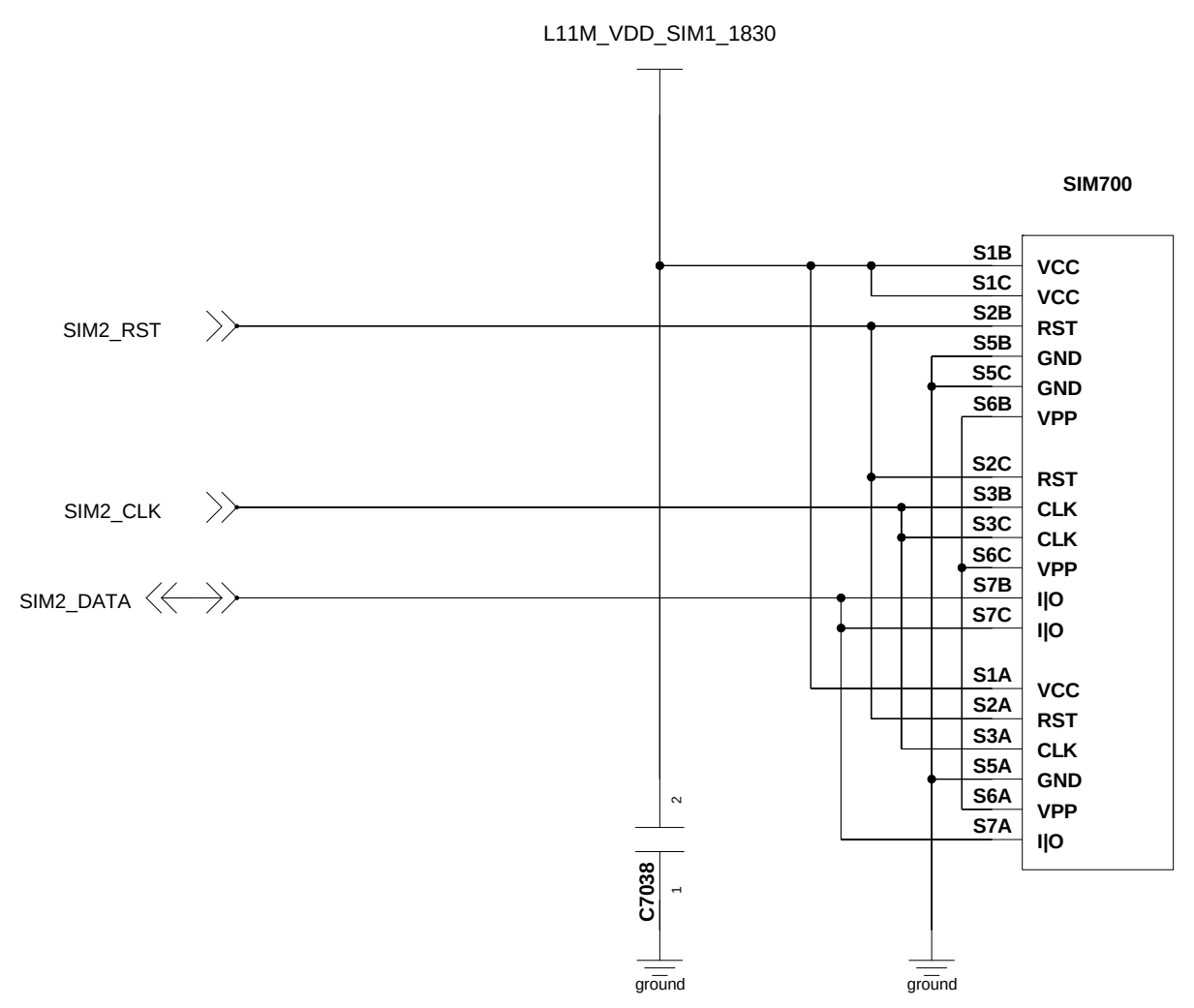
ACC&GYRO SENSOR

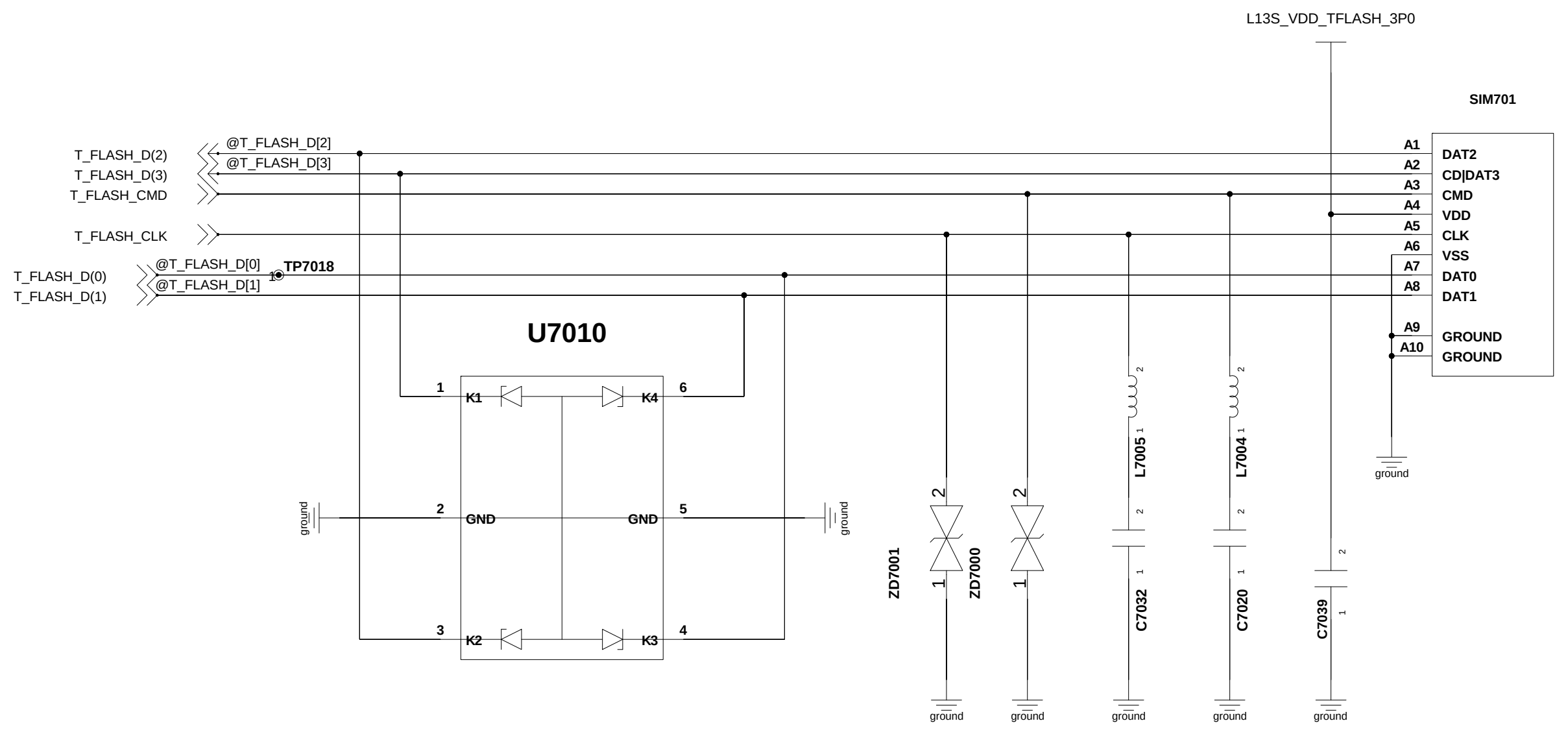

SRB4M_LDOIN_2P0
VDD_SENSOR_1P8
U7005
VIN
VOUT
SENSOR_LDO_EN
EN
GND
GND
C7003
C7005
ground

NA ONLY

DUAL SIM / T-FLASH (3in2)


L7M_VDDQ_1P8
L10M_VDD_SIM0_1830
VDD_NFC_SIM1_1P8_3P0
R7014
R7016
SIM702UP
SIM1_DATA
NFC_SWP1
SIM1_CLK
SIM1_RST
TP7020
TP7019
I|O
VPP
GND
CLK
RST
VCC
DETECT_LEVER
DETECT_SWITCH
SIM_DET
R7017
C7041
C7040
GND
ground


L11M_VDD_SIM1_1830
SIM700
SIM2_RST
SIM2_CLK
SIM2_DATA
VCC
VCC
RST
GND
GND
VPP
RST
CLK
CLK
VPP
I|O
I|O
VCC
RST
CLK
GND
VPP
I|O
C7038
ground


L13S_VDD_TFLASH_3P0
SIM701
T_FLASH_D(2)
T_FLASH_D(3)
T_FLASH_CMD
T_FLASH_CLK
T_FLASH_D(0)
T_FLASH_D(1)
@T_FLASH_D[2]
@T_FLASH_D[3]
@T_FLASH_D[0]
@T_FLASH_D[1]
TP7018
DAT2
CD|DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
GROUND
GROUND
U7010
K1
K4
GND
GND
K2
K3
ZD7001
ZD7000
L7005
L7004
C7032
C7020
C7039
ground

Stacked Dual SIM SOCKET (Reverse)

Engineer:
Drawn by:
R&D CHK:
DOC CTRL CHK:
MFG ENGR CHK:
A54
TITLE:
Size:
Changed by: | Date Changed: | Time Changed: | QA CHK: | REV: | Drawing Number: | Page:

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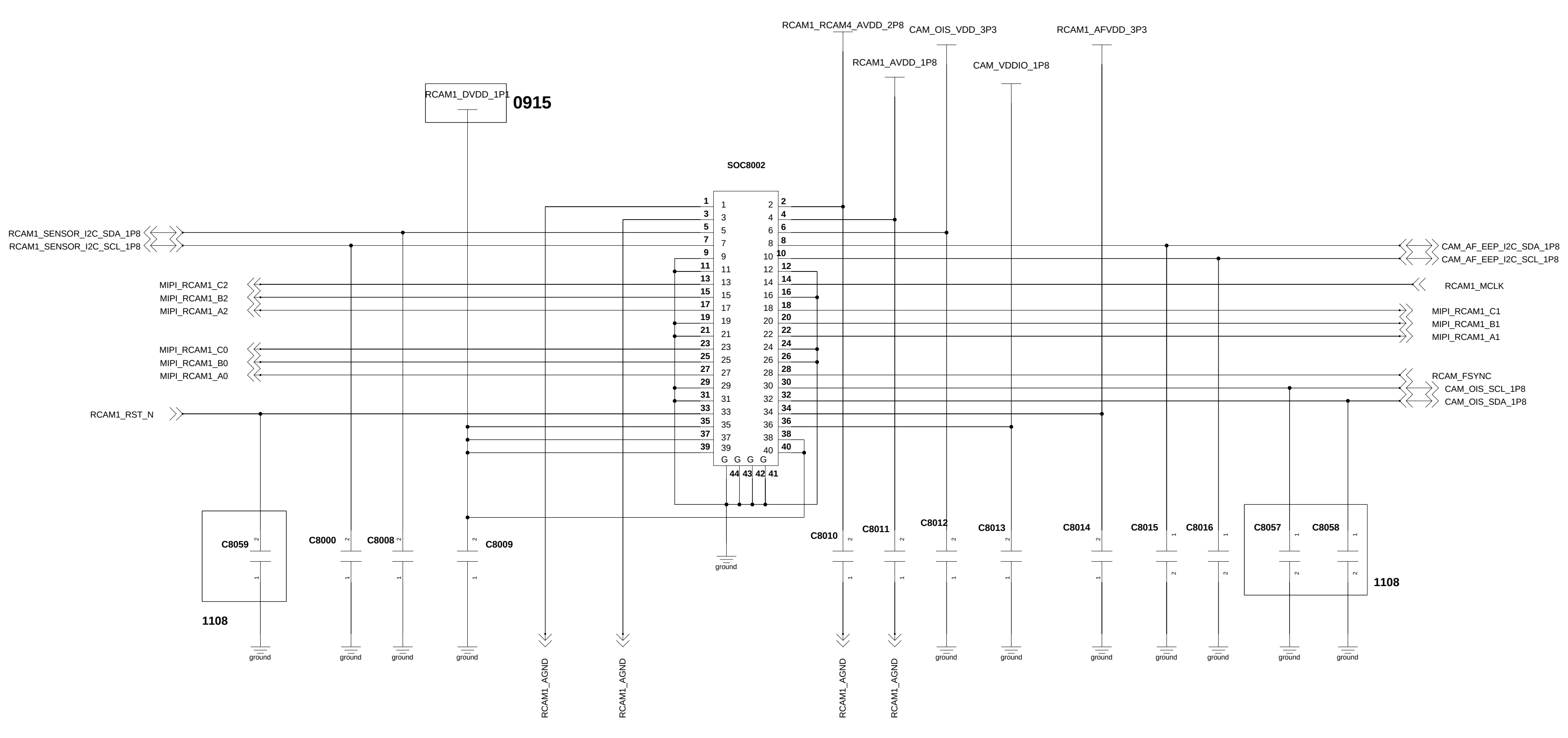
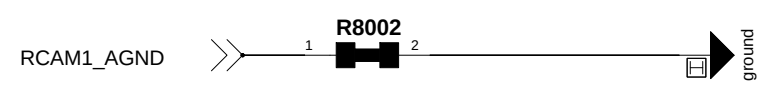
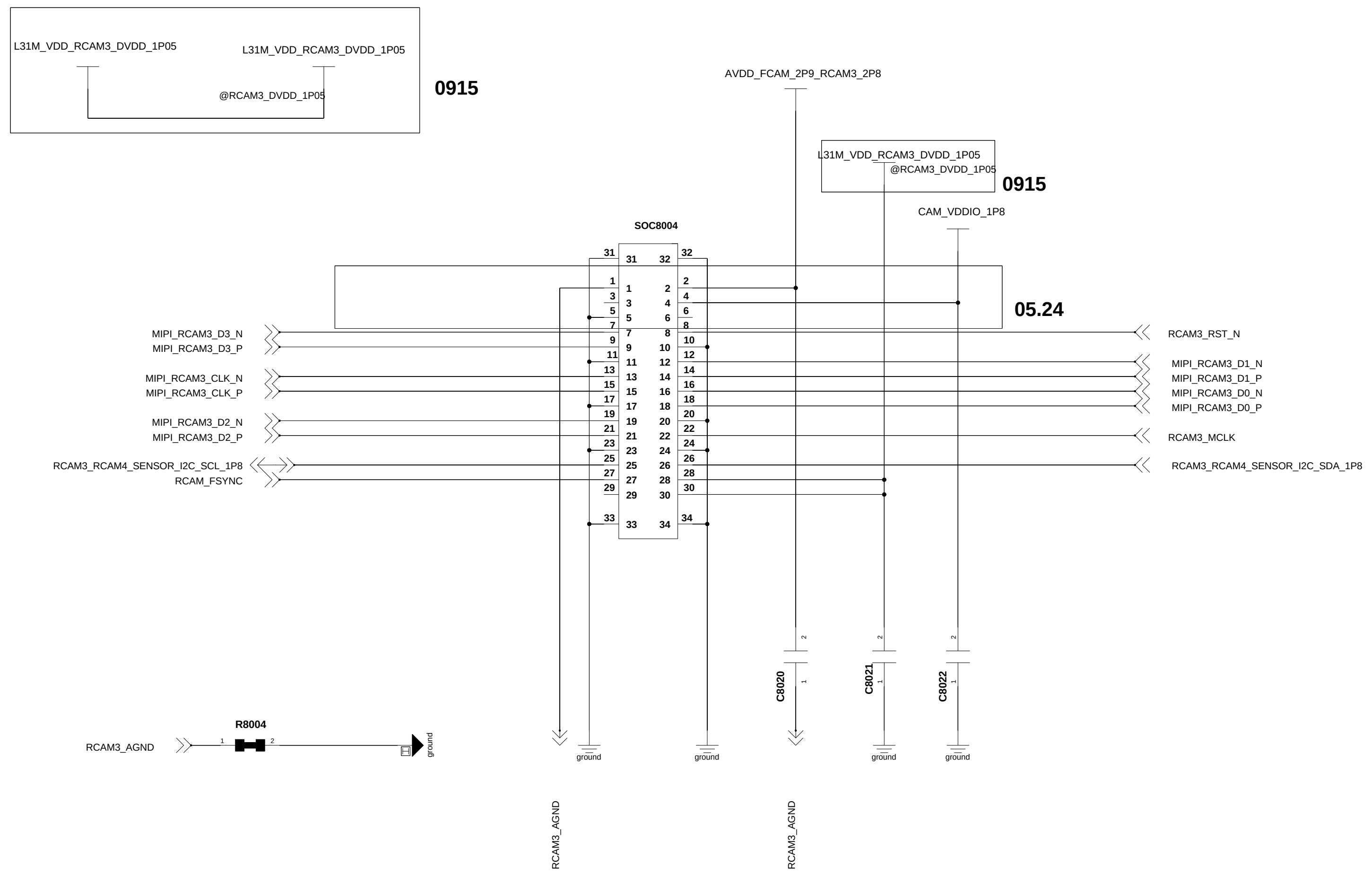
CAMERA

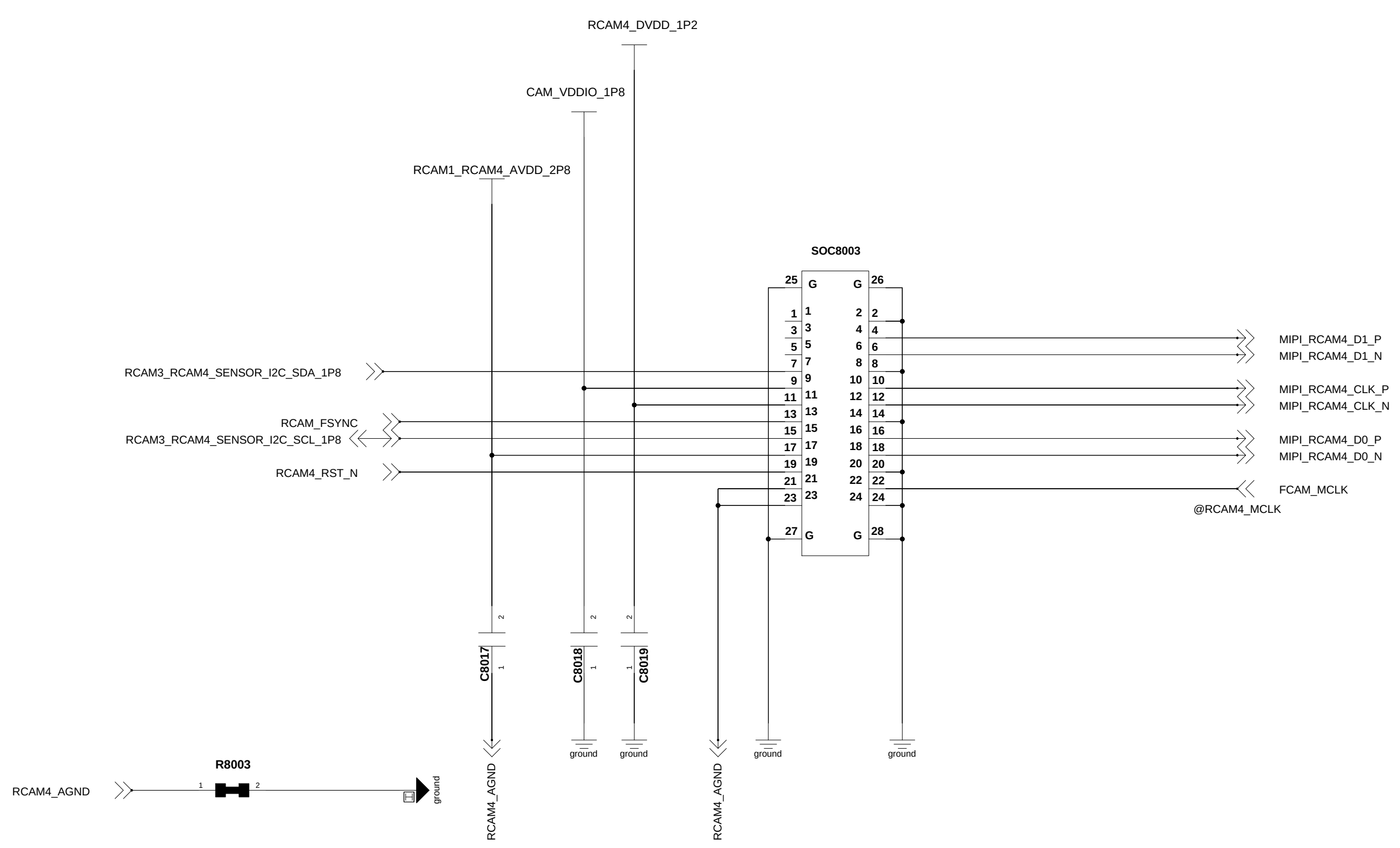

RCAM1_DVDD_1P1
0915
RCAM1_RCAM4_AVDD_2P8
CAM_OIS_VDD_3P3
RCAM1_AVDD_1P8
CAM_VDDIO_1P8
RCAM1_AFVDD_3P3
SOC8002
RCAM1_SENSOR_I2C_SDA_1P8
RCAM1_SENSOR_I2C_SCL_1P8
MIPI_RCAM1_C2
MIPI_RCAM1_B2
MIPI_RCAM1_A2
MIPI_RCAM1_C0
MIPI_RCAM1_B0
MIPI_RCAM1_A0
RCAM1_RST_N
CAM_AF_EEP_I2C_SDA_1P8
CAM_AF_EEP_I2C_SCL_1P8
RCAM1_MCLK
MIPI_RCAM1_C1
MIPI_RCAM1_B1
MIPI_RCAM1_A1
RCAM_FSYNC
CAM_OIS_SCL_1P8
CAM_OIS_SDA_1P8
C8059
1108
C8000
C8008
C8009
C8010
C8011
C8012
C8013
C8014
C8015
C8016
C8057
C8058
1108
ground
RCAM1_AGND


RCAM1_AGND
R8002
ground

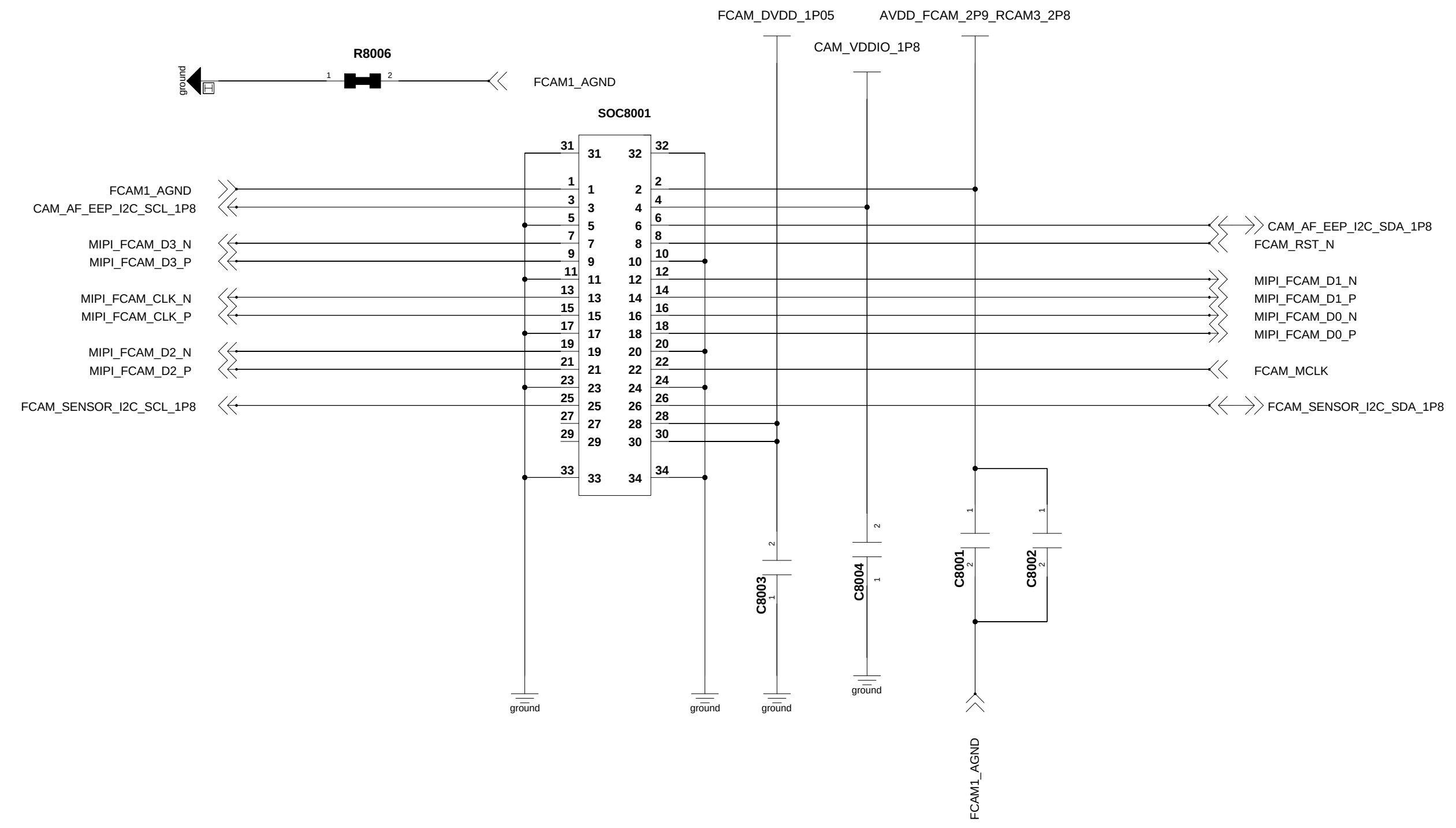
RCAM1_50M OIS(40PIN)


L31M_VDD_RCAM3_DVDD_1P05
L31M_VDD_RCAM3_DVDD_1P05
@RCAM3_DVDD_1P05
0915
AVDD_FCAM_2P9_RCAM3_2P8
L31M_VDD_RCAM3_DVDD_1P05
@RCAM3_DVDD_1P05
0915
CAM_VDDIO_1P8
SOC8004
05.24
MIPI_RCAM3_D3_N
MIPI_RCAM3_D3_P
MIPI_RCAM3_CLK_N
MIPI_RCAM3_CLK_P
MIPI_RCAM3_D2_N
MIPI_RCAM3_D2_P
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM_FSYNC
RCAM3_RST_N
MIPI_RCAM3_D1_N
MIPI_RCAM3_D1_P
MIPI_RCAM3_D0_N
MIPI_RCAM3_D0_P
RCAM3_MCLK
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
C8020
C8021
C8022
RCAM3_AGND
R8004
ground
RCAM3_AGND
RCAM3_AGND

RCAM3_SUPER WIDE_12M CAMERA(30pin)

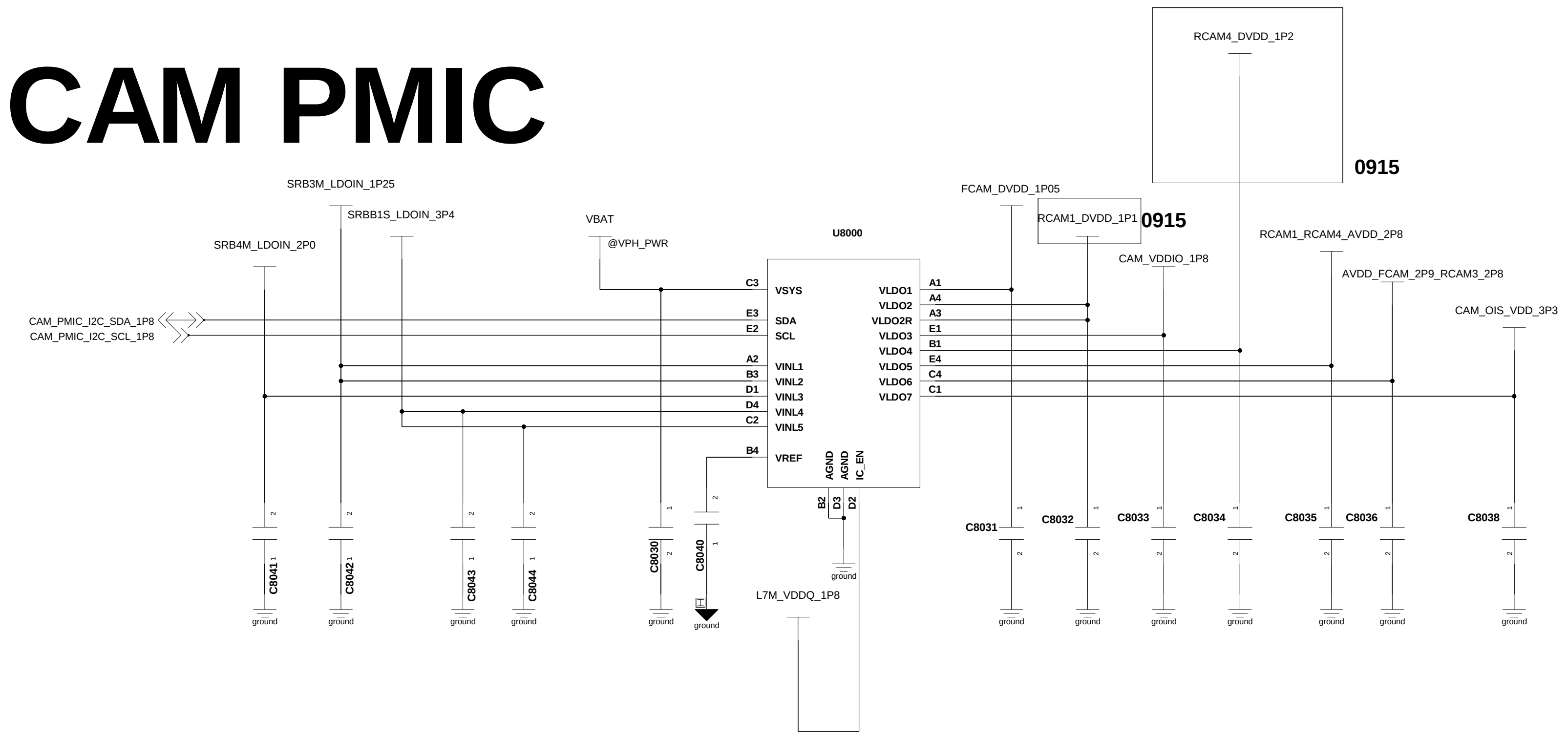

RCAM4_DVDD_1P2
CAM_VDDIO_1P8
RCAM1_RCAM4_AVDD_2P8
SOC8003
RCAM3_RCAM4_SENSOR_I2C_SDA_1P8
RCAM_FSYNC
RCAM3_RCAM4_SENSOR_I2C_SCL_1P8
RCAM4_RST_N
MIPI_RCAM4_D1_P
MIPI_RCAM4_D1_N
MIPI_RCAM4_CLK_P
MIPI_RCAM4_CLK_N
MIPI_RCAM4_D0_P
MIPI_RCAM4_D0_N
FCAM_MCLK
@RCAM4_MCLK
C8017
C8018
C8019
RCAM4_AGND
R8003
ground
RCAM4_AGND
RCAM4_AGND

RCAM4_5M_MACRO_CAMERA(24PIN)

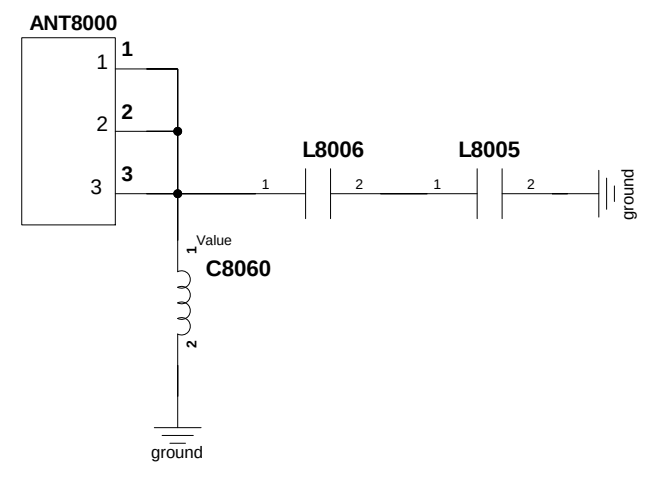

R8006
ground
FCAM1_AGND
FCAM_DVDD_1P05
AVDD_FCAM_2P9_RCAM3_2P8
CAM_VDDIO_1P8
SOC8001
FCAM1_AGND
CAM_AF_EEP_I2C_SCL_1P8
MIPI_FCAM_D3_N
MIPI_FCAM_D3_P
MIPI_FCAM_CLK_N
MIPI_FCAM_CLK_P
MIPI_FCAM_D2_N
MIPI_FCAM_D2_P
FCAM_SENSOR_I2C_SCL_1P8
CAM_AF_EEP_I2C_SDA_1P8
FCAM_RST_N
MIPI_FCAM_D1_N
MIPI_FCAM_D1_P
MIPI_FCAM_D0_N
MIPI_FCAM_D0_P
FCAM_MCLK
FCAM_SENSOR_I2C_SDA_1P8
C8003
C8004
C8001
C8002
ground
FCAM1_AGND

FRONT_32M CAMERA (30PIN)

CAM PMIC


RCAM4_DVDD_1P2
0915
SRB3M_LDOIN_1P25
SRBB1S_LDOIN_3P4
SRB4M_LDOIN_2P0
VBAT
@VPH_PWR
U8000
CAM_PMIC_I2C_SDA_1P8
CAM_PMIC_I2C_SCL_1P8
VSYS
SDA
SCL
VINL1
VINL2
VINL3
VINL4
VINL5
VREF
AGND
AGND
IC_EN
VLDO1
VLDO2
VLDO2R
VLDO3
VLDO4
VLDO5
VLDO6
VLDO7
FCAM_DVDD_1P05
RCAM1_DVDD_1P1
0915
CAM_VDDIO_1P8
RCAM1_RCAM4_AVDD_2P8
AVDD_FCAM_2P9_RCAM3_2P8
CAM_OIS_VDD_3P3
C8041
C8042
C8043
C8044
C8030
C8040
L7M_VDDQ_1P8
C8031
C8032
C8033
C8034
C8035
C8036
C8038
ground

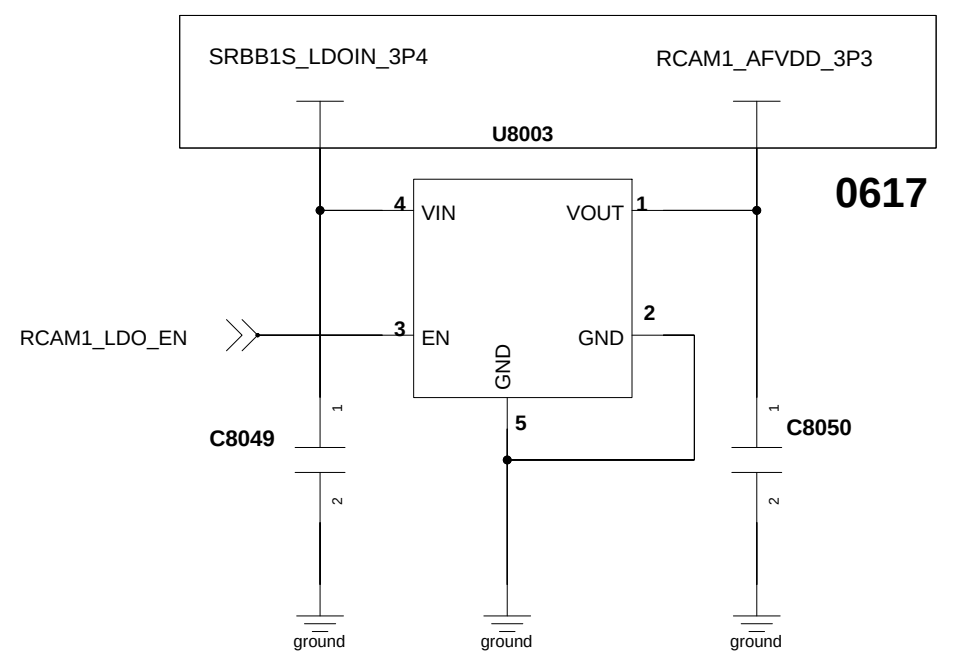
DECO CONTACT


ANT8000
L8006
L8005
Value
C8060
ground

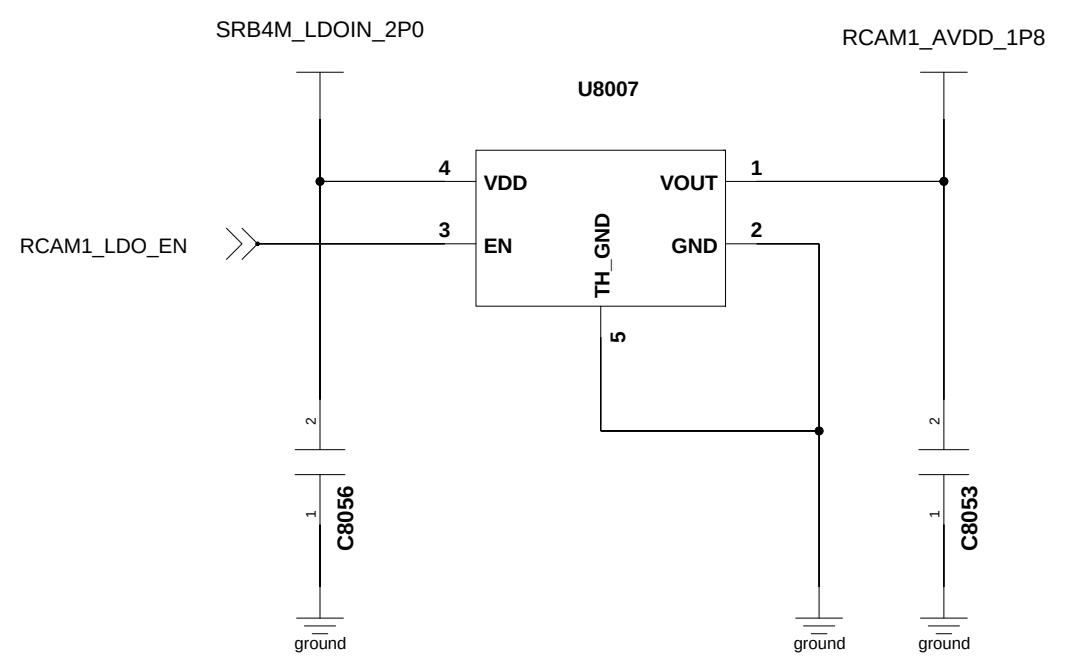
220915

RCAM4_AVDD

RCAM3_AVDD


SRBB1S_LDOIN_3P4
RCAM1_AFVDD_3P3
U8003
0617
VIN
VOUT
RCAM1_LDO_EN
EN
GND
GND
C8049
C8050
ground

RCAM1_AF


SRB4M_LDOIN_2P0
U8007
RCAM1_AVDD_1P8
VDD
VOUT
RCAM1_LDO_EN
EN
TH_GND
GND
C8056
C8053
ground

RCAM1_AVDD

CAM MCU_DVDD

0617

CAM MCU

RCAM4_DVDD

220915

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: CAMERA
DOC CTRL CHK:	Size:
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

Changed by: Date Changed: Time Changed:

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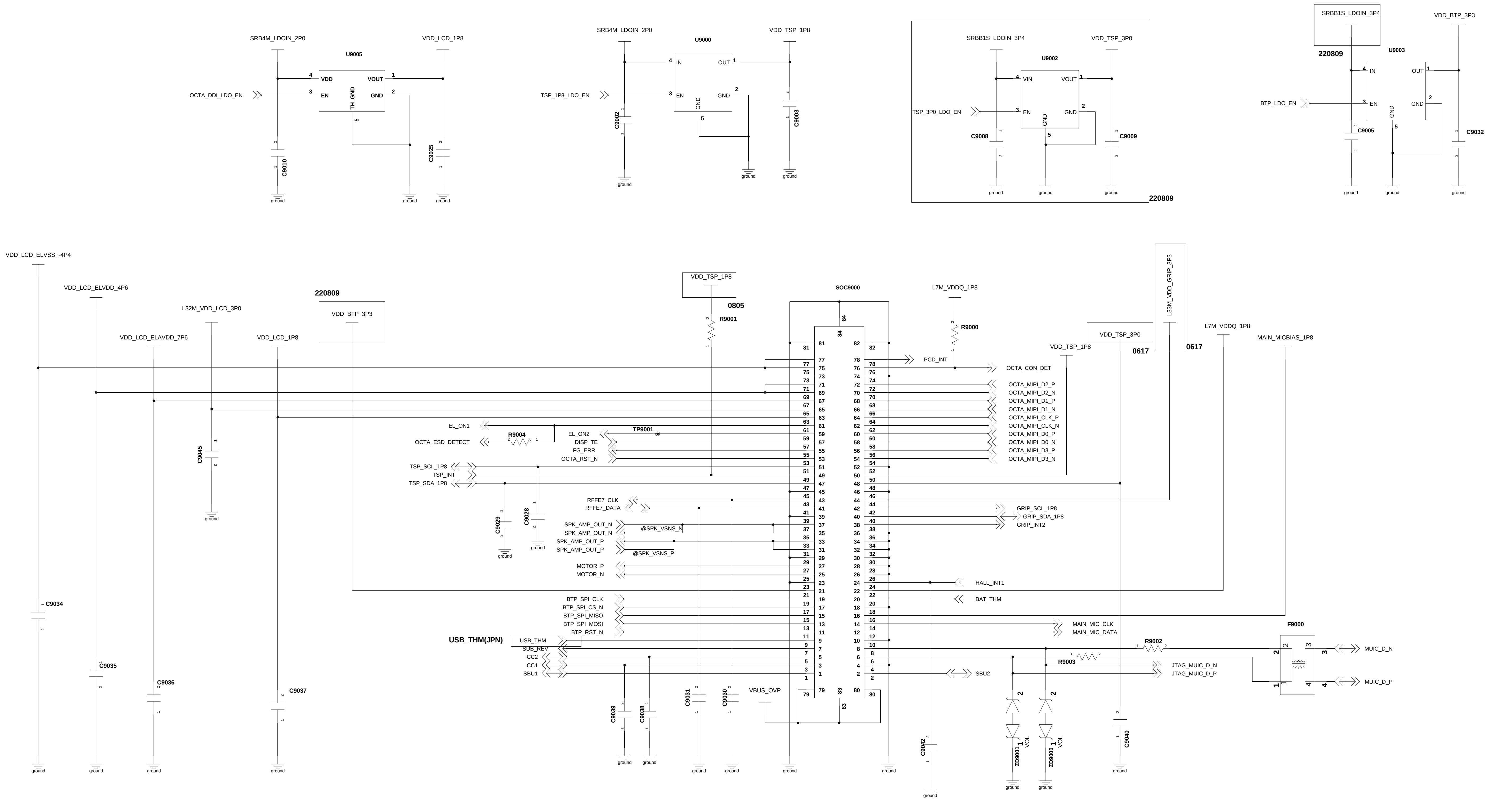
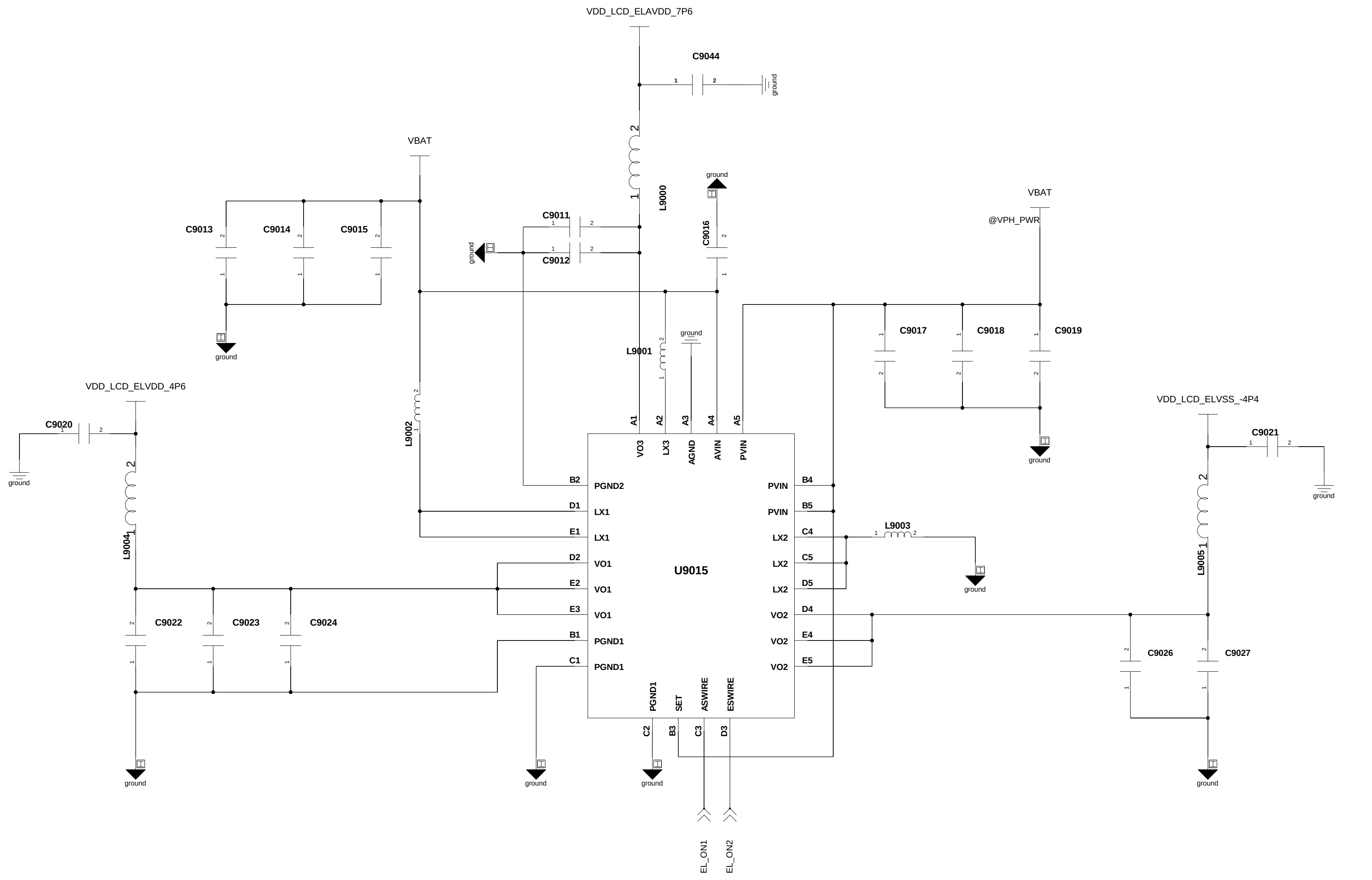
C2C (Conn to Conn FPCB)



SRB4M_LDOIN_2P0
VDD_LCD_1P8
U9005
VDD
VOUT
EN
GND
TH_GND
OCTA_DDI_LDO_EN
C9010
C9025
ground
SRB4M_LDOIN_2P0
U9000
IN
OUT
EN
GND
VDD_TSP_1P8
TSP_1P8_LDO_EN
C9002
C9003
SRBB1S_LDOIN_3P4
VDD_TSP_3P0
U9002
VIN
VOUT
TSP_3P0_LDO_EN
C9008
C9009
220809
SRBB1S_LDOIN_3P4
220809
U9003
VDD_BTP_3P3
BTP_LDO_EN
C9005
C9032
VDD_LCD_ELVSS_-4P4
VDD_LCD_ELVDD_4P6
L32M_VDD_LCD_3P0
VDD_LCD_ELAVDD_7P6
VDD_LCD_1P8
220809
VDD_BTP_3P3
VDD_TSP_1P8
0805
R9001
SOC9000
L7M_VDDQ_1P8
R9000
L33M_VDD_GRIP_3P3
VDD_TSP_3P0
0617
0617
VDD_TSP_1P8
L7M_VDDQ_1P8
MAIN_MICBIAS_1P8
PCD_INT
OCTA_CON_DET
OCTA_MIPI_D2_P
OCTA_MIPI_D2_N
OCTA_MIPI_D1_P
OCTA_MIPI_D1_N
OCTA_MIPI_CLK_P
OCTA_MIPI_CLK_N
OCTA_MIPI_D0_P
OCTA_MIPI_D0_N
OCTA_MIPI_D3_P
OCTA_MIPI_D3_N
EL_ON1
R9004
OCTA_ESD_DETECT
EL_ON2
TP9001
DISP_TE
FG_ERR
OCTA_RST_N
TSP_SCL_1P8
TSP_INT
TSP_SDA_1P8
C9045
C9029
C9028
RFFE7_CLK
RFFE7_DATA
SPK_AMP_OUT_N
SPK_AMP_OUT_N
SPK_AMP_OUT_P
SPK_AMP_OUT_P
@SPK_VSNS_N
@SPK_VSNS_P
MOTOR_P
MOTOR_N
GRIP_SCL_1P8
GRIP_SDA_1P8
GRIP_INT2
HALL_INT1
BAT_THM
BTP_SPI_CLK
BTP_SPI_CS_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_RST_N
MAIN_MIC_CLK
MAIN_MIC_DATA
USB_THM(JPN)
USB_THM
SUB_REV
CC2
CC1
SBU1
SBU2
R9002
R9003
JTAG_MUIC_D_N
JTAG_MUIC_D_P
F9000
MUIC_D_N
MUIC_D_P
VBUS_OVP
C9034
C9035
C9036
C9037
C9039
C9038
C9031
C9030
C9042
ZD9001
ZD9000
VOL
VOL
C9040

DISPLAY



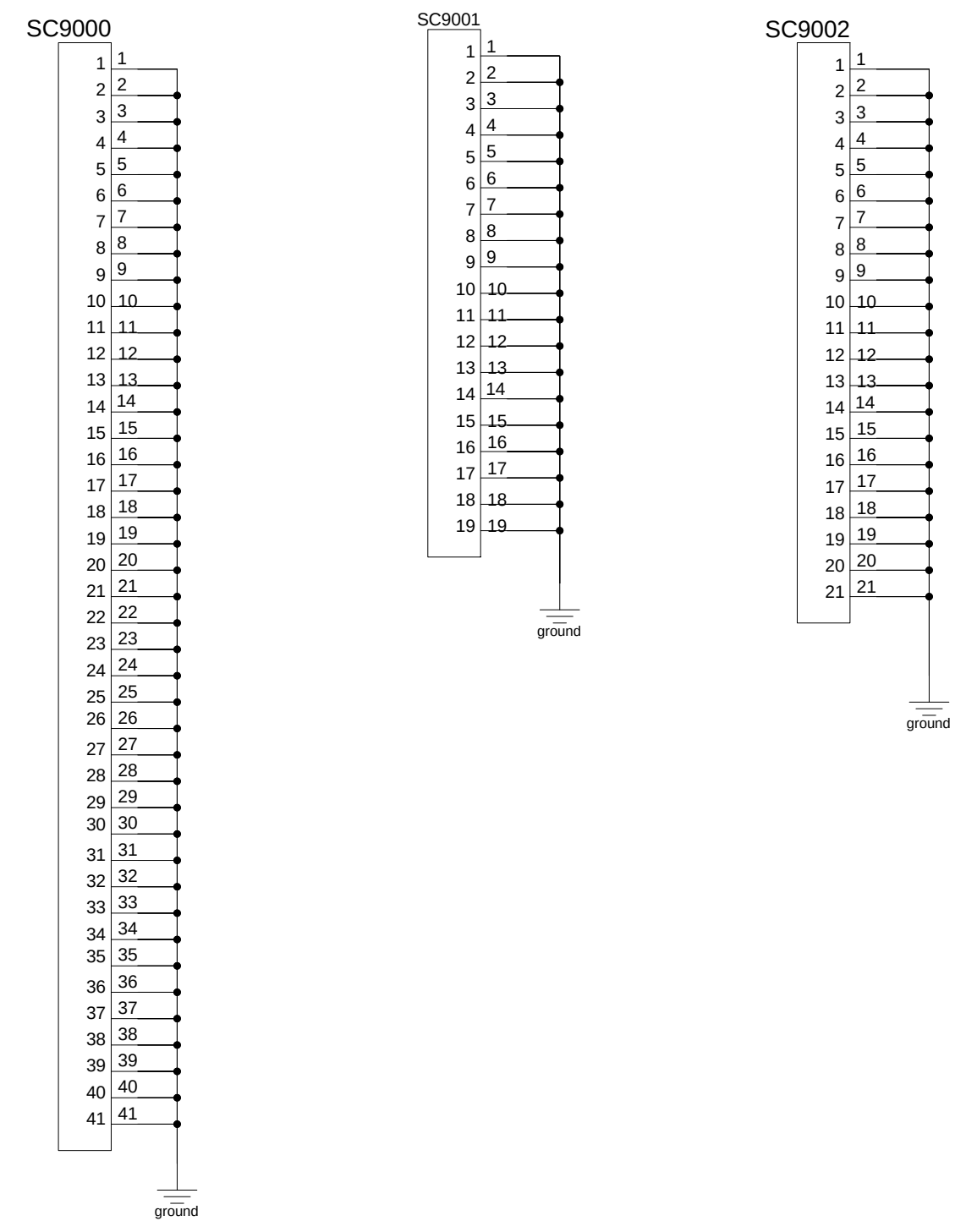
VDD_LCD_ELAVDD_7P6
C9044
VBAT
L9000
C9011
C9012
C9016
C9013
C9014
C9015
VBAT
@VPH_PWR
C9017
C9018
C9019
L9001
L9002
VDD_LCD_ELVDD_4P6
C9020
L9004
VDD_LCD_ELVSS_-4P4
C9021
L9005
U9015
VO3
LX3
AGND
AVIN
PVIN
PGND2
LX1
LX1
VO1
VO1
VO1
PGND1
PGND1
PVIN
PVIN
LX2
LX2
LX2
VO2
VO2
VO2
PGND1
SET
ASWIRE
ESWIRE
L9003
C9022
C9023
C9024
C9026
C9027
EL_ON1
EL_ON2
ground

DISPLAY PMIC

2D Barcode

CN9000
CN9001
221101
SUS9000
1
2
3
4
5
BOTH
SUS9002
1
2
3
4
5
BOTH
ground

SHIELD CAN



SC9000
SC9001
SC9002
ground

REGION		MEMORY		
R9013	B : EUR	R9019	6/128	
R9014	0 : HK CHN	R9015	8/128	
R9018	S : KOR SKT	R9017	8/256	
R9021	E : SEA, MEA, SWA E : AFR, LA, CIS			

mmW_ANT_CONN (Right)

NA ONLY

ES2.0 : OPEN
ES3.0 : 0 ohm

5G IFIC

NA ONLY

Engineer:	A54
Drawn by:	
R&D CHK:	TITLE: Size:
DOC CTRL CHK:	
MFG ENGR CHK:	
QA CHK:	REV: Drawing Number: Page:

Changed by: Date Changed: Time Changed: